FPT UNIVERSITY

# Capstone Project Report

## Report 3 – Software Requirement Specification

– Can Tho, October 2026 –

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# I. Record of Changes

| Date | A*<br>M, D | In charge | Change Description |
|---|---|---|---|
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**A - Added M - Modified D - Deleted*

# II. Software Requirement Specification

## 1. Product Overview

The Frevia project is a comprehensive freelancing platform that enables clients, freelancers, and administrators to build trusted professional relationships and efficiently manage the entire project lifecycle. Designed to connect businesses with skilled talent worldwide, Frevia provides intelligent job matching, structured contract workflows, and secure payment solutions to facilitate successful collaborations.

Through AI-powered recommendations, real-time communication, and advanced moderation mechanisms, the platform helps users discover suitable opportunities, deliver high-quality work, and maintain transparent interactions.

Frevia serves three primary stakeholder groups: clients seeking specialized expertise, freelancers showcasing their skills and services, and administrators responsible for maintaining platform integrity and operational efficiency. The platform offers an intuitive user experience, dynamic communication channels, and comprehensive management capabilities to support collaboration, professional growth, and secure transactions.

Key features of Frevia include:

- **Smart Job Matching and AI Recommendations:** The platform analyzes freelancer skills, experience, and preferences to recommend relevant job opportunities. AI-powered profile scoring provides quality assessments, while job alerts and saved searches help users stay informed about new opportunities.
- **End-to-End Contract and Milestone Management:** Clients and freelancers can negotiate project terms, define milestones and tasks, monitor progress, and securely manage payments. Integrated contract signing, activity logs, and file-sharing capabilities ensure transparency throughout the project lifecycle.
- **Community and Forum Engagement:** Frevia includes a community forum where users can discuss industry topics, exchange knowledge, and seek professional advice. Features such as categories, posts, comments, reactions, and reporting mechanisms encourage active participation and self-moderation.
- **Identity Verification and Content Moderation:** Freelancers can verify their identities through document submission, while administrators utilize moderation tools, banned keyword management, and AI-assisted violation detection to maintain a secure and trustworthy environment.

- **Wallet, Payments, and Gas Fee Subsidies:** Users can manage account balances, withdraw earnings, and review transaction histories. The platform also supports gas fee subsidy mechanisms for blockchain-based transactions, improving payment accessibility and reducing operational costs.
- **Administrative Management Tools:** Administrators have access to comprehensive management functionalities, including user administration, role and permission control, system configuration, advertisement management, activity monitoring, and dispute resolution, ensuring fair and reliable platform operations.

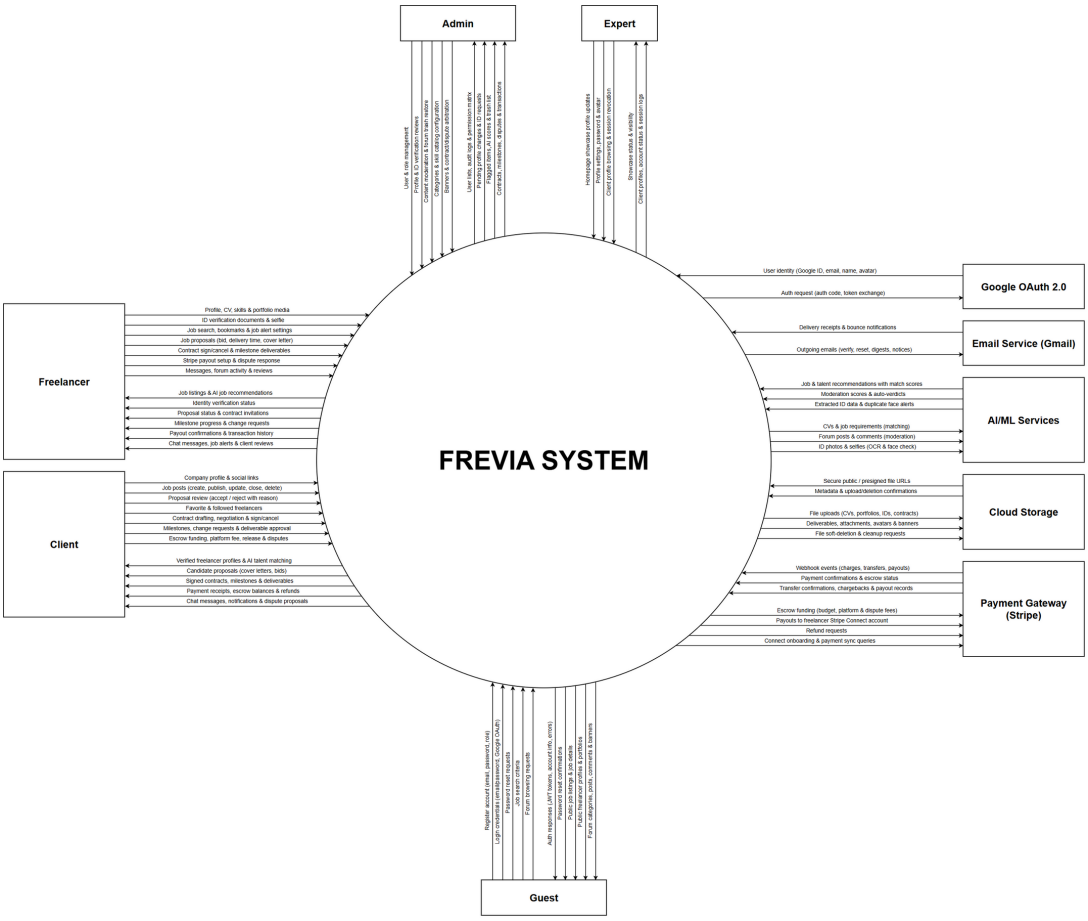


**Figure 1:** Context diagram

## 2. User Requirements

### 2.1 Actors

**Table 1:** Actors of the Frevia Platform

| No | Actor | Description |
|---|---|---|
| 1 | Guest | Guests are unregistered visitors who have not yet created an account on the platform. They can browse publicly available information such as job listings, freelancer profiles, forum content, and platform features. Guests may also register a new account (via email/password or Google OAuth) or authenticate using supported login methods to access platform services. |
| 2 | Client | Clients are individuals or organizations seeking professional services from freelancers. They can create and manage job postings, review proposals, hire freelancers, establish contracts, manage escrow payments, monitor project progress, initiate disputes, and provide reviews upon project completion. |
| 3 | Freelancer | Freelancers are service providers who offer professional skills and expertise through the platform. They can create and manage profiles, portfolios, and skills; search and apply for jobs; submit proposals; collaborate with clients; complete project tasks and milestones; and receive payments after successful project delivery. |
| 4 | Admin | Administrators are responsible for managing and maintaining the platform's operations, security, and compliance. Their responsibilities include user management, role and permission management, identity verification, content moderation, dispute resolution, system configuration, and monitoring platform activities to ensure a secure and reliable environment. |
| 5 | Expert | Experts are verified, high-ranking domain specialists featured in the platform's public expert showcase. They possess verified credentials, manage specialized public showcase profiles, consult on high-value client projects, and manage their security sessions and credentials. |

| No | Actor | Description |
|---|---|---|
| 6 | System | The internal automated execution engine and background scheduler of the platform. The System autonomously executes scheduled background jobs, manages deadline expirations, triggers automatic state transitions for contracts and proposals, processes automated dispute resolution payouts upon administrative decisions, and dispatches automated notifications. |
| 7 | AI Service | The artificial intelligence and machine learning microservice providing automated cognitive capabilities. It analyzes resumes to compute CV-to-job matching scores, evaluates forum posts and comments for toxicity and spam to assist content moderation, and performs optical character recognition (OCR) and face-matching validation on identity verification documents. |
| 8 | Payment Gateway (Stripe) | The third-party financial processor (Stripe Checkout and Stripe Connect) powering platform monetary operations. It securely processes client escrow funding, holds funds in escrow during project execution, handles automated payouts to freelancer bank accounts, processes refunds and dispute arbitration fees, and transmits asynchronous webhook events to synchronize payment statuses. |
| 9 | Email Service (Gmail / SMTP) | The external transactional email delivery provider (SMTP / Nodemailer). It delivers transactional and security communications to users, including email verification links, multi-factor authentication one-time passwords (OTP), password reset instructions, contract status updates, and dispute notifications. |
| 10 | Google OAuth 2.0 | Third-party federated identity provider. Enables seamless and secure single sign-on (SSO) authentication, allowing guests to register and log in to the Frevia platform using their Google accounts without requiring separate password management. |

| No | Actor | Description |
|---|---|---|
| 11 | Cloud Storage (Cloudinary) | Third-party media storage and content delivery network (CDN) service. Securely stores, optimizes, and delivers user-uploaded digital assets, including user avatars, portfolio media, project deliverable files, and identity verification documents. |

## 2.2 Use Cases

### 2.2.1 Diagram(s)

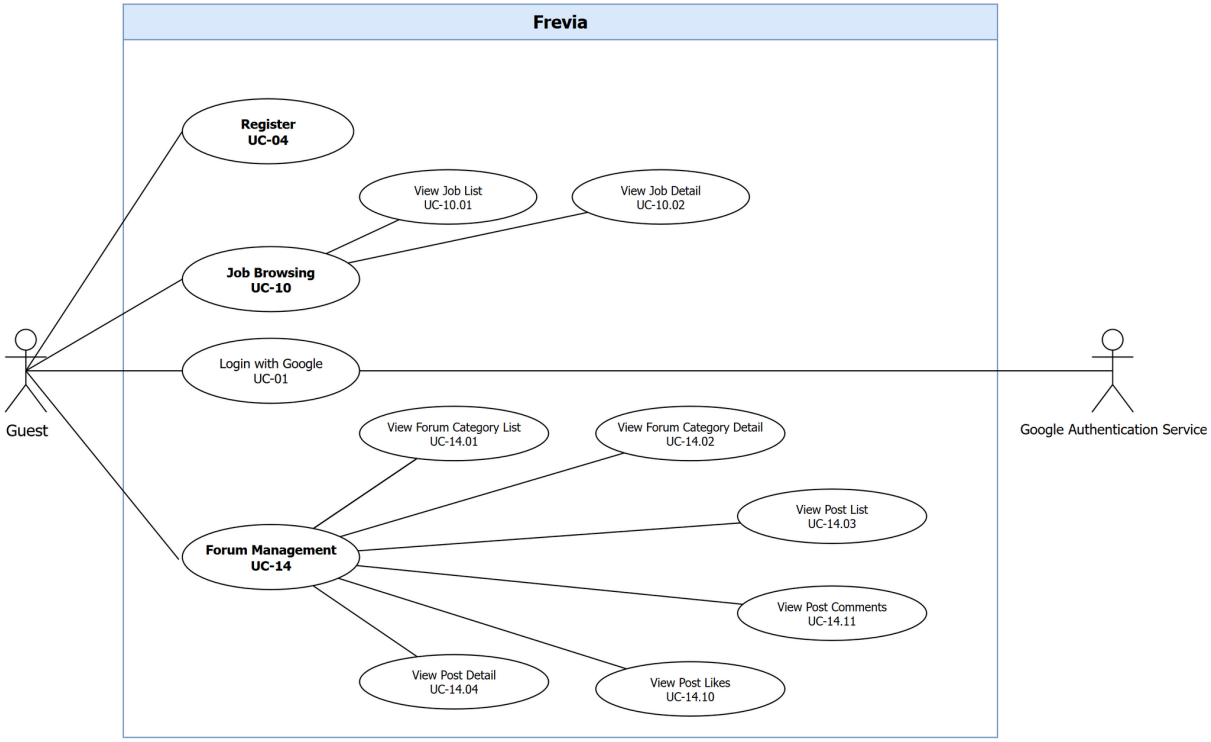


**Figure 2:** Guest Usecase

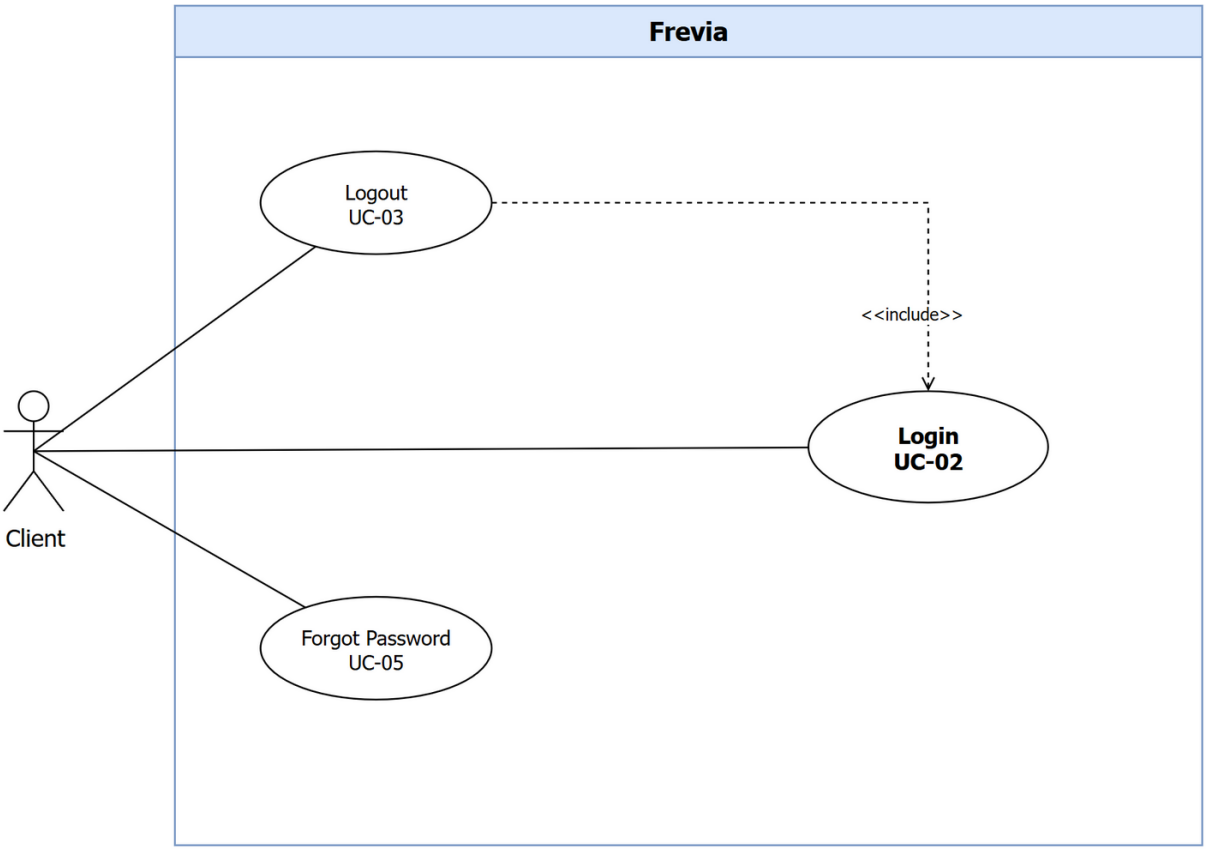


**Figure 3:** Client Usecase 1/6

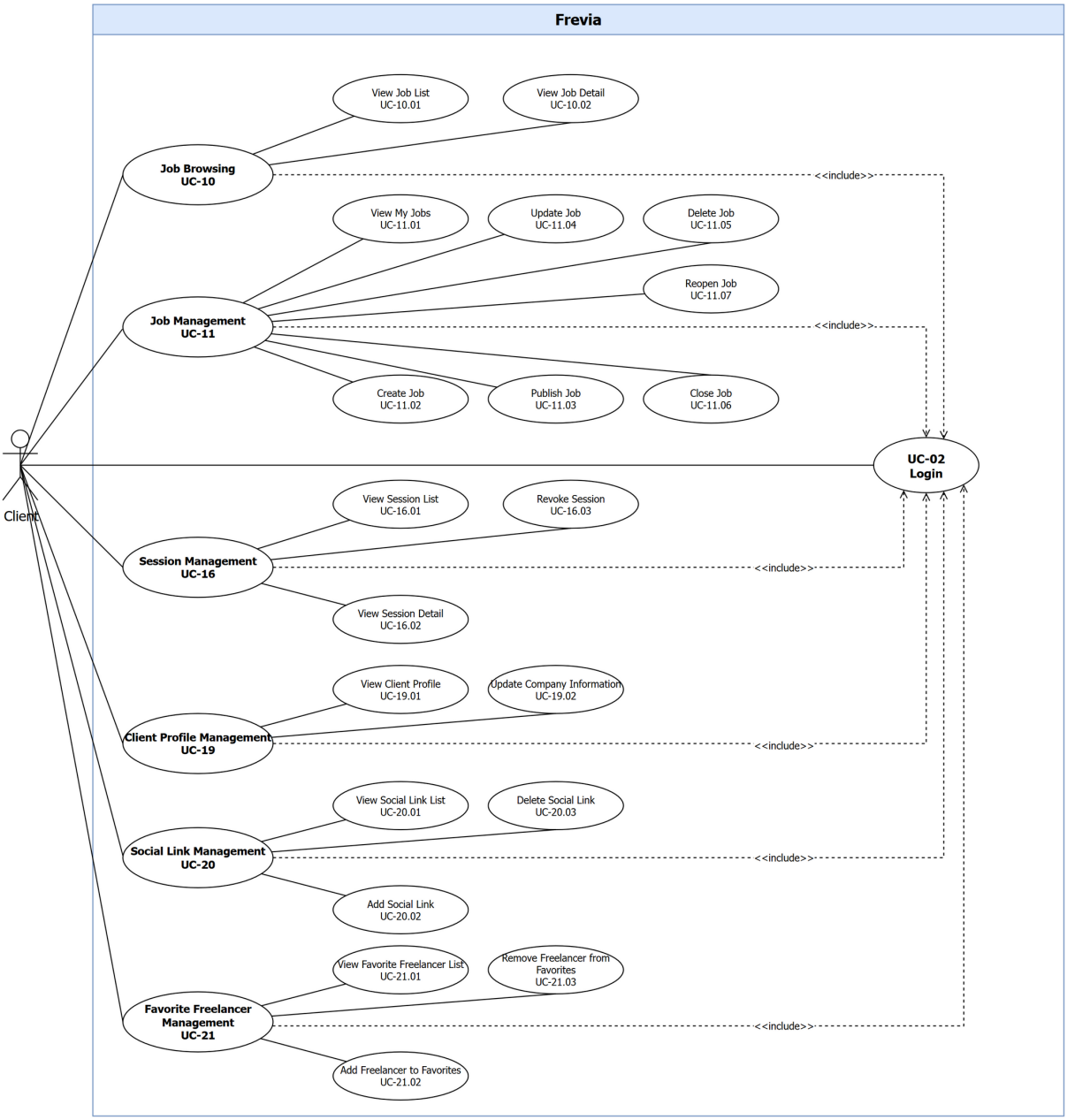


**Figure 4:** Client Usecase 2/6

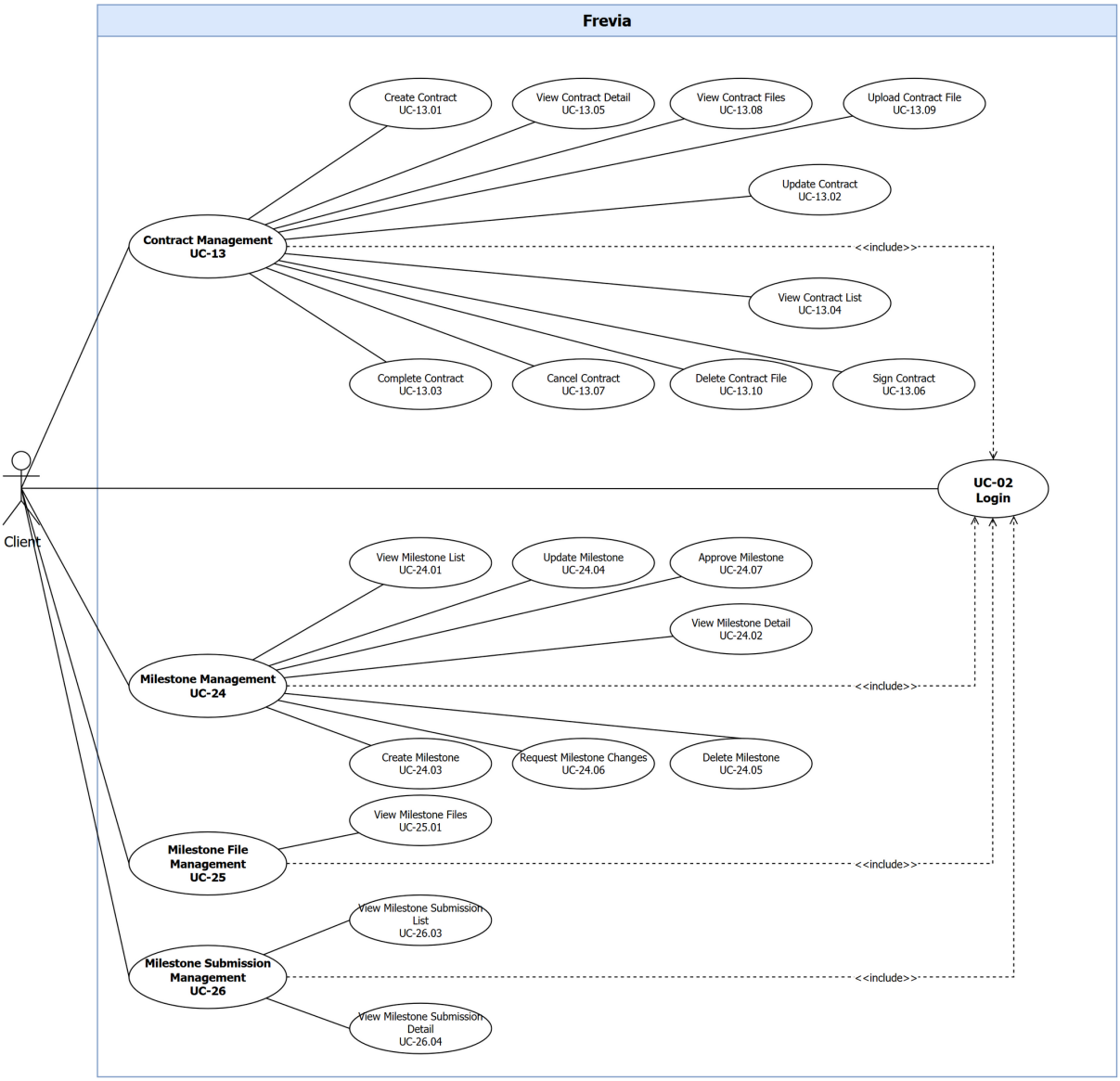


**Figure 5:** Client Usecase 3/6

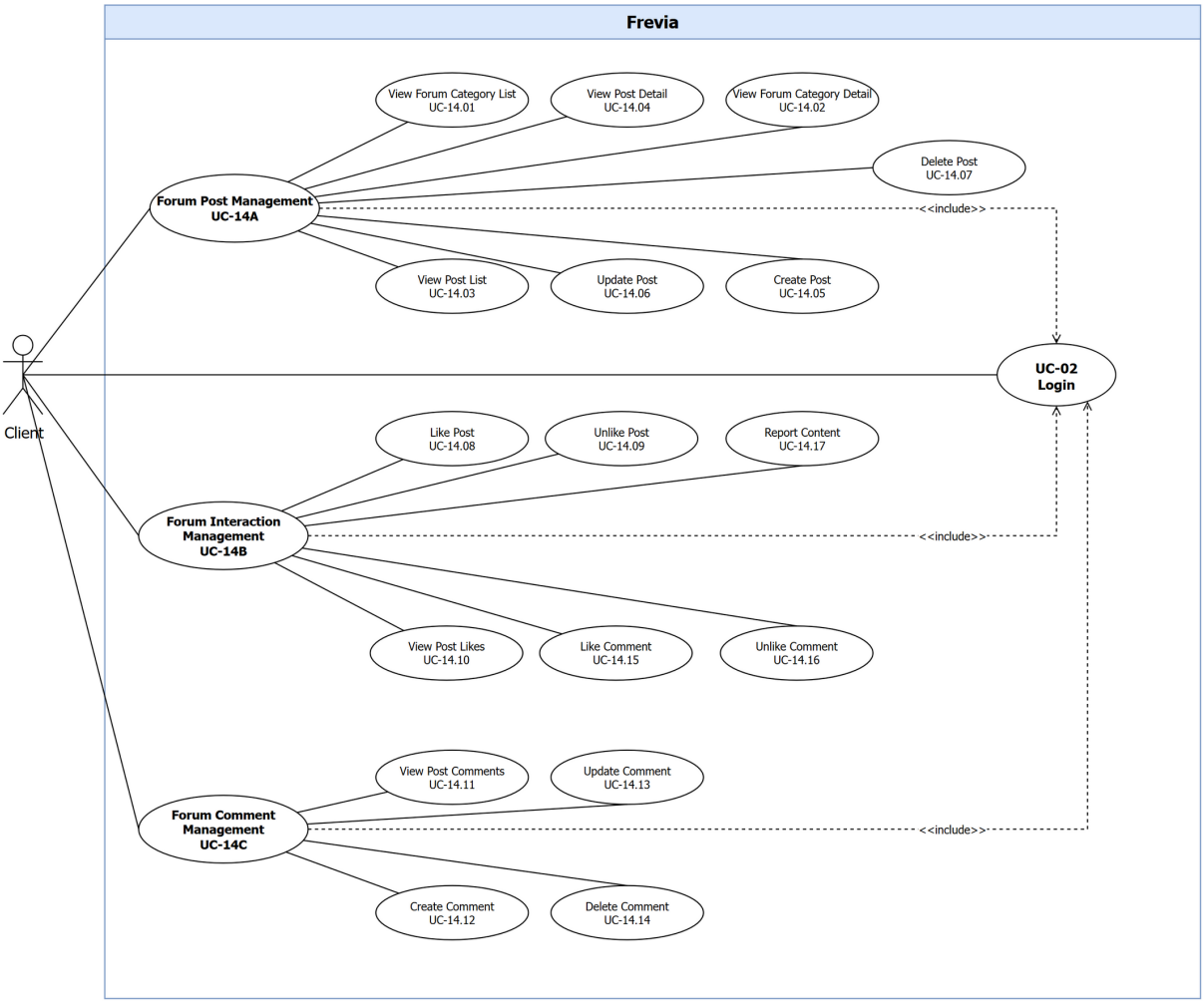


**Figure 6:** Client Usecase 4/6

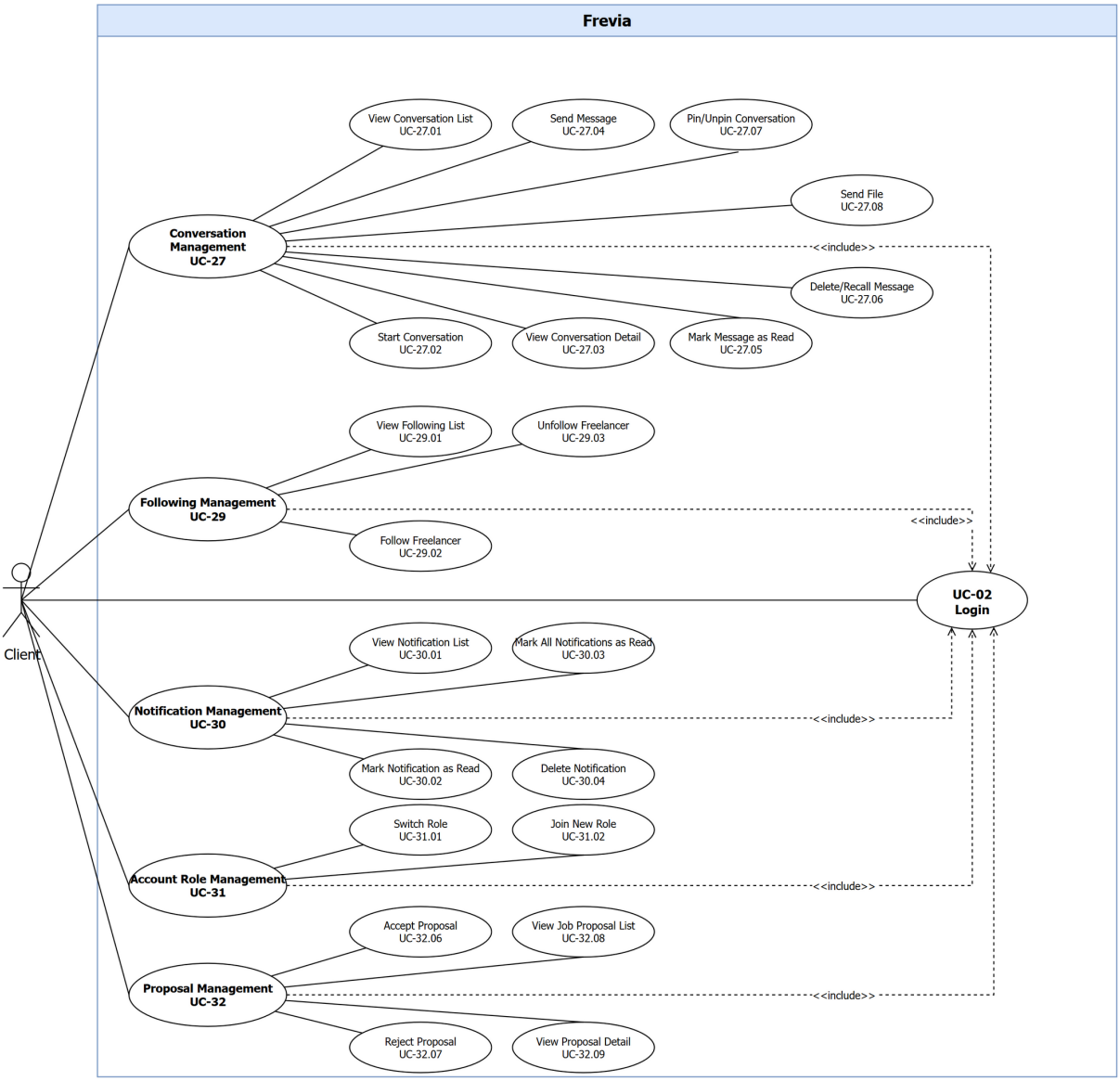


**Figure 7:** Client Usecase 5/6

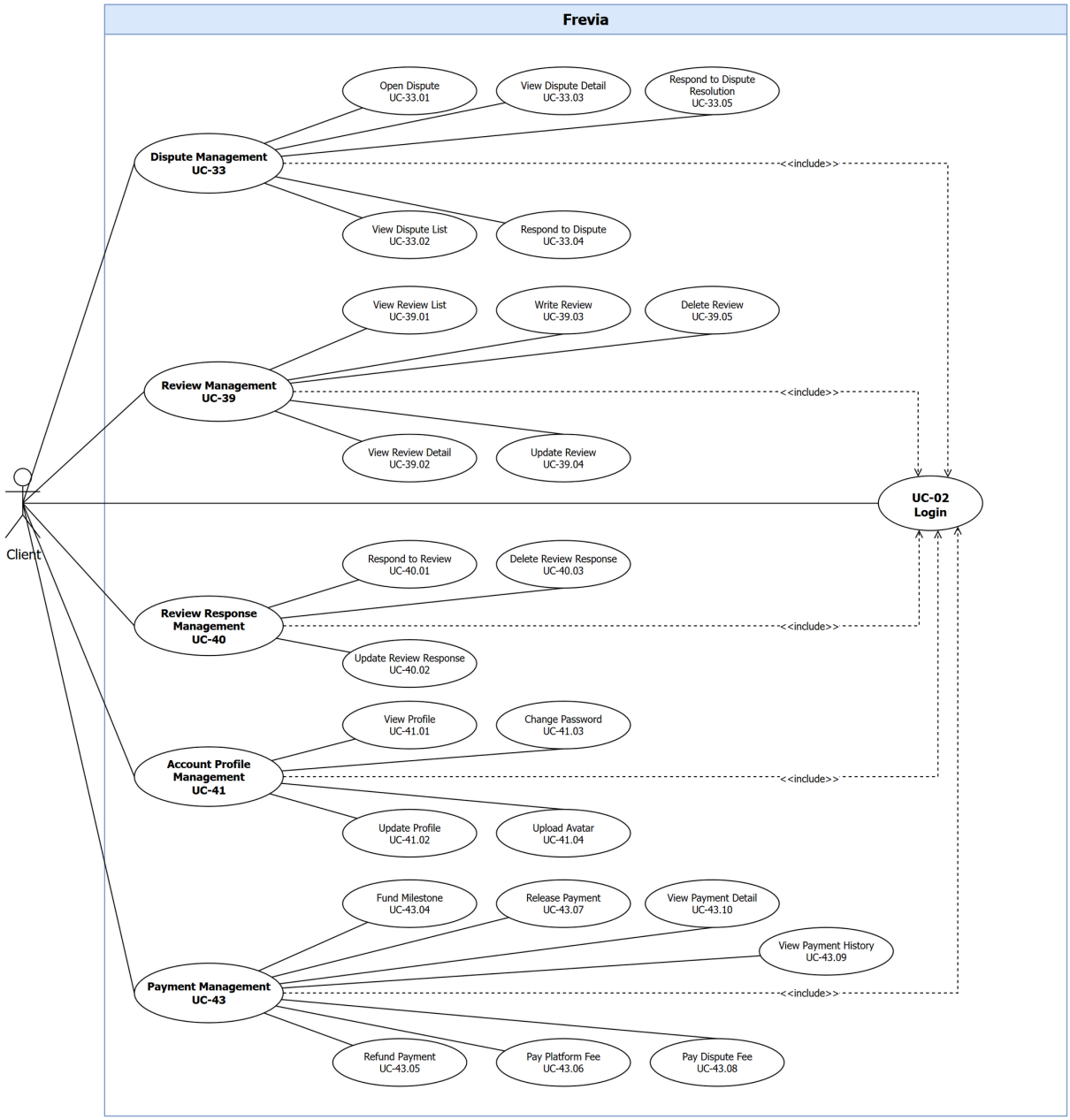


**Figure 8:** Client Usecase 6/6

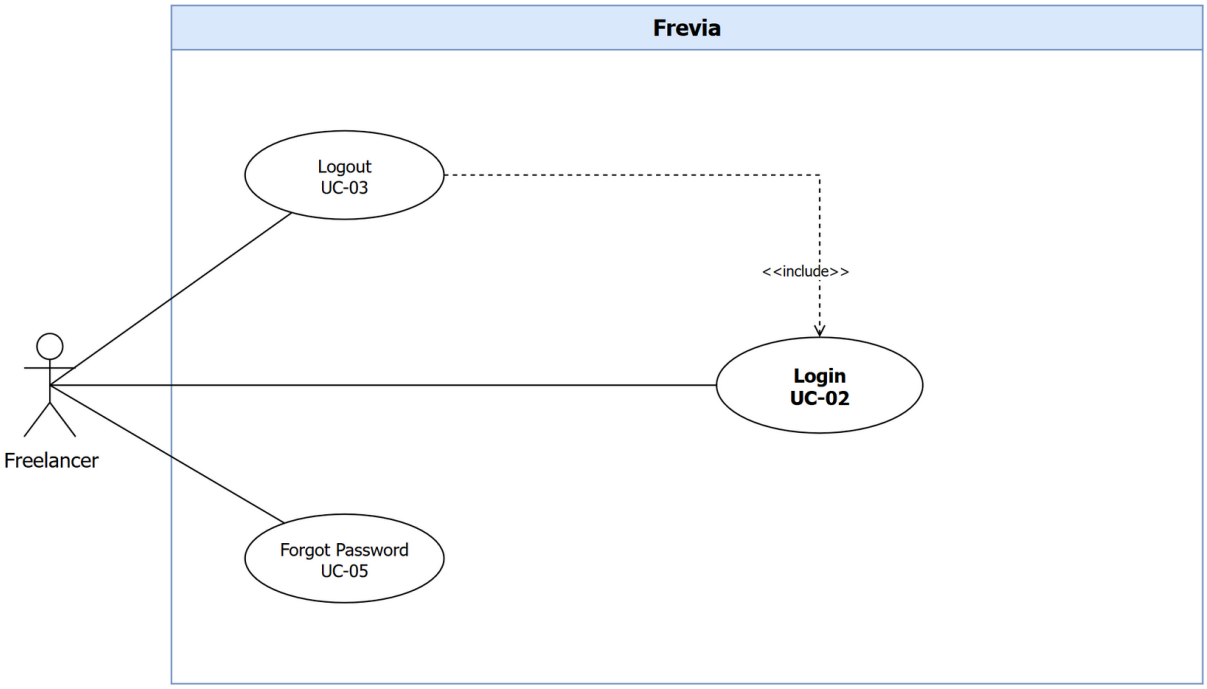


**Figure 9:** Freelancer Usecase 1/7

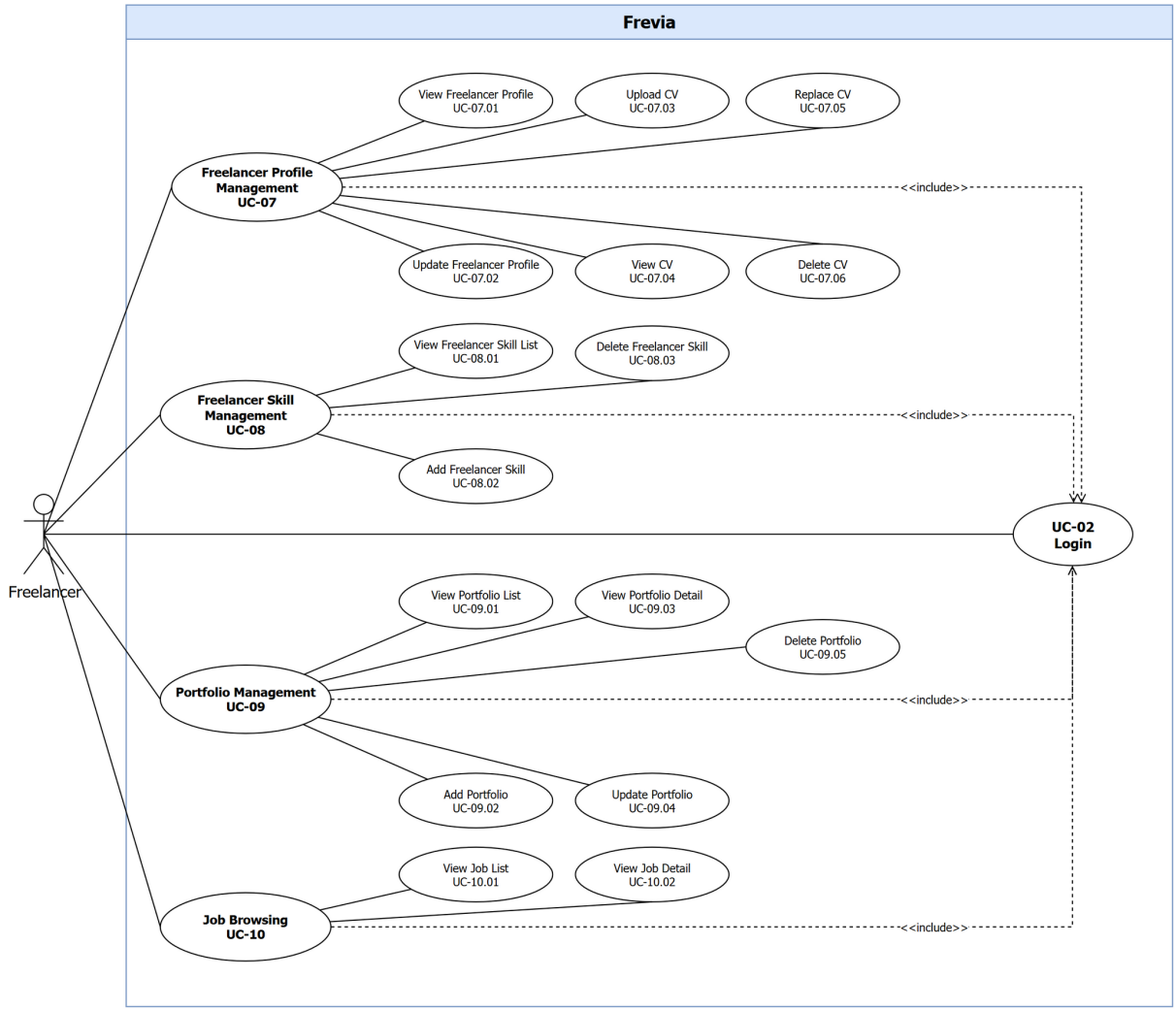


**Figure 10:** Freelancer Usecase 2/7

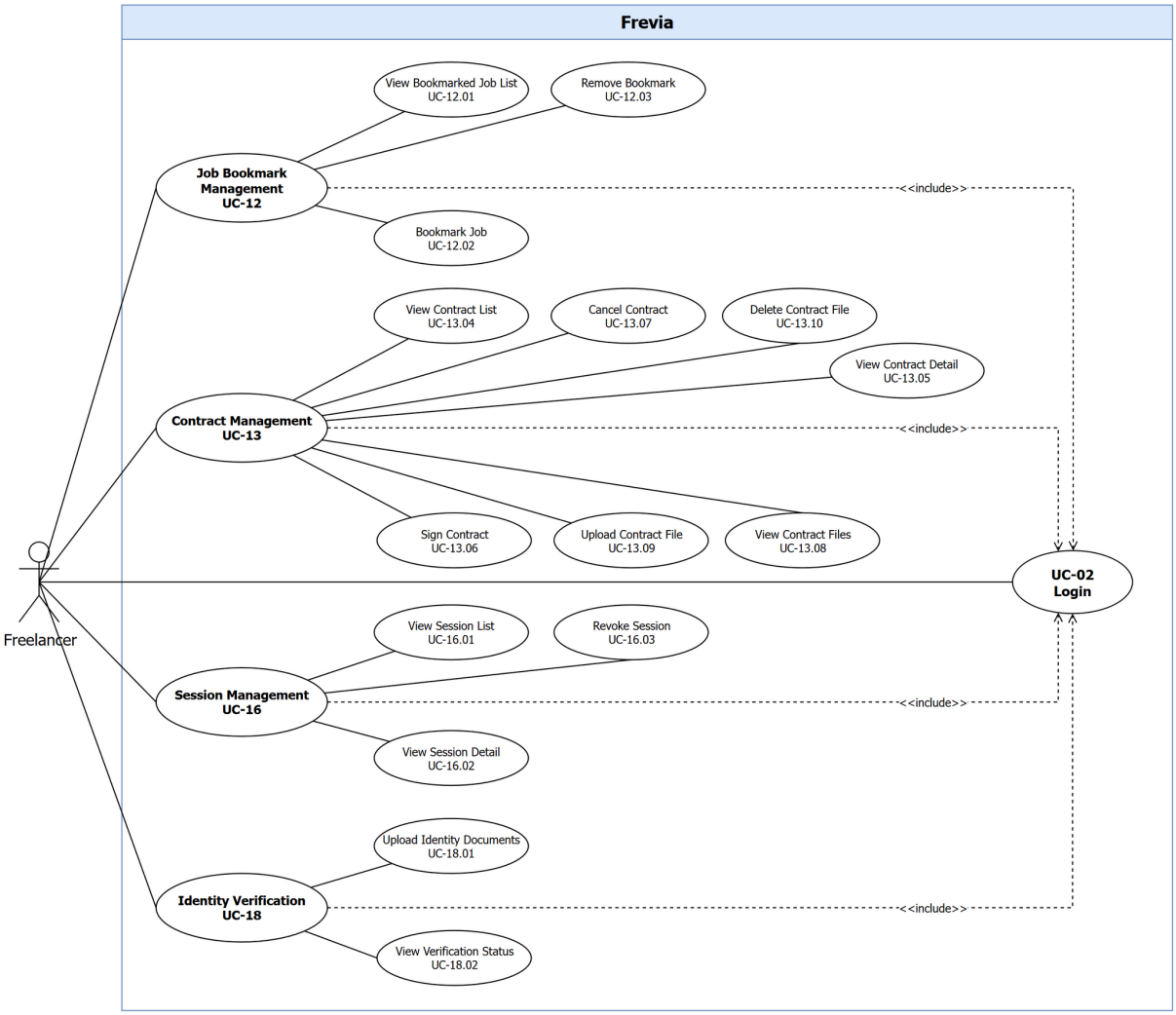


**Figure 11:** Freelancer Usecase 3/7

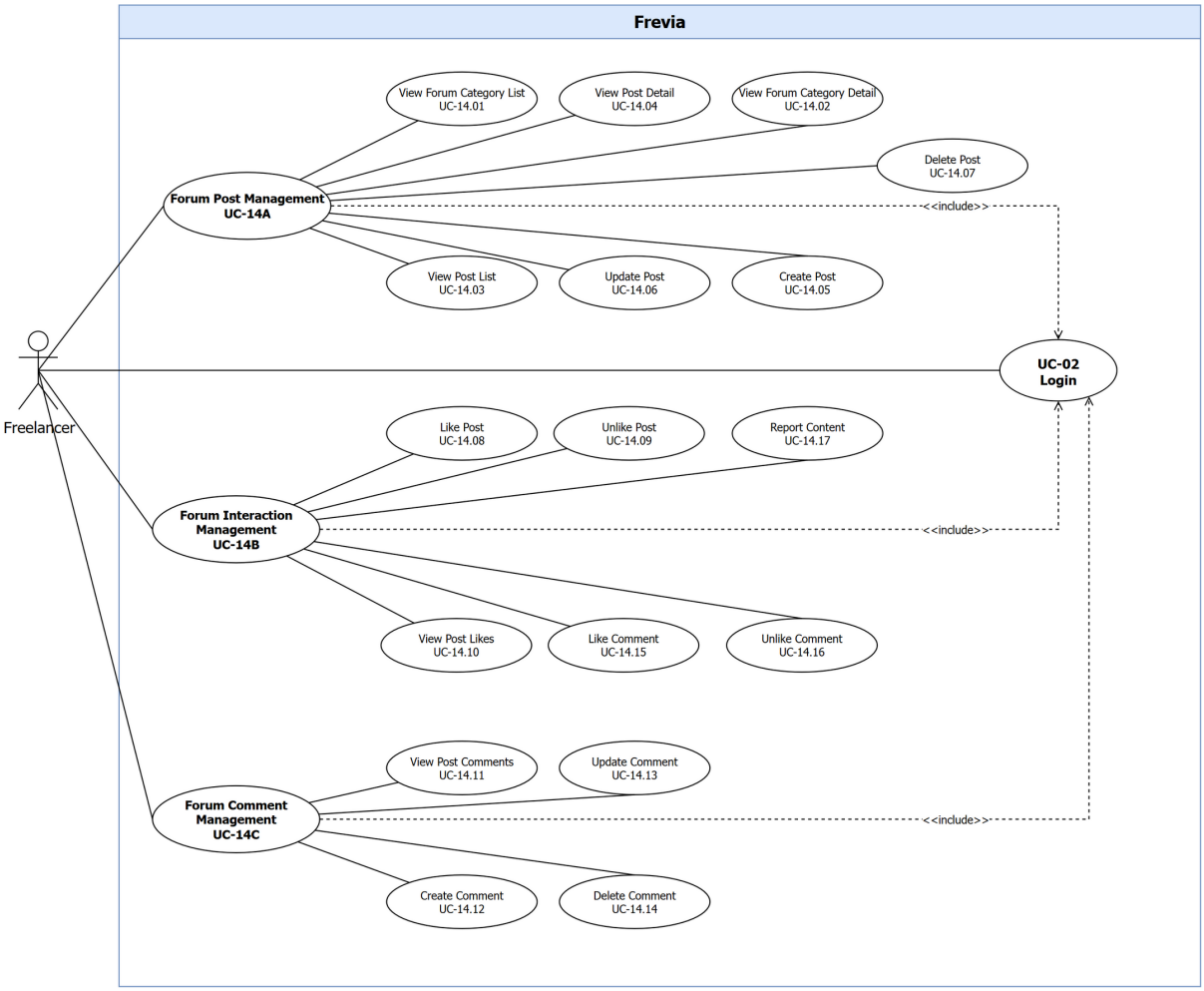


**Figure 12:** Freelancer Usecase 4/7

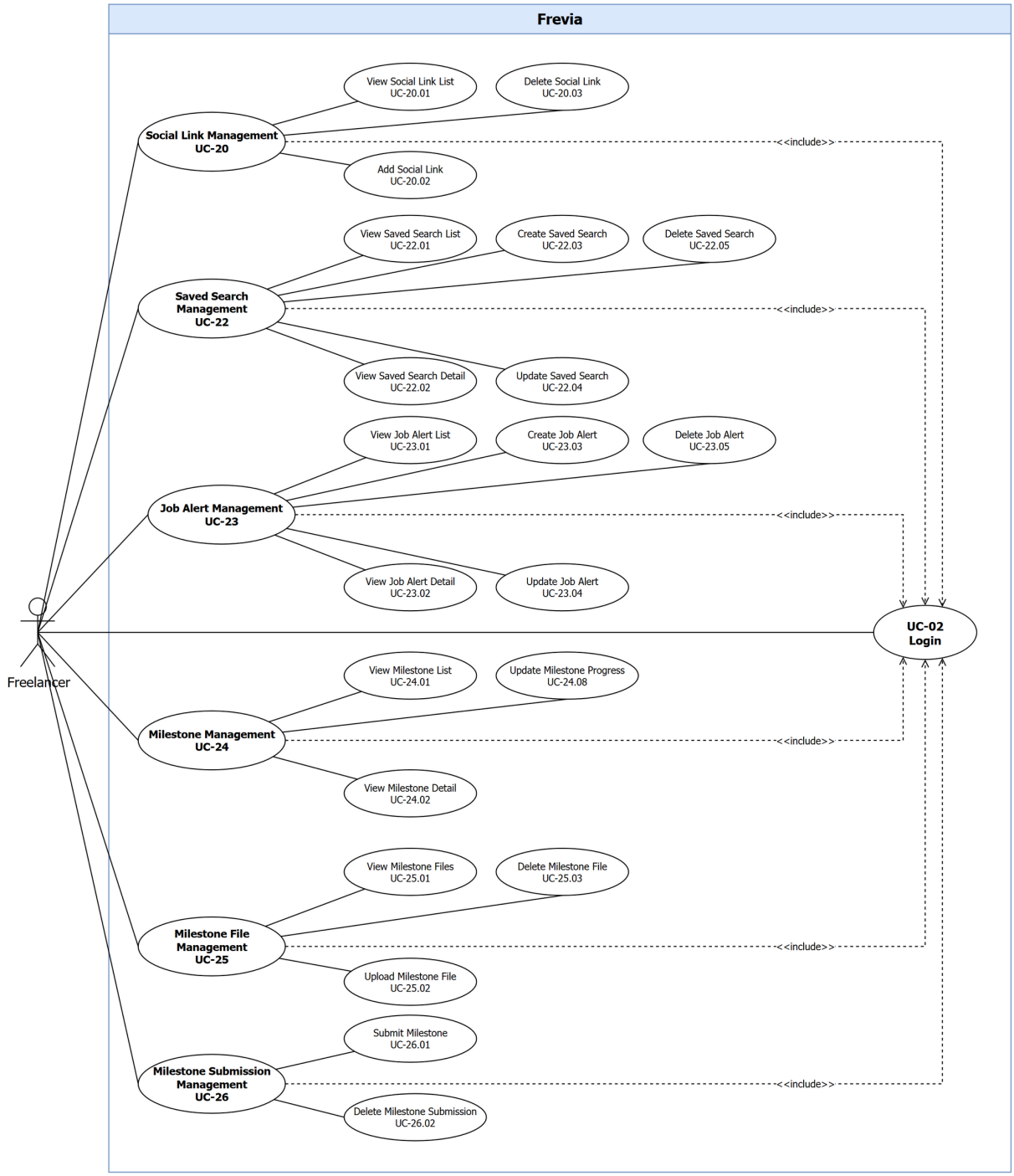


**Figure 13:** Freelancer Usecase 5/7

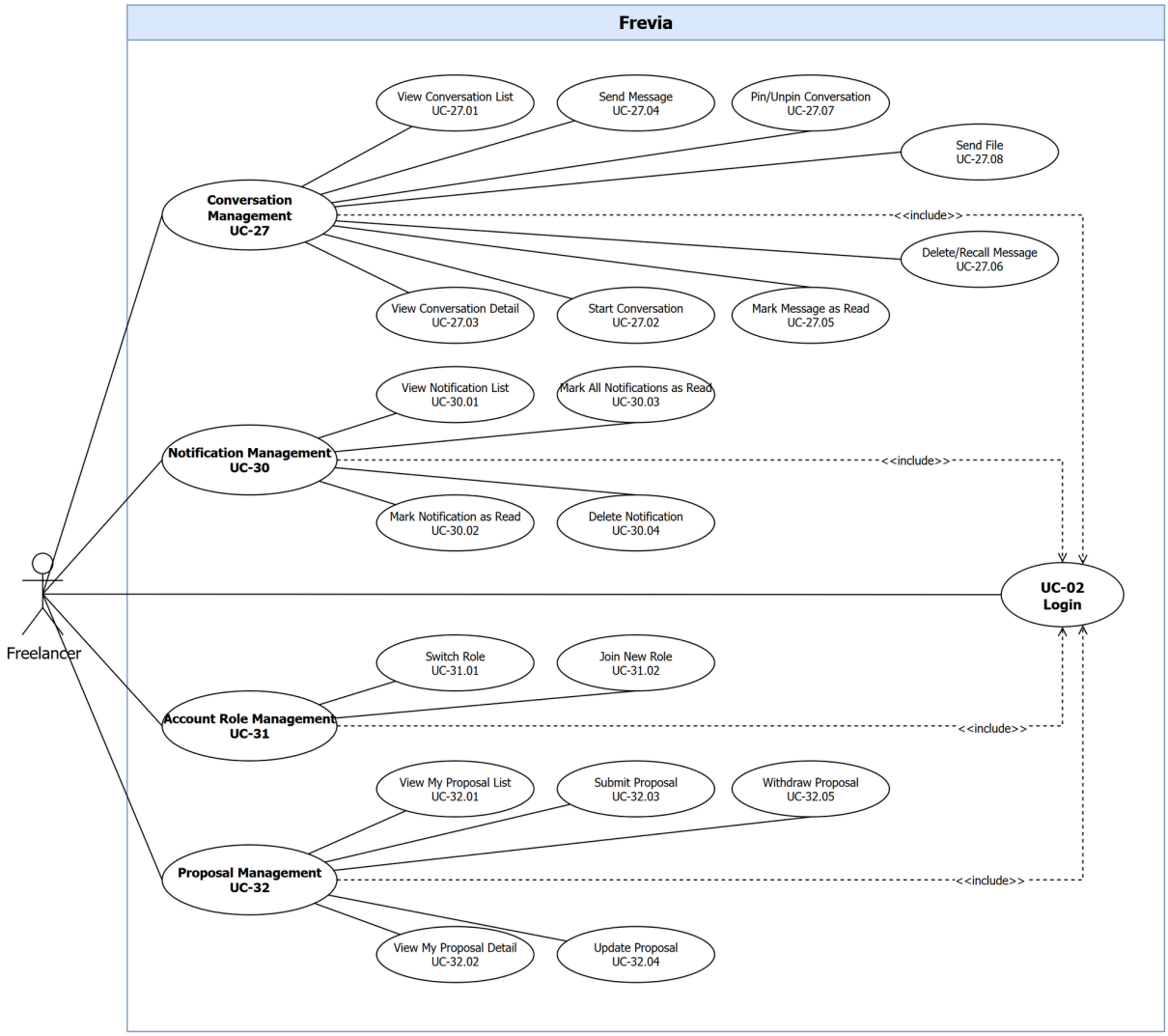


**Figure 14:** Freelancer Usecase 6/7

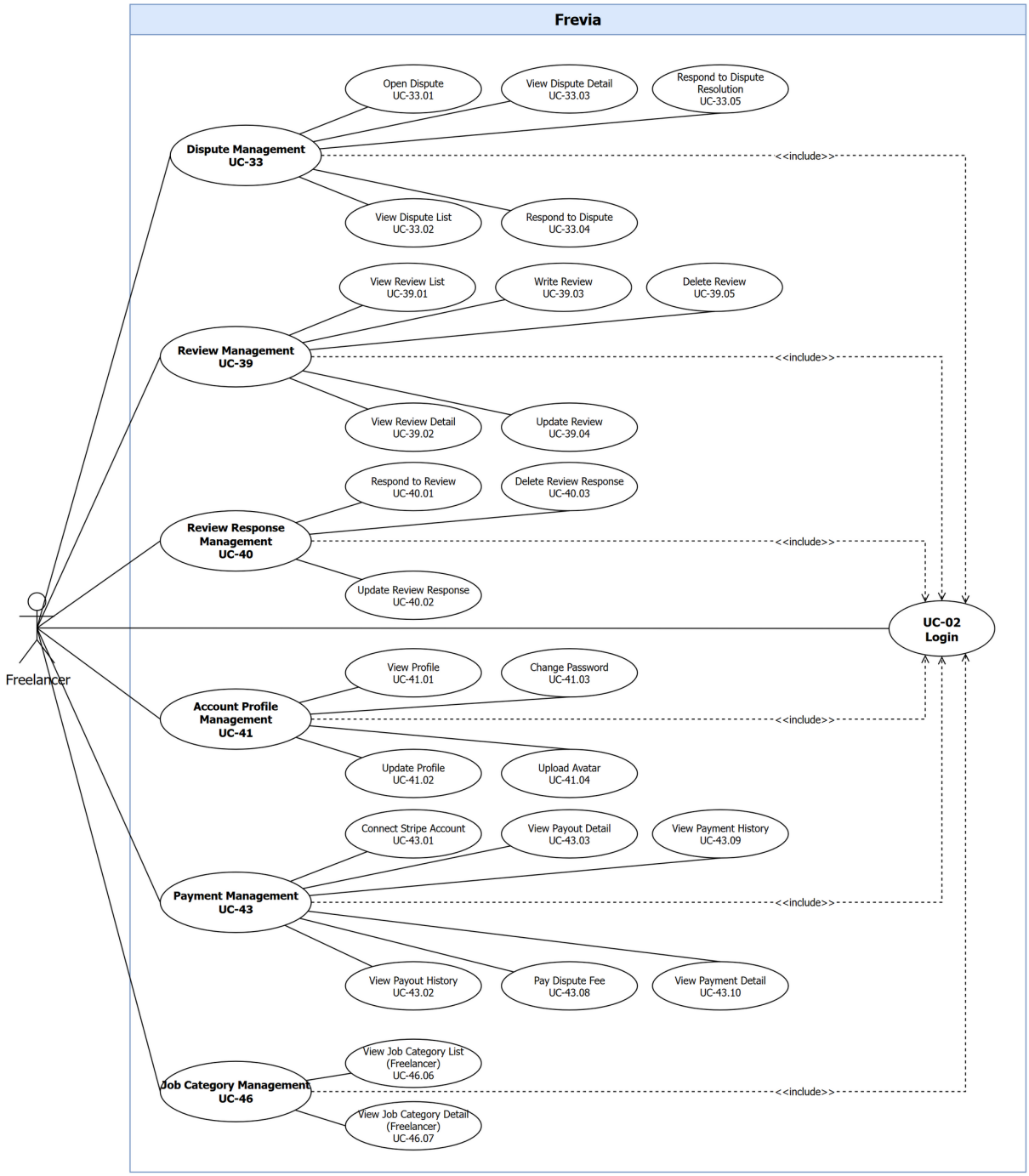


**Figure 15:** Freelancer Usecase 7/7

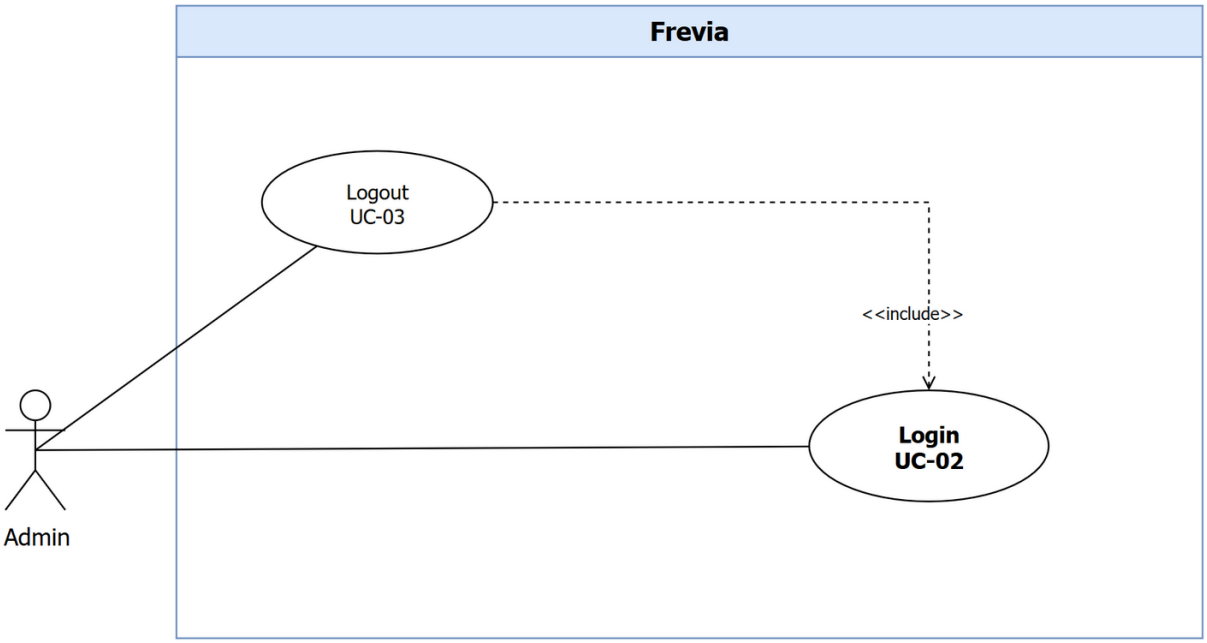


**Figure 16:** Admin Usecase 1/4

**Figure 17:** Admin Usecase 2/4

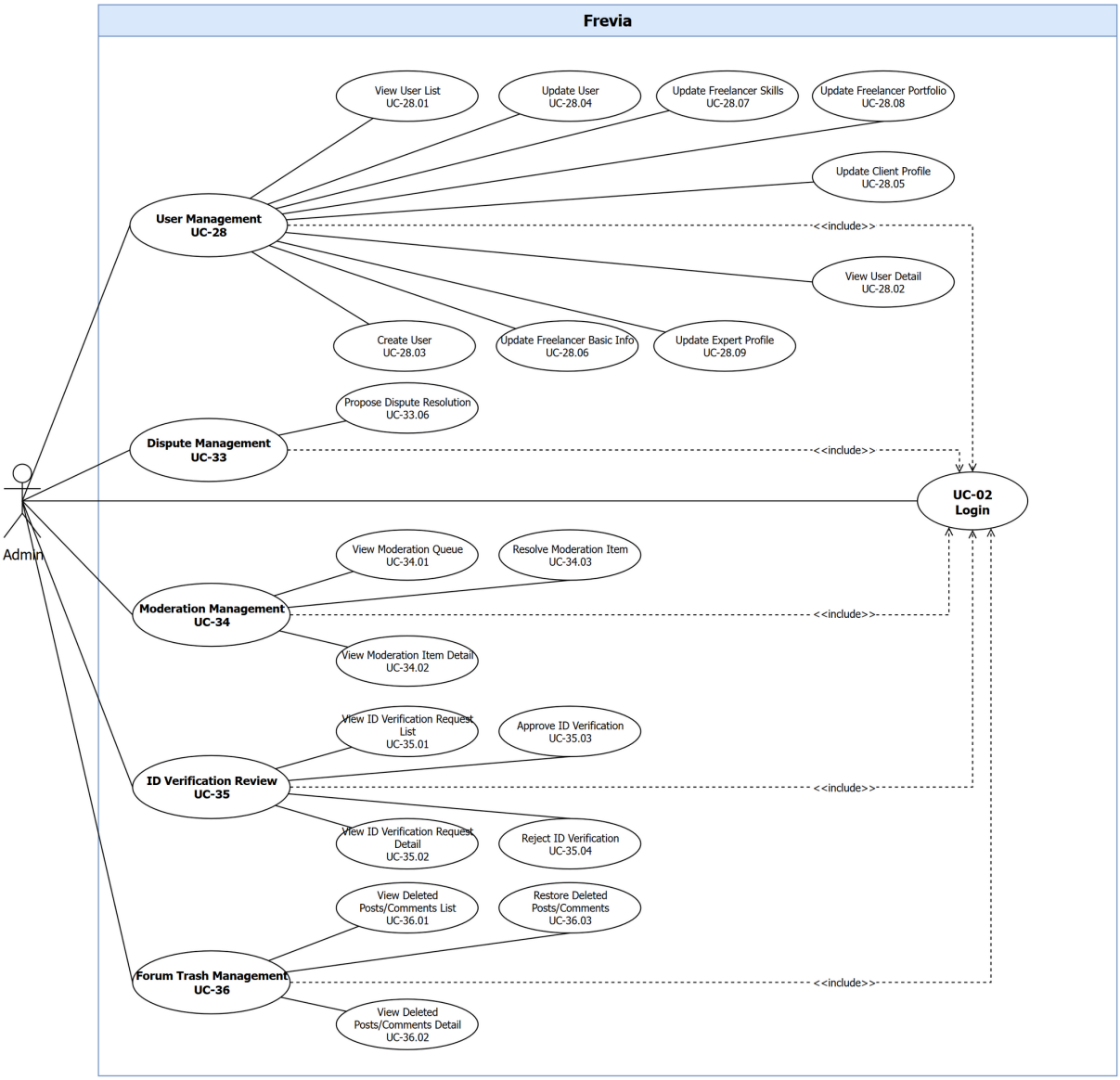


**Figure 18:** Admin Usecase 3/4

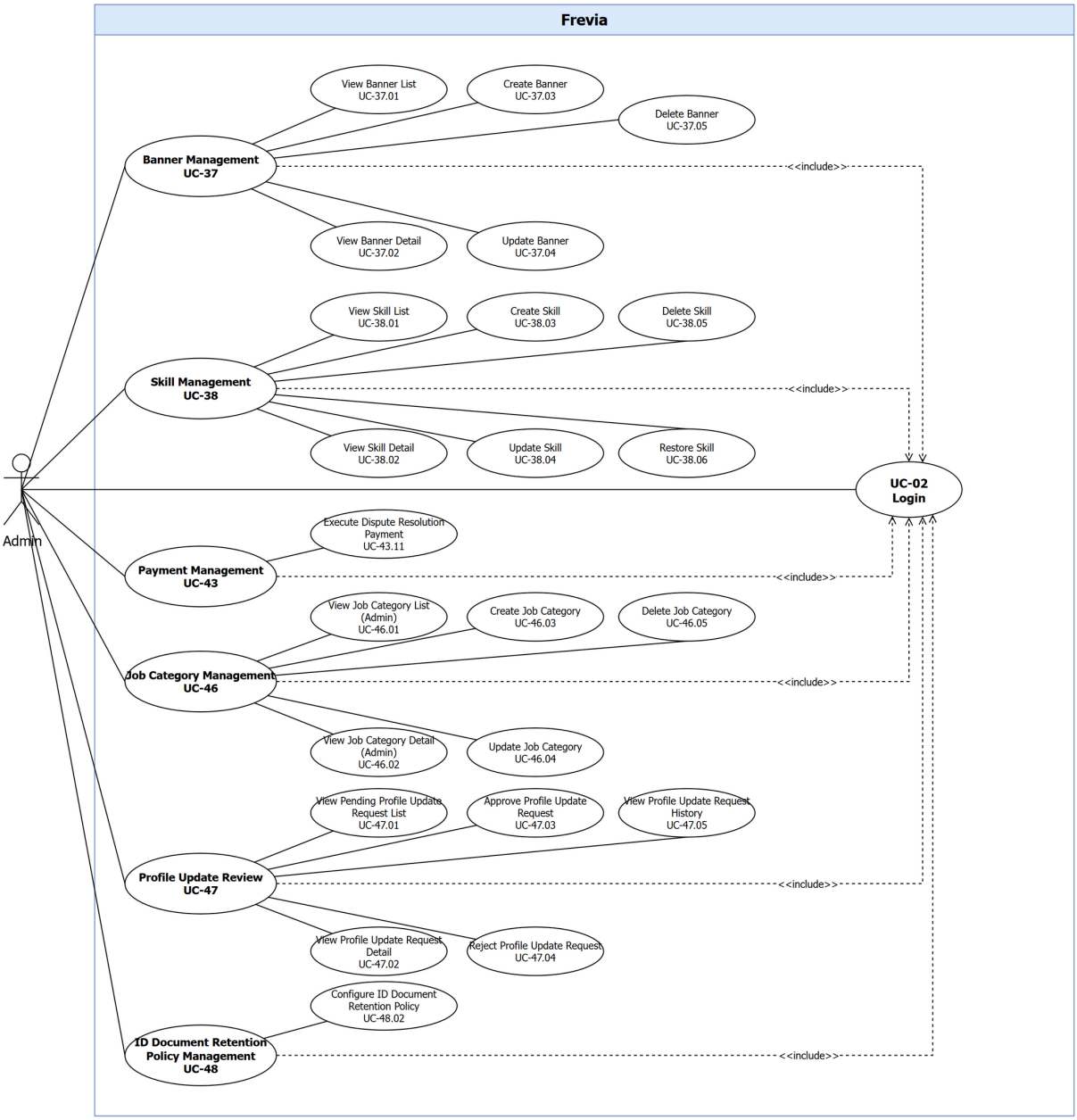


**Figure 19:** Admin Usecase 4/4

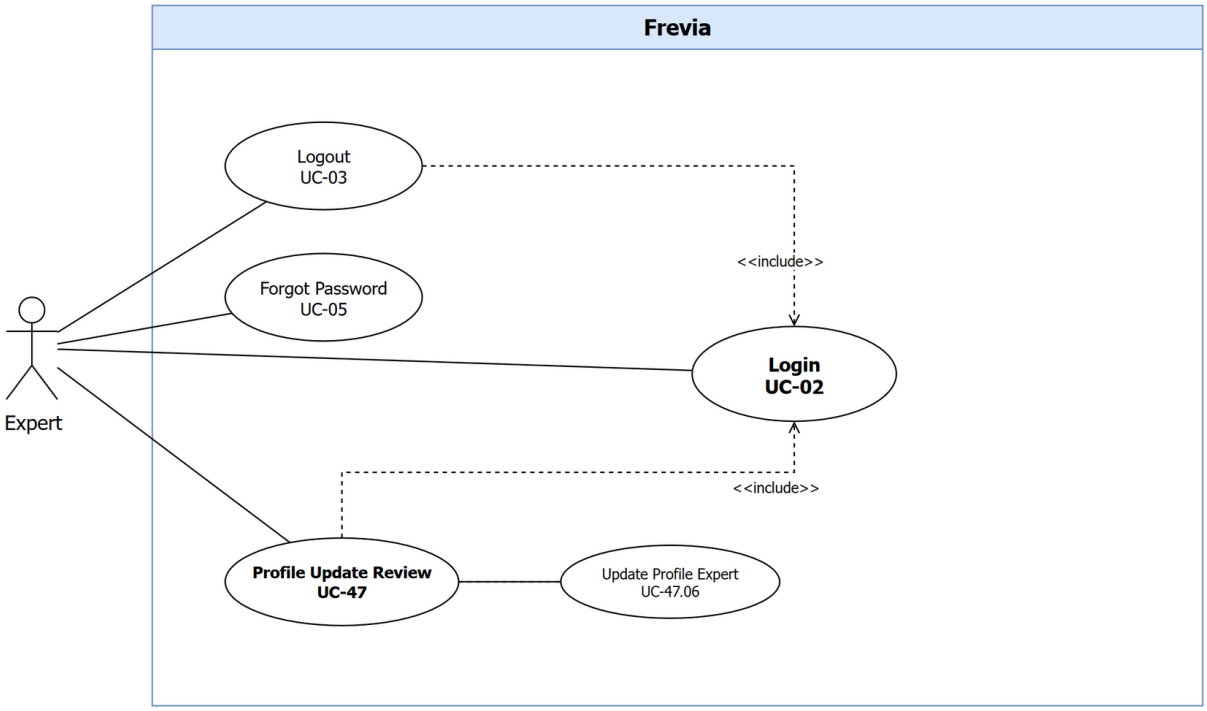


**Figure 20:** Expert Usecase

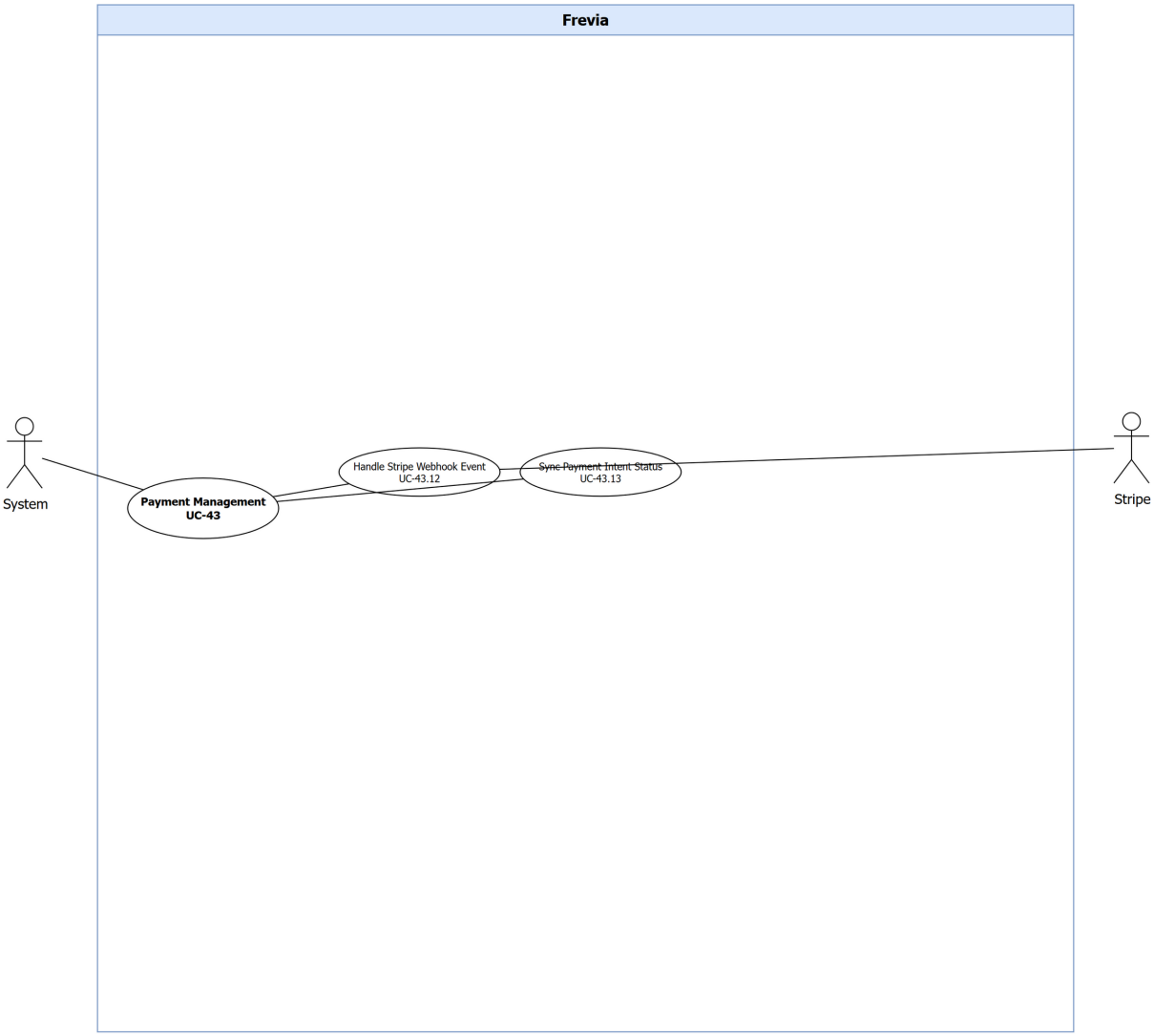


**Figure 21:** System Usecase

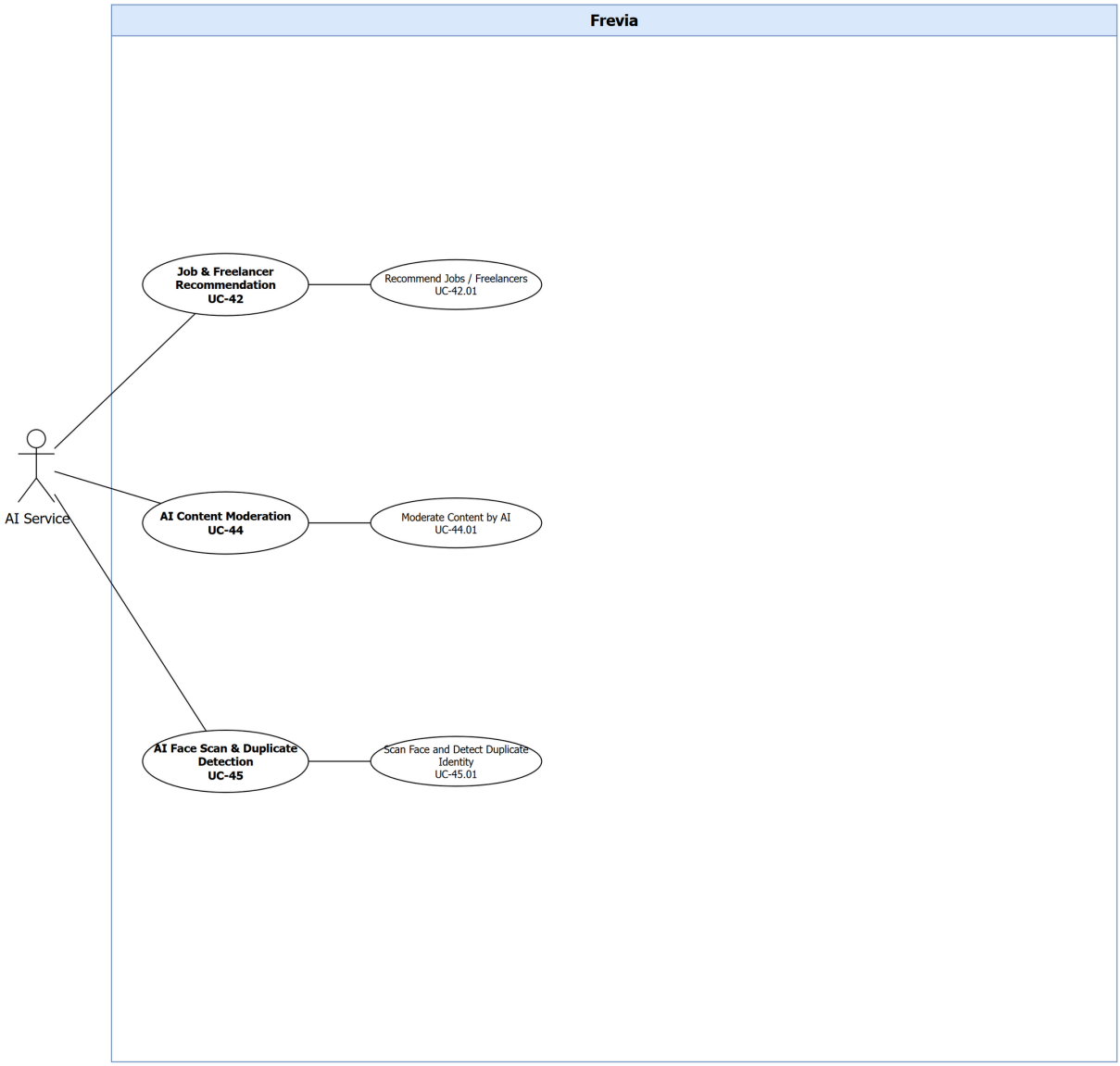


**Figure 22:** AI Service Usecase

### 2.2.2 Descriptions

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-01 | Login with Google | Guest | Allows users to login/register using Google OAuth. |
| UC-02 | Login | Guest | Allows users to log in using email/password and JWT authentication. |
| UC-03 | Logout | Client, Expert, Freelancer, Admin | Allows users to logout and terminate authentication session. |
| UC-04 | Register | Guest | Allows users to create new accounts using email and password. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-05 | Forgot Password | Guest | Allows users to request password reset via email verification link. |
| UC-06.01 | View Role List | Admin | Retrieve list of all roles along with their assigned permissions. |
| UC-06.02 | View Role Detail | Admin | View detailed information of a specific role including its permissions. |
| UC-06.03 | Create Role | Admin | Create a new role with name, description, and a list of permission IDs. |
| UC-06.04 | Update Role | Admin | Update a role's name, description, and replace all assigned permissions. |
| UC-06.05 | Delete Role | Admin | Delete a role from the system. |
| UC-06.06 | Update Role Permissions | Admin | Assign route access rights to roles. |
| UC-15.01 | View Permission List | Admin | Retrieve all permissions (id, name, path, method, module) for admin to assign to roles. |
| UC-15.02 | View Permission Detail | Admin | View detailed information of a specific permission. |
| UC-16.01 | View Session List | Client, Expert, Freelancer, Admin | View a list of active login sessions. |
| UC-16.02 | View Session Detail | Client, Expert, Freelancer, Admin | View login session information (device, IP address, login time, approximate location, etc.). |
| UC-16.03 | Revoke Session | Client, Expert, Freelancer, Admin | Log out of a specific device remotely. |
| UC-17.01 | Create Forum Category | Admin | Allows admins to create a new forum category. |
| UC-17.02 | Update Forum Category | Admin | Allows admins to update an existing forum category. |
| UC-17.03 | Delete Forum Category | Admin | Allows admins to soft-delete a forum category. |
| UC-14.01 | View Forum Category List | Admin | Allows admins to view the list of forum categories. |
| UC-14.02 | View Forum Category Detail | Admin | Allows admins to view the details of a specific forum category. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-17.06 | Restore Forum Category | Admin | Restore a deleted Category forum by setting deletedAt back to null. |
| UC-03 | Job Browsing | Guest, Client, Freelancer | Enables guests to browse and search for available job listings. |
| UC-03.1 | View List Job | Guest, Client, Freelancer | Retrieve paginated job list with filters (category, budget, skills, status, keywords) and sorting options. |
| UC-03.2 | View Job Detail | Guest, Client, Freelancer | Get full job details: title, description, budget, skills, client info, proposals count, attachments, timeline. |
| UC-04 | Manage Forum | Guest, Client, Freelancer, Admin | Manages forum operations including viewing, creating, editing, and moderating posts and comments. |
| UC-04.1 | View Forum Categories | Guest, Client, Freelancer, Admin | Displays all available forum categories for browsing. |
| UC-04.2 | View Forum Category Detail | Guest, Client, Freelancer, Admin | Shows detailed information of a specific forum category and its sub-forums. |
| UC-04.3 | View Post List | Guest, Client, Freelancer, Admin | Retrieves paginated list of posts within a forum category with filters and sorting. |
| UC-04.4 | View Post Detail | Guest, Client, Freelancer, Admin | Displays full post content, author info, creation date, and associated tags. |
| UC-04.5 | View Comments | Guest, Client, Freelancer, Admin | Retrieves all comments related to a specific forum post. |
| UC-04.6 | View Comment Detail | Guest, Client, Freelancer, Admin | Shows detailed information of a specific comment, including content, author, and timestamps. |
| UC-04.7 | Create Post | Client, Freelancer | Allows freelancers to create new discussion posts or questions in the forum. |
| UC-04.8 | Edit Post | Client, Freelancer | Allows freelancers to edit their own forum posts. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-04.9 | Delete Post | Client, Freelancer, Admin | Allows post owners or admins to delete a forum post. |
| UC-04.10 | Like Post | Client, Freelancer | Allows freelancers to like a forum post. |
| UC-04.11 | Unlike Post | Client, Freelancer | Allows freelancers to remove their like from a forum post. |
| UC-04.12 | Add Comment | Client, Freelancer | Allows freelancers to add comments to a forum post. |
| UC-04.13 | Edit Comment | Client, Freelancer | Allows freelancers to edit their own comments. |
| UC-04.14 | Delete Comment | Client, Freelancer | Allows comment owners or admins to delete a comment. |
| UC-04.15 | Like Comment | Client, Freelancer | Allows freelancers to like a comment. |
| UC-04.16 | Unlike Comment | Client, Freelancer | Allows freelancers to remove their like from a comment. |
| UC-04.17 | Report Content | Client, Freelancer | Allows freelancers to report inappropriate content in forum posts or comments. |
| UC-06 | Manage Job | Client, Freelancer | Manages job postings including creation, editing, deletion, and status changes. |
| UC-06.1 | View List Job | Guest, Client, Freelancer | Retrieve paginated job list with filters and sorting options. |
| UC-06.2 | Post New Job | Client | Allows clients to create and publish new job postings. |
| UC-06.3 | Edit Job | Client | Allows clients to edit their own job postings. |
| UC-06.4 | Delete Job | Client | Allows clients to delete their own job postings. |
| UC-06.5 | Change Job Status | Client | Allows clients to update job status (Open, In Progress, Closed, Cancelled). |
| UC-06.6 | View List Bookmarked Job | Freelancer | Retrieves list of jobs bookmarked by the freelancer. |
| UC-06.7 | Bookmark Job | Freelancer | Allows freelancers to save/bookmark a job for later reference. |
| UC-06.8 | Remove Bookmark | Freelancer | Allows freelancers to remove a bookmarked job from their list. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-13 | Manage Contract | Client, Freelancer, Admin | Manages contractual agreements, milestones, deliverables, and file exchanges. |
| UC-13.01 | Create Contract | Client | Allows clients to create and initiate a new contract with a freelancer. |
| UC-13.02 | Update Contract | Client | Allows clients to update contract terms, budget, milestones, or deadlines before signing. |
| UC-13.03 | Complete Contract | Client, Freelancer | Marks an active contract as completed after all milestones are approved and settled. |
| UC-13.04 | View Contract List | Client, Freelancer, Admin | Retrieves paginated list of contracts with status filters and sorting. |
| UC-13.05 | View Contract Detail | Client, Freelancer, Admin | Displays full contract details including terms, parties, status, and milestones. |
| UC-13.06 | Sign Contract | Freelancer, Client | Allows users to digitally accept and sign a contract to activate it. |
| UC-13.07 | Cancel Contract | Client, Freelancer | Allows either party to cancel or terminate a contract prior to completion. |
| UC-13.08 | View Contract Files | Client, Freelancer, Admin | Manages and accesses all files and documentation associated with a contract. |
| UC-13.09 | Upload Contract File | Client, Freelancer | Allows users to upload documents and deliverables to the contract workspace. |
| UC-13.10 | Delete Contract File | Client, Freelancer, Admin | Allows the uploader or admin to delete files from a contract workspace. |
| UC-10 | Manage Profile | Freelancer | Manages freelancer profile information and settings. |
| UC-10.1 | View Detailed Freelancer Profile | Freelancer | Displays full freelancer profile including skills, portfolio, experience, and ratings. |
| UC-10.2 | Update Your Detailed Freelancer Profile | Freelancer | Allows freelancers to edit and update their profile information. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-11 | Manage Skills | Freelancer | Manages freelancer skill set and proficiency levels. |
| UC-11.1 | View List Freelancer Skill | Freelancer | Retrieve the list of skills added by the freelancer. |
| UC-11.2 | Add Freelancer Skill | Freelancer | Allows freelancers to add new skills to their profile. |
| UC-11.3 | Delete Freelancer Skill | Freelancer | Allows freelancers to remove skills from their profile. |
| UC-12 | Manage Portfolio | Freelancer | Manages freelancer portfolio projects and work samples. |
| UC-12.1 | View Portfolio List | Freelancer | Retrieve the list of portfolios created by the freelancer. |
| UC-12.2 | Add New Portfolio | Freelancer | Allows freelancers to add new portfolio items with images and descriptions. |
| UC-12.3 | View Portfolio Detail | Freelancer | Displays full details of a portfolio item including media, description, and technologies used. |
| UC-12.4 | Update Portfolio | Freelancer | Allows freelancers to edit and update their portfolio items. |
| UC-12.5 | Delete Portfolio | Freelancer | Allows freelancers to remove portfolio items. |
| UC-15 | Identity Verification | Freelancer | Manages freelancer identity verification document submission and review. |
| UC-15.1 | Upload Identity Documents | Freelancer | Allows freelancers to upload ID card, passport, or license documents. |
| UC-15.2 | View Verification Status | Freelancer | Displays current identity verification status (Pending, Approved, Rejected). |
| UC-16 | View Client Profile | Guest, Client, Freelancer | Manages client company profile details and public showcase. |
| UC-16.1 | View Client Profile Detail | Guest, Client, Freelancer | Displays client company details, bio, website, and job history. |
| UC-16.2 | Update Company Information | Client | Allows clients to update company name, bio, website URL, and industry sector. |
| UC-17 | Social Links | Guest, Client, Freelancer | Manages external social and portfolio links on user profiles. |
| UC-17.1 | View Social Links | Guest, Client, Freelancer | Retrieves social links associated with a user profile. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-17.2 | Add Social Link | Client, Freelancer | Allows users to add external social links to their profile. |
| UC-17.3 | Delete Social Link | Client, Freelancer | Allows users to remove social links from their profile. |
| UC-18 | Favorites | Client | Manages saved favorite freelancer profiles for clients. |
| UC-18.1 | View Favorite Freelancers | Client | Retrieves paginated list of client's favorite freelancers. |
| UC-18.2 | Add to Favorites | Client | Allows clients to save a freelancer profile to their favorites. |
| UC-18.3 | Remove from Favorites | Client | Allows clients to remove a freelancer from their favorites list. |
| UC-19 | Following | Client, Freelancer | Manages followed freelancer network connections. |
| UC-19.1 | View Following List | Client, Freelancer | Retrieves list of freelancers currently followed by the user. |
| UC-19.2 | Follow Freelancer | Client, Freelancer | Allows users to follow a freelancer for activity updates. |
| UC-19.3 | Unfollow Freelancer | Client, Freelancer | Allows users to stop following a freelancer. |
| UC-20 | Manage Notifications | Client, Freelancer, Admin | Manages user in-app notification center and read states. |
| UC-20.1 | View Notification List | Client, Freelancer, Admin | Retrieves user notifications list with unread counter. |
| UC-20.2 | Mark Notification as Read | Client, Freelancer, Admin | Sets read status of a specific notification to true. |
| UC-20.3 | Mark All Notifications as Read | Client, Freelancer, Admin | Bulk marks all user notifications as read. |
| UC-20.4 | Delete Notification | Client, Freelancer, Admin | Soft-deletes a notification from user's notification list. |
| UC-21 | Account Management | Client, Freelancer | Manages user role switching and complementary role activation. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-21.1 | Switch Role | Client, Freelancer | Allows dual-role users to switch working context between Client and Freelancer. |
| UC-21.2 | Join New Role | Client, Freelancer | Allows single-role users to activate complementary secondary role. |
| UC-22 | Reviews | Guest, Client, Freelancer, Admin | Manages post-contract ratings and detailed feedback reviews. |
| UC-22.1 | View Reviews List | Guest, Client, Freelancer | Retrieves reviews list for a user profile or contract. |
| UC-22.2 | View Review Detail | Guest, Client, Freelancer | Displays rating breakdown scores and detailed comment text. |
| UC-22.3 | Write Review | Client, Freelancer | Allows contract participants to review each other upon project completion. |
| UC-22.4 | Edit Review | Client, Freelancer | Allows review authors to update ratings or feedback text. |
| UC-22.5 | Delete Review | Client, Freelancer, Admin | Soft-deletes a review record. |
| UC-23 | Manage Response Review | Client, Freelancer, Admin | Manages recipient responses to reviews. |
| UC-23.1 | Respond to Review | Client, Freelancer | Allows review recipients to post a public response. |
| UC-23.2 | Edit Response | Client, Freelancer | Allows response authors to modify response text. |
| UC-23.3 | Delete Response | Client, Freelancer, Admin | Soft-deletes a review response. |
| UC-24 | Manage Profile | Client, Freelancer, Admin | Manages account credentials, profile details, and avatar. |
| UC-24.1 | Get Profile | Client, Freelancer, Admin | Retrieves general account profile information. |
| UC-24.2 | Update Profile | Client, Freelancer, Admin | Updates general profile details (name, phone, bio, location). |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-24.3 | Change Password | Client, Freelancer, Admin | Allows authenticated users to change password securely. |
| UC-24.4 | Upload Avatar | Client, Freelancer, Admin | Allows users to upload and crop profile avatar image. |
| UC-25 | Profile Update Review | Admin | Manages manual admin review of profile modification requests. |
| UC-25.1 | View Pending Profile Update Requests | Admin | Retrieves list of profile update requests awaiting review. |
| UC-25.2 | View Profile Update Request Detail | Admin | Displays side-by-side diff comparison for profile changes. |
| UC-25.3 | Approve Profile Update Request | Admin | Approves profile modification request and updates live profile. |
| UC-25.4 | Reject Profile Update Request | Admin | Rejects profile request with mandatory reason note. |
| UC-25.5 | View Profile Update Request History | Admin | Views history of processed profile update requests. |
| UC-25.6 | Update Profile Expert | Expert, Admin | Updates expert consultant profile and homepage showcase. |
| UC-47.01 | View Pending Profile Update Request List | Admin | Retrieves profile update requests awaiting manual review. |
| UC-47.02 | View Profile Update Request Detail | Admin | Displays a request and compares current profile data with proposed changes. |
| UC-47.03 | Approve Profile Update Request | Admin | Approves a request and applies its proposed changes to the live profile. |
| UC-47.04 | Reject Profile Update Request | Admin | Rejects a request with a reason while preserving the current profile data. |
| UC-47.05 | View Profile Update Request History | Admin | Displays previously approved and rejected profile update requests. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-47.06 | Update Profile Expert | Expert | Allows an Expert to edit the profile used in the public expert experience. |
| UC-07.03 | Upload CV | Freelancer | Uploads a supported CV file and associates it with the Freelancer profile. |
| UC-07.04 | View CV | Freelancer | Views or downloads the CV currently associated with the Freelancer profile. |
| UC-07.05 | Replace CV | Freelancer | Replaces the existing CV with a newly uploaded supported file. |
| UC-07.06 | Delete CV | Freelancer | Removes the CV associated with the Freelancer profile. |
| UC-33 | Manage Dispute | Client, Freelancer, Admin | Manages formal dispute arbitration on funded contract milestones. |
| UC-33.01 | Open Dispute | Client, Freelancer | Allows a client or freelancer to initiate a formal dispute on a funded milestone. |
| UC-33.02 | View Dispute List | Client, Freelancer, Admin | Retrieves paginated overview of disputes with status filtering and deadline badges. |
| UC-33.03 | View Dispute Detail | Client, Freelancer, Admin | Displays full dispute case records, claims, evidence, fees, and resolutions. |
| UC-33.04 | Respond to Dispute | Client, Freelancer | Allows the respondent to submit a rebuttal statement and counter-evidence. |
| UC-33.05 | Respond to Dispute Resolution | Client, Freelancer | Allows involved parties to accept or reject the arbitrator's proposed decision. |
| UC-33.06 | Propose Dispute Resolution | Admin | Allows an administrator/arbitrator to evaluate claims and propose a fund split decision. |
| UC-43 | Manage Payment | Client, Freelancer, Admin | Handles platform financial transactions, escrow deposits, payouts, and fees via Stripe. |
| UC-43.01 | Connect Stripe Account | Freelancer | Connects a freelancer's bank account via Stripe Connect Express to receive payouts. |
| UC-43.02 | View Payout History | Freelancer, Admin | Retrieves paginated history of all payout transfers sent to the freelancer's bank. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-43.03 | View Payout Detail | Freelancer, Admin | Displays detailed ledger breakdown, fees, and arrival estimates for a payout transfer. |
| UC-43.04 | Fund Milestone | Client | Deposits milestone funds into platform escrow using credit card or saved payment methods. |
| UC-43.05 | Refund Payment | Client, Admin | Returns escrowed milestone funds to the client upon contract cancellation or dispute. |
| UC-43.06 | Pay Platform Fee | Client | Processes the platform engagement fee upon contract initiation via Stripe. |
| UC-43.07 | Release Payment | Client | Authorizes the release of escrowed milestone funds to the freelancer upon work approval. |
| UC-43.08 | Pay Dispute Fee | Client, Freelancer | Processes arbitration fee payment within the 24-hour dispute window via Stripe. |
| UC-43.09 | View Payment History | Client, Freelancer, Admin | Displays a comprehensive, paginated ledger of all account financial transactions. |
| UC-43.10 | View Payment Detail | Client, Freelancer, Admin | Shows itemized receipt details, fees, and downloadable invoice for a transaction. |
| UC-43.11 | Execute Dispute Resolution Payment | Admin, System | Executes split payouts (client refund and freelancer transfer) upon dispute finalization. |
| UC-43.12 | Handle Stripe Webhook Event | Stripe, System | Listens for and verifies asynchronous Stripe webhook events to reconcile database state. |
| UC-43.13 | Sync Payment Intent Status | Client, System | Synchronizes real-time payment intent and checkout status directly with the Stripe API. |
| UC-46 | Job Category Management | Admin, Freelancer | Provides administrative category management and freelancer categorization for jobs. |
| UC-46.01 | View Job Category List | Admin | Allows an administrator to view, search, and filter job categories with pagination. |
| UC-46.02 | View Job Category Detail | Admin | Displays detailed information of a specific job category including associated jobs. |
| UC-46.03 | Create Job Category | Admin | Allows an administrator to create a new job category with name, description, and status. |

| ID | Use Case | Actors | Description |
|---|---|---|---|
| UC-46.04 | Update Job Category | Admin | Allows an administrator to modify existing job category details and status. |
| UC-46.05 | Delete Job Category | Admin | Allows an administrator to soft-delete a job category after verifying active job dependencies. |
| UC-46.06 | Restore Job Category | Admin | Allows an administrator to restore a soft-deleted job category. |
| UC-46.07 | View Job Category List | Freelancer | Allows freelancers to browse active job categories when filtering available jobs. |
| UC-46.08 | View Job Category Detail | Freelancer | Allows freelancers to view category information and open jobs under that category. |

# 3. Functional Requirements

## 3.1 System Functional Overview

### 3.1.1 Screens Flow

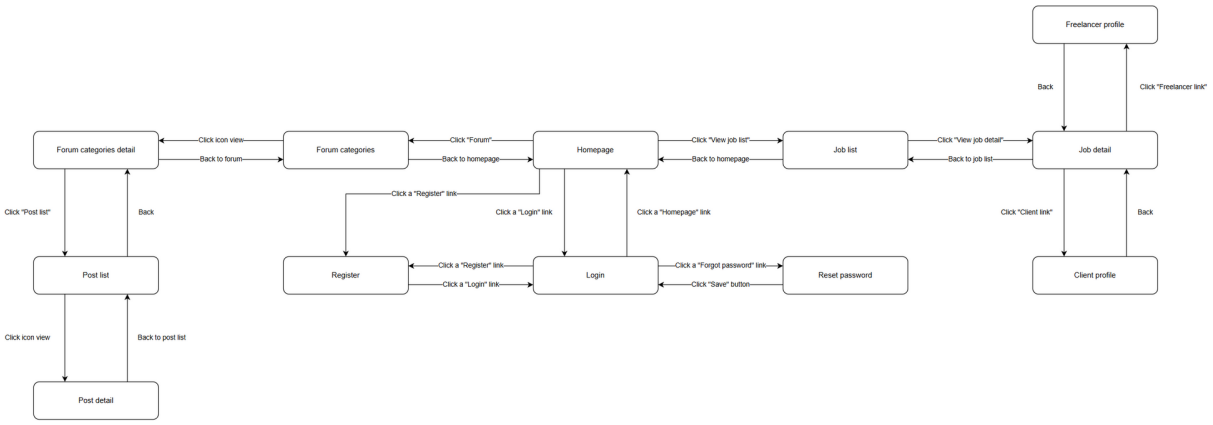


**Figure 23:** Screen Flow of Guest

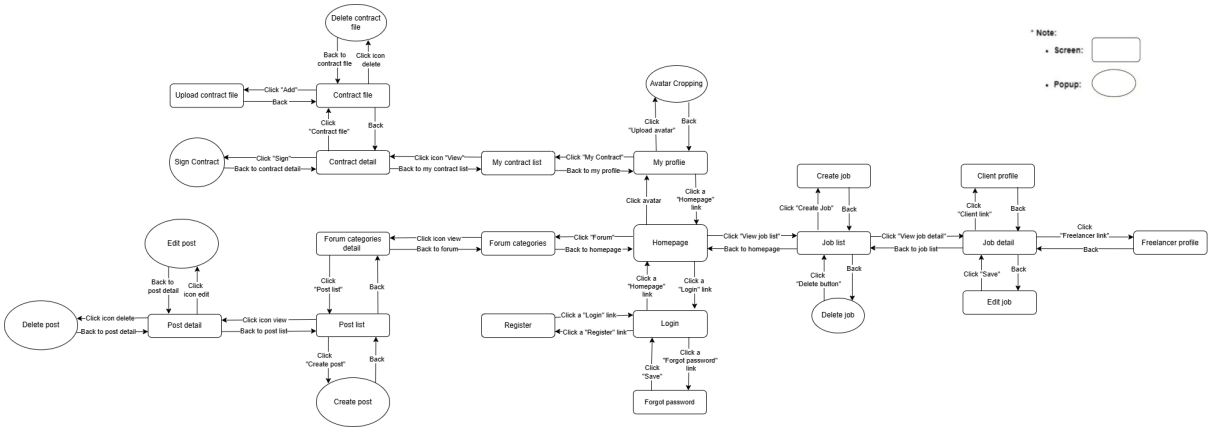


**Figure 24:** Screen Flow of Client

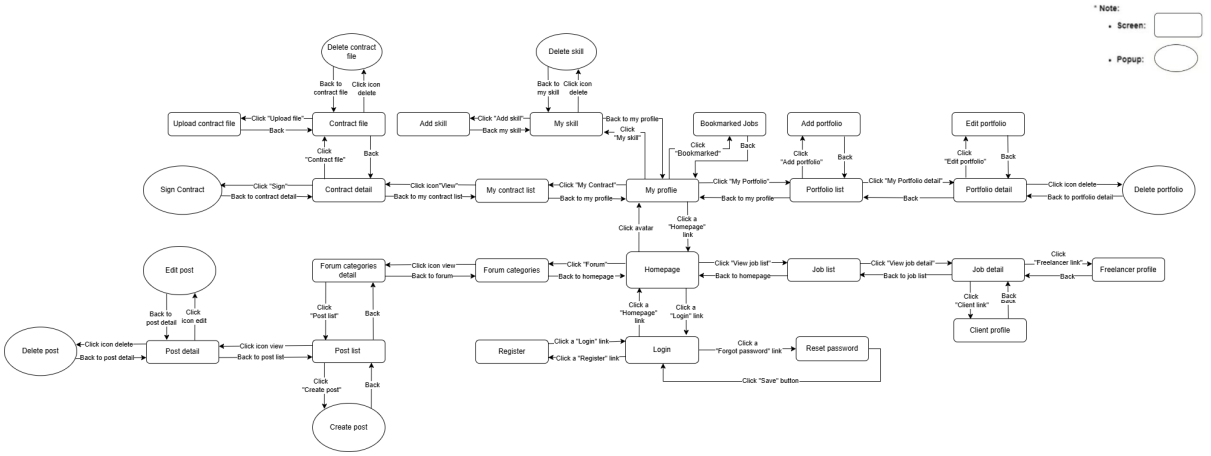


**Figure 25:** Screen Flow of Freelancer

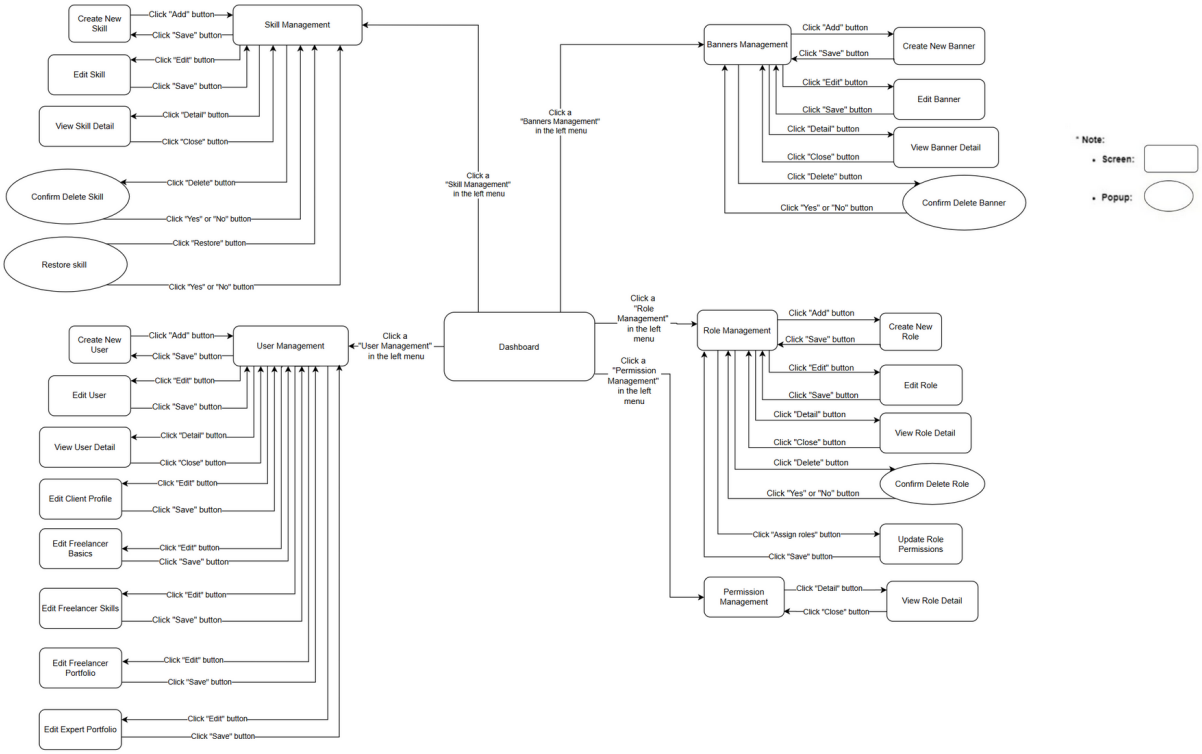


**Figure 26:** Screen Flow of Admin (Part 1)

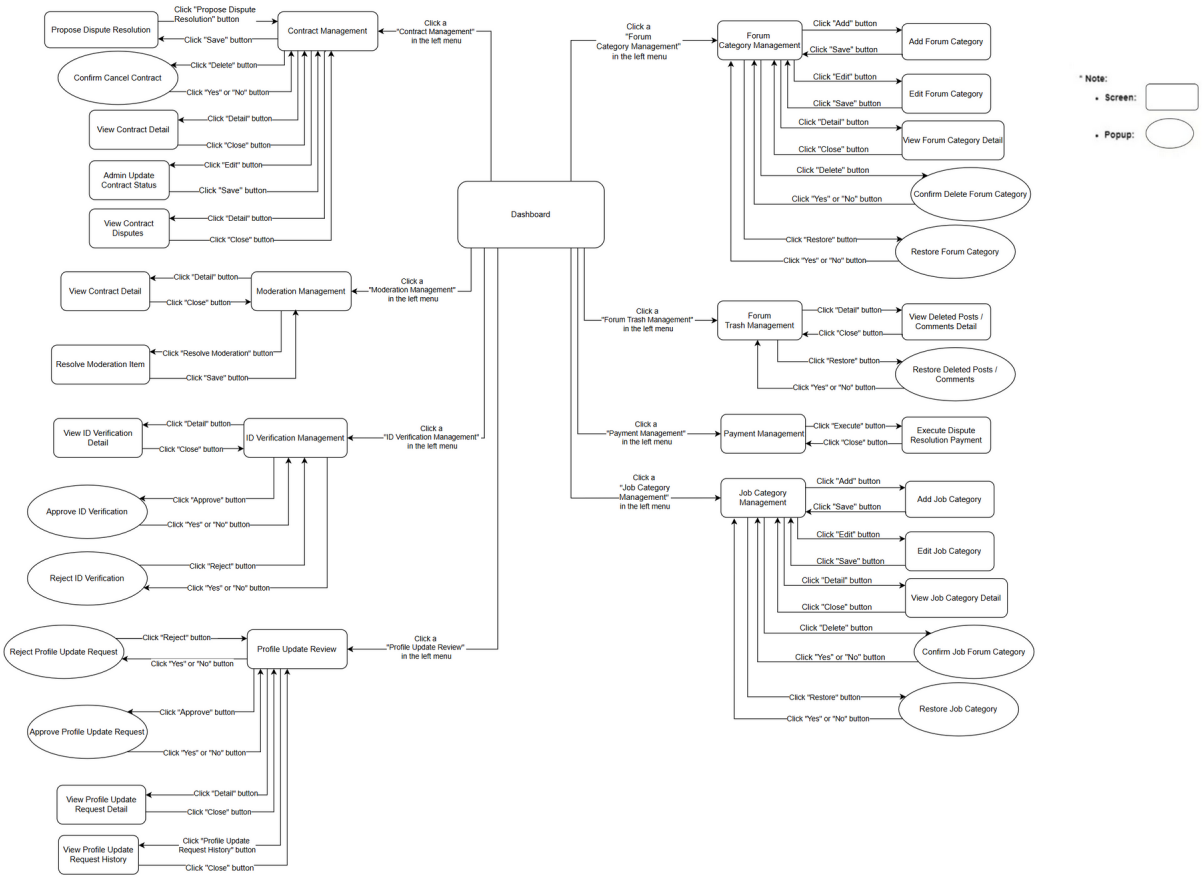


**Figure 27:** Screen Flow of Admin (Part 2)

### 3.1.2 Screen Descriptions

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 1 | Guest | Homepage | The main landing page introducing the platform, top job categories, featured job opportunities, and top-rated freelancers. |
| 2 | Guest | Login | Allows users to sign in using their registered email and password with secure JWT access and refresh token authentication. |
| 3 | Guest | Google Login | OAuth 2.0 third-party authentication interface allowing users to sign in or register instantly using their Google account. |
| 4 | Guest | Register | Allows new visitors to create an account by providing basic credentials and choosing their primary role (Client or Freelancer). |
| 5 | Guest | Forgot Password | Allows users to request a password reset email containing a secure, time-limited verification token. |
| 6 | Guest | Reset Password | Allows users to set a new account password after successfully verifying the reset link received in their email. |
| 7 | Guest | Job List | Public job directory displaying all active job postings with keyword search, category filtering, and budget sorting. |
| 8 | Guest | Job Detail | Displays full details of a specific job posting including title, description, budget, required skills, and client statistics. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 9 | Guest | Freelancer Profile | Public profile screen displaying a freelancer's professional title, bio, hourly rate, skills, reviews, and portfolio. |
| 10 | Guest | Client Profile | Public profile screen displaying a client's company description, posted jobs history, total spend, and feedback rating. |
| 11 | Guest | Forum Categories | Main forum overview screen displaying all discussion categories, total threads, and community activity statistics. |
| 12 | Guest | Forum Category Detail | Displays the list of discussion threads and topics contained within a selected forum category. |
| 13 | Guest | Post List | Displays all active discussion posts within a category with options to filter by popularity, latest activity, or keywords. |
| 14 | Guest | Post Detail | Displays the full text of a forum post, author information, timestamps, and the interactive threaded comments section. |
| 15 | Freelancer | My Profile | Central freelancer profile dashboard displaying bio, professional title, education, languages, hourly rate, and profile strength. |
| 16 | Freelancer | Edit Freelancer Profile | Form screen allowing freelancers to update their bio, professional title, education, availability status, and rate. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 17 | Freelancer | My Skills | Displays the full list of skills currently attached to the freelancer's profile along with proficiency levels. |
| 18 | Freelancer | Add Skill | Searchable modal dialog to select skills from the master catalog and attach them to the freelancer profile. |
| 19 | Freelancer | Delete Skill | Confirmation dialog allowing the freelancer to remove an attached skill from their profile. |
| 20 | Freelancer | Upload CV | Form screen allowing the freelancer to upload a curriculum vitae (CV/resume) document in PDF or DOCX format. |
| 21 | Freelancer | View CV | Document previewer interface allowing the freelancer to view and download their currently uploaded CV document. |
| 22 | Freelancer | Replace CV | Screen allowing the freelancer to select and upload a new CV document to overwrite their existing resume. |
| 23 | Freelancer | Delete CV | Confirmation dialog prompting the freelancer to confirm permanent deletion of their attached CV file. |
| 24 | Freelancer | Portfolio List | Displays a portfolio gallery of project showcases, case studies, and prior work samples created by the freelancer. |
| 25 | Freelancer | Add Portfolio | Form screen allowing the freelancer to create a new portfolio item with title, description, skills used, and media files. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 26 | Freelancer | Portfolio Detail | Modal/screen showing detailed case study information, full-resolution media, and live project URLs of a portfolio entry. |
| 27 | Freelancer | Edit Portfolio | Form screen allowing the freelancer to modify description, skills, links, or attachments of an existing portfolio item. |
| 28 | Freelancer | Delete Portfolio | Confirmation dialog prompting the freelancer to confirm the soft deletion of a portfolio project. |
| 29 | Freelancer | Identity Verification Upload | Screen allowing the freelancer to upload government-issued ID card/passport images and capture a selfie for eKYC verification. |
| 30 | Freelancer | Verification Status | Screen displaying the current status of the eKYC review (Pending Review, Approved, or Rejected with reason). |
| 31 | Freelancer | Bookmarked Jobs | Screen displaying all job postings that the freelancer has bookmarked for future review or proposal submission. |
| 32 | Freelancer | Saved Searches | Screen allowing the freelancer to view, execute, and manage custom saved job search filters and keyword queries. |
| 33 | Freelancer | Job Alerts | Screen allowing the freelancer to configure notification alerts for newly posted jobs matching specific skill criteria. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 34 | Freelancer | My Proposals List | Screen displaying all proposals submitted by the freelancer with current statuses (Pending, Accepted, Rejected). |
| 35 | Freelancer | Submit Proposal | Form screen allowing the freelancer to apply to a job with a cover letter, bid amount, delivery timeline, and milestones. |
| 36 | Freelancer | Proposal Detail | Detailed view of a submitted proposal showing proposed milestone breakdown, attachments, and client responses. |
| 37 | Freelancer | Edit Proposal | Form screen allowing the freelancer to adjust bid pricing, timeline, or cover letter of a pending proposal. |
| 38 | Freelancer | Withdraw Proposal | Confirmation dialog allowing the freelancer to withdraw an active proposal before client acceptance. |
| 39 | Freelancer | My Contract List | Screen displaying all contracts associated with the freelancer categorized by status (Active, Completed, Cancelled). |
| 40 | Freelancer | Contract Detail | Workspace displaying contract terms, milestone schedules, deadlines, escrow funding status, and activity timeline. |
| 41 | Freelancer | Sign Contract | Digital signature screen allowing the freelancer to formalize acceptance of the contract terms and deliverables. |
| 42 | Freelancer | Contract Files | Document repository displaying all project assets, specifications, and files shared within the contract workspace. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 43 | Freelancer | Upload Contract File | Form screen allowing the freelancer to upload deliverable files or reference documents to the contract repository. |
| 44 | Freelancer | Delete Contract File | Confirmation dialog allowing the freelancer to delete a file they uploaded to the contract workspace. |
| 45 | Freelancer | Milestone List | Displays all agreed milestone deliverables of the contract along with individual payment amounts and deadlines. |
| 46 | Freelancer | Milestone Detail | Screen displaying detailed instructions, attached files, progress logs, and submission history for a specific milestone. |
| 47 | Freelancer | Update Milestone Progress | Form screen allowing the freelancer to post progress notes, code links, or work-in-progress samples for a milestone. |
| 48 | Freelancer | Submit Milestone | Screen allowing the freelancer to formally submit finished deliverables and request client approval and payout. |
| 49 | Freelancer | Delete Milestone Submission | Confirmation dialog allowing the freelancer to cancel or retract a pending milestone submission. |
| 50 | Freelancer | Milestone Files | Dedicated file viewer showing assets and work items uploaded specifically for a contract milestone. |
| 51 | Freelancer | Open Dispute | Form screen allowing the freelancer to open a formal dispute regarding milestone deliverables or payment non-release. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 52 | Freelancer | Dispute Detail | Screen displaying dispute claims, evidence files, mediator messages, and resolution proposals from both parties. |
| 53 | Freelancer | Respond to Dispute | Form screen allowing the freelancer to submit rebuttal statements and supporting proof during arbitration. |
| 54 | Freelancer | Review List | Displays the list of feedback ratings and performance reviews received from clients upon contract completion. |
| 55 | Freelancer | Review Detail | Modal/screen displaying specific multi-category ratings, detailed client comments, and response timestamps. |
| 56 | Freelancer | Write Review | Form screen allowing the freelancer to rate and review the client after a contract has been completed. |
| 57 | Freelancer | Edit Review | Form screen allowing the freelancer to modify their submitted rating and comments within the allowed review window. |
| 58 | Freelancer | Respond to Review | Form screen allowing the freelancer to post an official public response to a client review on their profile. |
| 59 | Freelancer | Stripe Connect Onboarding | External redirect interface allowing the freelancer to set up their Stripe Express payout account and banking info. |
| 60 | Freelancer | Payout History | Screen displaying historical escrow earnings releases, transaction fees, and payout deposit statuses. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 61 | Freelancer | Payout Detail | Detailed receipt view for an individual payout transaction showing date, gross earnings, net transfer, and bank reference. |
| 62 | Freelancer | Conversation List | Messaging inbox displaying active direct chats with clients, unread counters, and snippet previews. |
| 63 | Freelancer | Direct Chat | Real-time messaging workspace supporting instant text chat, deliverable file attachments, and delivery receipts. |
| 64 | Freelancer | Notification List | Screen displaying user notifications for job invites, proposal updates, milestone payments, and forum interactions. |
| 65 | Freelancer | Switch Role | Quick switch control allowing the user to seamlessly transition between Freelancer and Client workspace views. |
| 66 | Freelancer | Join New Role | Registration screen allowing an existing single-role freelancer to activate the complementary Client role. |
| 67 | Client | Client Dashboard | Main workspace dashboard for clients showing active job postings, proposal counts, active contracts, and spent funds. |
| 68 | Client | Client Profile | Company profile screen displaying corporate name, website URL, industry description, and past hire ratings. |
| 69 | Client | Edit Company Information | Form screen allowing the client to update their business profile, logo, headquarters address, and company bio. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 70 | Client | Post Job | Multi-step job creation wizard allowing the client to define job title, category, required skills, budget, and scope. |
| 71 | Client | My Jobs | Screen displaying all job listings posted by the client with status filters (Draft, Open, In Progress, Closed, Cancelled). |
| 72 | Client | Edit Job | Form screen allowing the client to modify description, budget range, deadline, or skill tags of an open job posting. |
| 73 | Client | Delete Job | Confirmation modal prompting the client to confirm soft deletion or cancellation of an existing job posting. |
| 74 | Client | Change Job Status | Control action allowing the client to pause, close, or reopen an existing job posting. |
| 75 | Client | Job Proposals List | Screen displaying all incoming proposals submitted by freelancers for a specific job, with sorting and filtering. |
| 76 | Client | Proposal Detail (Client View) | Detailed evaluation screen displaying a freelancer's proposal, cover letter, rate, and portfolio match. |
| 77 | Client | Accept Proposal | Action modal to accept a freelancer's proposal and transition automatically into contract creation. |
| 78 | Client | Reject Proposal | Action dialog allowing the client to decline a proposal with optional constructive feedback. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 79 | Client | Create Contract | Form screen allowing the client to generate a formal contract with defined fixed or milestone payment structures. |
| 80 | Client | My Contract List (Client) | Screen displaying all current and archived contracts managed by the client with status indicators. |
| 81 | Client | Contract Detail (Client) | Central management interface for monitoring project milestones, communication, deliverables, and payments. |
| 82 | Client | Update Contract Terms | Form screen allowing the client to propose amendments to contract deadlines, scope, or milestone budgets. |
| 83 | Client | Sign Contract | Screen allowing the client to digitally sign and ratify the contract agreement. |
| 84 | Client | Cancel Contract | Screen allowing the client to request mutual contract termination and configure escrow refund allocations. |
| 85 | Client | Contract Files (Client) | Document repository displaying shared project specifications, requirements documents, and assets. |
| 86 | Client | Upload Contract File | Modal allowing the client to upload additional reference materials, NDAs, or briefs to the contract repository. |
| 87 | Client | Delete Contract File | Confirmation dialog prompting the client to remove an uploaded document from the contract repository. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 88 | Client | Milestone Management | Interface displaying the contract's milestone timeline, deposit status, and deliverable review actions. |
| 89 | Client | Fund Milestone | Secure checkout screen allowing the client to fund an upcoming milestone into escrow using Stripe. |
| 90 | Client | View Milestone Submissions | Review screen displaying work files, source repositories, or documentation submitted by the freelancer. |
| 91 | Client | Approve Milestone | Action screen enabling the client to approve accepted deliverables and authorize release of escrow funds. |
| 92 | Client | Request Milestone Changes | Form modal allowing the client to request revisions from the freelancer with detailed change instructions. |
| 93 | Client | Open Dispute (Client) | Form screen allowing the client to initiate formal arbitration when milestone deliverables fail to satisfy terms. |
| 94 | Client | Dispute Detail (Client) | Screen displaying the active dispute case, uploaded evidence, mediator commentary, and ruling proposals. |
| 95 | Client | Respond to Dispute | Form screen allowing the client to submit rebuttal evidence, project specifications, and timeline records. |
| 96 | Client | Pay Dispute Fee | Payment screen enabling the client to pay the arbitration fee deposit required for administrative dispute mediation. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 97 | Client | Payment History | Screen displaying comprehensive billing transactions, monthly statements, invoices, and platform service fees. |
| 98 | Client | Payment Detail | Detailed invoice view showing breakdown of milestone amount, processing fees, taxes, and payment method details. |
| 99 | Client | Saved Payment Methods | Wallet management screen allowing clients to securely add, view, and set default payment cards via Stripe. |
| 100 | Client | Favorite Freelancers | Screen displaying bookmarked freelancer profiles for rapid direct hiring, re-engagement, or job invitations. |
| 101 | Client | Following List | Screen showing all freelancers followed by the client to receive updates on newly published portfolio projects. |
| 102 | Client | Write Review (Client) | Form screen allowing the client to submit a performance rating and written evaluation for the hired freelancer. |
| 103 | Client | Edit Review (Client) | Form screen allowing the client to update their submitted rating or review text within the permissible editing window. |
| 104 | Client | Respond to Review (Client) | Form screen allowing the client to publish a formal response to feedback left by the freelancer. |
| 105 | Client | Social Links Management | Screen allowing the client to manage connected social media links and external business references. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 106 | Admin | Admin Dashboard | Executive dashboard displaying key system health metrics, daily active users, transaction volume, and alerts. |
| 107 | Admin | Admin Profile | Screen allowing the system administrator to view and edit administrative profile credentials and preferences. |
| 108 | Admin | Admin Change Password | Screen allowing the administrator to update their security password with multi-factor verification. |
| 109 | Admin | Role Management | Screen displaying all defined system access roles (Admin, Moderator, Client, Freelancer) and assigned permissions. |
| 110 | Admin | Create New Role | Form screen allowing administrators to create a new RBAC role with customized permission sets. |
| 111 | Admin | Edit Role | Form screen allowing administrators to rename a role, update its description, or modify its scope of authority. |
| 112 | Admin | View Role Detail | Screen showing detailed permissions, system capabilities, and user counts associated with a specific role. |
| 113 | Admin | Confirm Delete Role | Confirmation dialog prompting the administrator to confirm the permanent removal of a role. |
| 114 | Admin | Update Role Permissions | Matrix configuration screen allowing granular assignment of system capabilities and endpoints to a role. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 115 | Admin | Permission Management | Screen displaying the comprehensive system permission catalog organized by functional domain module. |
| 116 | Admin | View Permission Detail | Modal showing underlying API endpoints, scope parameters, and roles currently granting a specific permission. |
| 117 | Admin | User Management | Screen displaying all registered users with filters by role, email verification status, and account state. |
| 118 | Admin | Create New User | Form screen allowing administrators to manually provision a user account with specific roles and attributes. |
| 119 | Admin | Edit User | Form screen allowing administrators to modify user email, account status (Active, Suspended, Banned), and roles. |
| 120 | Admin | View User Detail | Comprehensive profile view showing user activity history, linked contracts, billing records, and audit logs. |
| 121 | Admin | Edit Client Profile (Admin) | Form screen allowing administrators to update company profile attributes, verification badges, and notes. |
| 122 | Admin | Edit Freelancer Basics | Form screen allowing administrators to update a freelancer's headline, bio, hourly rate, and profile status. |
| 123 | Admin | Edit Freelancer Skills | Screen allowing administrators to review, add, or prune skills associated with a freelancer's profile. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 124 | Admin | Edit Freelancer Portfolio | Screen allowing administrators to inspect, edit, or remove inappropriate or copyrighted portfolio items. |
| 125 | Admin | Edit Expert Profile | Form screen allowing administrators to manage verified expert status, endorsements, and homepage spotlight details. |
| 126 | Admin | Banners Management | Screen displaying promotional and announcement banners with active schedules, impressions, and target links. |
| 127 | Admin | Create New Banner | Form screen allowing administrators to configure a new banner campaign with image asset, placement, and dates. |
| 128 | Admin | Upload Banner Image | Media upload interface with image cropping for hero, sidebar, and notification banner placements. |
| 129 | Admin | Edit Banner | Form screen allowing administrators to modify copy, destination link, display order, or active state of a banner. |
| 130 | Admin | View Banner Detail | Screen displaying detailed engagement analytics, click-through rates, and live preview of a banner. |
| 131 | Admin | Confirm Delete Banner | Confirmation dialog prompting the administrator to verify permanent deletion of a banner. |
| 132 | Admin | Skill Management | Screen displaying the master skill catalog utilized by the platform for freelancer profiles and job tags. |
| 133 | Admin | Create New Skill | Form screen allowing administrators to register a new standardized skill name and categorize it. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 134 | Admin | Edit Skill | Form screen allowing administrators to rename an existing skill or reassign it to a different category. |
| 135 | Admin | View Skill Detail | Screen displaying skill metadata, category relations, and total freelancer and job associations. |
| 136 | Admin | Confirm Delete Skill | Confirmation dialog prompting the administrator to soft-delete a skill from the active selection list. |
| 137 | Admin | Restore Skill | Action dialog allowing administrators to restore a previously soft-deleted skill back to active availability. |
| 138 | Admin | Job Category Management | Screen displaying the hierarchical job category classification tree and industry taxonomies. |
| 139 | Admin | Add Job Category | Form screen allowing administrators to create a new job category with title, slug, icon, and parent node. |
| 140 | Admin | Edit Job Category | Form screen allowing administrators to update the title, description, or hierarchy of an existing category. |
| 141 | Admin | View Job Category Detail | Screen displaying category statistics including active job postings, freelancer counts, and revenue. |
| 142 | Admin | Confirm Delete Job Category | Confirmation dialog prompting the administrator to soft-delete an existing job category. |
| 143 | Admin | Restore Job Category | Action dialog allowing administrators to restore a previously soft-deleted job category. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 144 | Admin | Forum Category Management | Screen displaying all community forum categories, permission rules, and moderation assignments. |
| 145 | Admin | Add Forum Category | Form screen allowing administrators to configure a new discussion section with name, description, and icon. |
| 146 | Admin | Edit Forum Category | Form screen allowing administrators to update forum category titles, display ordering, or access rules. |
| 147 | Admin | View Forum Category Detail | Screen displaying detailed forum category metrics, thread volume, active participants, and post frequency. |
| 148 | Admin | Confirm Delete Forum Category | Confirmation dialog prompting the administrator to soft-delete a forum category. |
| 149 | Admin | Restore Forum Category | Action dialog allowing administrators to reactivate a previously soft-deleted forum category. |
| 150 | Admin | Moderation Queue | Screen displaying flagged user reports, reported offensive posts, spam comments, and policy violation cases. |
| 151 | Admin | View Moderation Item Detail | Screen displaying original reported content, reporter comments, automated AI toxicity scores, and context. |
| 152 | Admin | Resolve Moderation Item | Action screen allowing administrators to dismiss flags, issue user warnings, delete content, or suspend accounts. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 153 | Admin | Forum Trash Management | Screen displaying all soft-deleted forum threads and comments awaiting administrative review or permanent purge. |
| 154 | Admin | View Deleted Posts/Comments Detail | Screen showing original deleted post text, deletion metadata, reason, and author history. |
| 155 | Admin | Restore Deleted Posts/Comments | Action dialog allowing administrators to recover and reinstate accidentally or incorrectly removed forum posts. |
| 156 | Admin | ID Verification Management | Screen displaying queue of pending eKYC identity submissions awaiting official verification review. |
| 157 | Admin | View ID Verification Detail | Screen displaying government ID photos, automated face match score, OCR data, and liveness pose validation. |
| 158 | Admin | Approve ID Verification | Action dialog allowing administrators to verify identity documents and assign the Verified Freelancer badge. |
| 159 | Admin | Reject ID Verification | Action dialog allowing administrators to reject an eKYC submission with specific rejection guidance to the user. |
| 160 | Admin | Profile Update Review | Screen displaying incoming requests from freelancers to update verified identity and sensitive profile fields. |
| 161 | Admin | View Profile Update Request Detail | Side-by-side comparison screen highlighting changes between current profile data and requested modifications. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 162 | Admin | View Profile Update Request History | Audit log screen showing previously processed profile update requests, administrative decisions, and notes. |
| 163 | Admin | Approve Profile Update Request | Action dialog allowing administrators to accept changes and immediately publish them to the live profile. |
| 164 | Admin | Reject Profile Update Request | Action dialog allowing administrators to decline requested revisions with feedback provided to the applicant. |
| 165 | Admin | Admin Contract Oversight | Screen displaying all platform contracts across all users with filters for status, date, and dispute flags. |
| 166 | Admin | Admin View Contract Detail | Comprehensive audit screen displaying full contract history, escrow transactions, milestones, and dispute logs. |
| 167 | Admin | Admin Update Contract Status | Action screen enabling administrators to override contract status in exceptional administrative scenarios. |
| 168 | Admin | View Contract Disputes | Screen displaying all escalated disputes requiring third-party administrative mediation and binding rulings. |
| 169 | Admin | Propose Dispute Resolution | Form screen allowing administrator mediator to propose binding escrow fund distribution between the parties. |

| No | Actor | Screen Name | Description |
|---|---|---|---|
| 170 | Admin | Confirm Cancel Contract | Confirmation dialog enabling administrators to terminate stalled or abandoned contracts and refund escrow. |
| 171 | Admin | Execute Dispute Resolution Payment | Screen allowing administrators to release escrow balances and execute final payouts per the dispute ruling. |
| 172 | Admin | Session Management (Admin) | Screen displaying active user login sessions across devices with client IP, user agent, and force-logout controls. |

### 3.1.3 Screen Authorization

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 1 | Homepage | ✓ | ✓ | ✓ | |
| 2 | Login | ✓ | | | |
| 3 | Google Login | ✓ | | | |
| 4 | Register | ✓ | | | |
| 5 | Forgot Password | ✓ | ✓ | ✓ | ✓ |
| 6 | Reset Password | ✓ | ✓ | ✓ | ✓ |
| 7 | Job List | ✓ | ✓ | ✓ | |
| 8 | Job Detail | ✓ | ✓ | ✓ | |
| 9 | Freelancer Profile | ✓ | ✓ | ✓ | |
| 10 | Client Profile | ✓ | ✓ | ✓ | |
| 11 | Forum Categories | ✓ | ✓ | ✓ | ✓ |
| 12 | Forum Category Detail | ✓ | ✓ | ✓ | ✓ |
| 13 | Post List | ✓ | ✓ | ✓ | ✓ |
| 14 | Post Detail | ✓ | ✓ | ✓ | ✓ |
| 15 | My Profile | | | ✓ | |

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 16 | Edit Freelancer Profile | | | ✓ | |
| 17 | My Skills | | | ✓ | |
| 18 | Add Skill | | | ✓ | |
| 19 | Delete Skill | | | ✓ | |
| 20 | Upload CV | | | ✓ | |
| 21 | View CV | | | ✓ | |
| 22 | Replace CV | | | ✓ | |
| 23 | Delete CV | | | ✓ | |
| 24 | Portfolio List | | | ✓ | |
| 25 | Add Portfolio | | | ✓ | |
| 26 | Portfolio Detail | ✓ | ✓ | ✓ | |
| 27 | Edit Portfolio | | | ✓ | |
| 28 | Delete Portfolio | | | ✓ | |
| 29 | Identity Verification Upload | | | ✓ | |
| 30 | Verification Status | | | ✓ | |
| 31 | Bookmarked Jobs | | | ✓ | |
| 32 | Saved Searches | | | ✓ | |
| 33 | Job Alerts | | | ✓ | |
| 34 | My Proposals List | | | ✓ | |
| 35 | Submit Proposal | | | ✓ | |
| 36 | Proposal Detail | | ✓ | ✓ | |
| 37 | Edit Proposal | | | ✓ | |
| 38 | Withdraw Proposal | | | ✓ | |
| 39 | My Contract List | | | ✓ | |
| 40 | Contract Detail | | ✓ | ✓ | ✓ |
| 41 | Sign Contract | | ✓ | ✓ | |
| 42 | Contract Files | | ✓ | ✓ | ✓ |
| 43 | Upload Contract File | | ✓ | ✓ | |

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 44 | Delete Contract File | | ✓ | ✓ | ✓ |
| 45 | Milestone List | | ✓ | ✓ | ✓ |
| 46 | Milestone Detail | | ✓ | ✓ | ✓ |
| 47 | Update Milestone Progress | | | ✓ | |
| 48 | Submit Milestone | | | ✓ | |
| 49 | Delete Milestone Submission | | | ✓ | |
| 50 | Milestone Files | | ✓ | ✓ | ✓ |
| 51 | Open Dispute | | ✓ | ✓ | |
| 52 | Dispute Detail | | ✓ | ✓ | ✓ |
| 53 | Respond to Dispute | | ✓ | ✓ | |
| 54 | Review List | ✓ | ✓ | ✓ | ✓ |
| 55 | Review Detail | ✓ | ✓ | ✓ | ✓ |
| 56 | Write Review | | ✓ | ✓ | |
| 57 | Edit Review | | ✓ | ✓ | |
| 58 | Respond to Review | | ✓ | ✓ | |
| 59 | Stripe Connect Onboarding | | | ✓ | |
| 60 | Payout History | | | ✓ | |
| 61 | Payout Detail | | | ✓ | |
| 62 | Conversation List | | ✓ | ✓ | |
| 63 | Direct Chat | | ✓ | ✓ | |
| 64 | Notification List | | ✓ | ✓ | ✓ |
| 65 | Switch Role | | ✓ | ✓ | |
| 66 | Join New Role | | ✓ | ✓ | |
| 67 | Client Dashboard | | ✓ | | |
| 68 | Client Profile | ✓ | ✓ | ✓ | |
| 69 | Edit Company Information | | ✓ | | |
| 70 | Post Job | | ✓ | | |
| 71 | My Jobs | | ✓ | | |

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 72 | Edit Job | | ✓ | | |
| 73 | Delete Job | | ✓ | | |
| 74 | Change Job Status | | ✓ | | |
| 75 | Job Proposals List | | ✓ | | |
| 76 | Proposal Detail (Client View) | | ✓ | | |
| 77 | Accept Proposal | | ✓ | | |
| 78 | Reject Proposal | | ✓ | | |
| 79 | Create Contract | | ✓ | | |
| 80 | My Contract List (Client) | | ✓ | | |
| 81 | Contract Detail (Client) | | ✓ | | |
| 82 | Update Contract Terms | | ✓ | | |
| 83 | Sign Contract | | ✓ | ✓ | |
| 84 | Cancel Contract | | ✓ | ✓ | ✓ |
| 85 | Contract Files (Client) | | ✓ | | |
| 86 | Upload Contract File | | ✓ | ✓ | |
| 87 | Delete Contract File | | ✓ | ✓ | ✓ |
| 88 | Milestone Management | | ✓ | | |
| 89 | Fund Milestone | | ✓ | | |
| 90 | View Milestone Submissions | | ✓ | ✓ | ✓ |
| 91 | Approve Milestone | | ✓ | | |
| 92 | Request Milestone Changes | | ✓ | | |
| 93 | Open Dispute (Client) | | ✓ | | |
| 94 | Dispute Detail (Client) | | ✓ | | |
| 95 | Respond to Dispute | | ✓ | ✓ | |
| 96 | Pay Dispute Fee | | ✓ | ✓ | |
| 97 | Payment History | | ✓ | ✓ | |
| 98 | Payment Detail | | ✓ | ✓ | |
| 99 | Saved Payment Methods | | ✓ | | |

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 100 | Favorite Freelancers | | ✓ | | |
| 101 | Following List | | ✓ | | |
| 102 | Write Review (Client) | | ✓ | | |
| 103 | Edit Review (Client) | | ✓ | | |
| 104 | Respond to Review (Client) | | ✓ | | |
| 105 | Social Links Management | | ✓ | ✓ | |
| 106 | Admin Dashboard | | | | ✓ |
| 107 | Admin Profile | | | | ✓ |
| 108 | Admin Change Password | | | | ✓ |
| 109 | Role Management | | | | ✓ |
| 110 | Create New Role | | | | ✓ |
| 111 | Edit Role | | | | ✓ |
| 112 | View Role Detail | | | | ✓ |
| 113 | Confirm Delete Role | | | | ✓ |
| 114 | Update Role Permissions | | | | ✓ |
| 115 | Permission Management | | | | ✓ |
| 116 | View Permission Detail | | | | ✓ |
| 117 | User Management | | | | ✓ |
| 118 | Create New User | | | | ✓ |
| 119 | Edit User | | | | ✓ |
| 120 | View User Detail | | | | ✓ |
| 121 | Edit Client Profile (Admin) | | | | ✓ |
| 122 | Edit Freelancer Basics | | | | ✓ |
| 123 | Edit Freelancer Skills | | | | ✓ |
| 124 | Edit Freelancer Portfolio | | | | ✓ |
| 125 | Edit Expert Profile | | | | ✓ |
| 126 | Banners Management | | | | ✓ |
| 127 | Create New Banner | | | | ✓ |

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 128 | Upload Banner Image | | | | ✓ |
| 129 | Edit Banner | | | | ✓ |
| 130 | View Banner Detail | | | | ✓ |
| 131 | Confirm Delete Banner | | | | ✓ |
| 132 | Skill Management | | | | ✓ |
| 133 | Create New Skill | | | | ✓ |
| 134 | Edit Skill | | | | ✓ |
| 135 | View Skill Detail | | | | ✓ |
| 136 | Confirm Delete Skill | | | | ✓ |
| 137 | Restore Skill | | | | ✓ |
| 138 | Job Category Management | | | | ✓ |
| 139 | Add Job Category | | | | ✓ |
| 140 | Edit Job Category | | | | ✓ |
| 141 | View Job Category Detail | | | | ✓ |
| 142 | Confirm Delete Job Category | | | | ✓ |
| 143 | Restore Job Category | | | | ✓ |
| 144 | Forum Category Management | | | | ✓ |
| 145 | Add Forum Category | | | | ✓ |
| 146 | Edit Forum Category | | | | ✓ |
| 147 | View Forum Category Detail | | | | ✓ |
| 148 | Confirm Delete Forum Category | | | | ✓ |
| 149 | Restore Forum Category | | | | ✓ |
| 150 | Moderation Queue | | | | ✓ |
| 151 | View Moderation Item Detail | | | | ✓ |
| 152 | Resolve Moderation Item | | | | ✓ |
| 153 | Forum Trash Management | | | | ✓ |

| No | Screen / Page | Guest | Client | Freelancer | Admin |
|---|---|---|---|---|---|
| 154 | View Deleted Posts/Comments Detail | | | | ✓ |
| 155 | Restore Deleted Posts/Comments | | | | ✓ |
| 156 | ID Verification Management | | | | ✓ |
| 157 | View ID Verification Detail | | | | ✓ |
| 158 | Approve ID Verification | | | | ✓ |
| 159 | Reject ID Verification | | | | ✓ |
| 160 | Profile Update Review | | | | ✓ |
| 161 | View Profile Update Request Detail | | | | ✓ |
| 162 | View Profile Update Request History | | | | ✓ |
| 163 | Approve Profile Update Request | | | | ✓ |
| 164 | Reject Profile Update Request | | | | ✓ |
| 165 | Admin Contract Oversight | | | | ✓ |
| 166 | Admin View Contract Detail | | | | ✓ |
| 167 | Admin Update Contract Status | | | | ✓ |
| 168 | View Contract Disputes | | | | ✓ |
| 169 | Propose Dispute Resolution | | | | ✓ |
| 170 | Confirm Cancel Contract | | | | ✓ |
| 171 | Execute Dispute Resolution Payment | | | | ✓ |
| 172 | Session Management (Admin) | | | | ✓ |

### 3.1.4 Non-Screen Functions

| # | Feature | System Function | Description |
|---|---|---|---|
| 1 | **Manage Authentication** | [POST] /api/auth/register<br>[POST] /api/auth/otp<br>[POST] /api/auth/login<br>[POST] /api/auth/refresh-token<br>[POST] /api/auth/logout<br>[POST] /api/auth/forgot-password<br>[GET] /api/auth/google-link<br>[GET] /api/auth/google/callback<br>[GET] /api/auth/me<br>[POST] /api/auth/switch-role<br>[POST] /api/auth/join-role | The Authentication feature handles user identity verification and session establishment. It supports account registration with OTP verification, email/password login with JWT access and refresh tokens, token renewal, logout, and password reset through an email verification link. Users can also sign in with Google OAuth, retrieve the currently authenticated account, and switch or join a role (Client or Freelancer) on the same account. |
| 2 | **Manage Account Profile** | [GET] /api/account-profile<br>[PUT] /api/account-profile<br>[PUT] /api/account-profile/password<br>[POST] /api/account-profile/avatar<br>[GET] /api/account-profile/avatar/{userId}/file<br>[GET] /api/clients/{userId}<br>[PUT] /api/clients/me/profile<br>[GET] /api/social-links<br>[POST] /api/social-links<br>[DELETE] /api/social-links/{id}<br>[GET] /api/favorites/freelancers<br>[POST] /api/favorites/freelancers/{freelancerId}<br>[DELETE] /api/favorites/freelancers/{freelancerId}<br>[GET] /api/following/freelancers/discover<br>[GET] /api/following/freelancers<br>[POST] /api/following/freelancers/{freelancerId}<br>[DELETE] /api/following/freelancers/{freelancerId} | The Account Profile feature lets authenticated users manage their account-level information and social network. Users can view and update their account profile, change their password, upload and retrieve their avatar, and manage social links. Clients can view another client's public profile and update their own client profile. Users can also save favorite freelancers, discover and follow freelancers, and unfollow or remove them from the favorites list. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 3 | **Manage Freelancer Profile** | [GET] /api/profiles/{id}<br>[PUT] /api/profiles/{id}<br>[POST] /api/profiles/{id}/cv<br>[GET] /api/profiles/{id}/cv/file<br>[POST] /api/profiles/{id}/cv/analyze<br>[DELETE] /api/profiles/{id}/cv<br>[GET] /api/expert-profile/me<br>[PUT] /api/expert-profile/me<br>[PATCH] /api/users/{id}/expert-profile<br>[GET] /api/experts<br>[GET] /api/experts/{id} | The Freelancer Profile feature enables freelancers to build and maintain their professional identity, including bio, professional title, education, availability status and profile strength. Freelancers can upload, download, delete and AI-analyze their CV. The feature also exposes the expert (verified freelancer) profile endpoints, the public expert directory, and allows a freelancer to view and update their own expert profile while administrators can update it on behalf of a user. |
| 4 | **Manage Freelancer Skills** | [GET] /api/profiles/{id}/skills<br>[GET] /api/profiles/skills/suggestions<br>[POST] /api/profiles/{id}/skills<br>[DELETE] /api/profiles/skills/{id} | The Freelancer Skills feature allows freelancers to define and showcase their professional capabilities. Users can view the skills attached to a profile, search the skill catalog for suggestions while typing, add a skill together with its proficiency level, and remove a skill that is no longer relevant. This keeps each freelancer's skill set accurate for the client's talent evaluation process. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 5 | **Manage Freelancer Portfolio** | [GET] /api/profiles/{id}/portfolios<br>[POST] /api/profiles/{id}/portfolios<br>[GET] /api/profiles/portfolios/{id}<br>[PUT] /api/profiles/portfolios/{id}<br>[DELETE] /api/profiles/portfolios/{id} | The Freelancer Portfolio feature enables freelancers to demonstrate their past work visually. Freelancers can list every portfolio item of a profile, create a new item with project details and media, view the detail of a specific item, update its content or media, and soft-delete an item so that it is no longer displayed publicly. A well-organized portfolio helps freelancers build credibility with prospective clients. |
| 6 | **Manage Job** | [GET] /api/jobs<br>[GET] /api/jobs/{slug}<br>[GET] /api/job-categories<br>[GET] /api/job-categories/{slug}<br>[GET] /api/manage-jobs/skills<br>[GET] /api/manage-jobs<br>[GET] /api/manage-jobs/{id}<br>[POST] /api/manage-jobs<br>[PATCH] /api/manage-jobs/{id}<br>[DELETE] /api/manage-jobs/{id}<br>[PATCH] /api/manage-jobs/{id}/status<br>[GET] /api/manage-jobs/bookmarks<br>[POST] /api/manage-jobs/jobs/{slug}/bookmark<br>[GET] /api/manage-jobs/jobs/{slug}/bookmark<br>[DELETE] /api/manage-jobs/bookmarks/{slug} | The Job Management feature is the core marketplace functionality that connects clients with freelancers. Guests and freelancers can browse and search the paginated job list, view a job detail by slug, and browse job categories. Clients can create a job posting with description, budget, required skills and timeline, list their own jobs, view a job detail, update or delete a job, and change its status (Open, In Progress, Closed, Cancelled); they can also look up skills while composing a job. Freelancers can list, add and remove bookmarked jobs and check whether a job is bookmarked. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 7 | **Manage Job Categories (Admin)** | [GET] /api/admin/job-categories<br>[GET] /api/admin/job-categories/{id}<br>[POST] /api/admin/job-categories<br>[PATCH] /api/admin/job-categories/{id}<br>[PATCH] /api/admin/job-categories/{id}/ restore<br>[DELETE] /api/admin/job-categories/{id} | The Job Category Management feature allows administrators to maintain the taxonomy used to classify job postings. Administrators can list all categories, view the detail of a category, create a new category, update its name or description, soft-delete a category, and restore a previously deleted category when it is needed again. |
| 8 | **Manage Saved Searches** | [GET] /api/saved-searches<br>[GET] /api/saved-searches/{id}<br>[POST] /api/saved-searches<br>[PATCH] /api/saved-searches/{id}<br>[DELETE] /api/saved-searches/{id} | The Saved Search feature allows authenticated users to store job search criteria for later reuse. Users can list their saved searches, view the detail of one saved search, create a new saved search from the current filters, update its name or filter conditions, and delete a saved search that is no longer needed. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 9 | **Manage Proposals** | [GET] /api/proposals/my<br>[GET] /api/proposals/jobs/{jobId}/mine<br>[GET] /api/proposals/jobs/{jobId}/submitted<br>[GET] /api/jobs/{jobId}/proposals<br>[GET] /api/proposals/{id}<br>[GET] /api/proposals/{id}/client<br>[POST] /api/proposals/jobs/{jobId}<br>[POST] /api/proposals/jobs/{jobId}/drafts<br>[PATCH] /api/proposals/{id}/draft<br>[PATCH] /api/proposals/{id}/submit<br>[PATCH] /api/proposals/{id}/withdraw<br>[PATCH] /api/proposals/{id}/interview<br>[PATCH] /api/proposals/{id}/reject<br>[PATCH] /api/proposals/{id}/accept | The Proposal Management feature supports the bidding process between freelancers and clients. Freelancers can submit a proposal to a job, save it as a draft, update and submit a draft, withdraw a submitted proposal, and view their own proposals for a job. Clients can list all proposals submitted to their job, view proposal details, and move a proposal through interview, acceptance or rejection, while acceptance can later be used to create a contract. |
| 10 | **Manage Contract** | [GET] /api/contracts<br>[GET] /api/contracts/{id}<br>[POST] /api/contracts<br>[PATCH] /api/contracts/{id}<br>[PATCH] /api/contracts/{id}/sign<br>[PATCH] /api/contracts/{id}/complete<br>[PATCH] /api/contracts/{id}/cancel<br>[GET] /api/contracts/{contractId}/files<br>[POST] /api/contracts/{contractId}/files<br>[DELETE] /api/contracts/{contractId}/files/{fileId} | The Contract Management feature manages the agreement between a client and a freelancer throughout the project lifecycle. Users can create a contract, list their contracts, view contract details, update contract terms, sign the contract to confirm participation, complete the contract and request cancellation. Contract participants can also list, upload and delete contract-level files that document the agreement. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 11 | **Manage Milestone** | [GET] /api/contracts/{contractId}/milestones<br>[GET] /api/contracts/{contractId}/milestones/{id}<br>[POST] /api/contracts/{contractId}/milestones<br>[PATCH] /api/contracts/{contractId}/milestones/{id}<br>[PATCH] /api/contracts/{contractId}/milestones/{id}/progress<br>[DELETE] /api/contracts/{contractId}/milestones/{id} | The Milestone Management feature breaks a contract into payable deliverables. Contract participants can list and view the milestones of a contract, create a new milestone with its amount and due date, update milestone information, report progress on a milestone, and delete a milestone that is no longer part of the agreement. |
| 12 | **Manage Milestone Files** | [GET] /api/contracts/{contractId}/milestones/{milestoneId}/files<br>[POST] /api/contracts/{contractId}/milestones/{milestoneId}/files<br>[DELETE] /api/contracts/{contractId}/milestones/{milestoneId}/files/{fileId} | The Milestone File feature allows contract participants to attach supporting documents and media to a specific milestone. Users can list every file of a milestone, upload a new file, and delete a file that was uploaded by mistake or is no longer relevant. |
| 13 | **Manage Milestone Submissions** | [POST] /api/contracts/{contractId}/milestones/{milestoneId}/submissions<br>[GET] /api/contracts/{contractId}/milestones/{milestoneId}/submissions<br>[GET] /api/contracts/{contractId}/milestones/{milestoneId}/submissions/{id}<br>[PATCH] /api/contracts/{contractId}/milestones/{milestoneId}/submissions/{id}/request-changes<br>[PATCH] /api/contracts/{contractId}/milestones/{milestoneId}/submissions/{id}/approve | The Milestone Submission feature supports the deliverable review workflow. A freelancer can submit work for a milestone, and both parties can list the submissions of a milestone and view the detail of one submission. The client can request changes so that the freelancer revises and resubmits the work, or approve the submission to complete the milestone and release the corresponding payment. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 14 | **Manage Payments** | [POST] /api/payments/connect/onboarding-link<br>[GET] /api/payments/connect/status<br>[GET] /api/payments/payment-methods<br>[POST] /api/payments/customer-portal<br>[POST] /api/payments/contracts/{contractId}/platform-fee/intent<br>[POST] /api/payments/contracts/{contractId}/platform-fee/checkout-session<br>[POST] /api/payments/contracts/{contractId}/platform-fee/saved-card<br>[POST] /api/payments/contracts/{contractId}/milestones/{milestoneId}/fund/intent<br>[POST] /api/payments/contracts/{contractId}/milestones/{milestoneId}/fund/checkout-session<br>[POST] /api/payments/contracts/{contractId}/milestones/{milestoneId}/fund/saved-card<br>[POST] /api/payments/contracts/{contractId}/milestones/{milestoneId}/release<br>[POST] /api/payments/contracts/{contractId}/milestones/{milestoneId}/refund<br>[POST] /api/payments/disputes/{disputeId}/fee/intent<br>[POST] /api/payments/disputes/{disputeId}/fee/checkout-session<br>[POST] /api/payments/disputes/{disputeId}/settle<br>[GET] /api/payments/transactions<br>[POST] /api/payments/intents/{paymentIntentId}/sync<br>[POST] /api/payments/checkout-sessions/{sessionId}/sync<br>[POST] /api/payments/webhook | The Payment Management feature integrates Stripe to provide secure escrow-style transactions. Users can connect a Stripe account and check its onboarding status, list saved payment methods, open the Stripe customer portal, and pay the platform fee or fund a milestone either through a PaymentIntent, a checkout session, or a saved card. Clients can release or refund milestone funds, pay a dispute fee and settle a dispute, and view their transaction history. The system also synchronizes PaymentIntent and CheckoutSession status and receives asynchronous Stripe webhook events. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 15 | **Manage Dispute** | [GET] /api/disputes<br>[POST] /api/disputes<br>[GET] /api/disputes/{id}<br>[POST] /api/disputes/{id}/fee<br>[POST] /api/disputes/{id}/response<br>[GET] /api/disputes/milestone/{milestoneId}<br>[POST] /api/disputes/{id}/review<br>[GET] /api/admin/disputes<br>[GET] /api/admin/disputes/{id}<br>[POST] /api/admin/disputes/{id}/decision<br>[POST] /api/admin/disputes/{id}/<br>final-decision | The Dispute Management feature handles conflicts that arise during a milestone. Participants can open a dispute for a milestone, list and view their disputes, pay the dispute fee, submit a response to the other party's claim, and review the proposed resolution. The dispute of a specific milestone can be looked up. Administrators can list all disputes, view dispute details, issue a decision, and record the final decision that settles the conflict. |
| 16 | **Manage Review** | [GET] /api/reviews/contracts/{contractId}<br>[POST] /api/reviews/contracts/{contractId}<br>[GET] /api/reviews/{reviewId}<br>[PATCH] /api/reviews/{reviewId}<br>[DELETE] /api/reviews/{reviewId}<br>[POST] /api/reviews/{reviewId}/response<br>[PATCH] /api/reviews/responses/{responseId}<br>[DELETE] /api/reviews/responses/<br>{responseId} | The Review Management feature lets contract participants rate and comment on each other after a contract is completed. Users can list the reviews of a contract, create a review, view, update and delete their own review, and respond to a review they received. Responses can also be updated or deleted by their author, helping both parties maintain a trustworthy reputation on the platform. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 17 | **Manage Forum** | [GET] /api/forums/categories<br>[GET] /api/forums/categories/top<br>[GET] /api/forums/categories/{id}<br>[GET] /api/forums/categories/by-slug/{slug}<br>[GET] /api/forums/posts<br>[GET] /api/forums/posts/top<br>[GET] /api/forums/posts/{id}<br>[GET] /api/forums/posts/by-slug/{slug}<br>[GET] /api/forums/users/top<br>[POST] /api/forums/posts<br>[DELETE] /api/forums/posts/{id}<br>[GET] /api/forums/posts/{postId}/comments<br>[POST] /api/forums/posts/{postId}/comments<br>[PATCH] /api/forums/posts/{postId}/comments/{id}<br>[DELETE] /api/forums/posts/{postId}/comments/{id}<br>[POST] /api/forums/posts/{postId}/comments/{commentId}/like<br>[POST] /api/forums/posts/{id}/like<br>[GET] /api/forums/posts/{id}/like<br>[POST] /api/forums/posts/{postId}/reports<br>[POST] /api/forums/posts/{postId}/comments/{commentId}/reports<br>[GET] /api/forums/posts/{postId}/is-reported<br>[GET] /api/forums/posts/{postId}/comments/{commentId}/is-reported | The Forum Management feature enables community discussion and knowledge sharing within the platform. Guests and authenticated users can browse forum categories (including the most active categories), browse and search posts (including the most interacted posts and the most active users), and open a post by id or slug. Registered users can create a post, delete their own post, and manage comments (list, add, edit and delete). Posts and comments support a like/unlike toggle and a liker list, and users can report inappropriate posts or comments and check whether they have already reported them. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 18 | **Manage Forum Moderation** | [GET] /api/forums/admin/stats<br>[GET] /api/forums/admin/categories<br>[GET] /api/forums/admin/categories/{id}<br>[POST] /api/forums/admin/categories<br>[PATCH] /api/forums/admin/categories/{id}<br>[DELETE] /api/forums/admin/categories/{id}<br>[PATCH] /api/forums/admin/categories/{id}/restore<br>[GET] /api/forums/admin/comments<br>[GET] /api/forums/admin/pending-posts<br>[GET] /api/forums/admin/trash/posts<br>[GET] /api/forums/admin/trash/comments<br>[PATCH] /api/forums/admin/trash/posts/{id}/restore<br>[PATCH] /api/forums/admin/trash/comments/{id}/restore<br>[PATCH] /api/forums/admin/posts/{id}/approve<br>[PATCH] /api/forums/admin/posts/{id}/reject<br>[GET] /api/forums/reports<br>[PATCH] /api/forums/reports/{reportId}/status | The Forum Moderation feature gives administrators the tools to keep the community healthy. Administrators can view forum statistics, list, view, create, update, soft-delete and restore forum categories, and list all comments. New posts can be reviewed through the pending-post queue and either approved or rejected, while the trash lists allow deleted posts and comments to be reviewed and restored. The report queue lists content reported by users, and each report's status can be updated once it has been handled. |
| 19 | **Manage Messaging** | [GET] /api/conversations<br>[POST] /api/conversations<br>[GET] /api/conversations/{id}/messages<br>[POST] /api/conversations/{id}/messages<br>[POST] /api/conversations/{id}/files<br>[DELETE] /api/conversations/{id}/messages/{messageId}<br>[PATCH] /api/conversations/{id}/read<br>[PATCH] /api/conversations/{id}/hide<br>[PATCH] /api/conversations/{id}/pin | The Messaging feature provides real-time communication between clients and freelancers. Users can list their conversations, start a new conversation, list and send messages, and send files within a conversation. A message can be deleted or recalled, a conversation can be marked as read to reset its unread counter, hidden from the conversation list, and pinned or unpinned. Messages are delivered in real time through a WebSocket gateway. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 20 | **Manage Notification** | [GET] /api/notifications<br>[PATCH] /api/notifications/read-all<br>[PATCH] /api/notifications/{id}/read<br>[DELETE] /api/notifications/{id} | The Notification feature delivers in-app notifications about relevant events such as proposals, contracts, milestones, payments and forum moderation. Users can list their notifications with their read state, mark a single notification or all notifications as read, and delete a notification they no longer want to keep. |
| 21 | **Manage Roles** | [GET] /api/roles<br>[GET] /api/roles/{id}<br>[POST] /api/roles<br>[PATCH] /api/roles/{id}<br>[PUT] /api/roles/{id}/permissions<br>[DELETE] /api/roles/{id} | The Role Management feature allows administrators to define and control user permissions. Administrators can list all roles, view the detail of a role together with its permissions, create a new role, update its name or description, replace the permission set assigned to a role, and delete a role that is no longer needed. This provides fine-grained access control and proper segregation of duties across user types. |
| 22 | **Manage Permissions** | [GET] /api/permissions<br>[GET] /api/permissions/{id} | The Permission Management feature exposes the catalog of permissions used by the role-based access control system. Administrators can list every available permission and view the details of a single permission so that permissions can be assigned to roles accurately. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 23 | **Manage Sessions** | [GET] /api/sessions<br>[GET] /api/sessions/{id}<br>[DELETE] /api/sessions/{id} | The Session Management feature gives authenticated users visibility and control over their active login sessions. Users can list all sessions of their account, view the details of a specific session (such as device and login time), and revoke a session so that the corresponding refresh token can no longer be used. |
| 24 | **Manage Users (Admin)** | [POST] /api/users<br>[GET] /api/users<br>[GET] /api/users/{id}<br>[PATCH] /api/users/{id}<br>[PATCH] /api/users/{id}/client-profile<br>[PATCH] /api/users/{id}/freelancer-profile<br>[PUT] /api/users/{id}/freelancer-profile/skills<br>[POST] /api/users/{id}/freelancer-profile/portfolio-items<br>[PATCH] /api/users/{id}/freelancer-profile/portfolio-items/{itemId}<br>[DELETE] /api/users/{id}/freelancer-profile/portfolio-items/{itemId}<br>[GET] /api/users/skills-catalog | The User Management feature allows administrators to manage accounts from the administration side. Administrators can list users, view user details, create a new user, and update general user information. They can also edit a client profile, edit a freelancer's basic profile and replace the freelancer's skills, manage the freelancer's portfolio items, and read the skill catalog when performing these updates. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 25 | **Manage Identity Verification** | [POST] /api/identity-verifications/documents<br>[POST] /api/identity-verifications/ai-check<br>[POST] /api/identity-verifications/liveness-pose<br>[GET] /api/identity-verifications/status<br>[GET] /api/identity-verifications/documents/{id}/file<br>[GET] /api/admin/identity-verifications<br>[GET] /api/admin/identity-verifications/{id}<br>[GET] /api/admin/identity-verifications/{id}/file<br>[PATCH] /api/admin/identity-verifications/{id}/approve<br>[PATCH] /api/admin/identity-verifications/{id}/reject | The Identity Verification feature supports the eKYC process for user accounts. Users can upload identity documents, run an AI check on the submitted documents, capture liveness-pose images, check their verification status, and download an uploaded document file. Administrators can list verification requests, view request details and the attached file, then approve or reject the verification with a reason. |
| 26 | **Manage Profile Revisions** | [GET] /api/profile-revisions/me<br>[GET] /api/admin/profile-revisions<br>[GET] /api/admin/profile-revisions/{id}<br>[PATCH] /api/admin/profile-revisions/{id}/approve<br>[PATCH] /api/admin/profile-revisions/{id}/reject | The Profile Revision feature tracks changes made to freelancer profiles so that they can be moderated. A freelancer can view the latest revision of their own profile, while administrators can list all pending revisions, view the detail of a revision, and approve or reject it before the changes become publicly visible. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 27 | **Manage Banners** | [GET] /api/banners<br>[GET] /api/banners/image/{filename}<br>[GET] /api/admin/banners<br>[GET] /api/admin/banners/{id}<br>[POST] /api/admin/banners<br>[POST] /api/admin/banners/upload-image<br>[PATCH] /api/admin/banners/{id}<br>[DELETE] /api/admin/banners/{id} | The Advertisement Banner feature manages promotional banners displayed on the platform. The public endpoints return the active banners for a given position and serve the banner image files. Administrators can list and view all banners, create a new banner, upload its image, update its content, target position or schedule, and delete a banner. |
| 28 | **Manage Skills (Admin)** | [GET] /api/admin/skills<br>[GET] /api/admin/skills/{id}<br>[POST] /api/admin/skills<br>[PATCH] /api/admin/skills/{id}<br>[PATCH] /api/admin/skills/{id}/restore<br>[DELETE] /api/admin/skills/{id} | The Skill Management feature allows administrators to maintain the master skill catalog used by freelancer profiles and job postings. Administrators can list skills, view the detail of a skill, create a new skill, update its name or category, soft-delete a skill, and restore a previously deleted skill. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 29 | **AI Job & Freelancer Recommendation Engine** | [POST] /api/ai/recommendations/jobs<br>[POST] /api/ai/recommendations/freelancers<br>[GET] /api/ai/matching-score/{jobId}/{freelancerId} | The AI Recommendation Engine computes contextual matching scores between job postings and freelancer profiles using natural language embeddings and cosine similarity. It analyzes required skills, experience levels, budget constraints, and historical contract ratings to recommend optimal jobs to freelancers and qualified candidates to clients without manual keyword searching. |
| 30 | **AI Content Moderation** | [POST] /api/ai/moderation/check-text<br>[POST] /api/ai/moderation/filter-spam<br>[GET] /api/admin/moderation/ai-logs | The AI Content Moderation service processes user-generated text asynchronously across forum posts, thread comments, and direct chat messages. It analyzes submissions for profanity, toxic sentiment, hate speech, and spam. Content exceeding policy violation thresholds is automatically quarantined, flagged with high severity, and queued for administrator intervention while notifying the submitter. |

| # | Feature | System Function | Description |
|---|---|---|---|
| 31 | **AI Face Scan & Duplicate Identity Detection** | [POST] /api/identity-verifications/ai-check<br>[POST] /api/identity-verifications/liveness-pose<br>[POST] /api/ai/biometrics/duplicate-check | The AI Biometric Verification pipeline automates eKYC validation for freelancer identity onboarding. It extracts personal data from identity documents via optical character recognition (OCR), verifies head poses and face liveness via computer vision, and generates biometric facial feature vectors. The system cross-checks biometric signatures against existing verified profiles to prevent fraudulent account duplication. |
| 32 | **Automated Payment Processing & Webhook Synchroniza-tion** | [POST] /api/payments/webhook<br>[POST] /api/payments/intents/{paymentIntentId}/sync<br>[POST] /api/payments/checkout-sessions/{sessionId}/sync | The Automated Payment Processing service reliably handles asynchronous webhook callbacks emitted by Stripe. It processes payment intent confirmations, escrow funding authorizations, payout transfers, and dispute chargebacks. In addition, automated reconciliation routines synchronize transient payment intent statuses, resolve interrupted checkout sessions, and update contract ledger balances in real time. |

### 3.1.5 Entity Relationship Diagram

#### 3.1.5.1 Overall Entity Relationship Diagram

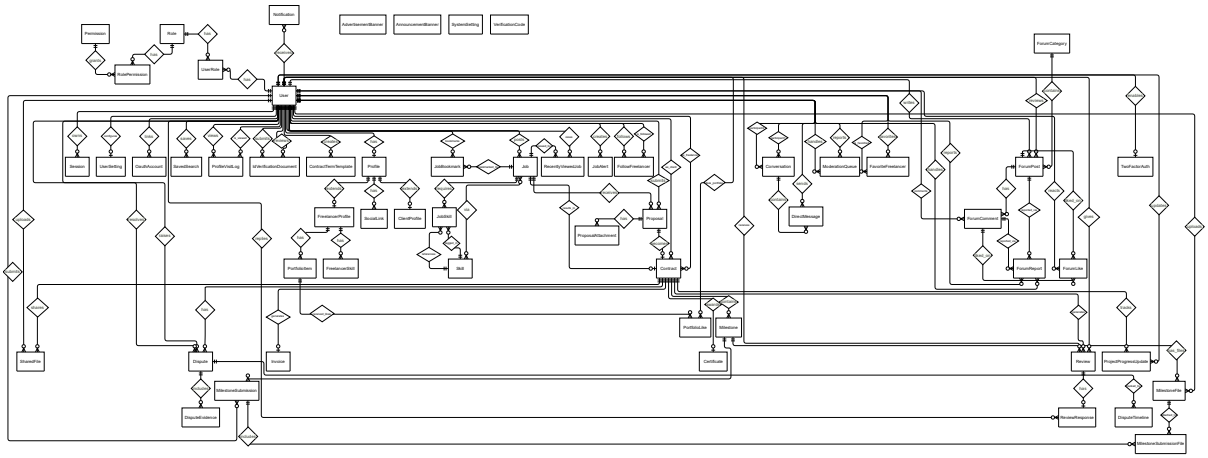

**Figure 28:** Overall Entity Relationship Diagram

#### 3.1.5.2 Authentication & Sessions

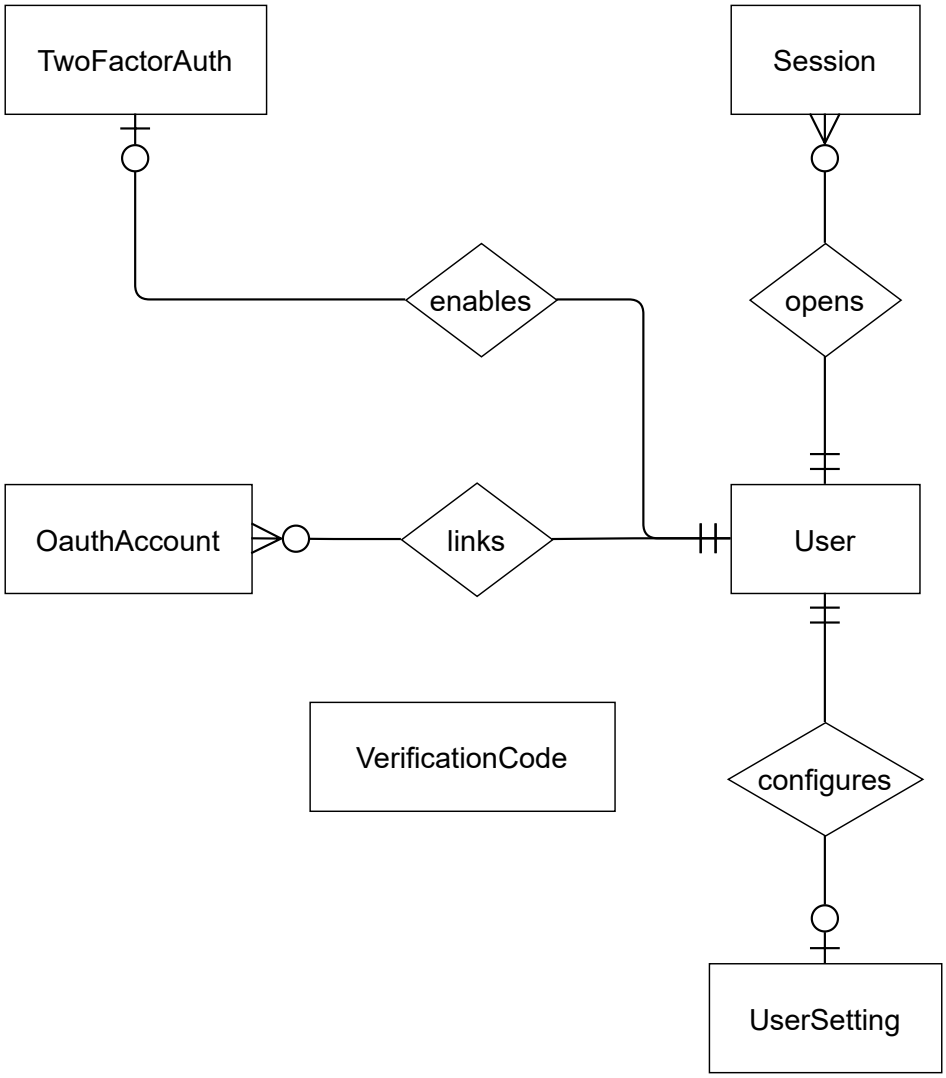


**Figure 29:** Authentication & Sessions ERD

#### 3.1.5.3 Role-Based Access Control

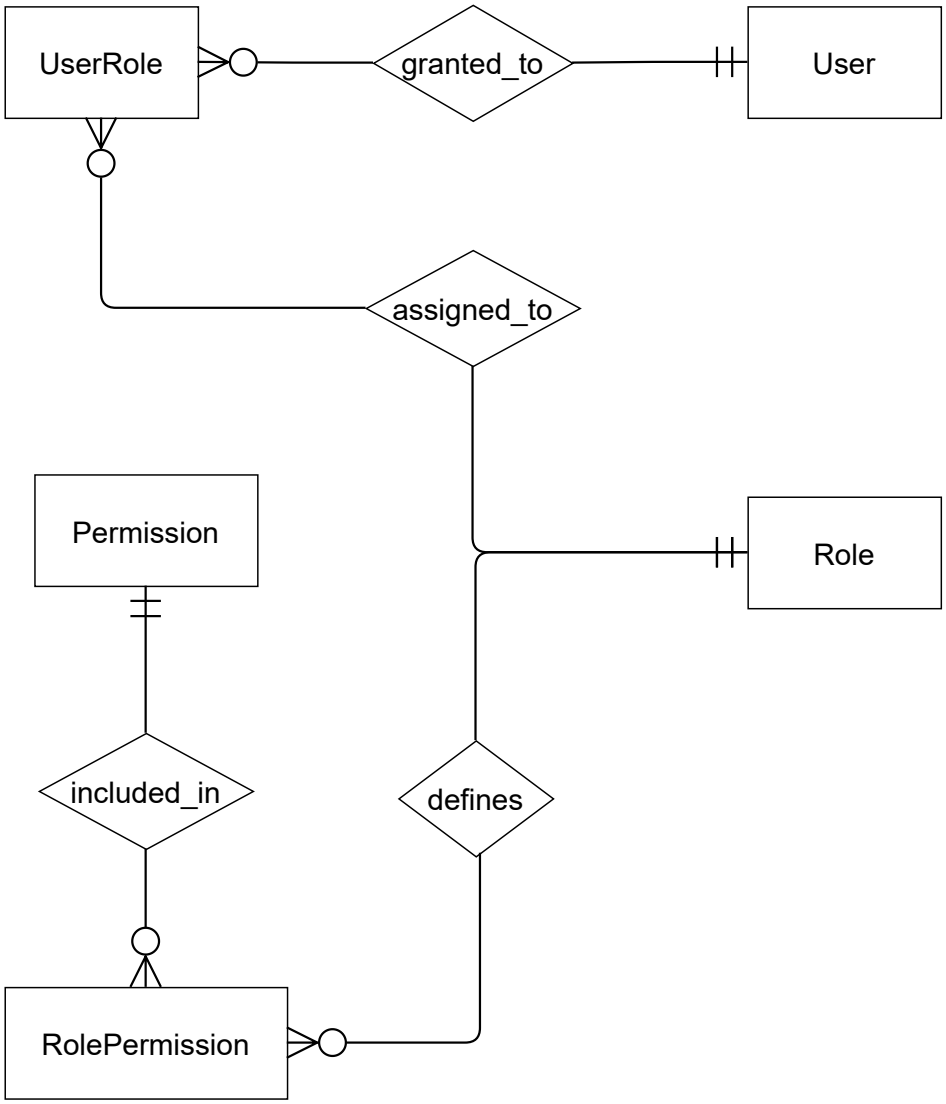


**Figure 30:** Role-Based Access Control ERD

#### 3.1.5.4 Profile Core

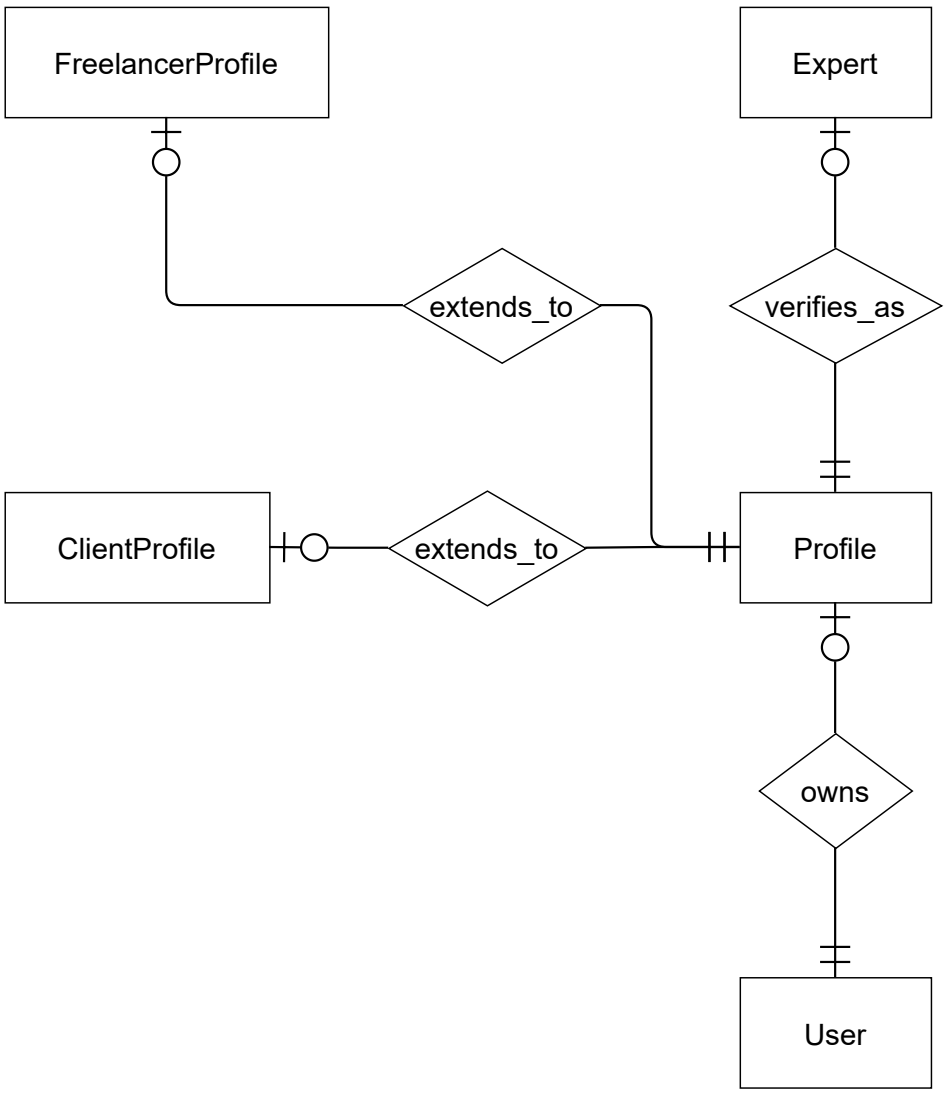


**Figure 31:** Profile Core ERD

#### 3.1.5.5 Profile History & Social Links

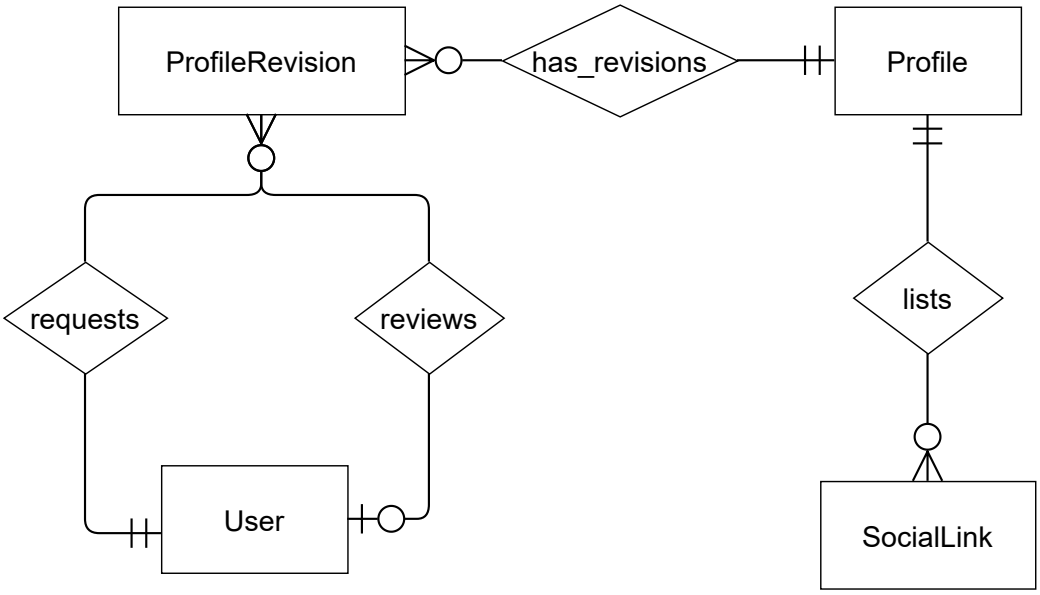


**Figure 32:** Profile History & Social Links ERD

#### 3.1.5.6 Identity Verification & Profile Visits

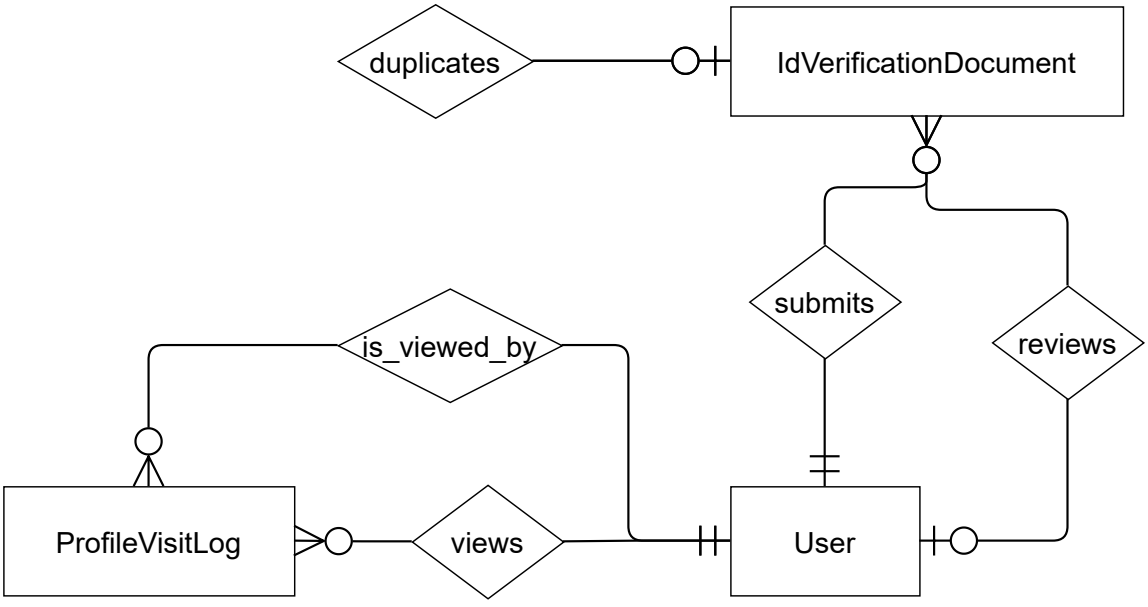


**Figure 33:** Identity Verification & Profile Visits ERD

#### 3.1.5.7 Portfolio & Skills

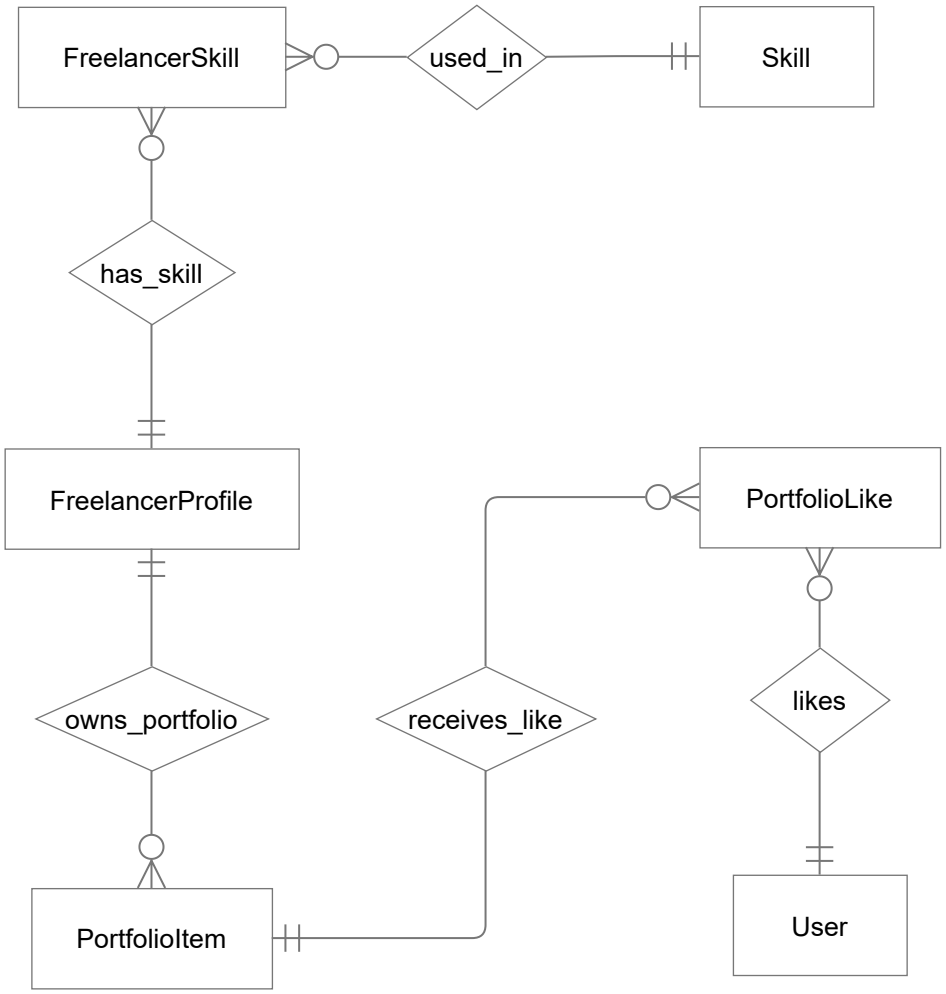


**Figure 34:** Portfolio & Skills ERD

#### 3.1.5.8 Favorites & Follows

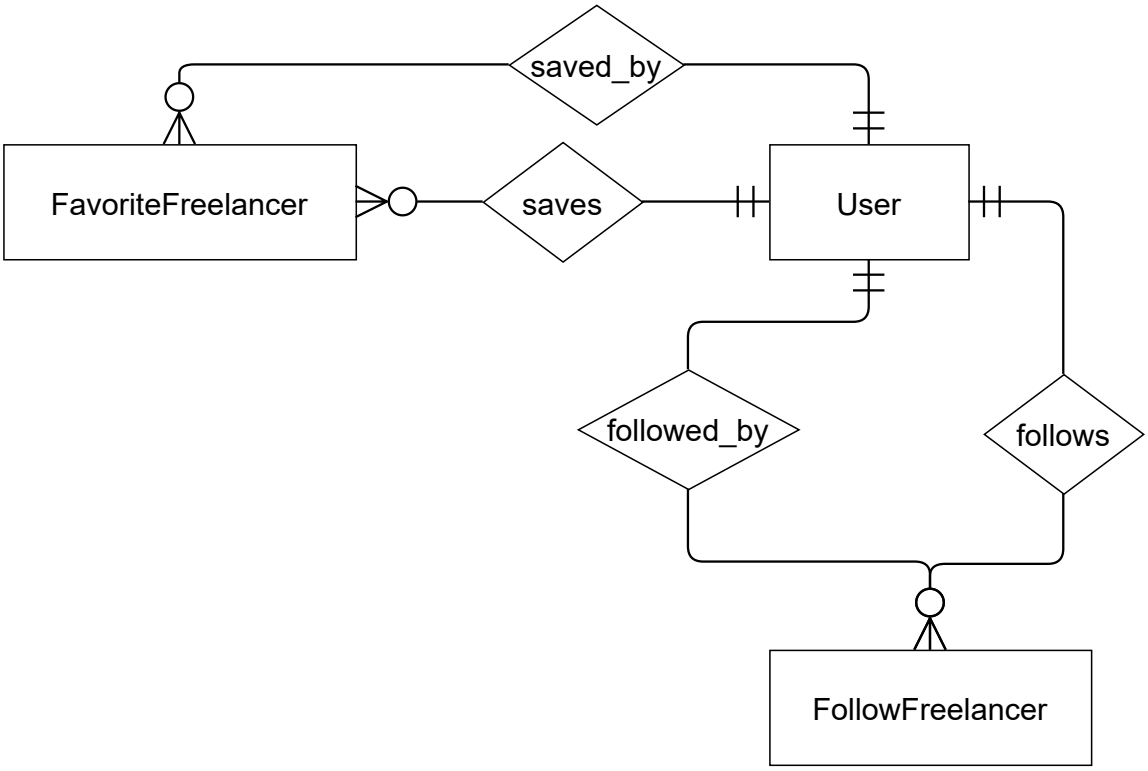


**Figure 35:** Favorites & Follows ERD

#### 3.1.5.9 Job Posting

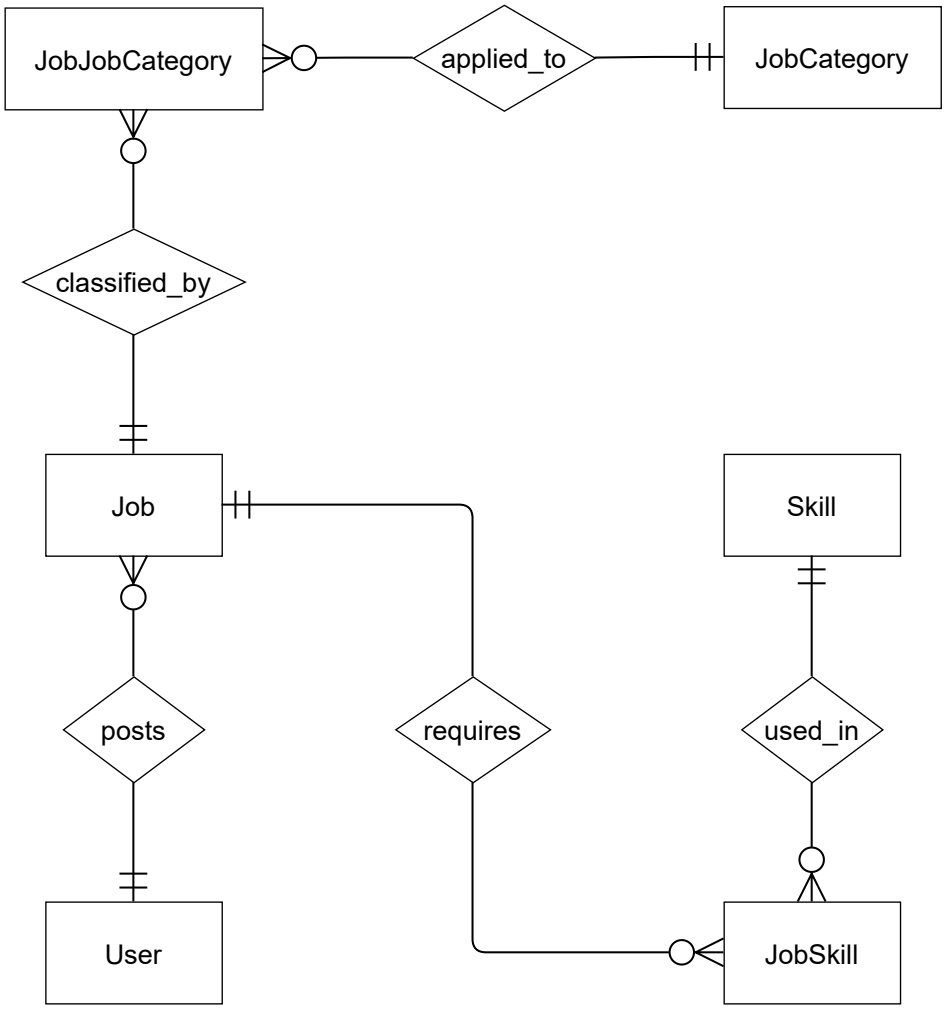


**Figure 36:** Job Posting ERD

#### 3.1.5.10 Job Discovery & Alerts

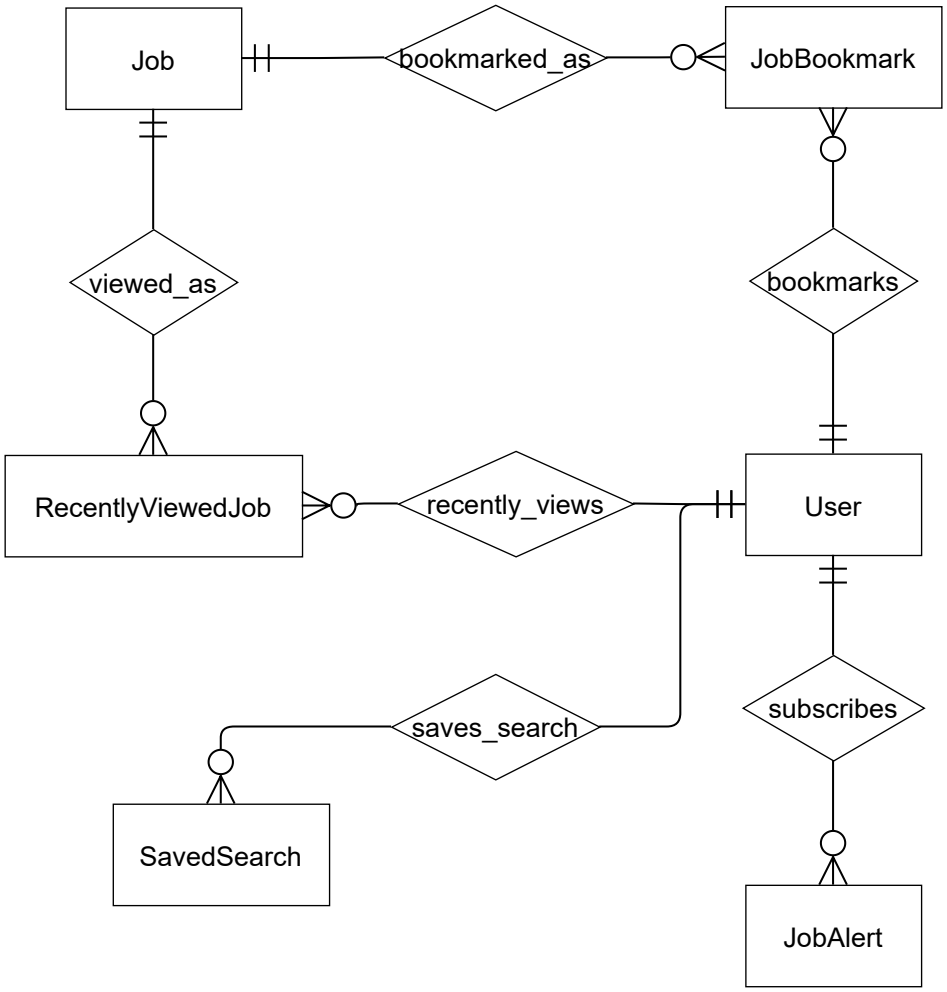


**Figure 37:** Job Discovery & Alerts ERD

#### 3.1.5.11 Proposal

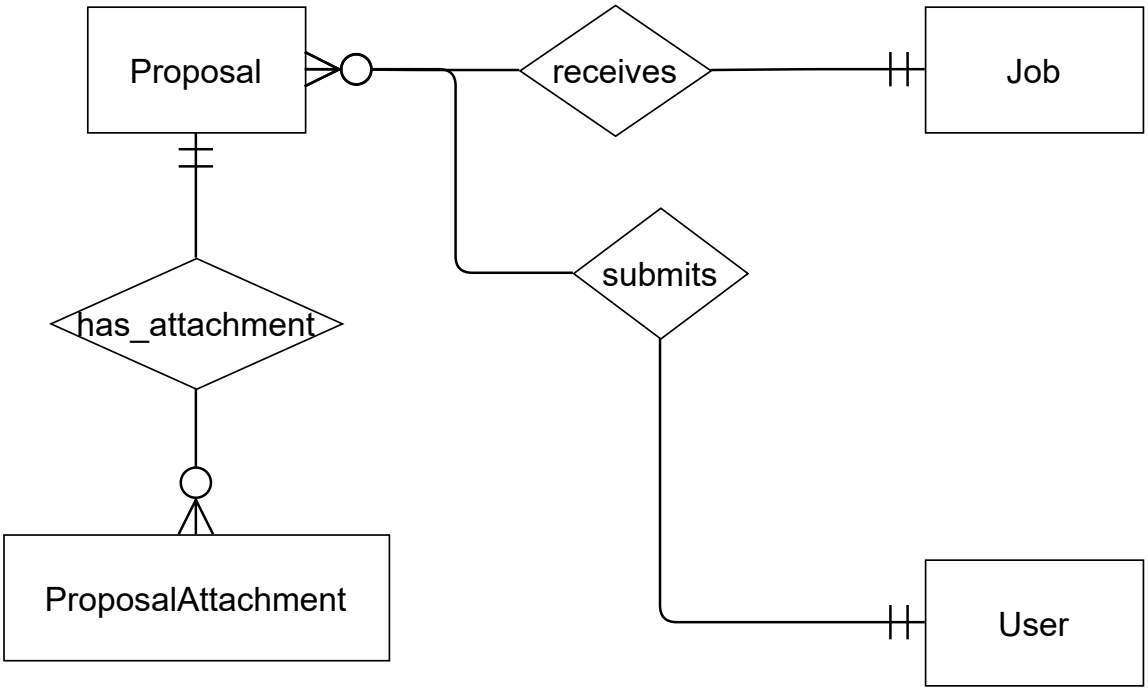


**Figure 38:** Proposal ERD

#### 3.1.5.12 Contract

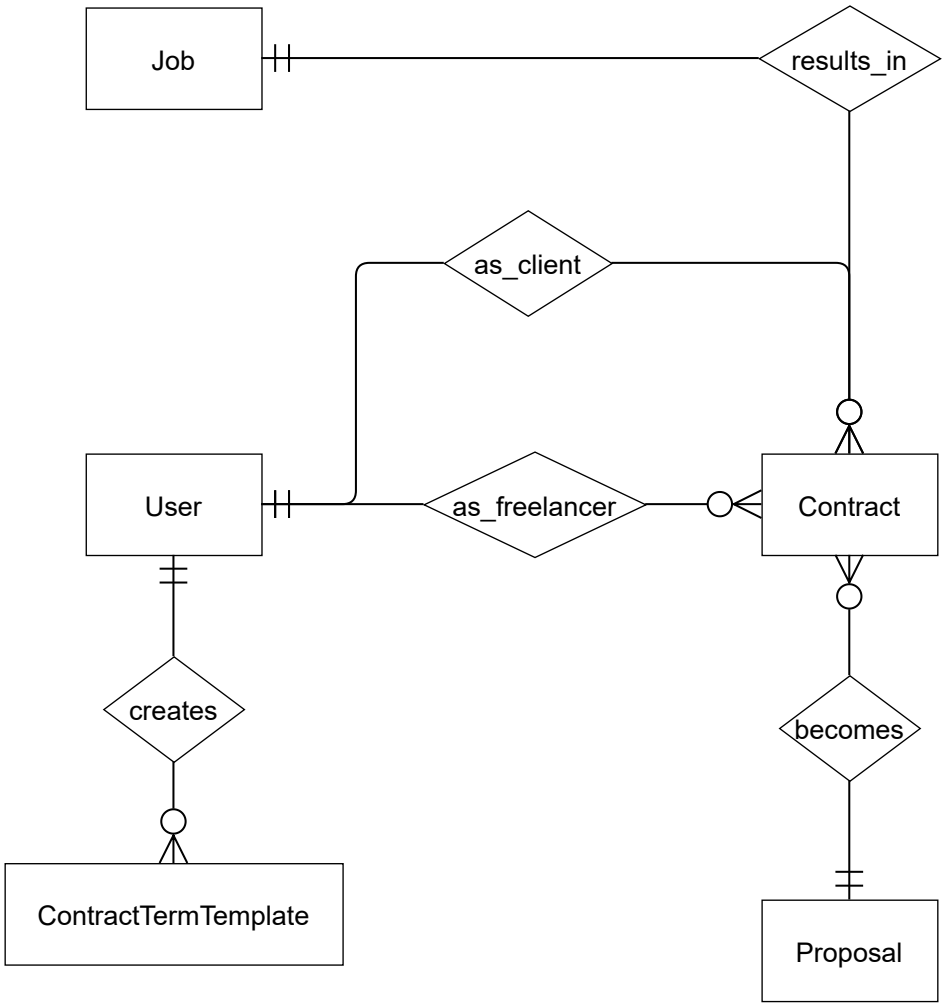


**Figure 39:** Contract ERD

#### 3.1.5.13 Milestone & Project Progress

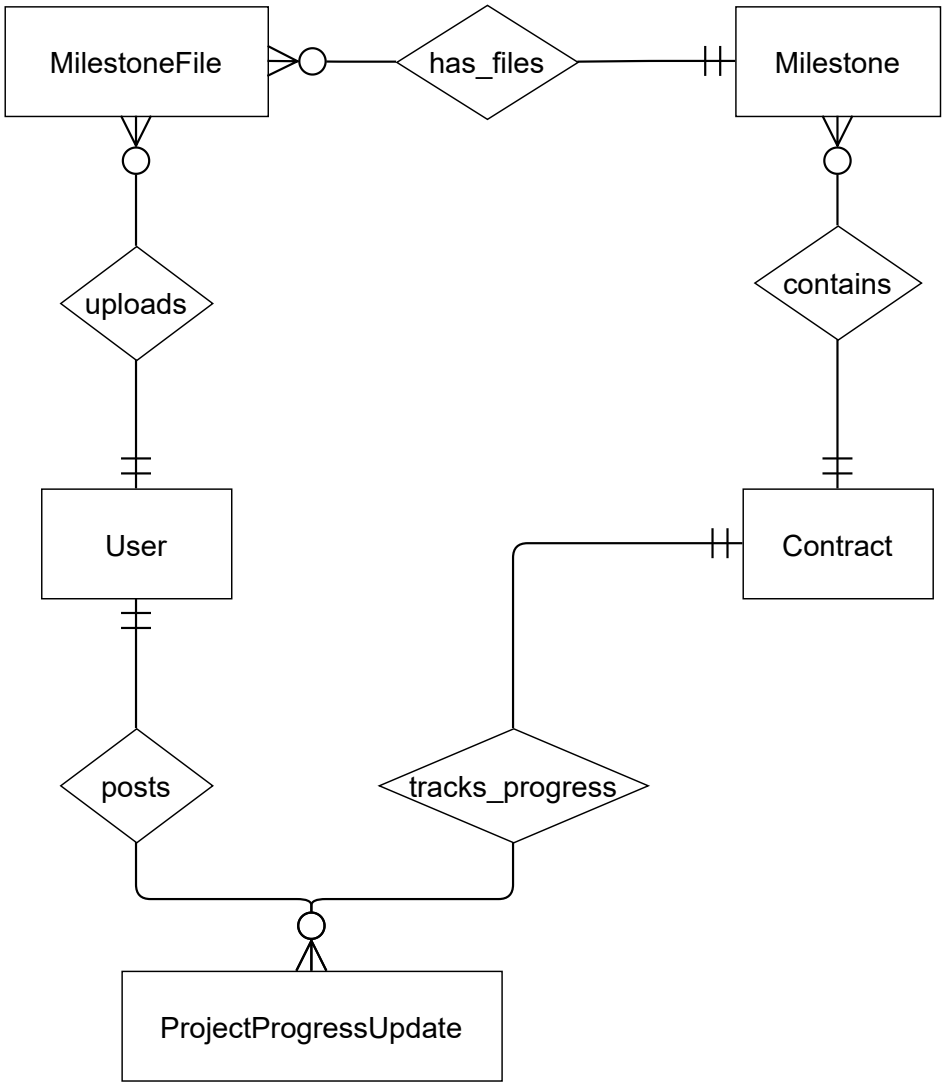


**Figure 40:** Milestone & Project Progress ERD

#### 3.1.5.14 Milestone Submission

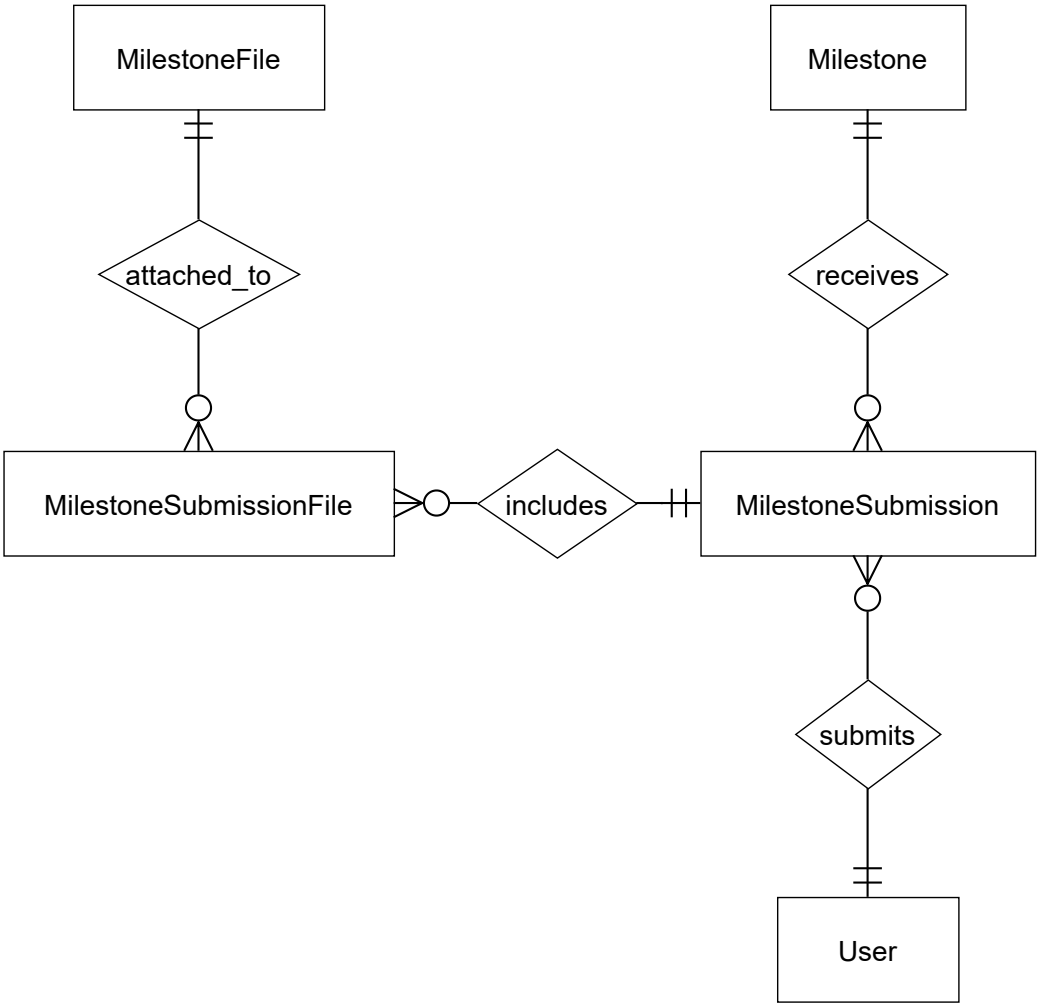


**Figure 41:** Milestone Submission ERD

#### 3.1.5.15 Escrow, Payments & Certificates

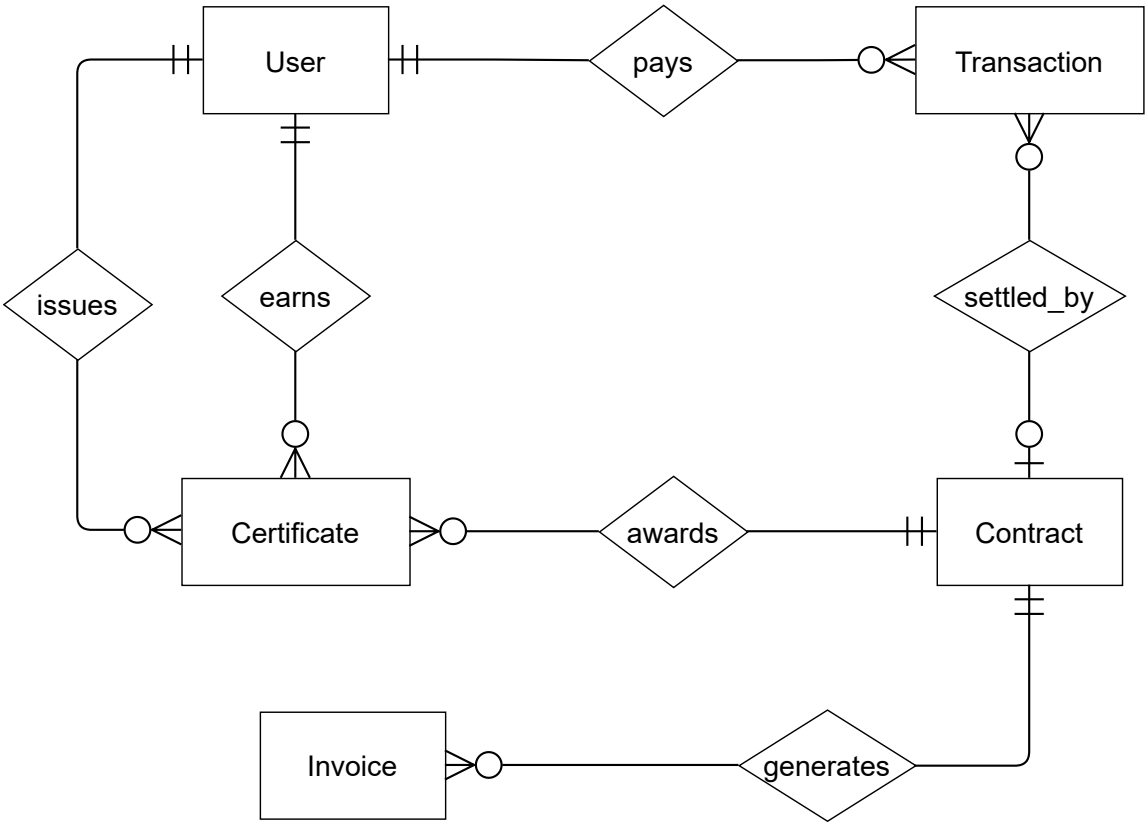


**Figure 42:** Escrow, Payments & Certificates ERD

#### 3.1.5.16 Dispute Core

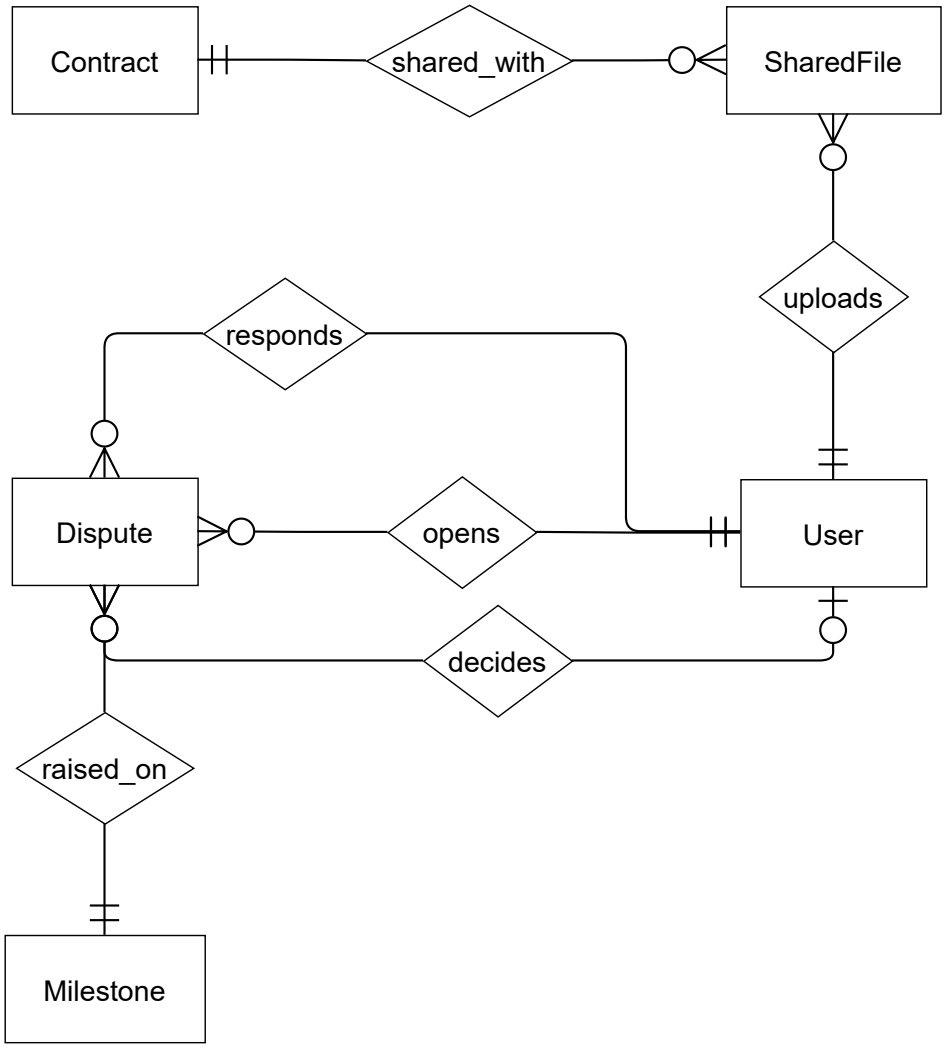


**Figure 43:** Dispute Core ERD

#### 3.1.5.17 Dispute Resolution

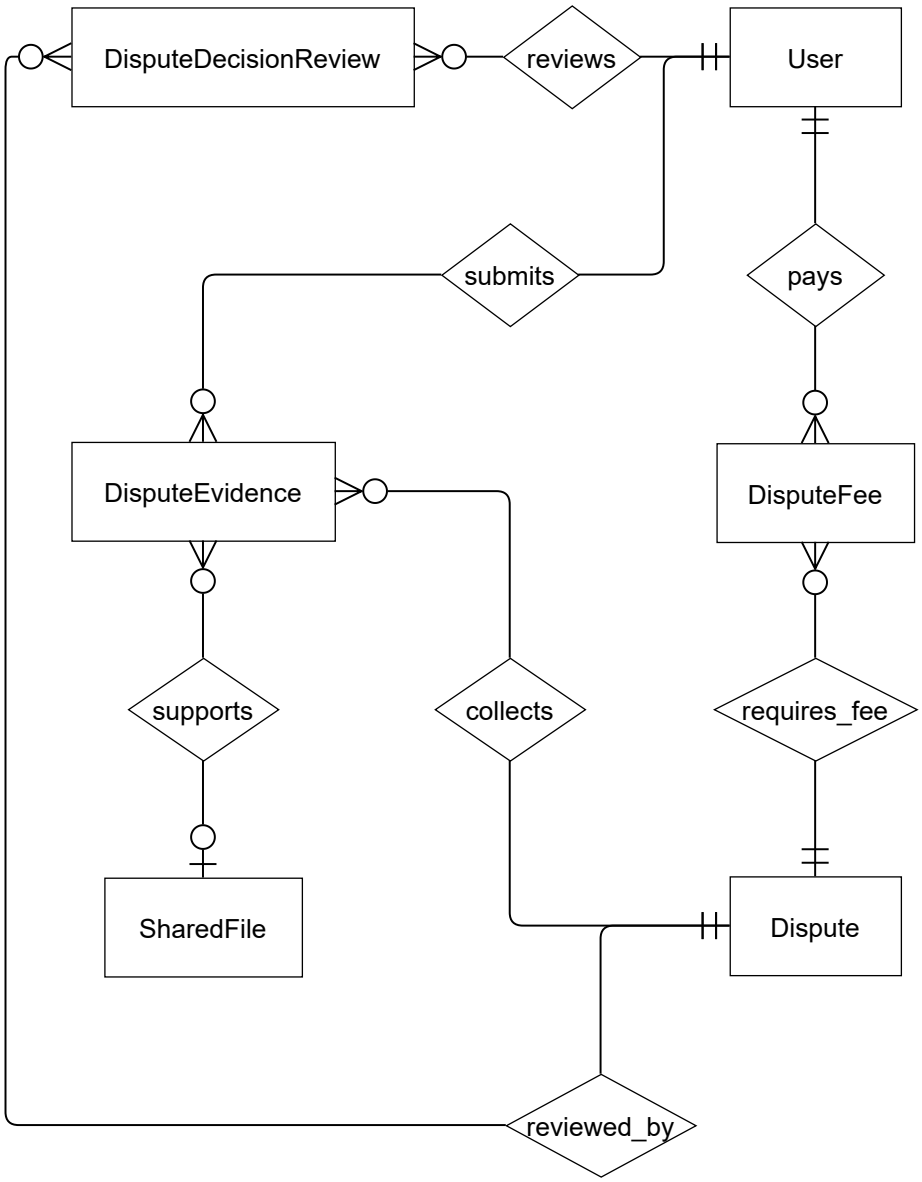


**Figure 44:** Dispute Resolution ERD

#### 3.1.5.18 Reviews & Reputation

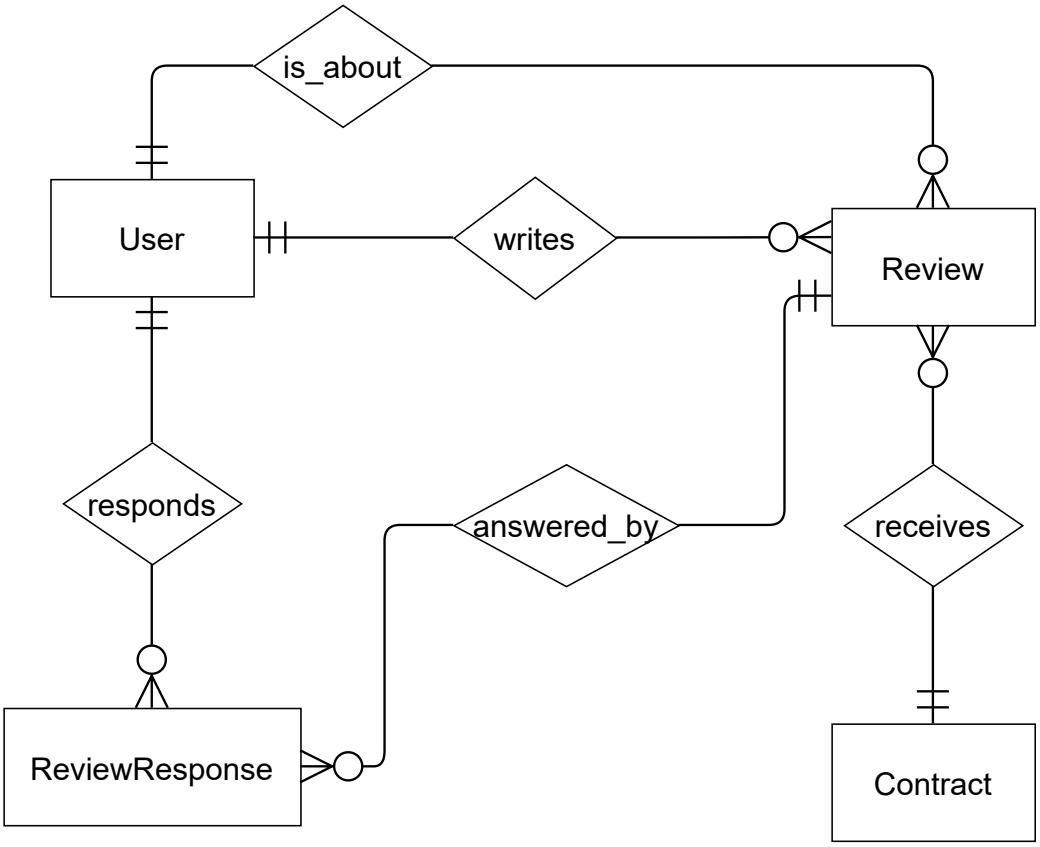


**Figure 45:** Reviews & Reputation ERD

#### 3.1.5.19 Forum Posts & Comments

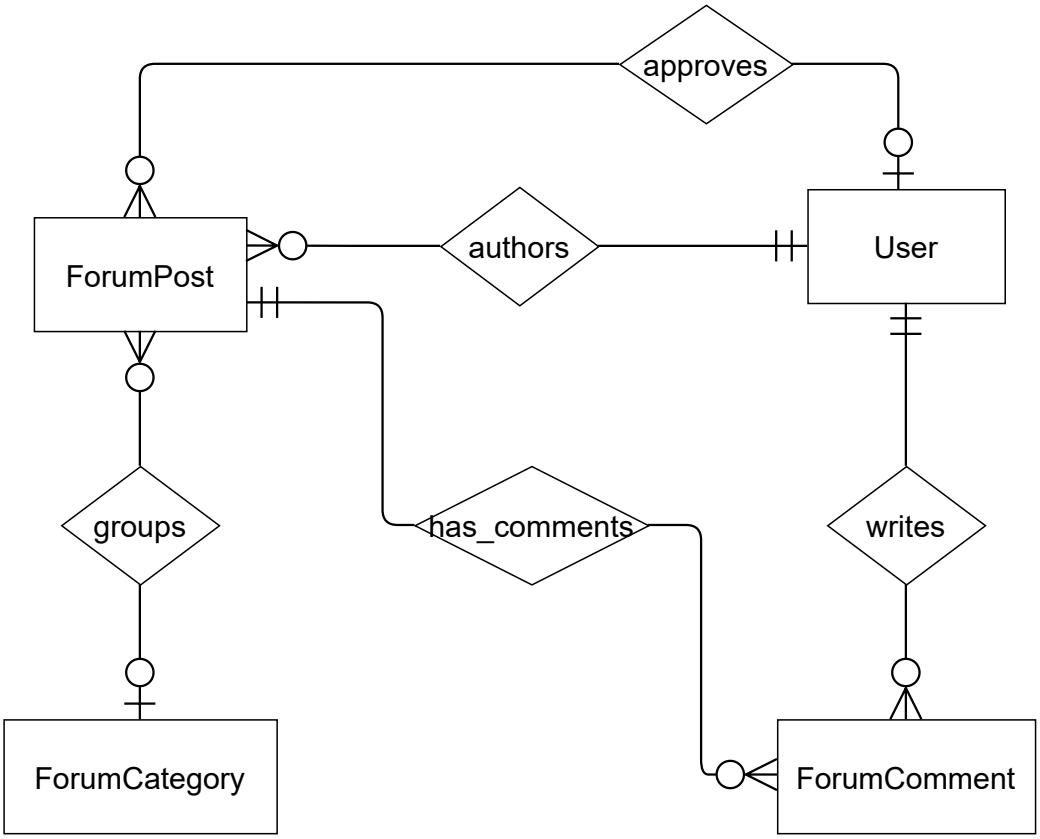


**Figure 46:** Forum Posts & Comments ERD

#### 3.1.5.20 Forum Likes

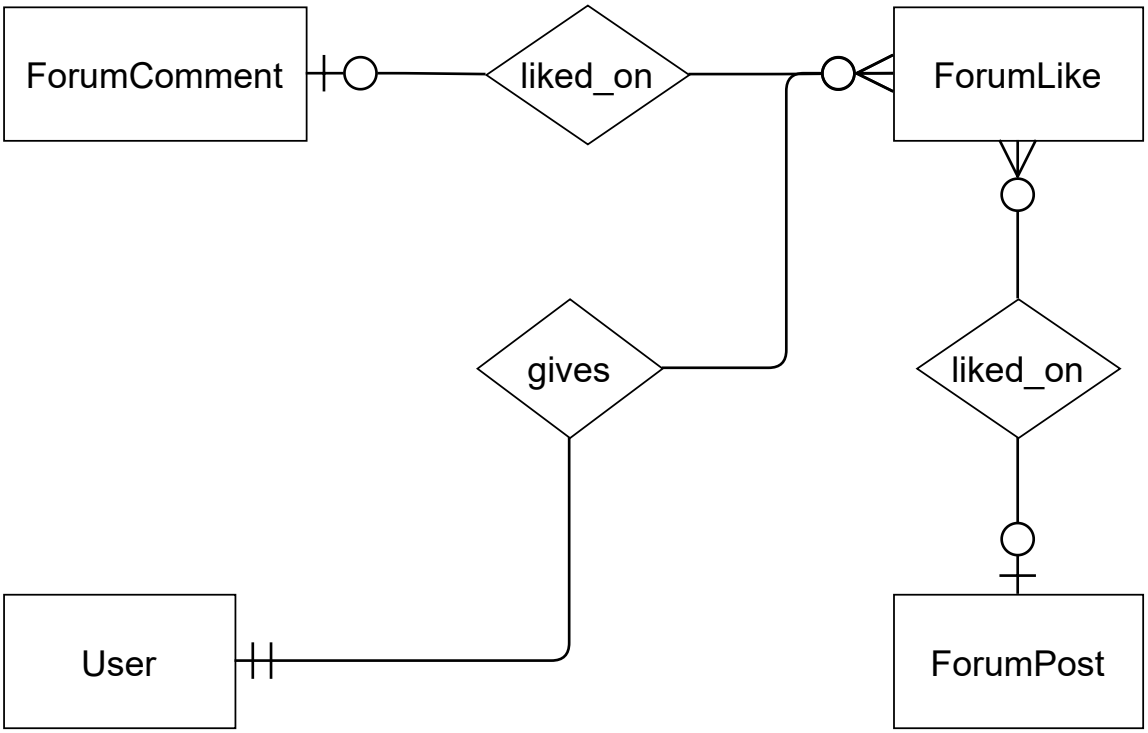


**Figure 47:** Forum Likes ERD

#### 3.1.5.21 Forum Reports & Moderation

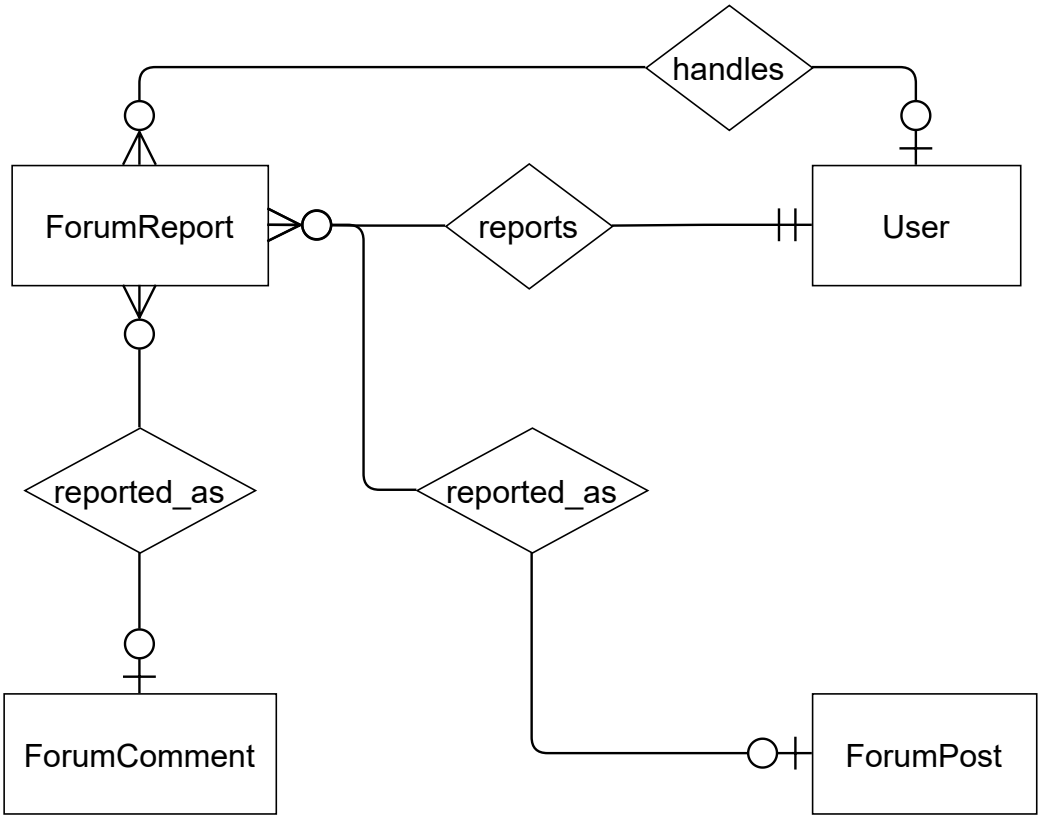


**Figure 48:** Forum Reports & Moderation ERD

#### 3.1.5.22 Messaging & Notifications

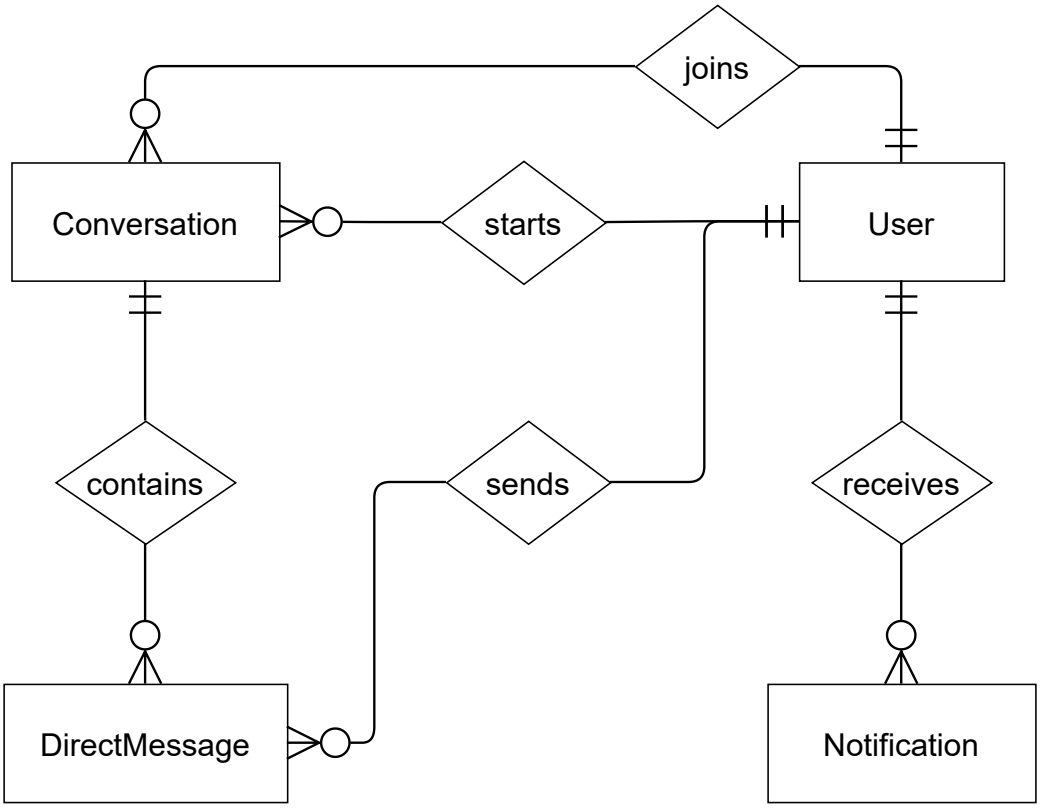


**Figure 49:** Messaging & Notifications ERD

#### 3.1.5.23 Administration & System

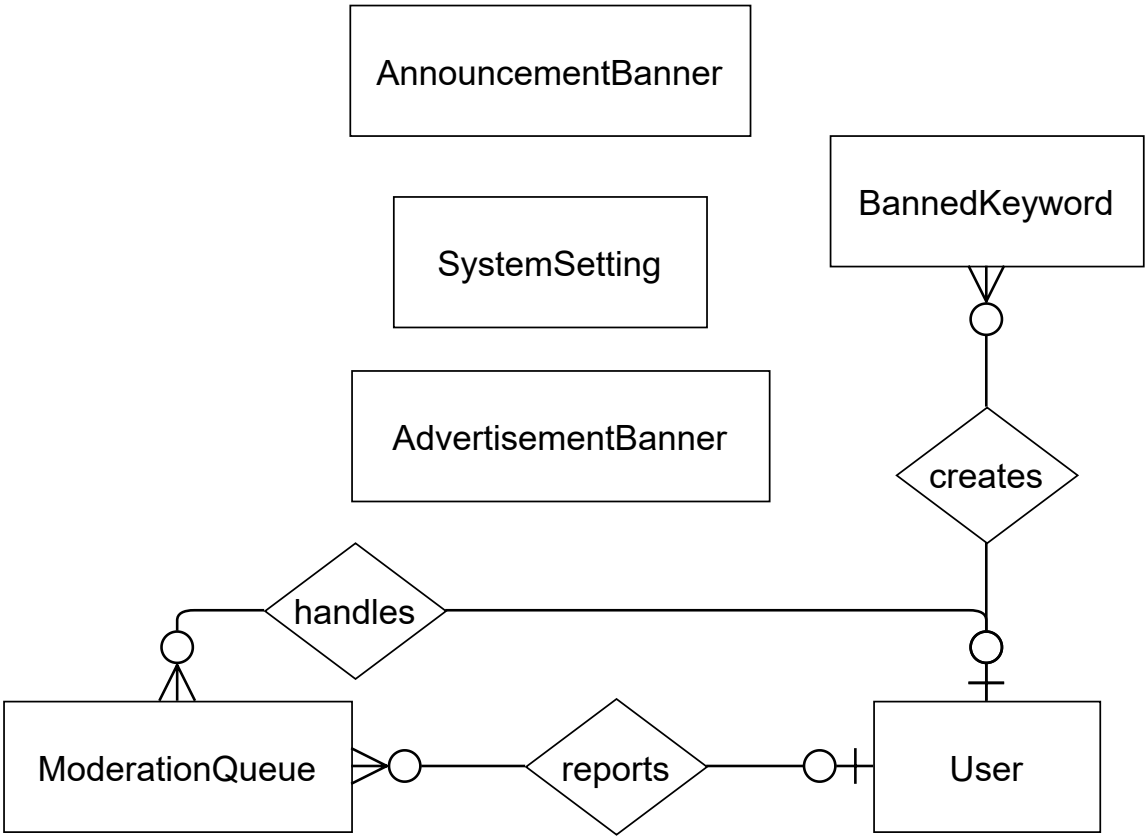


**Figure 50:** Administration & System ERD

## 3.2 User Authentication

### 3.2.1 Login with Google

**Function Trigger**

A guest clicks the "Google" authentication button on the sign-in screen or the registration screen.

**Navigation path:** Home → Login (`/login`) → Click "Google" button.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Guest (unauthenticated visitor).
- **Secondary Actor:** Google OAuth 2.0 Authorization Server.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia BFF authentication proxy (`/api/auth/google`), Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Allows a visitor to authenticate or register an account instantly using their existing Google account via OAuth 2.0. This eliminates the need to remember credentials or manually verify an email address, significantly reducing onboarding friction while ensuring verified identity data.

**Interface**

- Sign-in screen at `/login` presenting the prominent "Google" single sign-on action.
- External Google OAuth 2.0 consent window prompted by Google Identity services.
- Automatic redirect to role-specific workspace upon authentication completion.

**Data Processing Overview**

- The client initiates authentication by requesting `GET /api/auth/google-link`.
- The backend generates the Google OAuth consent URL configured with client ID, callback URI, and identity scopes (`openid email profile`).
- The guest grants permission on Google's consent dialog, and Google redirects to the callback endpoint with an authorization code.
- The backend exchanges the code for Google tokens and fetches profile information (email, full name, avatar URL).
- The system queries the database: if no account exists with the email, a new account is automatically provisioned with verified email status and assigned the default `Client` role.
- The system creates an active session, issues signed JWT access and refresh tokens, stores them in secure `httpOnly` cookies, and redirects the user to their designated dashboard.

### Screen Layout (Mock-up Description)

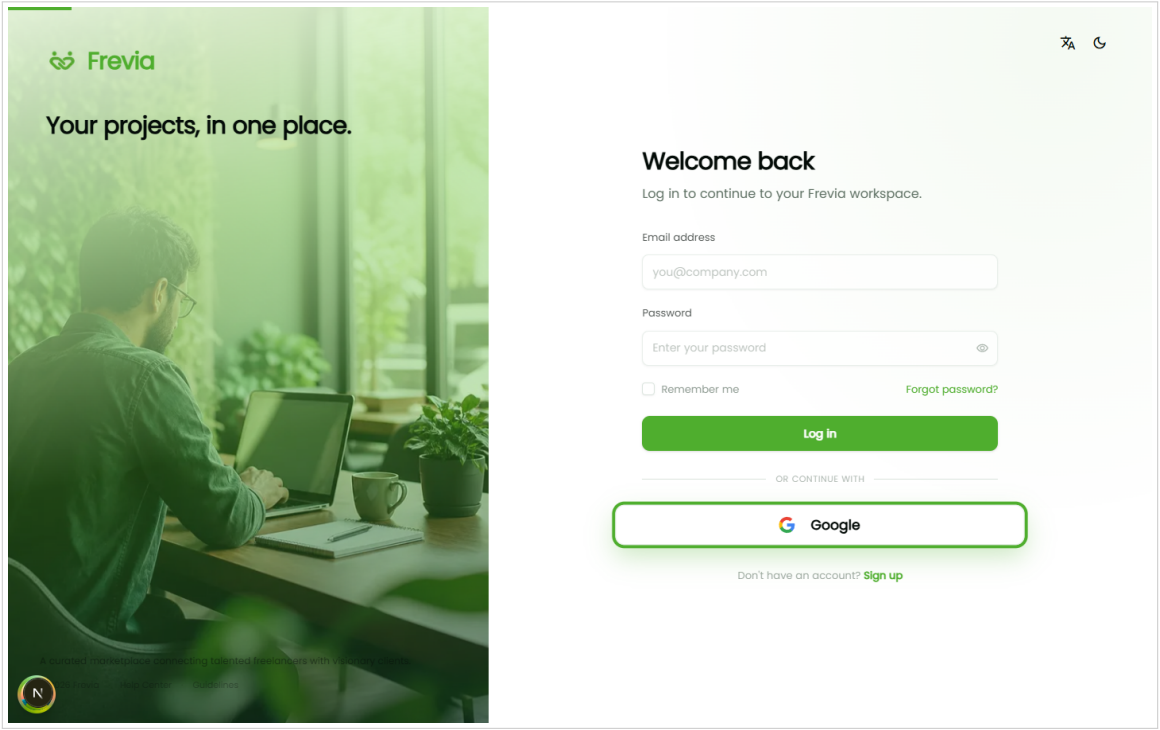


**Figure 51:** Login with Google interface

- The login container displays a greeting title “Welcome back” and an email/password form for standard authentication.
- A divider labeled “OR CONTINUE WITH” separates local credential inputs from social login providers.
- The “Google” button is prominent, featuring the official multi-colored Google logo, clean responsive styling, and clear action labeling.
- A footer link allows users without an account to switch to the registration screen.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| OAuth Code | Temporary authorization code returned by Google OAuth Server via redirect URI. |
| State Token | Cryptographic anti-CSRF state token validated by the authentication handler. |
| Session Headers | IP address and User-Agent captured from request headers for session tracking. |

**Validation Rules**

- The OAuth code must be valid, non-expired, and verifiable against Google OAuth endpoints.
- The email address retrieved from Google profile must be non-empty and marked as verified by Google.
- The matched user account must have `status = ACTIVE`; suspended or banned accounts are rejected.

**Business Rules**

- **BR-UA-01**: Google OAuth shall follow standard OAuth 2.0 authorization code grant with secure PKCE/state verification.
- **BR-UA-02**: New users authenticated via Google OAuth shall be automatically registered with the `Client` role and `isEmailVerified = true`.
- **BR-UA-03**: Suspended or banned accounts shall be denied access and notified with an administrative contact notice.
- **BR-UA-04**: Issued JWT tokens shall be stored exclusively in secure, `httpOnly`, `SameSite=Lax` cookies.

**Normal Flow**

1. The guest clicks the "Google" button on the login screen.
2. The browser receives the Google authorization URL from `/api/auth/google-link` and redirects to Google.
3. The guest authenticates with their Google account and authorizes Frevia.
4. Google redirects to `/api/auth/google/callback` with code and state parameters.
5. The backend exchanges the code for profile tokens and extracts identity details.
6. The backend finds or provisions the user record and initializes an active session.
7. The system writes authentication cookies and redirects the user to their role dashboard.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Guest denies permission or cancels the Google OAuth consent prompt. Google redirects with `error=access_denied`; the system redirects back to `/login` with notification "Google login was cancelled." |
| A2 | The authorization code has expired, was malformed, or was replayed. The token exchange fails; system displays "Authentication with Google failed. Please try again." |
| A3 | Anti-CSRF `state` token mismatch or missing session state. The backend rejects the callback with HTTP 400 Bad Request to protect against cross-site request forgery. |
| A4 | The account linked to the retrieved Google email has been suspended or banned by administrators. Access is rejected with HTTP 403: "Your account has been suspended. Please contact support." |
| A5 | Google OAuth service is temporarily unreachable due to network outage or timeout. The backend catches `FetchError` and prompts the user to try standard email login. |
| A6 | Database connection failure occurs during user auto-provisioning or session persistence. The transaction rolls back and returns HTTP 500: "Login failed due to a server error." |
| A7 | The browser blocks the external OAuth redirect due to strict popup/redirect blocker policies. The client displays an explicit fallback hyperlink to proceed with authentication. |

**Output Data**

| Data Name | Description |
|---|---|
| `accessToken` | Signed JWT containing user ID, email, active roles, and expiration timestamp (15 minutes). |
| `refreshToken` | Opaque or long-lived JWT token (7 days) persisted in the database session table. |
| User Object | User profile record including `id`, `email`, `name`, `avatarUrl`, and primary role. |

**Post-Conditions**

- The user is authenticated with active session cookies.
- An audit record is created in the database session log.
- The browser is navigated to the dashboard appropriate for the user's primary role.

### 3.2.2 Login

#### Function Trigger

A registered user enters their email and password into the sign-in form and submits.

**Navigation path:** Home → Login (`/login`).

#### Function Description

#### Actors / Roles

- **Primary Actor:** Guest (unauthenticated registered user).
- **Supporting Systems:** Frevia Web application (Next.js), Frevia BFF proxy (`/api/auth/login`), Frevia REST API (NestJS), and PostgreSQL database.

#### Purpose

Authenticates users using traditional email and password credentials, verifying identity against hashed passwords in the database, establishing an active session, and issuing JWT tokens.

#### Interface

- Login screen at `/login` with Email Address input and Password input.
- Toggle visibility button (eye icon) to show or hide entered password.
- "Remember me" checkbox and "Forgot password?" navigation link.
- Primary "Log in" button with loading spinner state during submission.

#### Data Processing Overview

- The web client validates input schema with Zod (`LoginBodySchema`) before sending.
- The client posts credentials to BFF `/api/auth/login`, which forwards to NestJS API.
- The API queries the user by email including associated roles and profiles.
- The system verifies the supplied password against the stored bcrypt hash.
- The system verifies that the account is active and the email address is confirmed.
- Upon success, a new session record is written into the database, JWT access/refresh tokens are generated and attached as `httpOnly` cookies, and user metadata is returned.

### Screen Layout (Mock-up Description)

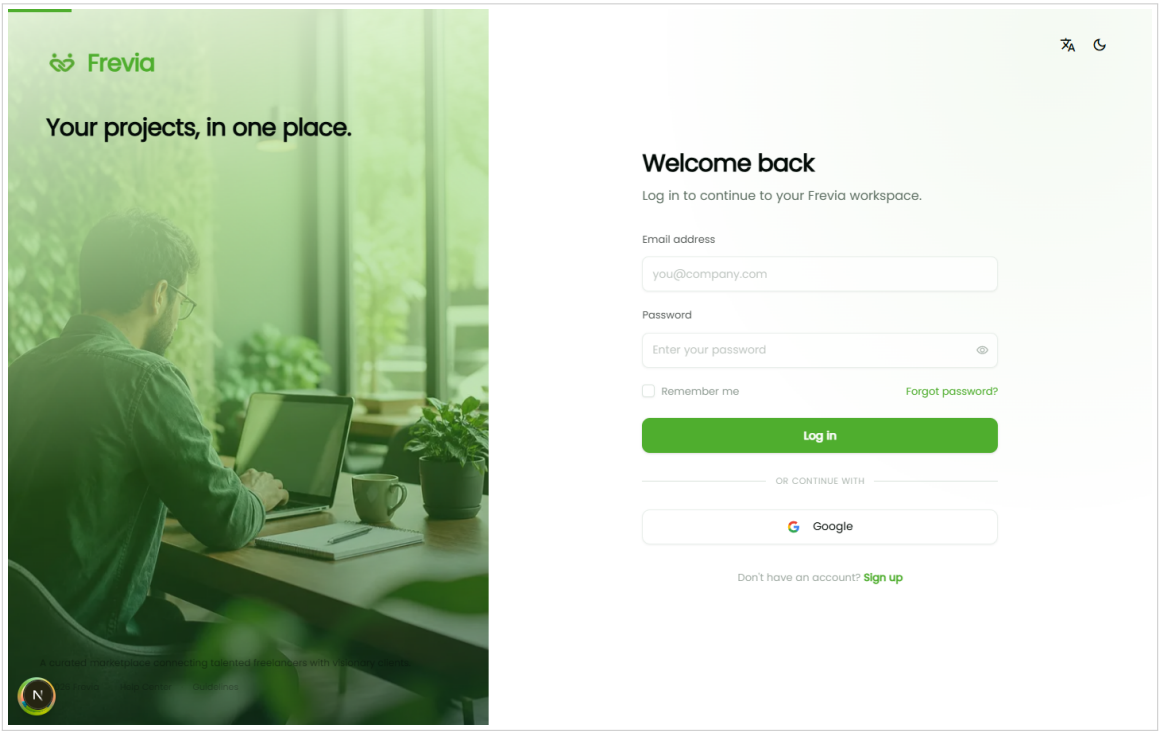


**Figure 52:** User login screen

- Left branded imagery panel showcases workspace lifestyle graphics with the Frevia slogan.
- Right form card provides fields for “Email address” and “Password” with subtle placeholder hints.
- Password input includes an embedded visibility toggle icon on the right edge.
- Action row features “Remember me” checkbox alongside “Forgot password?” link in primary green.
- Solid green “Log in” submit button spans full container width.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| email | Registered email address string; must be a valid RFC 5322 format. |
| password | User account password string; required, minimum 6 characters. |

**Validation Rules**

- email is trimmed and validated as a syntactically valid email address.
- password cannot be empty and must match the stored bcrypt hash.
- The account must not have deletedAt set and must have status = ACTIVE.
- The user must have verified their email address (isEmailVerified = true).

**Business Rules**

- **BR-UA-05**: Passwords must be verified using bcrypt comparison with salted hashes.
- **BR-UA-06**: Error responses shall use a generic message "Invalid email or password" to mitigate account enumeration.
- **BR-UA-07**: Unverified accounts shall be rejected with a notice prompting verification completion.
- **BR-UA-08**: Suspended accounts shall be denied access across all client interfaces.

**Normal Flow**

1. Guest navigates to /login and inputs valid registered email and password.
2. Guest clicks "Log in" button.
3. Form resolver executes schema validation and initiates HTTP POST request.
4. NestJS API retrieves user record, compares bcrypt hash, and verifies account active status.
5. The system records the session in the database and creates JWT tokens.
6. The BFF proxy serializes httpOnly cookies and responds with HTTP 200.
7. The frontend detects user roles and redirects to the appropriate dashboard (Admin, Client, or Freelancer).

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Required field is missing (empty email or password). Form resolver intercepts input and renders inline errors before sending any network request. |
| A2 | Email is not registered in the system. The API returns HTTP 401 with generic error `AuthMessage.INCORRECT_EMAIL` mapped to "Invalid email or password." |
| A3 | Incorrect password provided. Bcrypt comparison fails; API returns HTTP 422 with `AuthMessage.INCORRECT_PASSWORD` mapped to "Invalid email or password." |
| A4 | Account email is unconfirmed (`isEmailVerified = false`). System returns HTTP 403: "Please verify your email before logging in. Resend verification email?" |
| A5 | Account is marked as `status = BANNED` or `SUSPENDED`. System throws `UserBannedException` (HTTP 403): "Your account has been suspended. Please contact support." |
| A6 | Excessive consecutive failed login attempts detected. Rate limiter triggers HTTP 429: "Too many login attempts. Please wait 15 minutes before trying again." |
| A7 | Database connection times out or pool is exhausted. API returns HTTP 500; frontend alerts "Login failed. Please try again later." |
| A8 | Network connectivity drops during form submission. Client request fails with network error; frontend displays offline alert toast. |

**Output Data**

| Data Name | Description |
|---|---|
| `accessToken` | 15-minute validity JWT stored in `accessToken` cookie. |
| `refreshToken` | 7-day validity token stored in `refreshToken` cookie. |
| `user` | Sanitized user object containing ID, email, name, avatar, and roles list. |

**Post-Conditions**

- User authentication state is initialized.
- A new session row is created in the database containing client device metadata.
- Client redirects to the dashboard.

### 3.2.3 Logout

### Function Trigger

An authenticated user clicks the “Log out” button from the main navigation sidebar or user profile menu.
**Navigation path:** Any authenticated screen → Sidebar bottom → Click “Log out”.

### Function Description

### Actors / Roles

- **Primary Actor:** Client, Expert, Freelancer, Administrator (all authenticated users).
- **Supporting Systems:** Frevia Web application (Next.js), Frevia BFF proxy (`/api/auth/logout`), Frevia REST API (NestJS), and PostgreSQL database.

### Purpose

Terminates the current active session on both the client application and server database, revoking the validity of issued tokens and protecting user accounts from unauthorized session reuse.

### Interface

- Sidebar bottom control labeled “Log out” with logout icon.
- Header profile popover menu containing secondary sign-out trigger.
- Immediate redirection back to the public sign-in screen upon termination.

### Data Processing Overview

- Client dispatches a POST request to `/api/auth/logout`.
- The backend extracts the session token from the request cookies or authorization headers.
- The API locates the matching session row in the `Session` table and soft-deletes or purges it.
- The BFF proxy expires the `accessToken` and `refreshToken` cookies by setting their `Max-Age` to 0.
- The web client clears local client-side caches (TanStack Query, session store) and redirects the visitor to `/login`.

### Screen Layout (Mock-up Description)

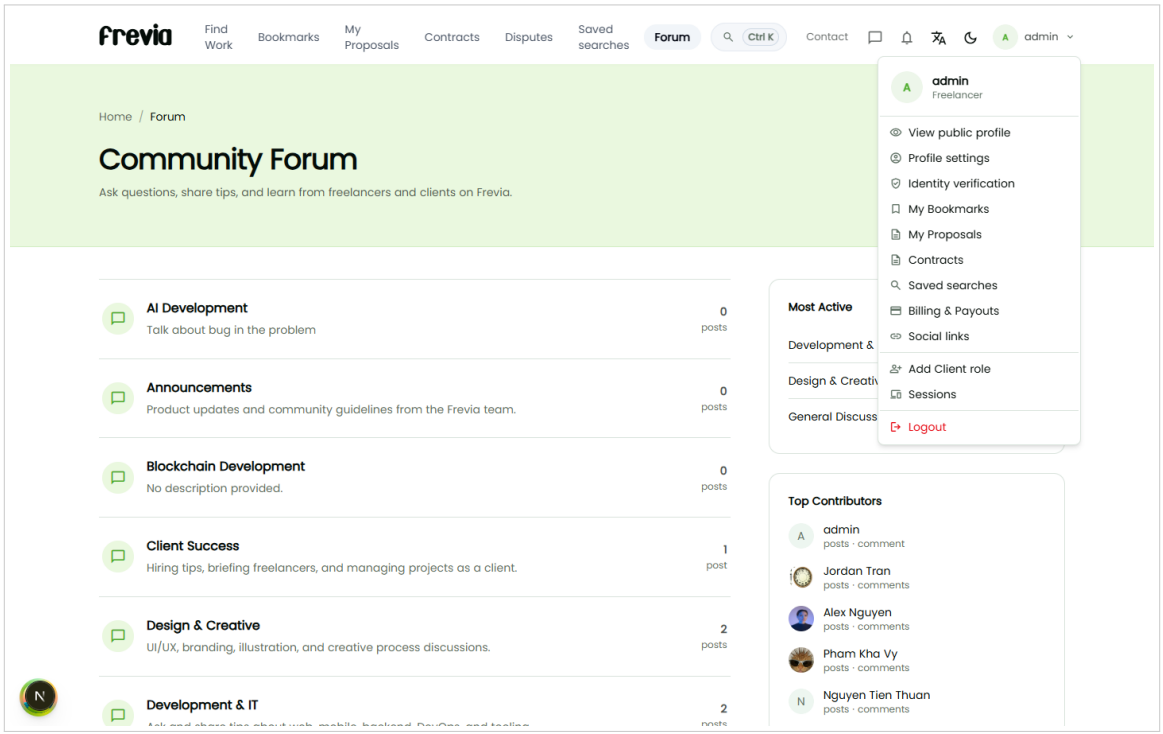


**Figure 53:** User logout interface

- Navigation sidebar includes workspace navigation links and administrative sections.
- The bottom section of the sidebar features the "Log out" command with a distinct exit glyph.
- Visual feedback indicates hover state, alerting the user to session termination.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| Session Cookies | Active `accessToken` and `refreshToken` supplied via HTTP request. |

### Validation Rules

- The request must supply a valid authentication cookie or bearer token.
- Session must correspond to an active record in the database.

### Business Rules

- **BR-UA-09**: Logout must invalidate the database session record before clearing browser cookies.

- **BR-UA-10**: Terminated sessions shall be rejected on all subsequent API requests.

**Normal Flow**

1. User clicks “Log out” in the application sidebar.
2. Frontend invokes `POST /api/auth/logout`.
3. Server deletes active session record from the database.
4. Response sets expired cookie headers (`Max-Age=0`).
5. Client resets in-memory cache and state.
6. Browser transitions to the public login page.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The session token in the cookie has already expired prior to clicking logout. The backend returns HTTP 200/204; the client cleans up remaining cookies and finishes redirect. |
| A2 | The session record was deleted remotely (e.g. via Revoke Session UC-16.03 on another device). System acknowledges logout request without error and purges local credentials. |
| A3 | Database failure occurs during session row removal. The BFF proxy forcefully clears client `httpOnly` cookies regardless, guaranteeing the user is locally signed out. |
| A4 | The user clicks the “Log out” button rapidly multiple times. The first call succeeds; subsequent calls encounter no active session and redirect gracefully. |
| A5 | Client network drops during the logout dispatch. The browser client detects the failure, purges in-memory auth tokens, and forces navigation to `/login`. |
| A6 | The token provided carries an invalid or tampered cryptographic signature. Backend rejects the token with 401, clears invalid cookies, and routes to login. |

**Output Data**

| Data Name | Description |
|---|---|
| Cookie Headers | Set-Cookie headers invalidating `accessToken` and `refreshToken`. |
| Response Body | JSON acknowledgment: `{ success: true, message: "Logged out successfully" }`. |

**Post-Conditions**

- Database session is terminated.
- Authentication cookies are eliminated from the browser.
- Protected routes are no longer accessible without re-authenticating.

### 3.2.4 Register

**Function Trigger**

A new visitor clicks the "Sign up" link and completes the registration form.

**Navigation path:** Home → Login (`/login`) → "Sign up" link (`/register`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** Guest.
- **Secondary Actor:** Mail Delivery Service (Nodemailer / SMTP Server).
- **Supporting Systems:** Frevia Web application (Next.js), Frevia BFF proxy, Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Allows a new user to create a Frevia account with credentials, select an initial platform role (Client or Freelancer), and verify their email via a single-use OTP code.

**Interface**

- Registration page at `/register` with Full Name, Email, Password, and Password Confirmation fields.
- Role selection cards allowing user to choose between "Client" (hire talent) or "Freelancer" (find work).
- Inline OTP verification section with "Send OTP" button and 6-digit verification code input.
- Checkbox for accepting Terms of Service and Privacy Policy.

- Primary submit button labeled “Continue to Profile”.

### Data Processing Overview

- User inputs email and clicks “Send OTP”. Backend validates email uniqueness and generates a 6-digit OTP code stored with a 10-minute expiration timestamp.
- The mailer sends the OTP to the specified email address.
- User fills credentials, chooses role, inputs OTP, checks terms, and submits `POST /api/auth/register`.
- Backend verifies OTP validity, hashes the password using bcrypt with 10 salt rounds, and writes user account to the database.
- A role profile (Client or Freelancer) is initialized, tokens are issued, and the user is redirected into onboarding.

### Screen Layout (Mock-up Description)

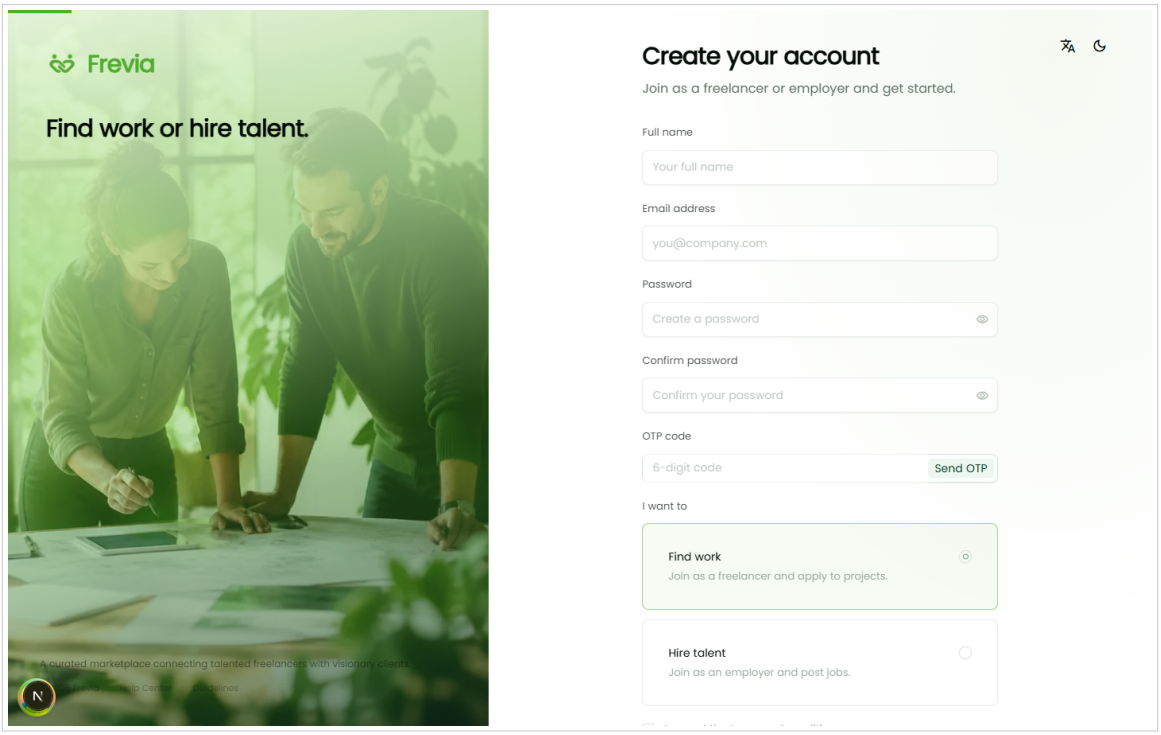


**Figure 54:** User registration screen

- Header displays “Create an account” and role choice selector.
- Two distinct selection cards allow choosing between “Join as Client” or “Join as Freelancer”.
- Form inputs collect Full Name, Email Address, Password, and Confirm Password with real-time strength feedback.
- OTP input field includes an integrated “Send OTP” trigger button.

- Terms of Service acceptance checkbox with active hyperlink references.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| `name` | User's full display name string (2 to 50 characters). |
| `email` | Valid email address string; must be unique across the platform. |
| `password` | Strong password (minimum 8 characters with letters, numbers, and symbols). |
| `role` | Selected initial platform role: `CLIENT` or `FREELANCER`. |
| `otp` | 6-digit verification code delivered to the user's email. |
| `agreeTerms` | Boolean flag indicating user accepted terms and policies. |

**Validation Rules**

- Email must conform to RFC 5322 format and not already exist in the database.
- Password must be at least 8 characters and match `confirmPassword`.
- OTP must match the unexpired code recorded for that email address.
- `agreeTerms` must be strictly `true`.

**Business Rules**

- **BR-UA-11**: Email addresses must be globally unique across all accounts.
- **BR-UA-12**: The user must explicitly choose an initial role; default assignment is forbidden.
- **BR-UA-13**: Passwords must be hashed with bcrypt prior to persistence.
- **BR-UA-14**: OTP codes expire after 10 minutes and can only be validated once.

**Normal Flow**

1. Visitor opens `/register` and inputs name, email, and password.
2. Visitor selects preferred role (Client or Freelancer).
3. Visitor clicks "Send OTP"; system sends verification email and starts countdown.
4. Visitor inputs 6-digit code received in email and checks terms agreement.
5. Visitor clicks "Continue to Profile".
6. System validates OTP, creates account and profile, issues tokens, and navigates to onboarding.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | One or more mandatory fields are omitted. Form resolver blocks request and surfaces inline error messages. |
| A2 | The email entered does not match standard RFC 5322 format. Form displays "Please enter a valid email address." |
| A3 | The email is already registered by another account. Backend returns HTTP 422 with `AuthMessage.EMAIL_ALREADY_EXISTS`: "This email is already registered." |
| A4 | The password does not satisfy complexity constraints (less than 8 chars, missing uppercase/number). Form displays requirement indicators. |
| A5 | Password confirmation does not match the entered password. Form highlights confirm field with "Passwords do not match." |
| A6 | The submitted OTP does not match the generated code. Backend throws `InvalidVerificationCodeException` (HTTP 422): "Invalid verification code." |
| A7 | The submitted OTP code has exceeded its 10-minute lifetime. Backend throws `OTPExpiredException` (HTTP 422): "OTP has expired. Please request a new one." |
| A8 | Guest repeatedly clicks "Send OTP" within a short timeframe. Rate limiter triggers `TooManyAttemptsException` (HTTP 422): "Too many attempts. Please try again later." |
| A9 | Mail server fails during OTP dispatch. Backend throws `FailedToSendOTPException` (HTTP 422): "Unable to send OTP. Please try again later." |
| A10 | Guest fails to check the Terms of Service box. System disables final submission with "You must accept the Terms of Service and Privacy Policy." |

**Output Data**

| Data Name | Description |
|---|---|
| `user` | Created user record with ID, email, name, role, and verification timestamp. |
| Auth Cookies | Newly issued `accessToken` and `refreshToken` stored in httpOnly cookies. |

**Post-Conditions**

- New user row and role relationship written to the database.
- User is immediately logged in and guided to role profile setup.

### 3.2.5 Forgot Password

**Function Trigger**

A user clicks "Forgot password?" on the login screen and submits their email address. **Navigation path:** Login (`/login`) → "Forgot password?" (`/forgot-password`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** Guest (registered user who lost access).
- **Secondary Actor:** Mail Delivery Service (Nodemailer / SMTP Server).
- **Supporting Systems:** Frevia Web application (Next.js), Frevia BFF proxy, Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Provides a secure self-service mechanism for users to reset their forgotten passwords via email verification links, preventing account lockout while safeguarding against credential enumeration attacks.

**Interface**

- Forgot password page at `/forgot-password` with email input and submit button.
- Confirmation feedback screen informing user to check their email inbox.
- Secure reset password screen accessed through the unique tokenized email link.

**Data Processing Overview**

- User enters email and submits `POST /api/auth/forgot-password`.

- Backend looks up user record; if found, generates a cryptographically random token with 1-hour expiration.
- An email containing the password reset URL is dispatched.
- Regardless of whether the email exists, the API returns the identical generic message to prevent account enumeration.
- When user opens the link and submits a new password, backend validates token, hashes new password, updates database, invalidates the token, and revokes all active sessions.

### Screen Layout (Mock-up Description)

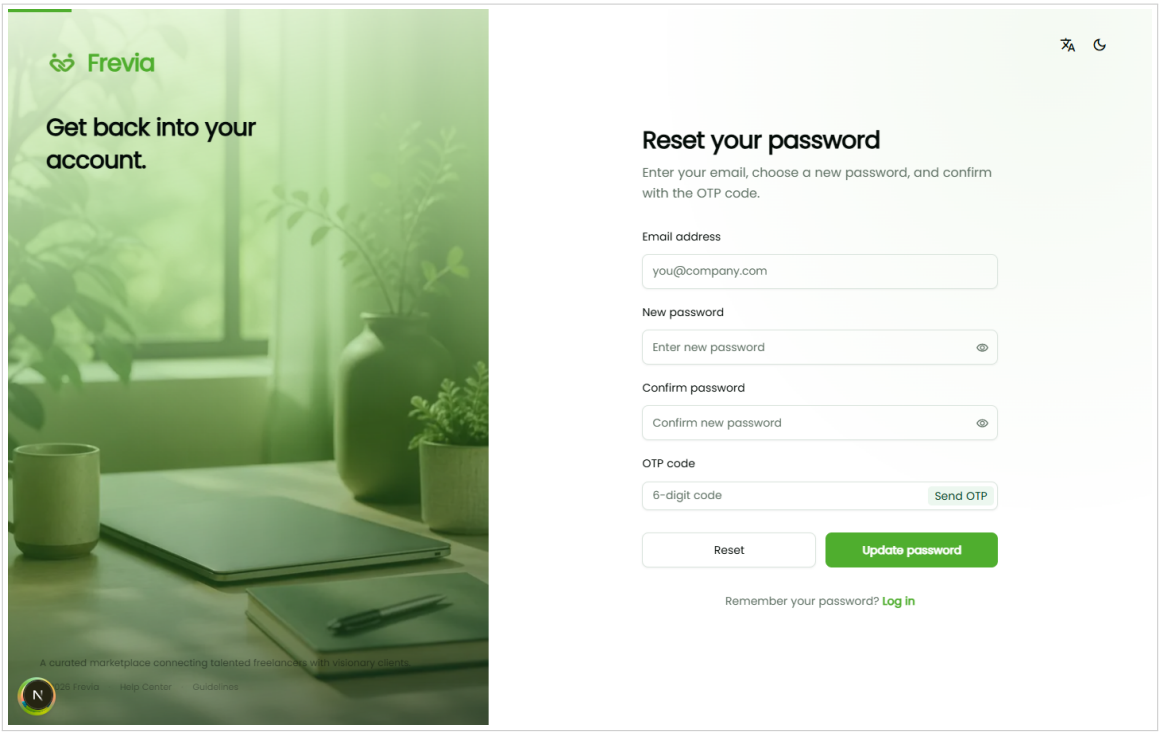


**Figure 55:** Forgot password screen

- Card heading reads “Forgot password?” with descriptive subtitle explaining the recovery process.
- Email input field with placeholder “you@company.com”.
- Solid green button “Send reset link”.
- Navigation link “Back to login” at the bottom of the card.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `email` | Registered email address string. |
| `token` | Secret reset token received in URL parameter (during password update). |
| `newPassword` | Replacement password meeting platform strength criteria. |

**Validation Rules**

- `email` must be a valid email string.
- Reset token must exist, not be expired, and have not been previously consumed.
- `newPassword` must meet minimum length (8 chars) and complexity rules.

**Business Rules**

- **BR-UA-15**: System must return identical confirmation text whether the email is registered or not (anti-enumeration).
- **BR-UA-16**: Reset tokens expire exactly 1 hour following issuance.
- **BR-UA-17**: Successful reset terminates all existing sessions for the target account.

**Normal Flow**

1. User navigates to `/forgot-password` and enters registered email.
2. User clicks "Send reset link".
3. Backend generates token, sends email, and returns generic confirmation.
4. User clicks the link in email to reach the reset screen.
5. User inputs new password and submits.
6. System updates password, revokes active sessions, and redirects to login with success toast.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Email field left blank or submitted with invalid syntax. Client validation displays "Please enter a valid email address." |
| A2 | The email entered is not associated with any account in the database. The system returns the identical success message to prevent user enumeration. |

| Case | Description |
|---|---|
| A3 | The account exists but was registered purely via Google OAuth (no local password). System sends an advisory email guiding the user to sign in via Google OAuth. |
| A4 | The password reset token in the link has exceeded its 1-hour expiration window. The reset page displays “This reset link has expired. Please request a new one.” |
| A5 | The token has already been consumed or carries an invalid cryptographic signature. System surfaces “This reset link is invalid or has already been used.” |
| A6 | The replacement password fails strength constraints or confirmation mismatch occurs. Form highlights the fields inline with guidance text. |
| A7 | Mail server failure during email dispatch. System logs error internally and returns standard response to protect user privacy. |
| A8 | Database connection drops during password update and session revocation. System returns HTTP 500; old password remains active. |

**Output Data**

| Data Name | Description |
|---|---|
| Email Dispatch | HTML email containing secure reset link with query token. |
| Status Message | Informational confirmation indicating reset link dispatch. |

**Post-Conditions**

- Account password is updated to new hashed value.
- Reset token is destroyed.
- Prior authentication sessions are revoked.

## 3.3 Role Management

### 3.3.1 View Role List

### Function Trigger

An administrator accesses the Role Management section from the admin portal sidebar. **Navigation path:** Admin Dashboard → Roles & Permissions → Roles (`/admin/roles`).

### Function Description

### Actors / Roles

- **Primary Actor:** Administrator.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API (NestJS), and PostgreSQL database.

### Purpose

Provides administrators with a centralized overview of all configured system access roles, displaying role names, descriptions, assigned permission sets, and action triggers for configuration.

### Interface

- Table view displaying roles with columns: ID, Name, Description, Permissions, Created At, and Actions.
- Action toolbar containing "Create Role" button.
- Row-level action buttons: Edit Role, Delete Role, and View Details.

### Data Processing Overview

- The client issues `GET /api/roles` upon rendering.
- The backend queries the database for all roles where `deletedAt = null`.
- Each role includes its associated permissions joined through the `RolePermission` relation.
- Results are returned ordered by creation date and rendered in the interactive table.

### Screen Layout (Mock-up Description)

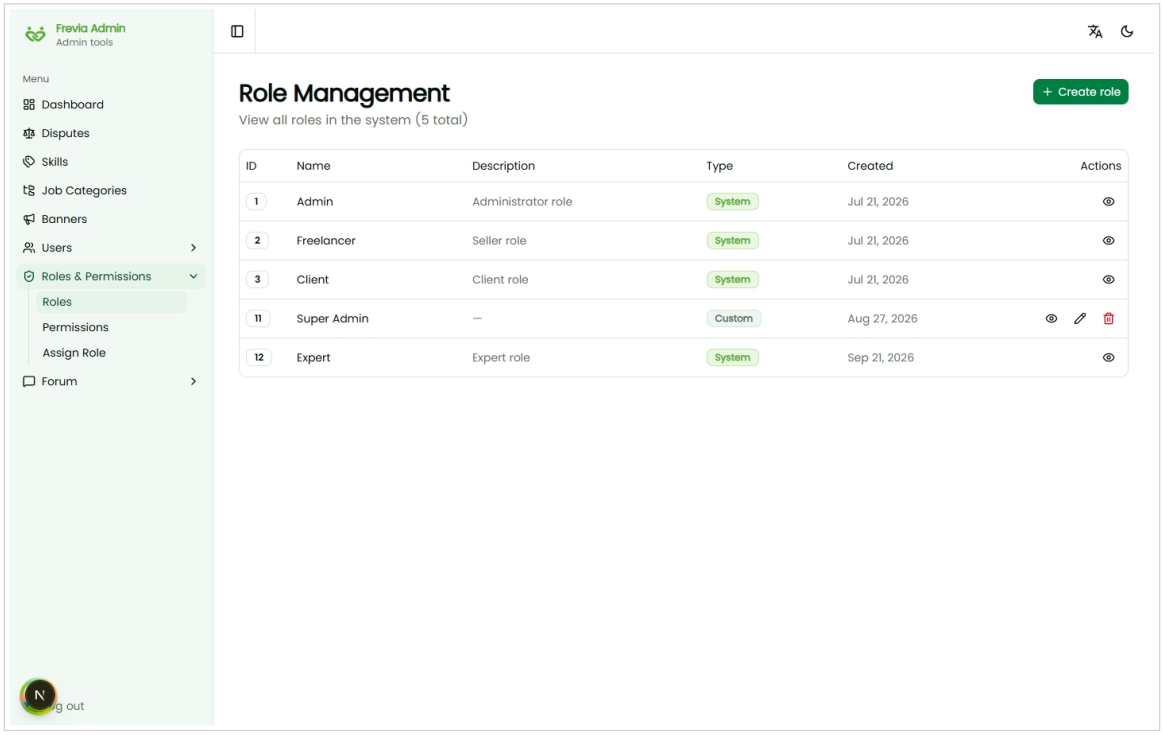


**Figure 56:** Role list screen

- Top header shows page title “Role Management” and total count subtitle.
- “Create Role” button positioned on the top right in green accent.
- Data table shows system roles (Admin, Client, Freelancer, Expert) and custom roles with permission badges.
- Actions column provides edit, delete, and detail view buttons for each record.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| Admin Session | Valid JWT authentication cookie with Administrator authorization. |

### Validation Rules

- User must possess the `Admin` role.
- Only active roles with `deletedAt = null` are returned.

### Business Rules

- **BR-RM-01**: Role management is strictly restricted to authenticated Administrators.
- **BR-RM-02**: Soft-deleted roles shall be excluded from the active list.

**Normal Flow**

1. Admin clicks “Roles” in the admin sidebar.
2. Browser sends `GET /api/roles`.
3. API queries active roles and joined permissions.
4. Table renders all roles with their assigned permission badges and action controls.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Unauthenticated or non-admin user navigates to the URL. The route middleware blocks access and redirects to `/login` or returns HTTP 403 Forbidden. |
| A2 | Database connection drops or query times out. The API throws `FailedToLoadRolesException` (HTTP 500); client displays “Unable to load role list. Please try again later.” |
| A3 | No active roles found in the database. The table renders a clean placeholder “No roles available” with a call-to-action to create a role. |
| A4 | Elevated network latency slows data arrival. TanStack Query displays skeleton placeholders across table rows, preventing layout jump. |
| A5 | Admin session token expires while viewing the list. A background query receives HTTP 401; the application clears cookies and redirects to sign-in. |
| A6 | A role title or description contains exceptionally long text. The table cell applies CSS truncation with hover tooltip, preventing column overflow. |

**Output Data**

| Data Name | Description |
|---|---|
| `roles[]` | Array of role records containing `id`, `name`, `description`, `permissions[]`, and `createdAt`. |

**Post-Conditions**

- Read-only operation; system data remains unchanged.

### 3.3.2 View Role Detail

**Function Trigger**

An administrator clicks the "View details" button (eye icon) on a role row or navigates to `/admin/roles/:id`.
**Navigation path:** `/admin/roles` → Click View Details icon on a row.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Administrator.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Presents detailed information regarding a specific role, including its identifier, name, description, assigned permissions, and user counts.

**Interface**

- Role detail screen showing Role ID, Name, Description, and Metadata cards.
- Permissions catalog panel showing assigned route permissions grouped by category.
- Navigation back-button to return to the role table.

**Data Processing Overview**

- Client dispatches `GET /api/roles/:id`.
- Backend fetches role matching `id` with `deletedAt = null`.
- Backend joins permissions through the relation model and returns role details.
- Client renders role attributes and permission badges.

### Screen Layout (Mock-up Description)

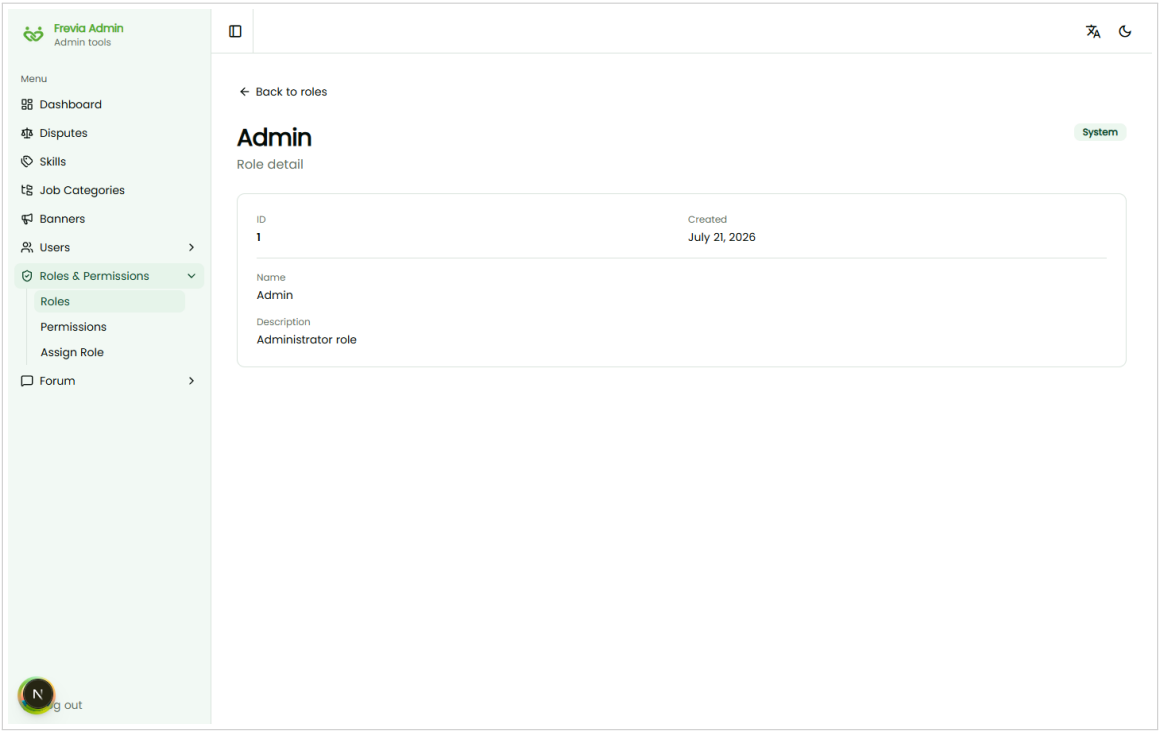


**Figure 57:** Role detail screen

- Header card presents role title with action buttons (Edit, Delete).
- Information section details Role ID, Creation Timestamp, and Description.
- Lower container displays full assigned permissions breakdown.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Numeric identifier of the target role passed in route path. |

### Validation Rules

- Target role must exist and not be soft deleted.

### Business Rules

- **BR-RM-03**: Soft-deleted roles cannot be viewed in detail mode.

### Normal Flow

1. Admin clicks view icon on a role in the role list.

2. Application requests `GET /api/roles/:id`.
3. API returns role object with permissions.
4. Screen displays full role specifications and assigned routes.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Non-numeric or invalid ID format in URL path (e.g., `/admin/roles/abc`). Router fails validation and shows 404 Not Found page. |
| A2 | Role ID does not exist or has already been soft-deleted. API throws `RoleNotFoundException` (HTTP 404); page displays "Role not found" with return button. |
| A3 | Database query failure during role retrieval. API throws `FailedToLoadRoleDetailException` (HTTP 500); client shows error notification toast. |
| A4 | The role currently has zero permissions assigned. The permissions section displays an informative empty card "No permissions assigned to this role yet." |
| A5 | Unauthenticated visitor attempts to access direct detail URL. Middleware intercepts and redirects to `/login`. |
| A6 | The role is deleted by another administrator concurrently while being viewed. Subsequent edits or page refreshes reflect the deletion and navigate back. |

**Output Data**

| Data Name | Description |
|---|---|
| Role Record | Comprehensive role object with permissions list. |

**Post-Conditions**

- Read-only operation.

### 3.3.3 Create Role

### Function Trigger

An administrator clicks the "Create Role" button on the Role Management page.

**Navigation path:** `/admin/roles` → Click "Create Role" button.

### Function Description

### Actors / Roles

- **Primary Actor:** Administrator.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API (NestJS), and PostgreSQL database.

### Purpose

Allows administrators to establish a new custom RBAC role with defined title, description, and initial permission bindings.

### Interface

- Modal dialog with fields for Role Name and Description.
- Cancel button and solid green "Create Role" submit button.

### Data Processing Overview

- Admin enters details and submits `POST /api/roles`.
- Backend checks for duplicate role names and prevents collision with reserved system roles.
- System persists new role record in PostgreSQL.
- Table view refreshes immediately to display the created role.

### Screen Layout (Mock-up Description)

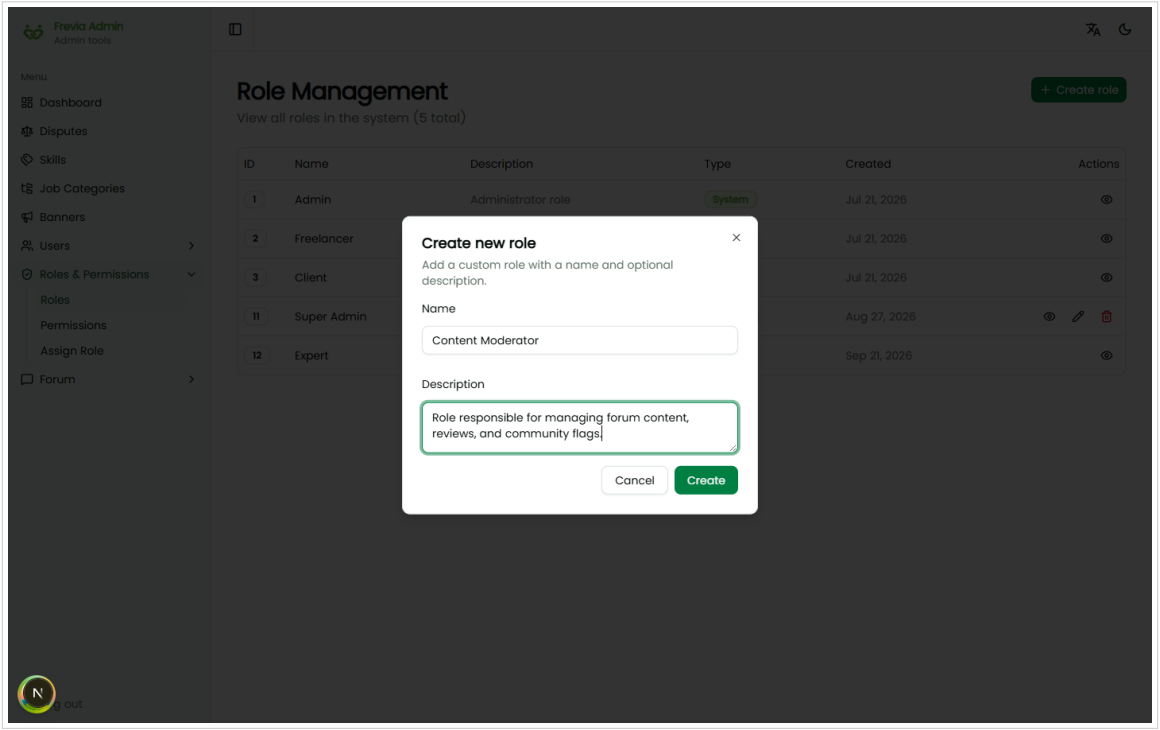


**Figure 58:** Create role dialog

- Dialog modal overlays the page with dark backdrop.
- Header reads “Create Role” with descriptive instruction text.
- Name input marked with required asterisk (*).
- Description textarea for functional documentation.
- Footer provides “Cancel” and “Create Role” actions.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `name` | Unique role name string (required, 2 to 50 characters). |
| `description` | Optional role purpose description string. |

### Validation Rules

- Role name must be non-empty and unique among active roles.
- Role name must not match system keywords: `Admin`, `Client`, `Freelancer`, `Expert`.

### Business Rules

- **BR-RM-04**: System role names are immutable and reserved.
- **BR-RM-05**: Duplicate active role names are strictly rejected.

**Normal Flow**

1. Admin clicks “Create Role” button.
2. Admin inputs role name and description in dialog.
3. Admin clicks “Create Role” button to submit.
4. Backend creates record and returns HTTP 201.
5. Dialog closes, success toast appears, and table updates.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Role name field submitted empty or consisting only of whitespace. Frontend validation halts request: “Name is required.” |
| A2 | Role name already in use by another active role. API throws `RoleAlreadyExistsException` (HTTP 409): “A role with this name already exists.” |
| A3 | Admin enters a reserved system role name (`Admin`, `Client`, `Freelancer`). API throws `CannotModifySystemRoleException` (HTTP 403): “Cannot use a reserved system role name.” |
| A4 | Name exceeds maximum allowed character length (50 characters). Client enforces limit and shows character limit indicator. |
| A5 | Database error occurs during insertion. API throws `FailedToCreateRoleException` (HTTP 500); system shows error toast and retains entered text in dialog. |
| A6 | Admin clicks “Cancel” or presses Escape key. Dialog closes immediately and unsaved data is discarded. |
| A7 | Rapid double-clicking on submit button. Button disables into loading state on first click, preventing duplicate records. |

**Output Data**

| Data Name | Description |
|---|---|
| Created Role | Newly persisted role entity with assigned ID and timestamps. |

**Post-Conditions**

- New role record exists in database and is visible in role list.

### 3.3.4 Update Role

**Function Trigger**

An administrator clicks the "Edit" icon (pencil) on a role row in the Role Management table.

**Navigation path:** `/admin/roles` → Click Edit icon on row.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Administrator.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Allows administrators to modify the display name and descriptive notes of an existing role.

**Interface**

- Edit Role dialog prefilled with the role's current name and description.
- Save Changes button and Cancel button.

**Data Processing Overview**

- Admin modifies fields and clicks "Save Changes".
- Request sent via `PATCH /api/roles/:id` or `PUT /api/roles/:id`.
- Backend validates name uniqueness against other active roles.
- Updates database row and returns updated entity.

### Screen Layout (Mock-up Description)

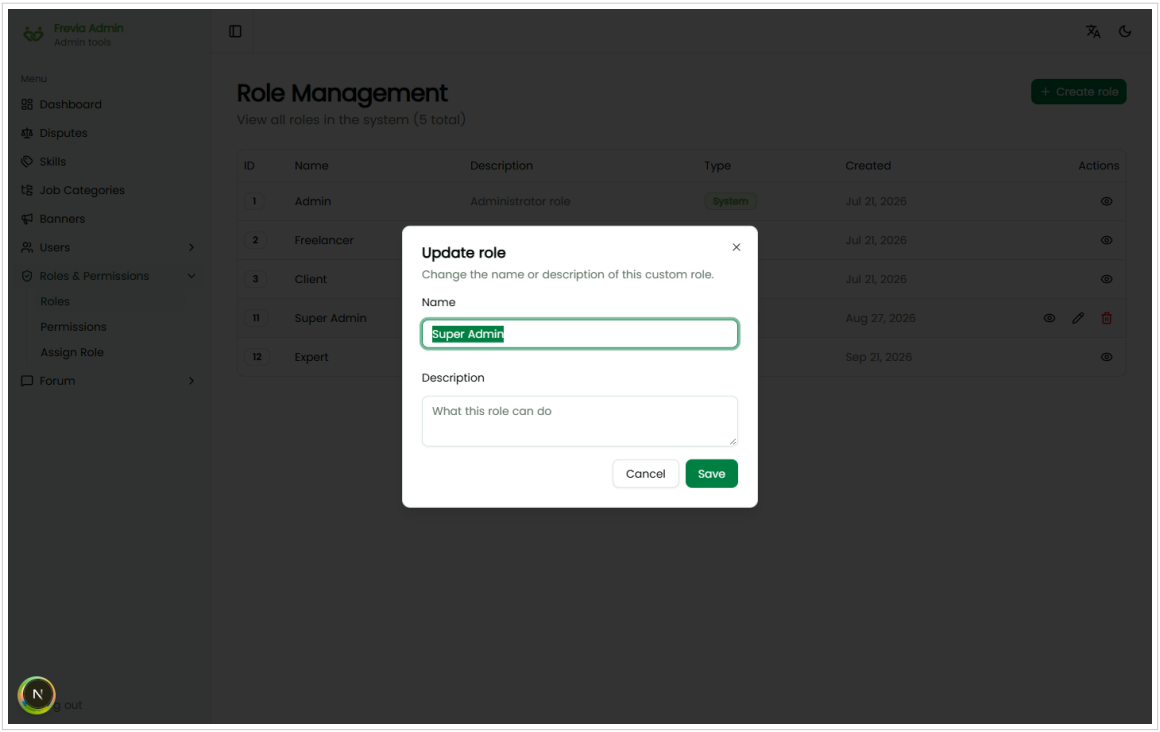


**Figure 59:** Update role dialog

- Dialog title states “Update Role” with explanation.
- Name input pre-populated with current role title.
- Description area pre-populated with existing text.
- Action buttons: Cancel and Save Changes.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Target role identifier. |
| `name` | Updated role title string. |
| `description` | Updated description string. |

### Validation Rules

- Name must be non-empty and unique among active roles.
- System role names cannot be reassigned to other roles.

### Business Rules

- **BR-RM-06**: Role ID remains unchanged throughout update.

**Normal Flow**

1. Admin clicks edit icon on desired role row.
2. Modal opens with existing values prefilled.
3. Admin updates fields and clicks “Save Changes”.
4. Server updates record, closes dialog, and refreshes view.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Target role was deleted or no longer exists. API throws `RoleNotFoundException` (HTTP 404): “Role not found.” |
| A2 | Admin erases name field. Client validation rejects submission: “Role name is required.” |
| A3 | Updated name conflicts with another existing active role. API throws `RoleAlreadyExistsException` (HTTP 409): “A role with this name already exists.” |
| A4 | Admin attempts to rename a protected system role. API throws `CannotModifySystemRoleException` (HTTP 403): “System roles cannot be renamed.” |
| A5 | Database update query fails. API throws `FailedToUpdateRoleException` (HTTP 500); error toast is shown and form retains modified values. |
| A6 | Admin clicks “Cancel” or backdrop. Dialog dismisses and values revert to database state without persisting. |
| A7 | Network disconnects while submitting. Form indicates network failure and allows retry without re-typing. |

**Output Data**

| Data Name | Description |
|---|---|
| Updated Entity | Modified role row with updated timestamp. |

**Post-Conditions**

- Updated role information is reflected across all administrative screens.

### 3.3.5 Delete Role

**Function Trigger**

An administrator clicks the "Delete" icon (trash can) on a custom role row.

**Navigation path:** `/admin/roles` → Click Delete icon on row.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Administrator.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Soft-deletes a custom role from the platform, verifying that it is not assigned to active users and protecting core system roles from accidental deletion.

**Interface**

- Delete confirmation alert dialog with warning message.
- "Cancel" action and destructive red "Delete Role" button.

**Data Processing Overview**

- Admin confirms deletion in dialog modal.
- Client dispatches `DELETE /api/roles/:id`.
- Backend verifies role is not a protected system role and is not currently held by active users.
- Backend sets `deletedAt = new Date()` to soft-delete the record.
- Table updates and excludes the deleted role from the active list.

### Screen Layout (Mock-up Description)

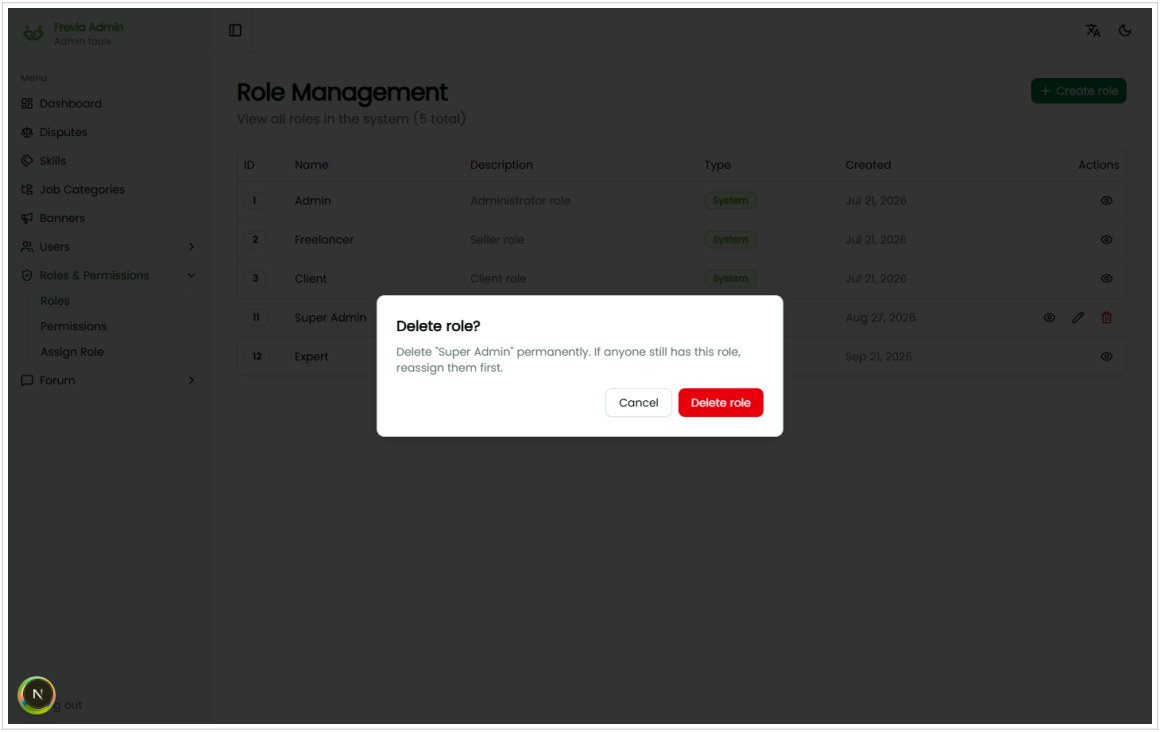


**Figure 60:** Delete role confirmation dialog

- Alert dialog heading “Delete Role” with role name.
- Confirmation warning text explaining the implications of removal.
- Action buttons: Cancel and Delete Role in red destructive styling.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Target role identifier to be soft-deleted. |

### Validation Rules

- Role must exist and not already be deleted.
- Role must not be a protected system role (`Admin`, `Client`, `Freelancer`).
- Role must not have active assigned users.

### Business Rules

- **BR-RM-07**: Deletion must be executed as a soft delete via `deletedAt`.
- **BR-RM-08**: Roles assigned to active users cannot be deleted until users are reassigned.

**Normal Flow**

1. Admin clicks delete icon on a custom role row.
2. Confirmation alert dialog appears.
3. Admin confirms deletion by clicking “Delete Role”.
4. Backend performs soft delete and returns HTTP 200.
5. Table view removes row and shows success toast.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Role does not exist or has already been soft-deleted. API throws `RoleNotFoundException` (HTTP 404): “This role has already been deleted.” |
| A2 | Role is currently assigned to one or more active users in `UserRole`. API throws `RoleInUseException` (HTTP 409): “This role cannot be deleted while assigned to active users.” |
| A3 | Admin attempts to delete a protected system role (`Admin`, `Client`, `Freelancer`, `Expert`). API throws `CannotModifySystemRoleException` (HTTP 403): “System roles cannot be deleted.” |
| A4 | Admin clicks “Cancel” or presses Escape. The dialog dismisses without dispatching any deletion request. |
| A5 | Database error during soft-delete execution. API throws `FailedToDeleteRoleException` (HTTP 500); client notifies “Unable to delete role. Please try again later.” |
| A6 | The role was concurrently deleted by another admin session. Table refreshes and informs user the role is already removed. |

**Output Data**

| Data Name | Description |
|---|---|
| Result Status | Success confirmation with soft-deleted timestamp. |

**Post-Conditions**

- Target role has `deletedAt != null`.
- Role is omitted from all standard role selectors and lists.

### 3.3.6 Update Role Permissions

**Function Trigger**

An administrator navigates to the Assign Role screen, selects a role, modifies permission checkboxes, and saves.
**Navigation path:** Admin Dashboard → Roles & Permissions → Assign Role (`/admin/roles/assign`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** Administrator.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API (NestJS), and PostgreSQL database.

**Purpose**

Configures granular API route access permissions for any system or custom role by selecting or unchecking individual permissions grouped by functional module.

**Interface**

- Role selector dropdown at the top of the interface.
- Permission matrix grouped by module cards (AUTH, ROLES, USERS, FORUMS, DISPUTES, etc.).
- Checkbox controls for individual permissions and module-level "Select All" toggles.
- Save permissions button fixed at bottom or header.

**Data Processing Overview**

- Selecting a role fetches its current permissions via `GET /api/roles/:id`.
- Checkboxes automatically pre-populate with existing assigned permissions.
- Admin modifies selections and clicks "Save permissions".

- Client dispatches `PUT /api/roles/:id/permissions` with array of selected permission IDs.
- Backend executes a transaction: purges existing `RolePermission` associations for this role and inserts new bindings for all selected IDs.
- Cache is refreshed and permission changes take effect immediately.

### Screen Layout (Mock-up Description)

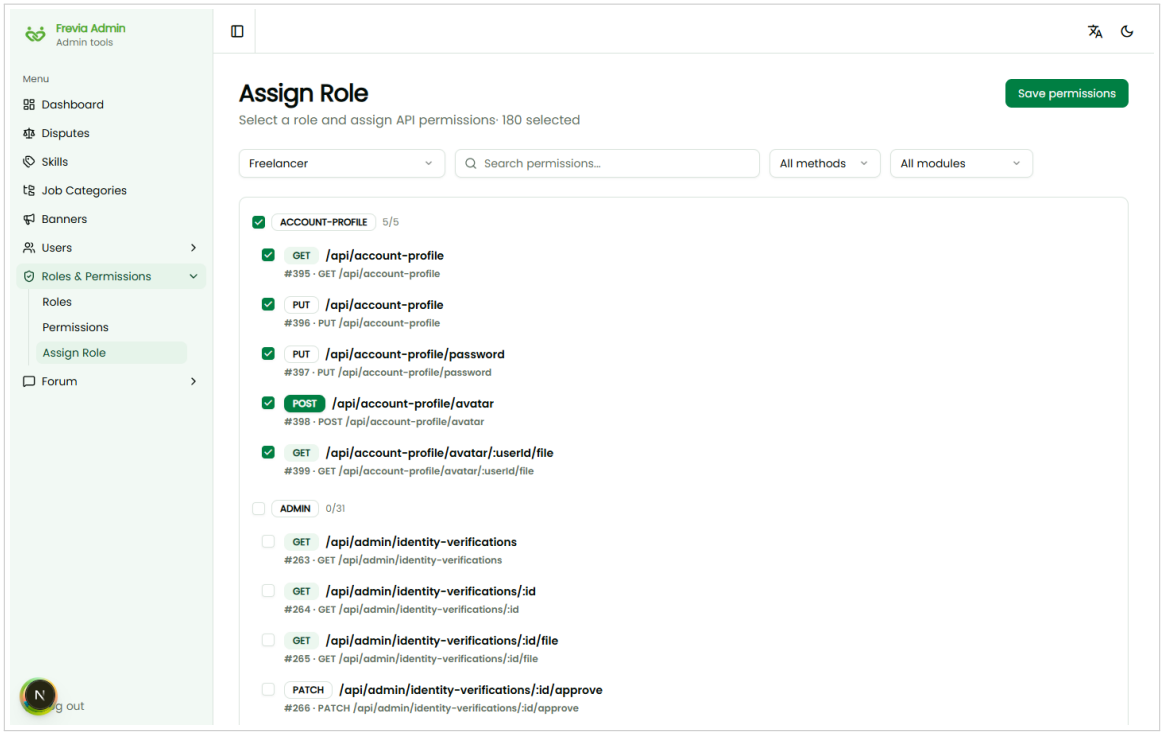


**Figure 61:** Assign permissions to role screen

- Header displays “Assign Role Permissions” and the role selection dropdown.
- Main area renders permission module cards with HTTP method badges (GET, POST, PUT, DELETE).
- Checkboxes next to each permission indicate active access rights.
- “Save permissions” button triggers atomic database updates.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `roleId` | Target role identifier. |
| `permissionIds[]` | Array of selected integer permission IDs. |

**Validation Rules**

- Target role must exist and not be soft deleted.
- All passed `permissionIds` must match existing active permissions in the database.

**Business Rules**

- **BR-RM-09**: Permission updates replace the role's entire permission mapping set.
- **BR-RM-10**: Permission changes take effect immediately across all active user sessions with that role.

**Normal Flow**

1. Admin opens the Assign Role page and selects a role from the dropdown.
2. System loads permissions and pre-checks currently assigned rights.
3. Admin checks or unchecks desired route permissions.
4. Admin clicks "Save permissions".
5. Backend replaces role permission mappings in an atomic transaction.
6. Success notification confirms update and new access rights are live.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | Admin attempts to save without selecting a role. UI disables button and prompts "Choose a role to start assigning permissions." |
| A2 | Target role does not exist or was soft-deleted. API throws `RoleNotFoundException` (HTTP 404): "Role not found." |
| A3 | Admin deselects all checkboxes. System prompts confirmation for empty permissions set, then clears all route access upon confirmation. |
| A4 | One or more permission IDs no longer exist in database. API throws `InvalidPermissionIdsException` (HTTP 400): "One or more selected permissions are invalid." |

| Case | Description |
|---|---|
| A5 | Database transaction failure during batch update. API throws `FailedToSetRolePermissionsException` (HTTP 500); prior bindings remain unchanged. |
| A6 | Admin attempts to revoke critical access rights from the core Administrator role. System performs security check to prevent total admin lockout. |
| A7 | Network disconnects during save request. Checkbox selection state is preserved in client memory, and an alert toast offers immediate retry. |

**Output Data**

| Data Name | Description |
|---|---|
| Updated Role | Role record containing updated list of assigned permission IDs. |

**Post-Conditions**

- `RolePermission` table rows are synchronized with the selected set.
- Authorization guards apply new permissions upon subsequent user requests.

## 3.4 Manage Forum

### 3.4.1 View Forum Category List

**Function Trigger**

Any visitor opens the Community Forum area from the main navigation, or follows a direct link to the forum landing page. The function is public, therefore it runs on every visit of the page, whether or not the visitor is signed in.
**Navigation path:** Home → Forum (`/forum`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** Guest, Freelancer, Client and Administrator. The screen is readable by every role, including visitors without an account.
- **Supporting Systems:** Frevia Web application (Next.js), Frevia BFF proxy (`/api/backend`), Frevia REST API (NestJS) and PostgreSQL accessed through Prisma ORM.

### Purpose

The function is the entry point of the community area. It presents the active forum categories together with the number of published posts that each one contains, and it highlights both the categories and the members with the highest activity, so a visitor can reach an ongoing discussion in one click and can immediately see which topics are alive.

### Interface

- Community Forum page at `/forum`. Guests see the public header and footer; a signed-in member sees the dashboard shell with the sidebar of their role (Client or Freelancer).
- Hero section with the breadcrumb "Forum", the language switcher, the title "Community Forum" and the description "Ask questions, share tips, and learn from freelancers and clients on Frevia.".
- Category search form with the placeholder "Search categories…", a search icon, a clear button and a "Search" action. The search is applied automatically after a short pause and can also be submitted with Enter.
- List of category rows; each row shows a message icon, the category name, its description and the number of posts, and opens the category detail screen.
- "Most Active" panel listing the three categories with the largest number of posts.
- "Top Contributors" panel listing the five members with the largest number of posts, with their avatar, display name, post counter and comment counter.
- Result summary "Showing x–y of z categories" followed by numbered pagination with ellipsis; the list displays five categories per page.
- Empty-state panel with the title "No categories yet" or "No categories found" depending on whether a search is active; the no-result state offers a "Clear search" action.

### Data Processing Overview

- The server component of the page reads the optional `search`, `page` and `limit` parameters from the query string.
- The page requests `GET /api/forums/categories` and `GET /api/forums/users/top?limit=5` in parallel, adding the access token when the visitor is signed in.
- The API keeps only the categories whose `deletedAt` is null, orders the result by `name` in ascending order, and counts only the posts that are not soft deleted and whose `moderationStatus` is `APPROVED`.
- When a search term is present, the page filters the categories by name or by description without case sensitivity before computing the pagination; the API receives no search parameter for categories.
- The page applies the page size locally: the default is five categories per page, the page

number is clamped between 1 and the last page, and the summary reports the displayed range and the filtered total.

- The “Most Active” panel is computed from the complete category list by sorting it by `postCount` in descending order and taking the first three entries; it does not issue a separate request.
- The “Top Contributors” panel uses the second response: the API selects the users that are neither deleted nor banned, counts their approved posts and their non-deleted comments, and orders them by post count in descending order.
- A category without a description is rendered with the fallback text “No description provided.”; an empty result set is rendered as “No categories yet” or “No categories found” when a search is active.

### Screen Layout (Mock-up Description)

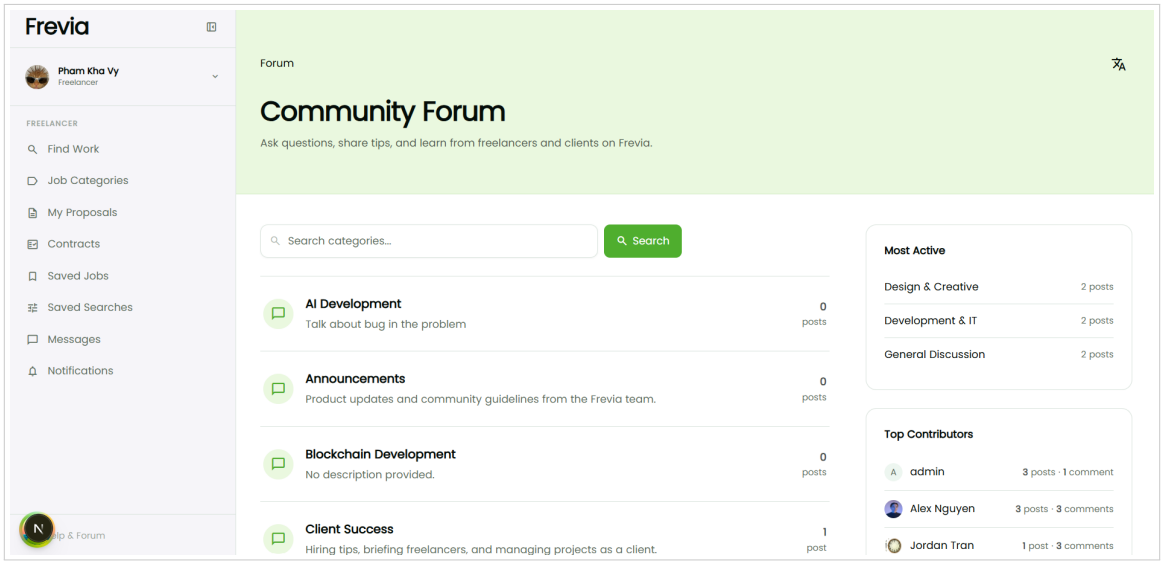


**Figure 62:** Forum category list screen

- The hero states the title “Community Forum” and a short description of the area, which introduces the forum to first-time visitors; the breadcrumb and the language switcher sit on the same line.
- The main area is split into two columns on large screens: the category list occupies the left column and the two ranking panels occupy the right column.
- Each category row is a link to the category detail screen. It displays a round message icon, the category name, the description limited to two lines and, on the right, the post counter with the word “posts”.
- The search form sits above the list. The clear button appears only when the field contains text, and the result summary and the pagination sit below the list.
- The “Most Active” panel ranks categories by post volume and gives a shortcut to the busiest

discussions.

- The “Top Contributors” panel shows the avatar, the display name, the post count and the comment count of the most active members.
- When the list is empty, the placeholder explains that categories will appear once they are created; when a search returns nothing, the placeholder invites the visitor to try another keyword or to clear the search.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| Page route | `/forum`. The page can also receive the optional `search`, `page` and `limit` parameters. |
| `search` | Optional keyword compared with the category name and description, without case sensitivity. |
| `page` | Page number of the category list; integer greater than or equal to 1, default 1. |
| `limit` | Number of categories per page; the page uses 5 by default and falls back to 5 when the parameter is not a positive integer. |
| `limit` (top users) | Number of members returned by the “Top Contributors” call; fixed to 5 by the page and validated by the API between 1 and 10. |
| Visitor session | Optional access token forwarded by the server request. A guest receives the same list and the same ranking panels. |

### Validation Rules

- The list endpoint takes no parameter, therefore it cannot be malformed; it always returns the complete list of visible categories.
- The `limit` of both ranking endpoints is coerced to an integer, must be at least 1 and at most 10, and defaults to 3 when it is absent.
- A category is visible only when its `deletedAt` column is null; a category moved to the trash is excluded from the list and from both rankings.
- A post contributes to a counter only when it is not soft deleted and its `moderationStatus` is `APPROVED`.

- A user appears in “Top Contributors” only when the account is not deleted and is not banned.

**Business Rules**

- **BR-FORUM-01**: Only categories with `deletedAt = NULL` shall be displayed.
- **BR-FORUM-02**: Categories shall be ordered by `name` in ascending order.
- **BR-FORUM-03**: `postCount` shall count only posts with `deletedAt = NULL` and `moderationStatus = APPROVED`.

**Normal Flow**

1. The visitor opens the Forum area from the navigation menu.
2. The web application renders the page shell and issues the category request.
3. The BFF proxy forwards the request to the REST API, adding the access token when the visitor is signed in.
4. The API queries the categories that are not soft deleted and counts their approved posts.
5. The API returns the ordered list of categories with `postCount`.
6. The client renders the category grid, applying the description fallback where needed.
7. The client issues the two ranking requests for the side panels.
8. The API returns the three most active categories and the five most active members.
9. The client renders both panels; the page is now fully displayed.
10. The visitor selects a category card and is routed to the category detail screen.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The database cannot be queried. The API returns a server error carrying `Error.FailedToLoadForumCategories` and the page shows a generic error instead of the grid. |
| A2 | No category exists or every category has been soft deleted. The page renders the empty state “No categories yet”; the ranking panels are hidden. |
| A3 | A category has no description. The card shows “No description provided.” instead of an empty area. |
| A4 | The ranking call fails while the list succeeds. The grid remains usable and only the affected panel is omitted, so navigation is never blocked. |
| A5 | The visitor loses the network connection. The client displays the generic error message and offers a retry by reloading the page. |

| Case | Description |
|---|---|
| A6 | The access token has expired. The BFF proxy refreshes the token once and repeats the call; if the refresh fails the visitor continues as a guest. |
| A7 | The account was banned while the page was open. The BFF proxy detects the ban response, clears the authentication cookies and the client redirects to the sign-in page. |

**Output Data**

| Data Name | Description |
|---|---|
| `categories[]` | Visible categories ordered by name, each with `id`, `name`, `slug`, `description`, `createdAt`, `updatedAt` and `postCount`. |
| `categories` (top) | Up to three categories ordered by post count in descending order, with the same fields as the list. |
| `users` (top) | Up to five members with `id`, `displayName`, `avatarUrl`, `postCount` and `commentCount`. |
| Empty collection | An empty array when no category is visible; the client renders the empty state described above. |

**Post-Conditions**

- The visitor sees every active category with the number of published posts it holds.
- No data is modified: the function is read-only.
- Deleted categories, deleted posts, rejected posts and pending posts are never exposed through this screen.

### 3.4.2 View Forum Category Detail and Post List

**Function Trigger**

The visitor selects a category row on the forum landing page, follows a shared category link, or searches for a post from the category screen.
**Navigation path:** Home → Forum → *Category* (`/forum/{category-slug}-{category-id}`).

### Function Description

### Actors / Roles

- **Primary Actor:** Guest, Freelancer, Client and Administrator.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

### Purpose

The function lists the published posts of one category so that members can read, search and filter the discussions of a topic. It also gives signed-in members the entry point to create a post, to restrict the list to their own posts, and to reach the most liked and commented posts published in the category during the last seven days.

### Interface

- Breadcrumb "Forum / *Category*" and the category header with its name, description and post count.
- Search box with the placeholder "Search posts…" and a clear-search button; the search is submitted with Enter and filters the posts of the current page by title.
- "My Posts" toggle, displayed only to signed-in members.
- "New Post" button, which opens the post creation dialog; a guest is routed to the sign-in page instead.
- Page-size selector labelled "10 / page" by default, offering 5, 10, 20 and 50 posts per page.
- Filter chips row introduced by "Filters:", with one removable chip for the search term, one removable chip for "My Posts" and a "Clear all" action.
- Post rows with the title, a two-line excerpt of the content, the author avatar and name, the creation date, the like count and the comment count.
- Summary "Showing x–y of z posts", which appends "(yours)" while "My Posts" is active, followed by numbered pagination with ellipsis.
- "Trending in {category}" panel, displayed in the left column on large screens and above the list on smaller screens, listing up to three posts of the last seven days.
- Empty states for no post ("No posts yet"), no search result ("No posts found") and no personal post ("You have no posts here").

### Data Processing Overview

- The page resolves the category from the slug in the URL and requests `GET /api/forums/posts?page=..`
- When the visitor submits the search box, the client filters the posts already loaded for the current page by title, without case sensitivity, and rewrites the URL with a `search`

parameter and page 1. The request to the API is not repeated with the keyword, so the search narrows the current page only.

- When "My Posts" is active, the page reads the `myPosts=1` flag from the URL and sends the identifier of the signed-in member as the `userId` parameter.
- The API filters the posts whose `deletedAt` is null and whose `moderationStatus` is `APPROVED`, applies the category, author and optional title filters, and sorts the result by `createdAt` in descending order.
- Lists and counts are read in one database transaction, then the API returns the page together with the pagination block.
- The trending panel requests the top posts of the category with a limit of three; the API keeps the approved posts published during the last seven days and orders them by like count, then comment count, then creation date.
- Changing the page, the page size, the search term or the "My Posts" filter rewrites the query string, so the state of the list can be shared and restored by reloading.

### Screen Layout (Mock-up Description)

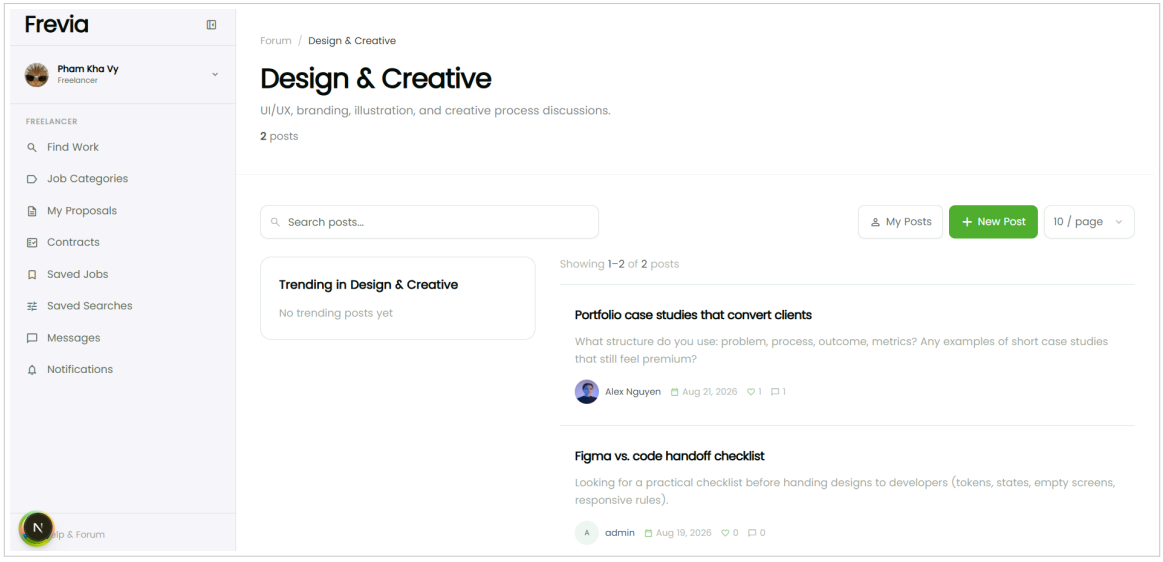


**Figure 63:** Category detail screen with the list of its posts

- The breadcrumb reads "Forum / *Category*", followed by the category name, its description and the number of approved posts it contains.
- The toolbar groups the search box, the "My Posts" toggle, the "New Post" button and the page-size selector; the search box submits on Enter and can be cleared with the cross icon.
- When a filter is active, a chips row labelled "Filters:" recalls the search term and the "My Posts" state, each chip removable, and offers "Clear all".
- Each post is a row rather than a card: it shows the title, the beginning of the content without the rich-text markup, the author avatar and name, the date, and the like and comment

counters.

- The summary “Showing x–y of z posts” sits above the list and appends “(yours)” when “My Posts” is active, so the member knows that the range is restricted to their own discussions.
- The “Trending in {category}” panel occupies the left column on large screens and is stacked above the list on smaller screens; it shows a numbered shortcut for each of the three most liked and commented posts of the last seven days, or “No trending posts yet” when the window is empty.
- The footer of the list shows numbered pagination with ellipsis, previous and next controls, and the range of records currently displayed.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `categoryId` | Identifier of the category taken from the URL; it restricts the list to one category. |
| `page` | Page number, integer greater than or equal to 1, default 1. |
| `limit` | Page size, integer greater than or equal to 1, default 10; the selector offers 5, 10, 20 and 50 and re-sends the request when it changes. |
| `search` | Optional search term compared with the post title, without case sensitivity. It is stored in the URL and applied by the client to the posts of the current page. |
| `myPosts` | Optional flag set to `1` by the “My Posts” toggle; the page converts it into the author filter described below. |
| `userId` | Optional author filter sent to the API when `myPosts=1` and a member is signed in; it always carries the identifier of that member. |

### Validation Rules

- `page` and `limit` are coerced to integers and must be at least 1; the page falls back to 1 and the page size to 10 when the parameters are absent or not numeric, while a negative value is rejected by the API validation pipe.
- `categoryId` and `userId` must be positive integers when they are sent.
- `search` is optional and is compared with the title only, without case sensitivity; leading and trailing spaces are trimmed by the client before the URL is rewritten.

- A requested page beyond the last page returns an empty list while the pagination block still reports the real total, so the client can render an empty state.
- Only approved and non-deleted posts are counted and displayed, therefore the totals never include posts that are pending, rejected or deleted.
- The `myPosts` flag is honoured only when a member is signed in; a guest who forces it in the URL receives the unfiltered list because no author identifier is sent.

**Business Rules**

- **BR-FORUM-06**: Only posts with `deletedAt = NULL` and `moderationStatus = APPROVED` shall be displayed.
- **BR-FORUM-07**: Posts shall be sorted by `createdAt` in descending order.
- **BR-FORUM-08**: The system shall support pagination and configurable page sizes.
- **BR-FORUM-09**: `commentCount` shall count only comments with `deletedAt = NULL`.

**Normal Flow**

1. The visitor selects a category row on the forum landing page.
2. The client routes to the category screen and reads the page, the page size, the search term and the “My Posts” flag from the query string.
3. The client requests the posts of the category from the API.
4. The API validates the filter, resolves the pagination offset and runs the list and count queries in one transaction.
5. The API applies the visibility rules and orders the posts by creation date from the newest to the oldest.
6. The API returns the page of posts with their like and comment counters plus the pagination block.
7. The client renders the rows, strips the rich-text markup for the excerpt and applies the empty states when no row is returned.
8. The client requests the trending posts of the category and renders the left panel.
9. The visitor optionally searches, changes the page size, moves to another page or activates “My Posts”. The search and the filters rewrite the URL, the page size and the pagination re-issue the request with the new parameters, and the summary follows the active state.
10. The visitor selects a post row and is routed to the post detail screen.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The category does not exist or was soft deleted. The page cannot resolve it and renders the not-found state with a link back to the forum. |
| A2 | The database query fails. The API returns `Error.FailedToLoadForumPosts` and the list area shows a generic error instead of the rows. |
| A3 | The requested page is larger than the last page. The API returns an empty list and the client renders “No posts yet” with the pagination intact. |
| A4 | The search term matches no title on the current page. The client reports “No posts found” with “No results for “{query}”. Try a different search.” and keeps the clear-filters action available. |
| A5 | “My Posts” is activated by a guest. The toggle is not rendered; if the flag is forced in the URL, the page sends no author identifier and the unfiltered list is displayed. |
| A6 | The page size or page number is not a positive integer. The page falls back to the default for missing or non-numeric values, while a negative value is rejected by the validation pipe and the client falls back to the first page. |
| A7 | The trending request fails. The list remains fully usable and the panel shows “No trending posts yet”. |
| A8 | The visitor loses the connection. The client shows the generic error message and the visitor can retry by reloading the current URL. |

**Output Data**

| Data Name | Description |
|---|---|
| `posts[]` | Visible posts of the requested page with `id`, `categoryId`, `userId`, `title`, `slug`, `content`, `moderationStatus`, `createdAt`, `updatedAt`, `likeCount`, `commentCount` and the author identity. |
| `pagination` | `page`, `limit`, `total` and `totalPages` computed from the filtered count. |
| `posts[]` (top) | Up to three trending posts of the category with the like count, the comment count and the interaction score used for the ranking. |

| Data Name | Description |
|---|---|
| Empty collection | An empty array when the page, the filter or the search matches no approved post. |

**Post-Conditions**

- The visitor sees one page of approved posts of the selected category, newest first.
- The search, the page size and the author filter are reflected in the URL and can be restored by reloading the page; the search itself narrows the posts already loaded for that page.
- No data is modified: the function is read-only.

### 3.4.3 View Post Detail

**Function Trigger**

The visitor selects a post row, follows a shared post link, or opens a related post from the side panel of another discussion.
**Navigation path:** Home → Forum → *Category* → *Post* (`/forum/{category}/{post-slug}-{post-id}`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** Guest, Freelancer, Client and Administrator.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

**Purpose**

The function displays a complete discussion: the post with its formatted content and its author, the engagement counters, the list of members who liked it and the comment thread. It is the screen from which every write action on the discussion is started, and it shows the author-specific actions only to the author.

**Interface**

- Breadcrumb “Forum / *Category* / *Post title*” and the line “Discussion in *Category*”.
- Post title, author avatar, display name, creation date and the “Edited {date}” marker when the post was modified after its creation.
- Rich-text body rendered with headings, lists and links.
- Action bar with the heart button and its counter, the comment counter and, for a signed-in member who is not the author, the “Report” action.

- “Edit” and “Delete” actions placed next to the post title and rendered only for the author.
- “N likes” trigger below the action bar, disabled while the post has no like, which opens the likes dialog described by a later function.
- Comment section below the action bar, described by the comment functions.
- “Related Posts” panel in the right column with up to three other discussions of the same category and the “View all posts →” link.

**Data Processing Overview**

- The client extracts the post identifier from the slug in the URL and requests `GET /api/forums/posts/{id}`.
- The API resolves the post and returns the detail projection with the category summary and the author identity.
- Only a post that is not soft deleted and whose status is `APPROVED` can be resolved; a pending or rejected post answers with a not-found error.
- When a member is signed in, the client requests the likes of the post to compute the counter and to derive whether that member already liked it. A guest skips this request, therefore the counter stays at zero and the likes trigger is disabled.
- The client requests the first fifty comments of the post to render the thread and the comment counter.
- The client requests four posts of the category for the related panel, removes the post that is currently open and keeps at most three of them.
- The page compares `updatedAt` with `createdAt` to decide whether the “Edited” marker is displayed, and compares the author identifier with the session identifier to decide whether the edit and delete actions are rendered.

### Screen Layout (Mock-up Description)

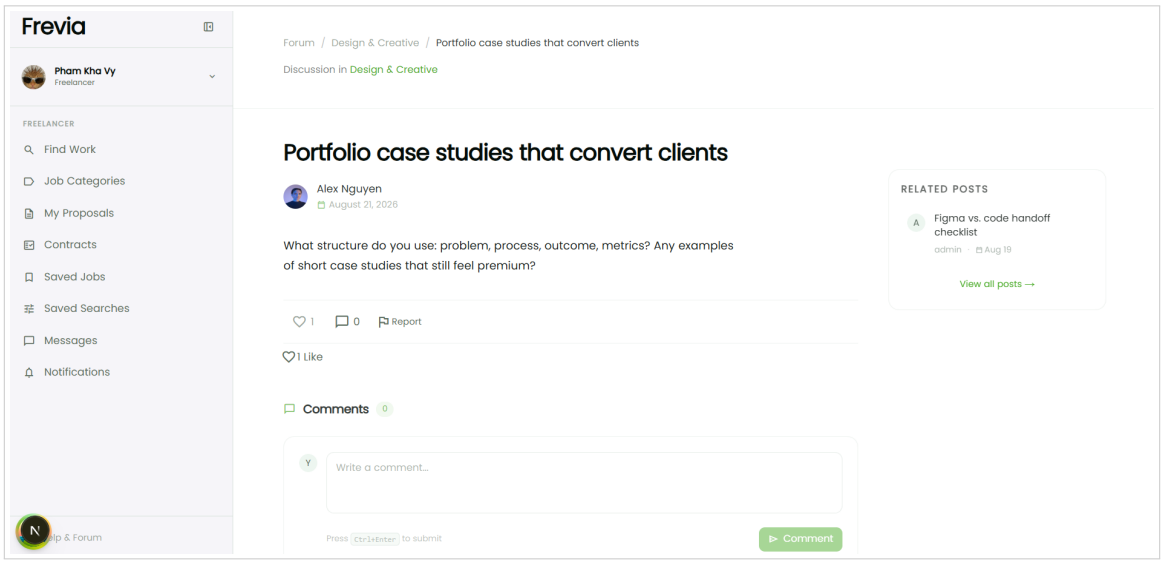


**Figure 64:** Post detail screen with the comment section

- The breadcrumb starts at “Forum”, continues with the category and ends with the post title, so the reader can walk back to the list at any time; the line “Discussion in *Category*” repeats the context.
- The title sits at the top of the article; the author’s “Edit” and “Delete” actions appear on the same line, at the right, and only when the reader is the author.
- The author block shows the avatar, the display name and the publication date; when the post was edited, the edition date is added next to the creation date.
- The body is rendered as formatted content, so headings, bullet lists and links written in the editor keep their appearance.
- The action bar holds the heart button with the like counter, the comment counter and the “Report” action for signed-in readers who are not the author; the heart is red and slightly enlarged once the member liked the post.
- The “N likes” trigger below the action bar opens the likes dialog and is disabled while the post has no like.
- The comment section below the action bar lists the thread and gives the signed-in member the comment editor.
- The right column proposes up to three related posts of the same category, each with its author and date, followed by the “View all posts →” link.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `slug` | Post slug taken from the URL; the client extracts the post identifier from its trailing segment. |
| `postId` | Identifier resolved from the slug and used by the detail, likes, comments and report calls of the same screen. |
| `page`, `limit` | Pagination of the comment thread; the detail screen requests the first page with a size of 50. |
| Session identity | Identifier of the signed-in member, used to decide whether the likes are loaded, to compute the liked state and to decide whether the author actions are displayed. |

### Validation Rules

- The identifier extracted from the slug must resolve to a post that is not soft deleted and whose `moderationStatus` is `APPROVED`; otherwise the API answers with `Error.ForumPostNotFound`.
- The detail projection must contain a title and a content, which are required fields of the post model.
- The comment pagination follows the same rules as the post list: integers greater than or equal to 1, default page 1 and default size 10; this screen explicitly asks for a size of 50.
- The liked state is derived only from the likes that belong to the post and match the session identifier. A guest always sees the unliked state with a zero counter because the likes query is not executed.
- The author actions are rendered only when the session identifier equals the post author identifier, never from a value supplied by the client.

### Business Rules

- **BR-FORUM-10**: Only posts with `deletedAt = NULL` and `moderationStatus = APPROVED` shall be displayed.
- **BR-FORUM-11**: Post details shall include title, content, author, category and created date.
- **BR-FORUM-09**: `commentCount` shall count only comments with `deletedAt = NULL`.
- **BR-FORUM-28**: The like detail shall include the user identifier and the profile of each member who liked the post.

**Normal Flow**

1. The visitor selects a post row in the category screen.
2. The client extracts the post identifier from the slug in the URL and requests the post detail.
3. The API resolves the post to a visible discussion and returns the detail projection.
4. The client renders the breadcrumb, the title, the author block and the formatted body.
5. When a member is signed in, the client requests the likes of the post; in every case it requests the first page of comments.
6. The API returns both collections with the profile of their authors.
7. The client computes the like counter, the liked state and the comment counter.
8. The client renders the action bar and exposes the author actions only to the author.
9. The client requests four posts of the category, removes the current post and renders at most three of them in the right panel.
10. The visitor reads the discussion and can like the post, open the likes list, write a comment, report the content or, as the author, edit or delete the post.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The identifier does not resolve to any post. The client displays the not-found screen with the message “This post may have been deleted or the link is incorrect.” and the action “Back to category”. |
| A2 | The post is still pending moderation or was rejected. The API hides it, therefore the same not-found screen is displayed and the post is never exposed before approval. |
| A3 | The post was soft deleted by its author or by an administrator. The detail call answers with `Error.ForumPostNotFound` and the not-found screen is displayed. |
| A4 | The likes request fails. The post body remains readable, the like counter falls back to zero and the likes trigger is disabled. |
| A5 | The comments request fails. The thread area falls back to the empty state while the post body and the action bar remain usable. |
| A6 | The session expires while the page is open. The BFF proxy refreshes the token once; when the refresh fails the visitor continues as a guest without author actions and without the likes list. |

| Case | Description |
|---|---|
| A7 | The visitor follows a link of a deleted category. The category summary is missing, so the client falls back to the “Category” label instead of a broken layout. |
| A8 | The connection drops while the page loads. The client shows the generic error message and the visitor can retry by reloading. |

**Output Data**

| Data Name | Description |
|---|---|
| Post detail | `id`, `categoryId`, `userId`, `title`, `slug`, `content`, `moderationStatus`, `moderationScore`, `createdAt` and `updatedAt`. |
| `category` | Identifier, name and slug of the category the post belongs to, or null when the post has no category. |
| `user` | Identifier and profile of the author, with the display name and the avatar. |
| `likes[]` | Members who liked the post, with their identifier, reaction type, creation date and profile. The collection is requested only for a signed-in member. |
| `comments[]` | First page of the thread with the content, the author, the dates, the like counter and the liked state of the current member. |
| `pagination` | Total number of visible comments and the pagination of the requested page. |

**Post-Conditions**

- The visitor reads the complete discussion with its engagement counters.
- The action bar exposes only the actions allowed by the role and the ownership of the post.
- No data is modified by the read itself; the counters reflect only visible rows, and a guest sees a zero like counter because the likes list is not loaded.

### 3.4.4 Create Post

### Function Trigger

A signed-in member selects the "New Post" button in the toolbar of a category screen, writes the post and confirms with the "Post" action. The button is rendered for Clients, Freelancers and Administrators; a guest who follows the button is routed to the sign-in page instead. **Navigation path:** Home → Forum → *Category* → New Post.

### Function Description

### Actors / Roles

- **Primary Actor:** Freelancer and Client. An Administrator uses the same dialog when browsing the forum.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, SensitiveAI moderation proxy, Frevia REST API and PostgreSQL through Prisma ORM.

### Purpose

The function lets a member start a new discussion inside a category. Because the content is user generated, every submission is screened by the moderation proxy before it reaches the forum, and the resulting post is stored with the moderation verdict so that only approved discussions become visible to the community.

### Interface

- Modal dialog titled "Create Post in *Category*" with the description "Share your thoughts with the community."
- "Title" input with the placeholder "What's on your mind?".
- "Content" rich-text editor with the placeholder "Write your post content here…" and the usual formatting tools.
- "Cancel" action, which discards the draft, and "Post" action, which submits it and shows a spinner while the request is running.
- Toast notification after the submission, chosen from three moderation outcomes: a success message and a redirection to the new post when the content is approved, an information message when it is queued for review, or an error message when it is rejected.

### Data Processing Overview

- The dialog trims the title and the content and blocks the submission while either of them is empty.
- The client posts the form to `/api/backend/forums/posts`.
- The BFF proxy recognises this exact route as the only forum route that must be screened

and forwards it to the SensitiveAI moderation proxy instead of the REST API.

- The moderation proxy scans the text, attaches the verdict (status, score and categories) to the request body and forwards the request to the REST API.
- The API validates the body, verifies that the category exists and is not soft deleted, and creates the post.
- When the request reaches the API without a verdict, the API calls its own `AIModerationService`: `ALLOW` maps to `APPROVED`, `REVIEW` to `PENDING` and `BLOCK` to `REJECTED`; a failure of that service is treated as `APPROVED` so publishing is not blocked by an outage.
- The slug is generated from the title, normalised to lower case without accents, and completed with a timestamp and a random suffix to keep it unique.
- The moderation status is taken from the verdict; when no verdict and no moderation result are available the post is stored as `PENDING`, which means it is invisible until an administrator approves it.
- The API returns the created post and the client invalidates the post lists of the category so the next visit shows the new discussion.
- The client branches on the returned status: `REJECTED` keeps the dialog open with the draft and shows an error toast; `PENDING` closes the dialog, clears the fields and shows an information toast without navigating; `APPROVED` closes the dialog, clears the fields, shows a success toast and navigates to the new post.

### Screen Layout (Mock-up Description)

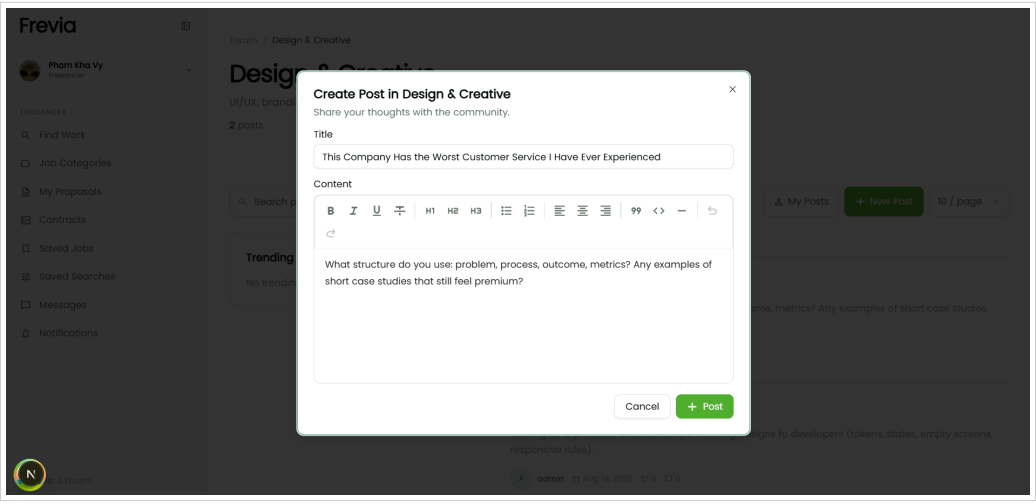


**Figure 65:** Create post dialog

- The dialog title names the category in which the post will be created, so the member cannot post in the wrong place by mistake.
- The title field is a single-line input; the content field is a rich-text editor that stores formatted HTML.

- The “Post” action stays disabled until both fields contain text, which prevents the most common validation error before the request is sent.
- The spinner replaces the icon of the action while the moderation proxy and the API are working, preventing a duplicate submission.
- The dialog closes and clears both fields when the post is approved or queued for review; it stays open with the draft when the content is rejected or the request fails, so the member can correct the text and retry.

### Function Details

#### Input Data

| Data Name | Description |
|---|---|
| `categoryId` | Identifier of the category in which the post is created; a positive integer sent by the dialog from the current category screen. |
| `title` | Post title, required, at most 255 characters in the database. |
| `content` | Post body, required; it is rich-text HTML produced by the editor. |
| `moderation` | Verdict attached by the moderation proxy: `status`, `score` between 0 and 1 and `categories`. It is an internal field, never typed by the member. |
| Access token | Bearer token forwarded by the BFF proxy; the API resolves the author from it, so the client cannot choose the author. |

#### Validation Rules

- The title and the content are required and must contain at least one character after trimming.
- `categoryId` is optional for the API but must be a positive integer when it is present; the dialog always sends the category of the current screen.
- The category must exist and must not be soft deleted, otherwise the request fails with `Error.ForumCategoryNotFound`.
- The author is taken from the access token, never from the request body, so a member cannot create a post on behalf of somebody else.
- The moderation verdict must respect its own contract: a status among `PENDING`, `APPROVED` and `REJECTED`, and an optional score between 0 and 1.

- The generated slug must remain unique; the timestamp and random suffix guarantee that two posts with the same title never collide.

**Business Rules**

- **BR-FORUM-12**: Only authenticated users can create posts.
- **BR-FORUM-13**: Every post must belong to exactly one user and may belong to a category.
- **BR-FORUM-14**: Title and content are required fields.
- **BR-FORUM-15**: A new post shall default to `moderationStatus = PENDING` unless moderation data is provided.

**Normal Flow**

1. The member opens a category screen and selects “New Post”.
2. The client opens the dialog with the title of the target category.
3. The member writes a title and a content; the “Post” action becomes available.
4. The client trims both fields and sends the request through the BFF proxy.
5. The proxy routes the request to the moderation proxy, which scans the text.
6. The moderation proxy attaches its verdict and forwards the request to the REST API.
7. The API validates the body and checks that the category exists.
8. The API generates the slug, resolves the moderation status and inserts the post.
9. The API returns the created post to the client.
10. The client invalidates the cached lists of the category.
11. When the status is `APPROVED`, the client closes the dialog, clears the fields, shows the success message and navigates to the new post.
12. When the status is `PENDING`, the client closes the dialog, clears the fields and shows the message “Your post has been saved and is awaiting moderator review.” without navigating.
13. When the status is `REJECTED`, the client keeps the dialog open with the draft and shows the rejection message returned by the moderation service.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The title or the content is empty after trimming. The action is disabled, and a forced request is rejected by the validation pipe with `Error.ForumPostTitleRequired` or `Error.ForumPostContentRequired`. |

| Case | Description |
|---|---|
| A2 | The category was deleted while the dialog was open. The API answers `Error.ForumCategoryNotFound` and the dialog stays open with the draft so the member can choose another category. |
| A3 | The member is not signed in or the session expired. The BFF proxy refreshes the token once; if the refresh fails the API rejects the request with an authentication error and the client redirects to the sign-in page. |
| A4 | The moderation proxy is unavailable. The request fails before reaching the API, no post is created, and the client shows an error toast while keeping the draft. |
| A5 | The moderation verdict is `REJECTED`. The post is stored with that status, is not publicly visible, and the dialog stays open with an error toast so the member can rewrite the content. |
| A6 | The moderation verdict is `PENDING`. The post is stored but hidden, the dialog closes with an information toast, and the member is told that the post awaits moderator review. |
| A7 | The insert fails at database level. The API returns `Error.FailedToCreateForumPost` and the client keeps the dialog open so nothing written by the member is lost. |
| A8 | The member double clicks the action. The action is disabled while the mutation is pending, therefore only one request is sent. |
| A9 | The member chooses “Cancel”. The dialog closes, the draft is cleared and no request is sent. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the created post, used by the client to build the detail route. |
| `slug` | Generated slug, unique for every post. |
| `categoryId`, `userId` | Category and author of the post. |
| `title`, `content` | Stored title and stored rich-text content. |
| `moderationStatus` | `APPROVED`, `PENDING` or `REJECTED`; it decides whether the post is immediately reachable. |

| Data Name | Description |
|---|---|
| `moderationScore`, `moderationCategories` | Verdict details kept for the moderation queue; the score may be null when the proxy did not compute one. |
| `createdAt`, `updatedAt` | Timestamps of the insert. |

**Post-Conditions**

- One post is inserted and belongs to the signed-in member and to the selected category.
- The post carries a moderation status and a unique slug.
- An approved post becomes visible in the category list immediately after the cache is invalidated; a pending or rejected post stays hidden until an administrator resolves it.
- The post appears in the moderation queue of the administrator.

### 3.4.5 Edit Post

**Function Trigger**

The author of a post opens the post detail screen and selects the "Edit" action placed next to the title. The action is rendered only for the author, therefore other members never see it.
**Navigation path:** Home → Forum → *Category* → *Post* → Edit.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Freelancer or Client who authored the post.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

**Purpose**

The function allows the author to correct the title or the body of a discussion without losing the discussion itself, its comments and its likes. The post keeps its identity, its creation date and its engagement, and the screen marks it as edited so readers know that the content changed after publication.

**Interface**

- Modal dialog titled "Edit Post" with the description "Update your post title and content."
- Title input and rich-text editor pre-filled with the current values of the post.

- “Cancel” action, which leaves the post unchanged, and “Save Changes” action, which submits the new values.
- The post detail screen refreshes the title and the body in place after a successful save, and the author line starts showing “Edited {date}”.
- Because the new text is screened again, the dialog may also close with a post whose moderation status became `PENDING` or `REJECTED`; the detail screen keeps the new values until the page is reloaded, when the post becomes hidden if it is no longer approved.

**Data Processing Overview**

- The client pre-fills the dialog from the post already loaded in the detail screen, so no additional read is required to open it.
- The client sends `PATCH /api/forums/posts/{id}` with the new title and content.
- The BFF proxy forwards the request directly to the REST API; only post creation goes through the SensitiveAI moderation proxy.
- The API loads the post with `deletedAt` null and rejects the request when the post does not exist.
- The API compares the author of the post with the identifier carried by the access token and rejects the request when they differ.
- The API re-screens the new title and content with `AIModerationService` when the request carries no verdict: `ALLOW` maps to `APPROVED`, `REVIEW` to `PENDING` and `BLOCK` to `REJECTED`; a failure of the moderation service is treated as `APPROVED`.
- The API updates the title, the content, the slug and the moderation fields, then returns the updated post.
- The client closes the dialog, replaces the title and the body in place and invalidates the cached detail and the post lists of the category so the new values appear everywhere without a reload.

### Screen Layout (Mock-up Description)

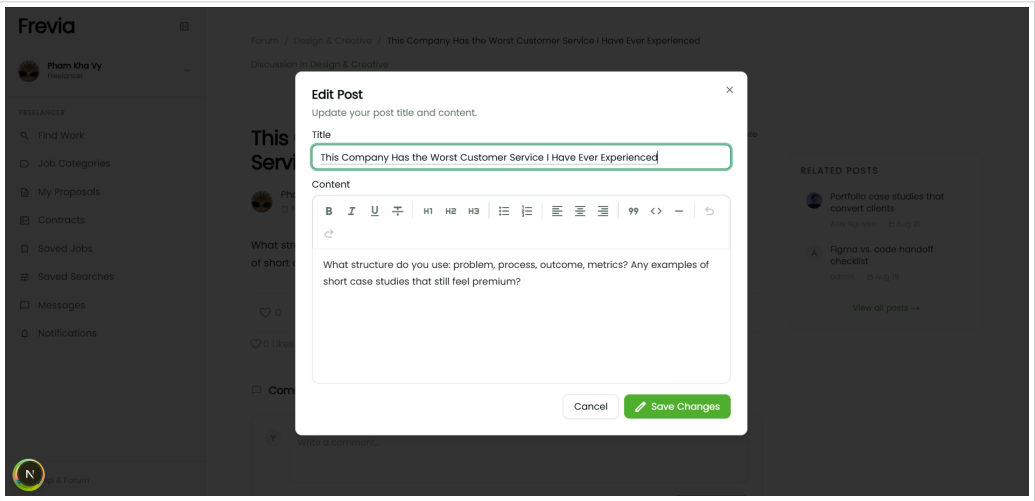


**Figure 66:** Edit post dialog with the current values

- The dialog reuses the layout of the creation dialog, which keeps the editing experience consistent with publishing.
- Both fields are pre-filled, so the member edits the existing text instead of rewriting it.
- The “Save Changes” action stays disabled while a field is empty, mirroring the requirement that both fields remain mandatory.
- A spinner is shown during the request; the dialog closes when the update succeeds.
- After the save, the detail screen shows the new title, the new body and the edition marker next to the publication date.
- The dialog does not display a moderation toast. When the new text is no longer approved, the post is hidden the next time the forum loads it, while the current screen keeps the saved values until it is refreshed.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Identifier of the post taken from the current route. |
| `title` | New title, required, at most 255 characters in the database. |
| `content` | New body, required; rich-text HTML produced by the editor. |
| Access token | Identifies the requester; the API compares its identifier with the author of the post. |

**Validation Rules**

- The post must exist and must not be soft deleted; otherwise the request fails with `Error.ForumPostNotFound`.
- The requester must be the author of the post; otherwise the request fails with `Error.ForumPostNotOwned`. The check is performed on the server, so hiding the action in the interface is only a convenience.
- The title and the content are required; an empty value is rejected by the validation pipe before the update is executed.
- The slug is regenerated from the new title on every update, which keeps the URL readable and unique while the post identifier stays unchanged.
- The new text is moderated again by the API; the returned status may be `PENDING` or `REJECTED` even when the post was previously approved.
- The category of the post is preserved unless a new category identifier is supplied; the dialog never changes it.

**Business Rules**

- **BR-FORUM-16**: Only the author of a post can edit that post.
- **BR-FORUM-17**: Title and content are required fields.
- **BR-FORUM-18**: The original post identifier remains unchanged after editing; the slug is regenerated when the title changes.

**Normal Flow**

1. The author opens the detail of a post they wrote.
2. The client renders the “Edit” and “Delete” actions next to the title.
3. The author selects “Edit” and the dialog opens with the current title and content.
4. The author changes one or both fields.
5. The client sends the update request with the access token.

6. The API loads the post and confirms that it exists and is not deleted.
7. The API confirms that the requester is the author.
8. The API re-screens the new text and resolves the new moderation status.
9. The API updates the title, the content, the slug and the moderation fields.
10. The API returns the updated post.
11. The client closes the dialog, refreshes the detail in place and invalidates the category lists.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The post no longer exists or was deleted in the meantime. The API returns `Error.ForumPostNotFound` and the client shows the error and closes the dialog. |
| A2 | The requester is not the author. The API returns `Error.ForumPostNotOwned`, the update is refused and the client displays the returned message. |
| A3 | The title or the content is cleared before saving. The action is disabled, and a forced request is rejected by the validation pipe. |
| A4 | The update fails at database level. The API returns `Error.FailedToUpdateForumPost` and the dialog stays open with the edited values. |
| A5 | The author cancels the dialog. No request is sent and the post keeps its previous values. |
| A6 | The session expires while the dialog is open. The BFF proxy refreshes the token once; when the refresh fails the request is rejected and the member is redirected to sign in. |
| A7 | Two editions are submitted in a row. The action is disabled while the mutation is pending, so the last saved values are the ones returned by the server. |
| A8 | The new text is flagged by moderation. The update still succeeds, but the post receives `PENDING` or `REJECTED` and disappears from the public lists on the next load; the dialog closes without a moderation toast. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the post, which never changes. |
| `title`, `content` | New values stored in the post. |
| `slug` | Slug regenerated from the new title. |
| `moderationStatus` | New verdict, `APPROVED`, `PENDING` or `REJECTED`. |
| `moderationScore`, `moderationCategories`, `moderationMessage` | Details of the new verdict, kept for the moderation queue. |
| `updatedAt` | New edition timestamp, used by the detail screen to display the "Edited" marker. |
| `createdAt` | Original publication timestamp, preserved. |

**Post-Conditions**

- The post contains the new title and the new content and keeps its author, category, comments and likes.
- The detail screen and the category list display the new values.
- The post is marked as edited because `updatedAt` differs from `createdAt`.
- If the new verdict is not `APPROVED`, the post leaves the public lists and waits in the moderation queue until an administrator resolves it.
- No other post and no comment is modified.

### 3.4.6 Delete Post

**Function Trigger**

The author of a post, or an Administrator, selects the "Delete" action on the post detail screen and confirms the operation in the confirmation dialog.
**Navigation path:** Home → Forum → *Category* → *Post* → Delete.

**Function Description**

**Actors / Roles**

- **Primary Actor:** the author of the post, or an Administrator acting on any post. The web interface renders the action for the author; the API also accepts an Administrator, who removes posts through the moderation and trash tools.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and

PostgreSQL through Prisma ORM.

### Purpose

The function removes a discussion from the public forum while preserving its record. Deletion is soft: the row receives a deletion timestamp and is filtered out of every public query, which keeps the data available for moderation, for the trash screen of the administrator and for any later audit.

### Interface

- "Delete" action with a bin icon, rendered for the author and for the Administrator.
- Confirmation dialog titled "Delete Post" with the warning "Are you sure you want to delete this post? This action cannot be undone.".
- "Cancel" action, which closes the dialog, and destructive "Delete" action, which performs the deletion.
- Redirect to the category screen after a successful deletion.

### Data Processing Overview

- The client sends `DELETE /api/forums/posts/{id}` with the access token.
- The API loads the post with `deletedAt` null to confirm that it exists and is still visible.
- The API compares the author with the requester and also reads the role of the requester from the token.
- When the requester is neither the author nor an Administrator, the deletion is refused.
- When the check succeeds, the API writes the current date and time in the `deletedAt` column of the post.
- The client invalidates the post lists and navigates back to the category, where the post is no longer listed.

### Screen Layout (Mock-up Description)

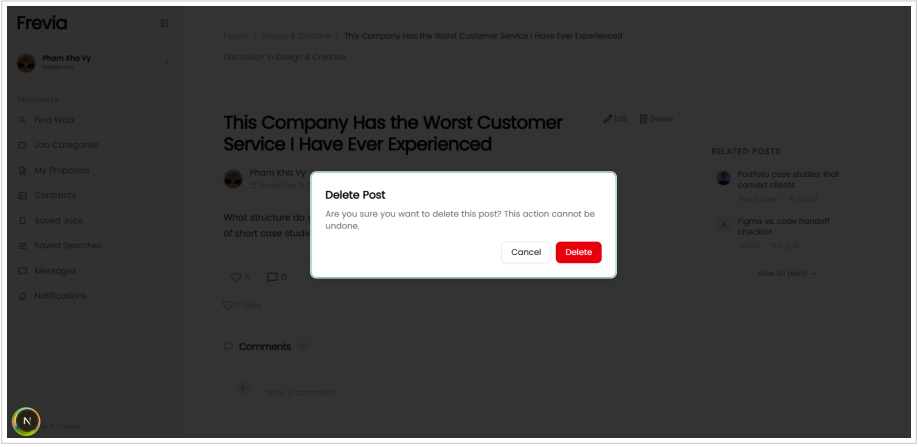


**Figure 67:** Delete post confirmation

- The confirmation dialog repeats the name of the object being deleted, which prevents deleting the wrong discussion.
- The warning states that the action cannot be undone from the member point of view, because the post disappears from the forum immediately.
- The destructive style of the confirmation action distinguishes it from the cancel action.
- After confirmation the screen returns to the category list, so the member sees the post removed from the discussion list.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Identifier of the post to delete, taken from the confirmation dialog. |
| Access token | Identifies the requester and carries the role used to authorize an Administrator. |
| `deletedAt` | Deletion timestamp written by the API; it is internal, not supplied by the client. |

### Validation Rules

- The post must exist and must not be already deleted; otherwise the API returns `Error.ForumPostNotFound`.
- The requester must be the author of the post or must hold the Administrator role; otherwise the API returns `Error.ForumPostNotOwned`.

- The deletion timestamp is produced by the server, so the client cannot backdate or falsify it.
- Only one row is affected per request; the comments and the likes of the post are preserved in the database and simply stop being reachable through the post.

**Business Rules**

- **BR-FORUM-19**: Only the author of a post or an Admin can delete that post.
- **BR-FORUM-20**: Post deletion shall be implemented as a soft delete using the `deletedAt` field.
- **BR-FORUM-21**: Soft-deleted posts shall not appear in public forum listings.

**Normal Flow**

1. The author or the Administrator opens the post detail screen.
2. The client renders the "Delete" action, which is available for this role.
3. The member selects the action and the confirmation dialog is displayed.
4. The member confirms the deletion.
5. The client sends the delete request with the access token.
6. The API loads the post and confirms that it exists and is not deleted.
7. The API confirms that the requester is the author or an Administrator.
8. The API writes the deletion timestamp in the post row.
9. The API returns the deleted post.
10. The client invalidates the lists, closes the dialog and routes back to the category.
11. The category list no longer contains the post and its counters are recomputed without it.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The post does not exist or was already deleted. The API returns `Error.ForumPostNotFound` and the client shows the error message. |
| A2 | A member tries to delete a post written by somebody else. The API returns `Error.ForumPostNotOwned` and nothing is deleted. |
| A3 | An Administrator deletes the post of another member. The operation is allowed and the post leaves the public forum immediately. |
| A4 | The database update fails. The API returns `Error.FailedToDeleteForumPost` and the post remains visible, so the member can retry. |

| Case | Description |
|---|---|
| A5 | The member cancels the confirmation. No request is sent and the post is unchanged. |
| A6 | The session expires before the confirmation. The request is rejected after the single token refresh attempt and the member is redirected to sign in. |
| A7 | Another member opens the same post while it is being deleted. The next request returns `Error.ForumPostNotFound` and their screen shows the not-found state. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the deleted post. |
| `userId`, `categoryId` | Author and category of the deleted post, returned for confirmation. |
| `title`, `slug` | Title and slug of the deleted post. |
| `deletedAt` | Server timestamp recorded in the row; the post is soft deleted from this moment. |

**Post-Conditions**

- The post is excluded from the category list, from the post detail and from the landing page counters.
- The row and its related comments and likes are retained for moderation and for the trash screen of the Administrator.
- The member is returned to the category screen, which reflects the new totals.

### 3.4.7 Like a Post

**Function Trigger**

A signed-in member selects the heart button in the action bar of a post. A guest who selects the button is routed to the sign-in page instead of producing a request.

**Navigation path:** Home → Forum → *Category* → *Post* → Like.

### Function Description

#### Actors / Roles

- **Primary Actor:** Freelancer and Client. An Administrator can also react while browsing.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

#### Purpose

The function lets a member express appreciation for a discussion and lets the community measure which posts are relevant. It is implemented as a toggle: the same action adds the reaction when the member has not liked the post yet and removes it when the member already liked it, so a member can never like the same post twice.

#### Interface

- Heart button with the like counter, placed in the action bar under the post body.
- Outline heart for a post that the member has not liked, filled heart for a post that the member liked.
- The button is disabled while the request is running, which prevents repeated toggles from a single intention.

#### Data Processing Overview

- The client sends `POST /api/forums/posts/{id}/like` to the BFF proxy, which forwards it to the REST API.
- The API loads the post with `deletedAt` null and answers with `Error.ForumPostNotFound` when it is missing.
- The API looks for a like row that matches the member and the post.
- When the row exists, the API removes the reaction and answers with `liked: false`.
- When the row does not exist, the API inserts a like row with the reaction type `like` and answers with `liked: true`.
- Before the request is sent, the client updates its cached likes list optimistically: it adds or removes the current member, so the heart and the counter react immediately.
- When the request fails, the client restores the previous likes list; when it settles, it invalidates the likes query and the post detail so the cache is reconciled with the server.
- The unique constraint of the like table on the pair member and post guarantees that a duplicate reaction cannot be stored, even if two requests are issued at the same time.

### Screen Layout (Mock-up Description)

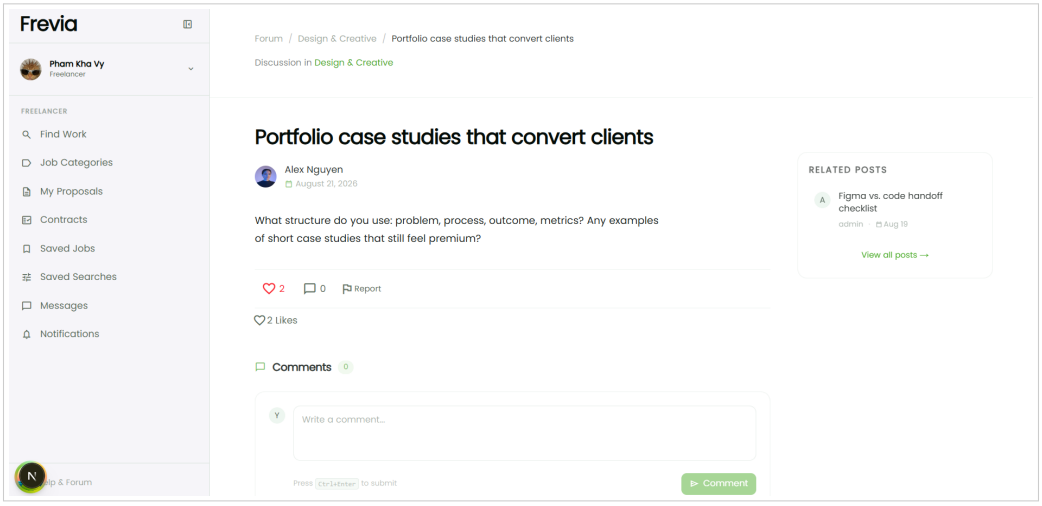


**Figure 68:** Post action bar after the post has been liked

- Before the reaction the heart is outlined and the counter shows the current number of likes.
- After the reaction the heart is filled, coloured in red and slightly enlarged, and the counter is incremented by one.
- The “N likes” trigger placed below the action bar shows the same total and opens the list of members who liked the post.
- The counter is the single source of truth for the likes list: the same number is shown on the detail screen, in the likes dialog and on the post rows of the category.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Identifier of the post taken from the current route. |
| Access token | Identifies the member who reacts; the API derives the like owner from it. |
| Reaction type | Constant value `like` stored in the reactivity row; the interface does not offer another reaction for the moment. |

### Validation Rules

- The requester must be authenticated; a request without a valid token is rejected with an authentication error.
- The post must exist and must not be soft deleted.

- A member can hold only one reaction per post, enforced by the unique constraint on the pair member and post.
- The reaction is always attached to the member of the token; the client cannot send a different member identifier.

**Business Rules**

- **BR-FORUM-22**: Only authenticated users can like posts.
- **BR-FORUM-23**: A user can have only one active like reaction per post.
- **BR-FORUM-24**: The reaction shall be stored in the `ForumLike` table.
- **BR-FORUM-25 to BR-FORUM-27**: removing a reaction is allowed only when the corresponding like record exists, and it deletes that record.

**Normal Flow**

1. The signed-in member opens a post they have not liked yet.
2. The client displays an outlined heart next to the current counter.
3. The member selects the heart and the client updates the cached likes list immediately.
4. The client sends the toggle request.
5. The API confirms that the post is visible.
6. The API finds no reaction of this member on this post.
7. The API inserts the like row.
8. The API answers with `liked: true`.
9. The client keeps the red filled heart and the incremented counter, then refetches the likes to reconcile the list with the server.
10. When the member selects the heart again, the same sequence removes the row and the interface returns to the outlined heart with the previous counter.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The post does not exist or was deleted. The API returns `Error.ForumPostNotFound` and the counter is left unchanged. |
| A2 | The member is not signed in. The client routes to the sign-in page before any request is sent, and a direct request is rejected with an authentication error. |
| A3 | The member already liked the post. The request removes the reaction instead of adding a second one, which is the intended toggle behaviour. |

| Case | Description |
|---|---|
| A4 | The insert fails at database level. The API returns `Error.FailedToLikeForumPost` and the interface keeps the previous state. |
| A5 | The member double clicks the heart. The button is disabled while the request runs, so only one reaction is created. |
| A6 | The likes query cannot be refreshed after a successful toggle. The optimistic state stays in the cache and the counter is reconciled on the next visit. |

**Output Data**

| Data Name | Description |
|---|---|
| `liked` | `true` when the reaction was added, `false` when it was removed. |
| Like row | Member, post, reaction type and creation date of the stored reaction. |
| Updated counter | The like counter of the post, recomputed by the client from the refreshed likes list. |

**Post-Conditions**

- The post holds exactly one reaction of the member, or none when the action removed it.
- The counter displayed on the post matches the number of likes stored for that post.
- The member appears in, or disappears from, the likes list of the post.

### 3.4.8 View Post Likes

**Function Trigger**

A visitor selects the "N Likes" trigger placed below the action bar of a post, which opens the list of the members who liked the post. The trigger is disabled while the counter is zero, so the dialog is only reachable once the post has at least one like. The endpoint itself is public, but a guest sees a zero counter because the detail screen derives the likes only for a signed-in member, so guests normally do not reach the dialog.
**Navigation path:** Home → Forum → *Category* → *Post* → Likes.

### Function Description

### Actors / Roles

- **Primary Actor:** Guest, Freelancer, Client and Administrator.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

### Purpose

The function answers the question “who liked this post”. It also feeds the state of the like button, because the detail screen derives both the counter and the liked state of the current member from the same list.

### Interface

- Dialog titled “Likes” with the description “People who liked this post.”.
- Row per member with the avatar, the display name and, when the profile is incomplete, a fallback name built from the member identifier.
- Close icon of the dialog, which dismisses it; the list itself has no footer action.
- Empty message “No likes yet.” when the dialog is opened while no reaction is loaded; the trigger itself is disabled at zero, so this state is normally not reachable.

### Data Processing Overview

- The client calls `GET /api/forums/posts/{id}/like` and caches the result under a key that contains the post identifier.
- The API reads every like row of the post and joins the profile of the member.
- The payload contains the identifier, the member, the reaction type and the creation date of the reaction.
- The detail screen uses the same payload to count the likes and to detect the reaction of the signed-in member by comparing identifiers; it requests the payload only for a signed-in member, while the dialog reuses the cached collection.
- The dialog renders one row per member with the avatar and the display name, and falls back to “User #{id}” when the profile is missing.

### Screen Layout (Mock-up Description)

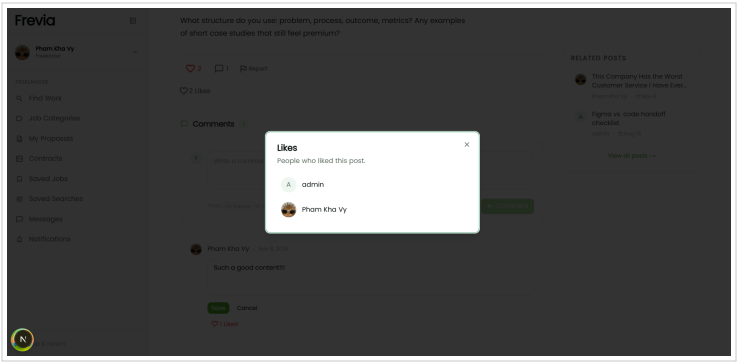


**Figure 69:** Likes dialog listing the members who liked the post

- The “N Likes” trigger placed below the post action bar opens the dialog and states how many members liked the discussion; it is disabled while the total is zero.
- Each row shows the avatar and the display name of a member, which links the reaction to a visible identity of the community.
- When no reaction is loaded, the dialog states “No likes yet.” instead of showing an empty panel.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `id` | Identifier of the post whose reactions are listed. |
| Session identity | Optional; the detail screen uses it to derive the liked state and the like counter, while the endpoint itself needs no token. |

### Validation Rules

- The post identifier must be a positive integer accepted by the route parameter pipe.
- The endpoint is public, so no token is required and no authorization rule applies.
- A post without reaction returns an empty array, which is a valid answer and not an error.
- Profile fields may be null; the interface then substitutes the fallback name and the initial of the display name for the avatar.

### Business Rules

- **BR-FORUM-28**: The like detail shall include the user identifier and the user profile information for each liker.

- **BR-FORUM-23**: A user can have only one active like reaction per post, therefore a member never appears twice in the list.

**Normal Flow**

1. The visitor opens a post detail screen.
2. For a signed-in member, the client requests the likes of the post to derive the counter and the liked state.
3. The API returns every reaction of the post with the profile of the member.
4. The client shows the “N Likes” trigger below the post action bar, next to the like and comment actions.
5. The visitor selects the trigger and the dialog opens with the list of members.
6. The visitor closes the dialog and returns to the discussion.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The post has no reaction. The trigger shows “0 Likes” and is disabled, so the dialog cannot be opened; a dialog opened from a stale cache states “No likes yet.”. |
| A2 | A member has no profile record. The row shows the fallback name “User #{id}” and the initial letter in place of the avatar. |
| A3 | The request fails. The dialog has no dedicated error panel and falls back to “No likes yet.”, while the trigger keeps the last counter derived from the detail screen. |
| A4 | The post was deleted while the dialog was open. The refreshed list returns no row and the dialog shows the empty state. |
| A5 | The visitor is offline. The dialog shows the same empty state and can be closed, which leaves the rest of the screen usable. |

**Output Data**

| Data Name | Description |
|---|---|
| `likes[]` | One entry per reaction with `id`, `userId`, `postId`, `commentId`, `reactionType` and `createdAt`. |
| `user` | Identifier, display name and avatar of the member who reacted. |
| Derived count | Length of the list, displayed as the likes counter of the post. |

**Post-Conditions**

- The visitor sees every member who liked the post, without duplicates.
- The "N Likes" trigger of the post matches the number of rows returned.
- No data is modified: the function is read-only.

### 3.4.9 Create Comment

**Function Trigger**

A signed-in member writes in the comment box placed under the post body and submits it with the "Comment" action or with the `Ctrl+Enter` shortcut. The text is screened by the AI moderation service before it becomes visible, so the comment is either published, held for review or rejected. A guest sees an invitation to sign in instead of the editor.
**Navigation path:** Home → Forum → *Category* → *Post* → Comments.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Freelancer and Client. An Administrator can also comment while browsing the forum.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

**Purpose**

The function lets a member answer a discussion, ask a question or share an experience inside the thread of a post. Comments are attached to one post and one author, can be edited and deleted by their owner, and carry their own like counter.

**Interface**

- Comment card between the post action bar and the thread, with the avatar of the member and a text area using the placeholder "Write a comment…".

- Hint “Press `Ctrl+Enter` to submit” below the text area.
- “Comment” action with a send icon, disabled while the field is empty or a request is running.
- Guest state: the editor is replaced by the sentence “Log in to join the discussion.”, where “Log in” links to the sign-in page.
- Moderation feedback through a toast: an error toast keeps the draft when the text is rejected, an informational toast confirms that the comment awaits review, and a success toast may confirm an approved comment.
- Empty state “No comments yet” with the sentence “Be the first to share a tip, question, or experience.” when the post has no approved comment.

**Data Processing Overview**

- The client trims the text and sends `POST /api/forums/posts/{postId}/comments` with the content.
- The BFF proxy forwards the request to the REST API, which validates the body.
- The API verifies that the post exists and is not soft deleted.
- The API screens the content with the shared AI moderation service, using the `forum-comment` context: a `BLOCK` verdict becomes `REJECTED`, a `REVIEW` verdict becomes `PENDING` and an `ALLOW` verdict becomes `APPROVED`. When the moderation service is unavailable, the content is approved as a fallback.
- The API inserts the comment with the author taken from the access token, the moderation status, score, categories, message and language.
- The response contains the new comment with its author profile, a like counter of zero and an unliked state, because a comment cannot be liked before it exists.
- On `REJECTED` the client keeps the draft and shows an error toast. On `PENDING` it clears the editor and shows an informational toast. On `APPROVED` it clears the editor and may show a success toast.
- In every outcome the client invalidates the comment queries of the post, which refreshes the thread and the comment counter. Only `APPROVED` comments are returned by the public listing, so a pending or rejected comment stays out of the thread.

### Screen Layout (Mock-up Description)

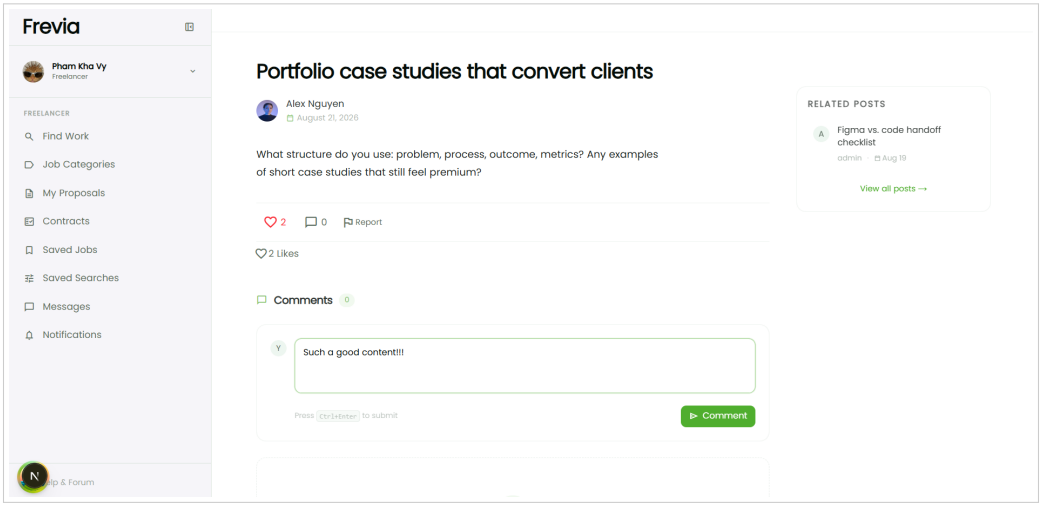


**Figure 70:** Comment editor under the post body

- The editor sits directly under the action bar, so the member writes without leaving the discussion; the text area shows the placeholder “Write a comment…”.
- The counter badge next to the “Comments” heading shows how many approved comments the post currently holds.
- The “Comment” action stays disabled while the field is empty, which prevents empty submissions, and the hint about `Ctrl+Enter` makes the shortcut discoverable.
- Every approved comment appears at the top of the thread, with the avatar, the display name, the date, the like action and, for the author, the edit and delete actions.
- A rejected draft remains in the editor with an error toast, while a pending comment leaves the editor empty and is not listed until a moderator approves it.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `postId` | Identifier of the post taken from the route; the comment is attached to it. |
| `content` | Comment text, required, at least one character after trimming. |
| Access token | Identifies the author of the comment; it is read by the API and never sent in the body. |

### Validation Rules

- The requester must be authenticated; the API rejects a request without a valid token.
- The content must not be empty; the validation pipe rejects an empty or whitespace-only body.
- The post must exist and must not be soft deleted, otherwise the request fails with `Error.ForumPostNotFound`.
- The author of the comment is always the member of the token, so a member cannot post a comment on behalf of somebody else.
- The content is screened by the AI moderation service before insert; the resulting status decides whether the comment is published, held for review or rejected.

**Business Rules**

- **BR-FORUM-32**: Only authenticated users can create comments.
- **BR-FORUM-33**: Comment content cannot be empty.
- **BR-FORUM-34**: Each comment shall be associated with a valid post and user.

**Normal Flow**

1. The signed-in member opens a post and scrolls to the comment section.
2. The client renders the editor with the placeholder and the hint.
3. The member writes the comment and submits it with the action or the shortcut.
4. The client trims the text and sends the request.
5. The API validates the content and loads the post.
6. The API screens the content with the AI moderation service.
7. The API inserts the comment with the author of the session and the moderation verdict.
8. The API returns the created comment with an empty like state and its moderation status.
9. The client handles the verdict: it keeps the draft on `REJECTED`, clears the editor on `PENDING` or `APPROVED`, and shows the matching toast.
10. The client refreshes the thread; an approved comment is displayed at the top of the list and the counter is incremented, while a pending comment stays hidden until approval.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The content is empty. The action is disabled, and a forced request is rejected by the validation pipe without touching the database. |

| Case | Description |
|---|---|
| A2 | The post does not exist or was deleted. The API returns `Error.ForumPostNotFound` and the client displays the error message while keeping the draft. |
| A3 | The member is not signed in. The editor is replaced by the sign-in invitation, and a direct request is rejected with an authentication error. |
| A4 | The insert fails at database level. The API returns `Error.FailedToCreateForumComment` and the editor keeps the text so it is not lost. |
| A5 | The member submits twice in a row. The action is disabled while the request runs, which prevents duplicated comments. |
| A6 | The session expires during the submission. The BFF proxy attempts one refresh; when it fails the request is rejected and the member is redirected to sign in. |
| A7 | The AI moderation returns `BLOCK`. The comment is stored with status `REJECTED`, is excluded from the public thread, and the editor keeps the draft with an error toast. |
| A8 | The AI moderation returns `REVIEW`. The comment is stored with status `PENDING`, is hidden until a moderator approves it, and the client clears the editor with an informational toast. |
| A9 | The AI moderation service is unavailable. The API falls back to `APPROVED`, so the comment is published and the client behaves as for a normal approval. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the created comment. |
| `postId` | Post the comment belongs to. |
| `content` | Stored comment text. |
| `user` | Identifier, display name and avatar of the author. |
| `likeCount`, `likedByMe` | Zero and false for a new comment. |
| `moderationStatus` | `APPROVED`, `PENDING` or `REJECTED` verdict of the AI moderation. |

| Data Name | Description |
|---|---|
| `moderationScore`, `moderationCategories` | Risk score and categories reported by the moderation service. |
| `moderationMessage`, `moderationLanguage` | Explanation shown to the author and the detected language, when available. |
| `createdAt`, `updatedAt` | Timestamps of the insert. |

**Post-Conditions**

- One comment is stored and belongs to the signed-in member and to the post.
- An approved comment is immediately visible to every visitor and increments the comment counter; a pending comment waits for moderation and a rejected comment stays out of the public thread.
- The author is informed of the moderation outcome through the matching toast, and the draft is preserved only when the text is rejected.

### 3.4.10 Update Comment

**Function Trigger**

The author of a comment selects the pencil action next to their comment, edits the text and confirms the change. The new text is re-screened by the AI moderation service, so an edit can change the visibility of the comment. The action is rendered only for the author of the comment.
**Navigation path:** Home → Forum → *Category* → *Post* → Comments → Edit.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Freelancer or Client who authored the comment.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

**Purpose**

The function lets a member correct a comment after publication, for example to fix a typo or to add a clarification, without deleting the reply and losing its position in the thread. The comment keeps its identifier, its author and its likes, and the new text replaces the previous one.

**Interface**

- Pencil action on the footer row of the comment, next to the like and delete actions.
- Inline editor pre-filled with the current text of the comment.
- “Save” action to confirm and “Cancel” action to discard the change.
- An “edited” marker next to the date as soon as the stored text differs from the original, because the update timestamp no longer matches the creation timestamp.
- Moderation feedback through a toast: a rejected edit keeps the editor open with an error, a pending edit closes it with an informational message, and an approved edit may close it with a success message.

**Data Processing Overview**

- The client sends `PATCH /api/forums/posts/{postId}/comments/{commentId}` with the new content.
- The API loads the comment with `deletedAt` null and rejects the request when it does not exist.
- The API verifies that the comment belongs to the post of the route, so a comment cannot be edited through the route of another post.
- The API compares the author of the comment with the member of the token and refuses the request when they differ.
- The API screens the new content with the AI moderation service, using the `forum-comment` context, and maps the verdict to `APPROVED`, `PENDING` or `REJECTED`.
- The API updates the content and the moderation status, score, categories, message and language, then returns the comment with its author profile.
- On `REJECTED` the client keeps the editor open with the text and shows an error toast; on `PENDING` or `APPROVED` it exits the edit mode, shows the matching toast and refreshes the thread.
- The client invalidates the comment queries of the post, and the comment is displayed with the “edited” marker when the update changed its timestamp.

### Screen Layout (Mock-up Description)

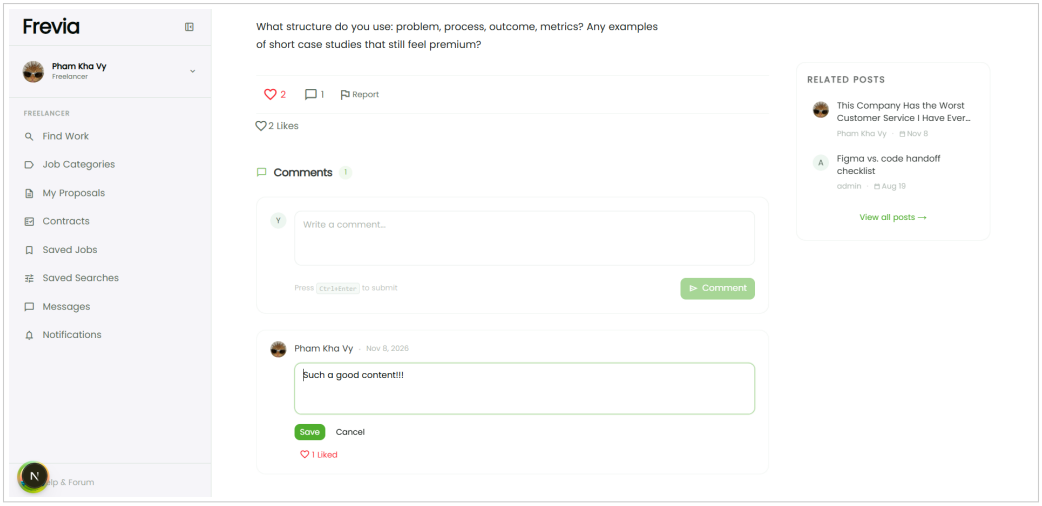


**Figure 71:** Inline editor of an existing comment

- The editor replaces the comment body in place, so the member keeps the context of the thread while editing; the avatar, name and date stay visible above the field.
- The existing text is pre-filled, which makes small corrections fast.
- The save action is disabled while the text is empty, because the content stays a required field, and the cancel action restores the previous text.
- After the save the comment shows the edited text and the “edited” marker next to the date, while the thread keeps the same order.
- The like action and its counter remain available below the editor, because the reaction belongs to the comment and not to its text.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `postId` | Identifier of the post from the route; it scopes the comment. |
| `commentId` | Identifier of the comment to update. |
| `content` | New comment text, required, at least one character. |
| Access token | Identifies the requester; the API compares it with the comment author. |

### Validation Rules

- The comment must exist and must not be soft deleted; otherwise the request fails with `Error.ForumCommentNotFound`.

- The comment must belong to the post of the route; a mismatch is reported as a missing comment, which hides the existence of the comment through another route.
- The requester must be the author of the comment; otherwise the request fails with `Error.ForumCommentNotOwner`.
- The content is required and must not be empty after trimming.
- The new content is screened again by the AI moderation service, so the status, score, categories and message of the comment may change after an edit.
- The update replaces the content and its moderation verdict: the author, the post, the creation date and the likes of the comment are preserved.

**Business Rules**

- **BR-FORUM-35**: Only the owner of a comment can edit it.
- **BR-FORUM-36**: Comment content cannot be empty.
- **BR-FORUM-37**: A comment must belong to the specified post to be edited.

**Normal Flow**

1. The author of a comment selects the pencil action on their own comment.
2. The client opens the inline editor with the current text.
3. The author modifies the text and confirms.
4. The client sends the update request with the access token.
5. The API loads the comment and confirms that it exists.
6. The API confirms that the comment belongs to the post of the route.
7. The API confirms that the requester is the author.
8. The API screens the new text with the AI moderation service.
9. The API stores the new content and its moderation verdict.
10. The API returns the updated comment.
11. The client handles the verdict, refreshes the thread and displays the new text with the "edited" marker when the comment remains public.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The comment does not exist or was deleted. The API returns `Error.ForumCommentNotFound`, the editor keeps the text and an error toast is shown. |

| Case | Description |
|---|---|
| A2 | The comment belongs to another post. The request is refused with the same not-found error, which prevents editing a comment through a foreign route. |
| A3 | The requester is not the author. The API returns `Error.ForumCommentNotOwner`, the text is not changed and an error toast is shown. |
| A4 | The new text is empty. The save action is disabled, and a forced request is rejected by the validation pipe. |
| A5 | The update fails at database level. The API returns `Error.FailedToUpdateForumComment` and the editor keeps the text. |
| A6 | The author cancels the edition. No request is sent and the previous text is restored. |
| A7 | The AI moderation returns `BLOCK`. The comment is stored with status `REJECTED`, is removed from the public thread, and the editor stays open with an error toast. |
| A8 | The AI moderation returns `REVIEW`. The comment is stored with status `PENDING`, is hidden until approval, and the client exits the edit mode with an informational toast. |
| A9 | The AI moderation service is unavailable. The API falls back to `APPROVED` and the edit is published normally. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the updated comment. |
| `postId` | Post the comment belongs to, unchanged. |
| `content` | New comment text. |
| `user` | Identifier and profile of the author, unchanged. |
| `moderationStatus` | New verdict of the AI moderation after the edit. |
| `moderationScore`, `moderationCategories` | Risk score and categories of the edited text. |
| `moderationMessage`, `moderationLanguage` | Explanation and detected language, when available. |

| Data Name | Description |
|---|---|
| `updatedAt` | New edition timestamp. |
| `createdAt` | Original creation timestamp, preserved. |

**Post-Conditions**

- The comment holds the new text and the verdict of the re-screening; its position, author and likes are preserved.
- An approved comment is displayed with the new text and the "edited" marker, while a pending or rejected comment is hidden from the public thread.
- No other comment and no post is modified.

### 3.4.11 Delete Comment

**Function Trigger**

The author of a comment, or an Administrator, selects the bin action next to a comment and confirms the deletion in the confirmation dialog.
**Navigation path:** Home → Forum → *Category* → *Post* → Comments → Delete.

**Function Description**

**Actors / Roles**

- **Primary Actor:** the author of the comment, or an Administrator acting on any comment. The web interface exposes the action to the author of the comment; the API also accepts an Administrator, who deletes comments through the moderation tools.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

**Purpose**

The function removes a reply from the thread. As for posts, the deletion is soft: the comment receives a deletion timestamp and is filtered out of every public query, so the record remains available for moderation and for the trash screen of the Administrator while the discussion stays coherent for the readers.

**Interface**

- Bin action on the footer row of the comment, next to the edit action.
- Confirmation dialog titled "Delete Comment" with the warning "Are you sure you want to delete this comment? This action cannot be undone.".

- “Cancel” action and destructive “Delete” action.
- The thread is refreshed after the deletion and the comment counter decreases.

### Data Processing Overview

- The client sends `DELETE /api/forums/posts/{postId}/comments/{commentId}`.
- The API loads the comment with `deletedAt` null and rejects the request when it is missing.
- The API verifies that the comment belongs to the post of the route.
- The API accepts the requester when it is the author or when the token carries the Administrator role.
- The API writes the deletion timestamp in the comment row.
- The client invalidates the comment queries and refreshes the thread and the counter.

### Screen Layout (Mock-up Description)

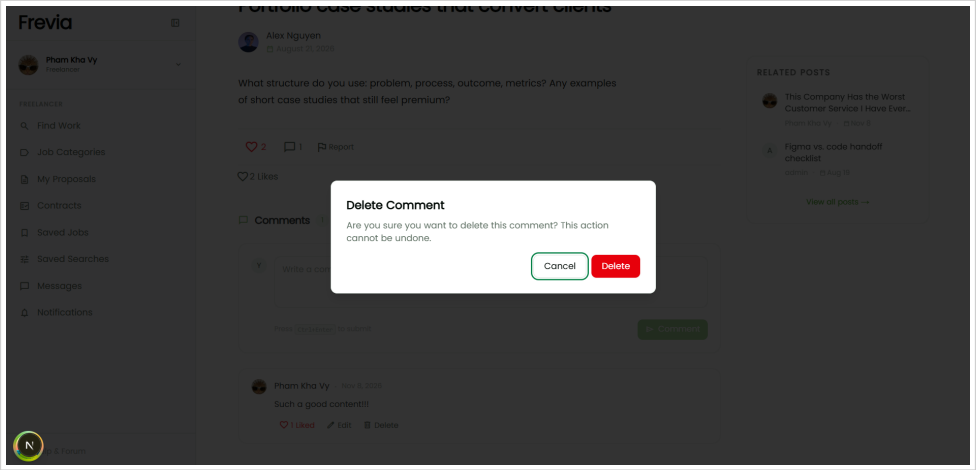


**Figure 72:** Delete comment confirmation

- The confirmation dialog names the object, so a member who owns several comments cannot delete the wrong one.
- The destructive style of the confirm action separates it from the cancel action and warns about the consequence.
- After the confirmation the comment disappears from the thread and the comment counter of the post is updated.
- The remaining comments keep their order, because the thread is sorted by creation date.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `postId` | Identifier of the post from the route; it scopes the comment. |
| `commentId` | Identifier of the comment to delete. |
| Access token | Identifies the requester and carries the role used to authorize an Administrator. |
| `deletedAt` | Deletion timestamp written by the server, not supplied by the client. |

### Validation Rules

- The comment must exist and must not be already deleted; otherwise the request fails with `Error.ForumCommentNotFound`.
- The comment must belong to the post of the route; a mismatch is reported as a missing comment.
- The requester must be the author of the comment or must hold the Administrator role; otherwise the request fails with `Error.ForumCommentNotOwner`.
- The deletion timestamp is produced by the server.
- The post itself is not deleted, and the likes of the comment are kept in the database.

### Business Rules

- **BR-FORUM-38**: Only the owner of a comment or an Admin can delete it.
- **BR-FORUM-39**: Comment deletion shall be implemented as a soft delete using the `deletedAt` field.
- **BR-FORUM-40**: Soft-deleted comments shall not be displayed in normal comment listings.

### Normal Flow

1. The author or the Administrator opens the post detail screen.
2. The client renders the bin action on the comments the member may delete.
3. The member selects the action and the confirmation dialog appears.
4. The member confirms the deletion.
5. The client sends the delete request with the access token.
6. The API loads the comment and confirms that it exists.

7. The API confirms that the comment belongs to the post of the route.
8. The API confirms that the requester is the author or an Administrator.
9. The API writes the deletion timestamp.
10. The API returns the deleted comment.
11. The client refreshes the thread, where the comment is no longer present.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The comment does not exist or was already deleted. The API returns `Error.ForumCommentNotFound` and the client shows the error message. |
| A2 | A member tries to delete a comment written by somebody else. The API returns `Error.ForumCommentNotOwner` and nothing is deleted. |
| A3 | An Administrator deletes a comment of another member. The operation is allowed and the comment disappears from the thread. |
| A4 | The comment belongs to another post. The request is refused with the not-found error, which prevents deleting through a foreign route. |
| A5 | The delete fails at database level. The API returns `Error.FailedToDeleteForumComment` and the comment remains visible. |
| A6 | The member cancels the confirmation. No request is sent and the comment is unchanged. |
| A7 | The session expires before the confirmation. The single refresh attempt decides between a successful retry and a redirect to the sign-in page. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the deleted comment. |
| `postId` | Post the comment belonged to. |
| `content` | Text of the deleted comment, returned for confirmation. |
| `deletedAt` | Server timestamp recorded in the row. |
| Updated counter | The comment counter of the post, recomputed without the deleted comment. |

**Post-Conditions**

- The comment is excluded from the thread and from every comment count.
- The row is retained for moderation and for the trash screen of the Administrator.
- The post remains published with its remaining comments.

### 3.4.12 Like a Comment

#### Function Trigger

A signed-in member selects the heart action on a comment of the thread. A guest does not see the action at all, because the comment footer renders it only for a signed-in member and the reaction requires an account.
**Navigation path:** Home → Forum → *Category* → *Post* → Comments → Like.

#### Function Description

#### Actors / Roles

- **Primary Actor:** Freelancer and Client. An Administrator can also react while browsing.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM.

#### Purpose

The function allows members to acknowledge a useful reply without writing another comment. It is a toggle, exactly like the post reaction: the first activation likes the comment and the second one removes the reaction, so one member can never like the same comment twice.

#### Interface

- Heart action in the footer of each comment, followed by the like counter when it is greater than zero and by the label “Like” or “Liked”.
- Outlined heart when the member has not liked the comment, filled and slightly enlarged red heart when the member liked it.
- The action is disabled while the request runs.
- The same footer holds the edit and delete actions for the author, and the report action for the other members, so a comment can be managed without leaving the thread.

#### Data Processing Overview

- The client sends `POST /api/forums/posts/{postId}/comments/{commentId}/like`.
- The API loads the comment with `deletedAt` null and rejects the request when it is missing.
- The API verifies that the comment belongs to the post of the route.

- The API looks for a reaction of the member on the comment.
- When the reaction exists, the API removes it and answers with `liked: false`.
- When the reaction does not exist, the API inserts it and answers with `liked: true`.
- Before the response arrives, the client updates the cached comment optimistically: it toggles the liked state and increments or decrements the counter of that comment only.
- When the request fails, the client restores the previous cache. When it settles, the client invalidates the comment queries of the post so the server becomes the source of truth again.
- The comment list computes the liked state per comment by matching the reactions of the signed-in member, so the states of the other comments are unaffected.

### Screen Layout (Mock-up Description)

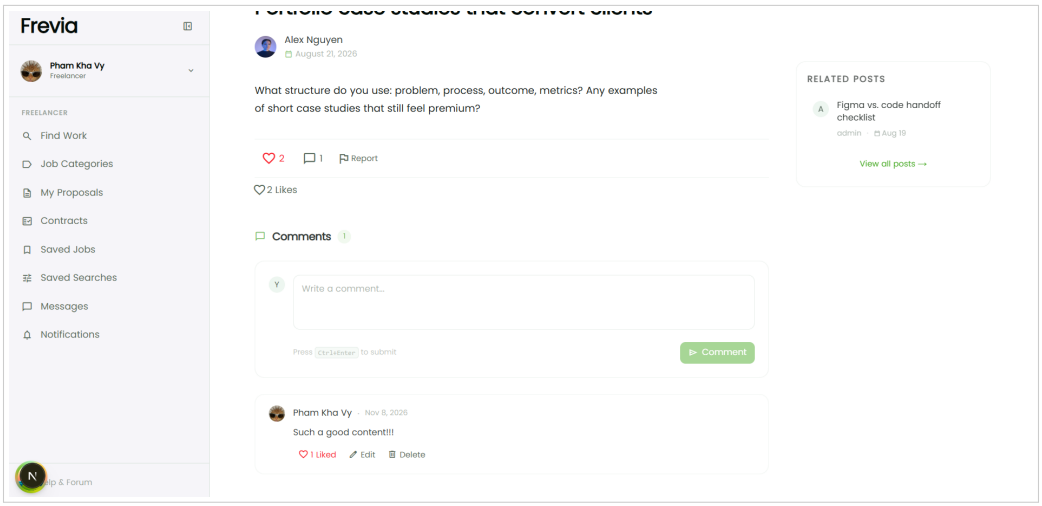


**Figure 73:** Comment liked by the current member

- The footer of the comment shows the heart, the counter and the label “Liked” once the member has reacted.
- After the reaction the heart is filled, slightly enlarged and coloured red, and the counter is incremented by one.
- The same footer offers the edit and delete actions to the author of the comment, while the report action is reserved to the other members.
- Only the selected comment changes; the reactions of the other comments and the likes of the post remain untouched.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| `postId` | Identifier of the post from the route; it scopes the comment. |
| `commentId` | Identifier of the comment to react to. |
| Access token | Identifies the member who reacts. |
| Reaction type | Constant value `like`. |

### Validation Rules

- The requester must be authenticated.
- The comment must exist, must not be soft deleted and must belong to the post of the route.
- A member can hold only one reaction per comment, enforced by the unique constraint of the like table on the pair member and comment.
- The reaction is attached to the member of the token, never to an identifier sent by the client.

### Business Rules

- **BR-FORUM-41**: Only authenticated users can like comments.
- **BR-FORUM-42**: A user can have only one active like reaction per comment.
- **BR-FORUM-43**: The reaction shall be stored in the `ForumLike` table.
- **BR-FORUM-44 to BR-FORUM-46**: removing a reaction is allowed only when the record exists, and it deletes that record.

### Normal Flow

1. The signed-in member finds a useful comment in the thread.
2. The client renders an outlined heart with the like counter when it is greater than zero.
3. The member selects the heart and the client sends the toggle request.
4. The API confirms that the comment exists and belongs to the post.
5. The API finds no reaction of this member on this comment.
6. The API inserts the reaction row.
7. The API answers with `liked: true`.
8. The client immediately fills the heart and increments the counter of that comment in the cache, then refetches the thread in the background.
9. Selecting the heart again removes the reaction and restores the previous counter.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The comment does not exist or was deleted. The API returns `Error.ForumCommentNotFound` and the counter is unchanged. |
| A2 | The comment belongs to another post. The request is refused with the same not-found error. |
| A3 | The member is not signed in. The heart action is not rendered for guests and no request can be sent from the interface. |
| A4 | The reaction already exists. The request removes it, which is the intended toggle behaviour. |
| A5 | The insert fails at database level. The API returns `Error.FailedToLikeForumComment`, the optimistic update is rolled back and the interface keeps the previous state. |
| A6 | The member clicks the heart twice quickly. The action is disabled while the request runs, so only one reaction is created. |

**Output Data**

| Data Name | Description |
|---|---|
| `liked` | `true` when the reaction was added, `false` when it was removed. |
| Like row | Member, post, comment, reaction type and creation date of the stored reaction. |
| Updated counter | The like counter of the comment, recomputed by the client. |

**Post-Conditions**

- The comment holds exactly one reaction of the member, or none after the second activation.
- The counter of the comment matches the number of stored reactions on that comment.
- The reactions of the post and of the other comments are unchanged.

### 3.4.13 Report Content

### Function Trigger

A signed-in member who is not the author selects the “Report” action in the post action bar or in the footer of a comment, writes a reason and submits the report. The action is hidden from the author of the content, because a member cannot report their own post or comment, and it is not rendered for guests.
**Navigation path:** Home → Forum → *Category* → *Post* → Report.

### Function Description

### Actors / Roles

- **Primary Actor:** Freelancer and Client.
- **Supporting Systems:** Frevia Web application, Frevia BFF proxy, Frevia REST API and PostgreSQL through Prisma ORM; the report is later handled by an Administrator in the moderation screen.

### Purpose

The function gives the community a way to flag content that violates the rules, either a discussion or a single reply. A report never changes the visibility of the content by itself: it creates a pending moderation entry that the Administrator reviews through the moderation module.

### Interface

- “Report” action with a flag icon in the action bar of the post and in the footer of each comment.
- Dialog titled “Report Content” with the prompt “Please provide a reason for reporting this content.”.
- “Reason” text area with the placeholder “Why are you reporting this content?”.
- “Cancel” action and destructive “Report” action, which stays disabled while the reason is empty or a request is running.
- “Reported” badge with a check icon replacing the action once the content has been reported by the member.
- Already-reported panel shown while the dialog is open and the status becomes reported: the description reads “You have already reported this content.” and the body “You have already reported this content. Our team will review it.”, with a single “Close” action and no reason field.
- Success toast “Report submitted successfully” after a successful submission.

### Data Processing Overview

- As soon as the report component mounts, the client checks whether the member already reported the content through `GET /api/forums/posts/{postId}/is-reported` or `GET /api/forums/posts/{postId}/comments/{commentId}/is-reported`, and keeps the answer in a query cache with an infinite stale time.
- The client sends `POST /api/forums/posts/{postId}/reports` for a post, or `POST /api/forums/posts/{postId}/comments/{commentId}/reports` for a comment, with the reason.
- The API verifies that the reported target exists and is not soft deleted.
- For a comment, the API also verifies that the comment belongs to the post of the route.
- The API looks for an existing report of the same member on the same target and refuses a duplicate.
- The API inserts the report with the reporter of the token, the target and the reason; the status defaults to `PENDING` and no administrator is assigned yet.
- The client clears the reason, shows the confirmation toast, marks the report as done in its cache and closes the dialog after a short delay.
- While the dialog is still open, the reported state replaces the form with the already-reported panel. Once the dialog closes, the action is replaced by the “Reported” badge, so the same item cannot be reported again from the interface.

### Screen Layout (Mock-up Description)

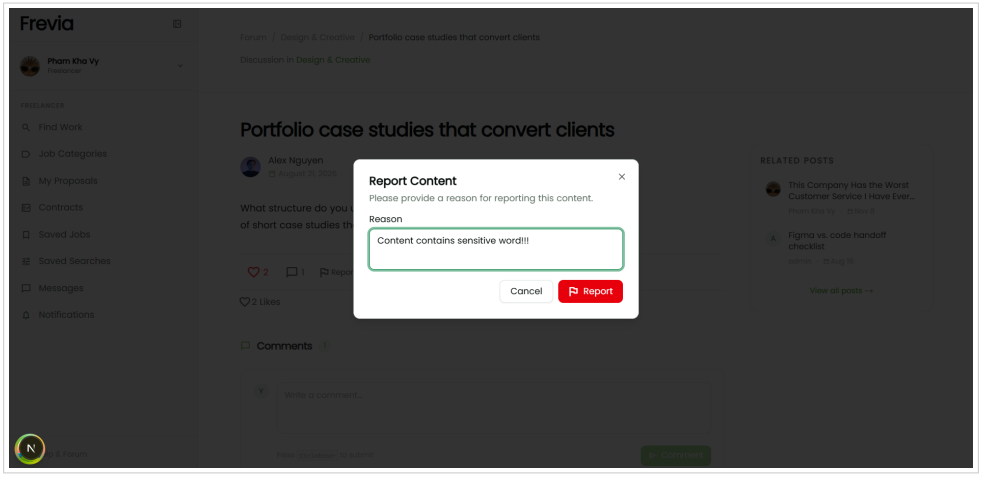


**Figure 74:** Report dialog with the reason field

- The action is available in the post action bar and in the footer of every comment, so a member can flag the exact item that is problematic.
- The dialog explains what is expected and gives a free-text reason, which the moderator reads before making a decision.
- The confirm action is destructive styled and disabled without a reason, which prevents empty reports.

- After a successful submission the toast “Report submitted successfully” confirms the action and the badge “Reported” prevents a second submission.
- When the status is already reported, the dialog displays the confirmation panel with “You have already reported this content. Our team will review it.” instead of the reason field.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| `postId` | Identifier of the reported post, or of the post that holds the reported comment. |
| `commentId` | Optional identifier of the reported comment; absent when the post itself is reported. |
| `reason` | Free-text justification of the report, required. |
| Access token | Identifies the reporter; it becomes `reporterId` in the stored report. |

**Validation Rules**

- The requester must be authenticated.
- The reason is required and must contain at least one character.
- The reported post must exist and must not be soft deleted, otherwise the request fails with `Error.ForumPostNotFound`.
- The reported comment must exist, must not be soft deleted and must belong to the post of the route, otherwise the request fails with `Error.ForumCommentNotFound`.
- Exactly one member may report a given target once; a second attempt fails with `Error.ForumReportAlreadyExists`.
- The status of a new report is `PENDING` and the administrator field stays empty until a moderator handles it.

**Business Rules**

- **BR-FORUM-47**: Only authenticated users can submit reports.
- **BR-FORUM-48**: Reports must be associated with either a post or a comment.
- **BR-FORUM-49**: New reports shall be created with status PENDING.
- **BR-FORUM-50**: A user may submit only one report for the same content item.

**Normal Flow**

1. The signed-in member reads a post or a comment that violates the community rules.
2. The client has already checked the reported status on mount and renders the “Report” action only when the member has not reported that item.
3. The member selects the “Report” action of that item.
4. The member writes a reason and confirms.
5. The client sends the report request with the access token.
6. The API validates the reason and loads the reported target.
7. The API confirms that the target exists and, for a comment, that it belongs to the post.
8. The API confirms that the member has not reported the same target yet.
9. The API inserts the report with status `PENDING`.
10. The API returns the created report.
11. The client clears the reason, shows the success toast, marks the cache as reported and closes the dialog after a short delay.
12. The “Reported” badge replaces the action for this member.

**Abnormal Flow / Exception Handling**

| Case | Description |
|---|---|
| A1 | The reason is empty. The confirm action is disabled, and a forced request is rejected by the validation pipe. |
| A2 | The post was deleted before the submission. The API returns `Error.ForumPostNotFound` and no report is stored. |
| A3 | The comment was deleted before the submission, or does not belong to the post. The API returns `Error.ForumCommentNotFound`. |
| A4 | The member already reported the same item. The API returns `Error.ForumReportAlreadyExists`, the client displays the translated error toast and the dialog stays open with the reason; the badge appears only once the cached status is reported. |
| A5 | An Administrator opens the moderation screen. Listing and updating reports require the Administrator role, otherwise the API returns `Error.ForumReportForbidden`. |
| A6 | The insert fails at database level. The API returns `Error.FailedToCreateForumReport` and the dialog stays open with the reason. |
| A7 | The member is not signed in. The action is not rendered for guests, and a direct request is rejected with an authentication error. |

| Case | Description |
|---|---|
| A8 | The member reports their own content through a crafted request. The record can be created, but the action is never offered in the interface, and the moderation screen treats it like any other report. |
| A9 | The reported-status check fails. The component treats the item as not reported, so the action is shown and a duplicate submission is refused by the API with the already-exists error. |

**Output Data**

| Data Name | Description |
|---|---|
| `id` | Identifier of the created report. |
| `reporterId` | Member who submitted the report. |
| `postId`, `commentId` | Reported targets; exactly one of them identifies the item, the other may be null for a comment report. |
| `reason` | Reason written by the reporter. |
| `status` | `PENDING` for a new report; later `REVIEWED`, `RESOLVED` or `DISMISSED` after the moderator decision. |
| `adminId`, `createdAt` | Administrator who handled the report, null at creation, and the creation timestamp. |

**Post-Conditions**

- One pending report exists for the reported item and the reporter.
- The reported content is not modified and stays visible until a moderator decides.
- The report appears in the moderation queue of the Administrator, where it can be reviewed, resolved or dismissed.
- The reporter sees the “Reported” badge and cannot submit the same report again from the interface.

## 3.5 Permission Management

### 3.5.1 View Permission List

### Function Trigger

An administrator accesses the Permission Management screen from the administrator console sidebar.
**Navigation path:** Admin Dashboard → Settings → Permissions (`/admin/permissions`).

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to retrieve and inspect the complete catalog of route-level system permissions defined in the application. This provides transparent visibility into available endpoints, HTTP methods, and module groupings necessary for configuring custom role-based access control (RBAC) policies.

### Interface

- Admin portal screen at `/admin/permissions`.
- Top filter bar featuring keyword search input, HTTP method filter dropdown, and module filter dropdown.
- Data table displaying permission ID, endpoint name, route path, HTTP method badge, module badge, creation date, and row action links.
- Pagination footer indicating total count and page selection controls.

### Data Processing Overview

- The client initiates an HTTP request: `GET /api/permissions` with query parameters (`page`, `limit`, `search`, `method`, `module`, `sortBy`, `order`).
- The backend guard verifies the caller's JWT access token and ensures the caller possesses `ADMIN` role privileges.
- `PermissionsController.getPermissions()` parses and validates the filter parameters through `PermissionFilterDto`.
- `PermissionsService.getPermissions()` invokes `PermissionsRepository.findAll(filter)`.
- The repository executes a database transaction retrieving active records (`deletedAt = null`), counting the matching records, and fetching distinct module names.
- The service formats the payload into a `PermissionListResponseDto` containing the

items array, pagination metadata, and active module list.

- The frontend renders the data table, applies color badges per HTTP method (GET: outline, POST: secondary, PUT/PATCH: outline, DELETE: destructive), and renders pagination buttons.

### Screen Layout (Mock-up Description)

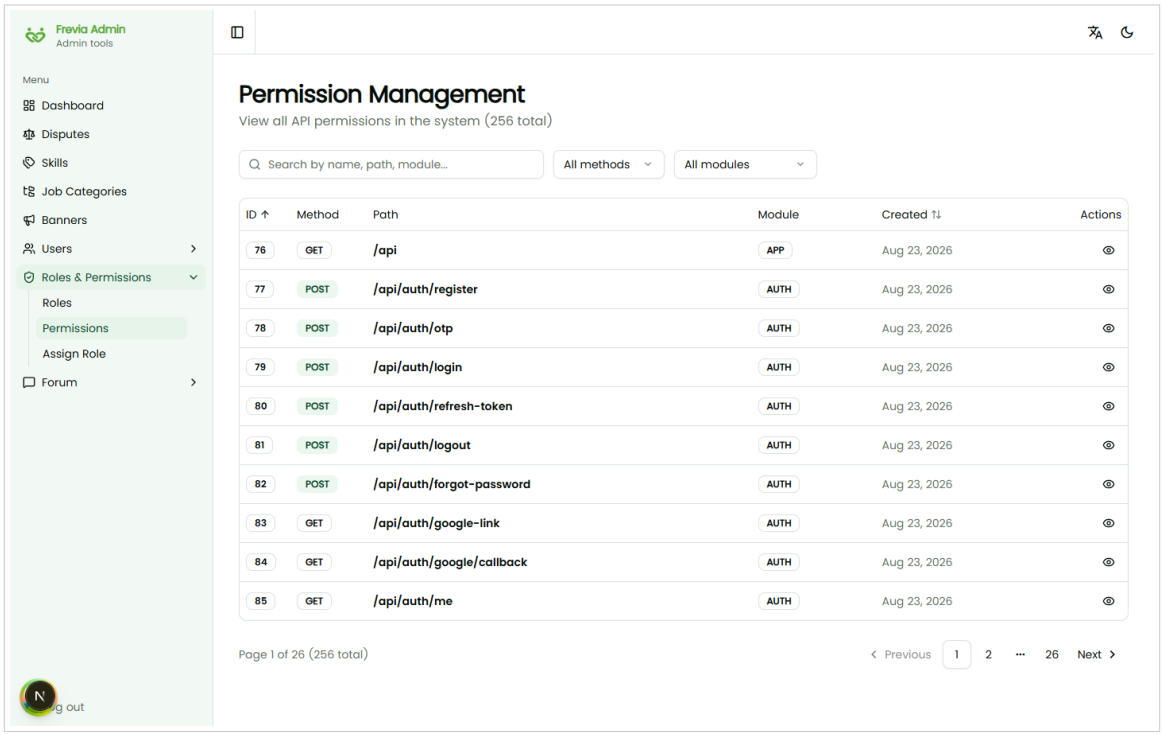


**Figure 75:** Permission list interface

- The top header displays the section title “Permissions” alongside a brief summary description.
- A filter bar contains a search input box with an icon, an HTTP method selector, and a module category dropdown.
- The table columns include: **ID**, **Name**, **Path**, **Method**, **Module**, **Created Date**, and **Actions**.
- Each HTTP method is visually differentiated with a dedicated colored badge for immediate visual recognition.
- A “View detail” icon button on each row directs the administrator to the detailed permission inspection view.
- The pagination toolbar displays the current page interval, total records found, and previous/next page navigation buttons.

**Function Details**

**Input Data**

| Data Name | Description |
| --- | --- |
| page | Target page number for paginated retrieval (defaults to 1). |
| limit | Maximum number of permission records returned per page (defaults to 10). |
| search | Optional text query matching permission name, API route path, or module name. |
| method | Optional filter matching HTTP method (`GET`, `POST`, `PUT`, `PATCH`, `DELETE`). |
| module | Optional filter matching functional module name (case-insensitive). |
| sortBy | Sorting column field name (`id` or `createdAt`, defaults to `id`). |
| order | Sorting sequence direction (`asc` or `desc`, defaults to `asc`). |

**Validation Rules**

| Data Name | Validation Rule |
| --- | --- |
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max length 100 characters; whitespace is automatically trimmed. |
| method | Must match one of the standard HTTP methods: `GET`, `POST`, `PUT`, `PATCH`, `DELETE`. |
| module | String, optional, max length 50 characters. |
| sortBy | Must strictly be one of: `id`, `createdAt`. |
| order | Must strictly be one of: `asc`, `desc`. |

**Business Rules**

- **BR-PERM-01**: Only active permissions with `deletedAt = null` shall be returned to the administrator.

- **BR-PERM-02**: Keyword search queries shall match `name`, `path`, or `module` case-insensitively using partial text matching.
- **BR-PERM-03**: The method filter shall filter records by exact HTTP verb equality.
- **BR-PERM-04**: The module filter shall perform case-insensitive exact matching against the permission module field.
- **BR-PERM-05**: The response must return distinct existing module names to dynamically populate client filter options.

**Normal Flow**

1. The administrator navigates to the "Permissions" page in the admin console.
2. The frontend issues `GET /api/permissions` with default query parameters (`page=1&limit=10&sortBy=`
3. The system verifies the administrator's JWT access token and validates query parameters.
4. `PermissionsService` queries `PermissionsRepository.findAll(filter)`.
5. The database returns matching permission rows, the total record count, and distinct module names.
6. The system returns HTTP 200 with the permission list, pagination metadata, and module list.
7. The frontend renders the data table, method badges, module badges, and pagination controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The API returns HTTP 401 Unauthorized. The client clears cached credentials and redirects the administrator to `/login`. |
| User lacks Admin privileges | The API returns HTTP 403 Forbidden. The interface displays an access denied toast: "You do not have permission to access this page." |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid pagination parameter (`page < 1` or non-integer) | The DTO validator rejects the request with HTTP 400 Bad Request. The frontend resets to page 1 automatically. |
| Invalid limit parameter (`limit > 100` or `limit < 1`) | The API rejects the request with HTTP 400 Bad Request. The client defaults to a safe limit value of 10. |
| Unsupported HTTP method filter string | The system returns HTTP 400 Bad Request. The UI clears the invalid filter selection and reloads the default list. |
| Database connection failure or query timeout | The backend logs the exception and throws `FailedToLoadPermissionsException` (HTTP 500). The UI displays: "Couldn't load permissions. Try again." |
| No permissions match active search/filters | The repository returns an empty collection (`permissions: []`, `total: 0`). The table displays a friendly empty-state notice: "No permissions match your filters." |

**Output Data**

| Output Field | Description |
|---|---|
| permissions | Array of permission objects containing `id`, `name`, `path`, `method`, `module`, and `createdAt`. |
| pagination.page | Current page number. |
| pagination.limit | Current page size limit. |
| pagination.total | Total count of matching permissions. |
| pagination.totalPages | Total calculated pages available. |
| modules | Array of unique module names available across active permissions. |

**Post-Conditions**

1. The permission catalog table is displayed with accurate HTTP badges and module tags.
2. The module filter dropdown is populated with distinct module names retrieved from the server.
3. The administrator can sort, filter, or select any permission row to view its details.

### 3.5.2 View Permission Detail

**Function Trigger**

An administrator clicks "View detail" on a permission row in the permission table or navigates directly to a permission detail URL.

**Navigation path:** Admin Dashboard → Settings → Permissions → Click "View detail" icon on a row (`/admin/permissions/:id`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to view comprehensive technical details of a specific system permission, including its full route path, allowed HTTP method verb, associated functional module, system identifier, and exact creation timestamp.

**Interface**

- Detailed permission overview page at `/admin/permissions/:id`.
- Top navigation breadcrumb with “Back to permissions” button.
- Header displaying the permission endpoint name and an HTTP verb badge.
- Information card containing structured field entries: ID, Creation timestamp, HTTP Method, Module name, and Endpoint Route Path.

### Data Processing Overview

- The client sends an HTTP request: `GET /api/permissions/:id` with the numeric permission ID.
- The NestJS `ParseIntPipe` validates that `id` is an integer.
- The permission guard verifies valid Admin credentials.
- `PermissionsController.getPermissionById(id)` delegates execution to `PermissionsService.get`
- `PermissionsRepository.findById(id)` queries `prisma.permission.findFirst` where `id = id` and `deletedAt = null`.
- If no record is found, the repository throws `PermissionNotFoundException` (HTTP 404).
- The server responds with HTTP 200 and the complete permission detail object.
- The web application renders the information card and formats dates using localized formatters.

### Screen Layout (Mock-up Description)

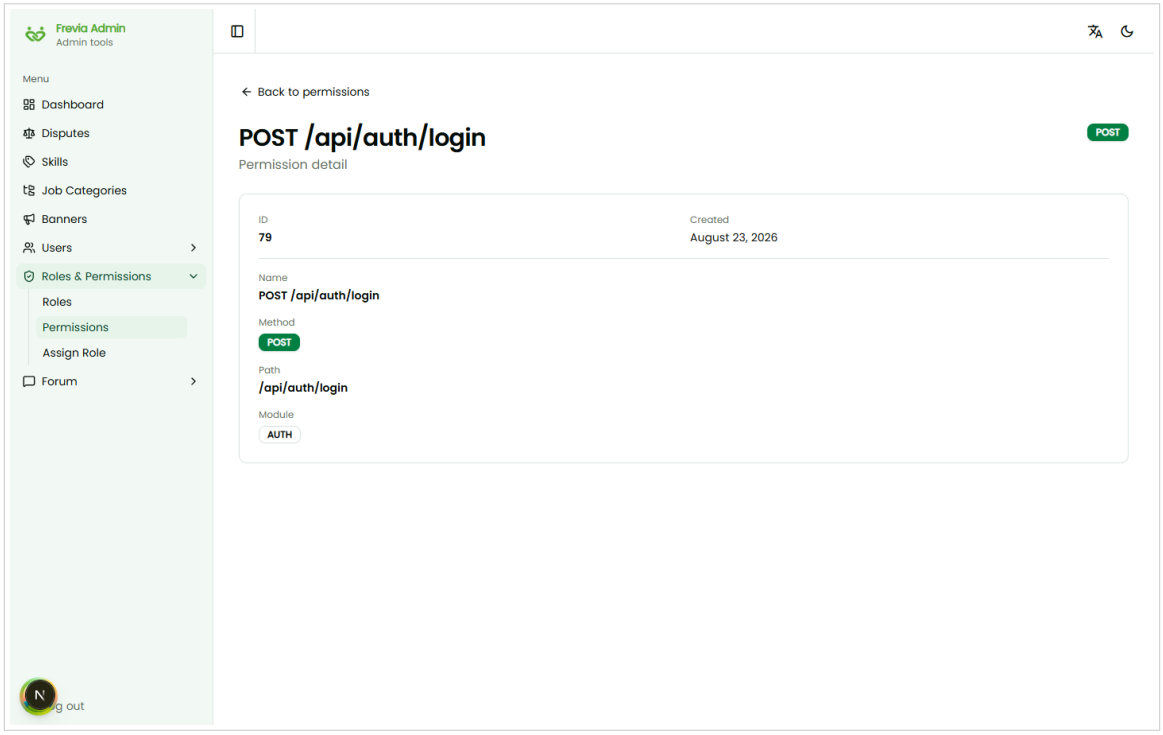


**Figure 76:** Permission detail interface

- A top “Back to permissions” navigation button with an arrow icon allows returning to the permission catalog.
- The header section displays the permission name in prominent monospace typography alongside the colored HTTP method badge (e.g., `POST`).
- A structured card contains labeled sections: **ID**, **Created Date**, **Method**, **Module**, and **Path**.
- The API route path is formatted in a distinct highlighted monospace container for easy inspection.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the permission passed in the URL path. |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0; non-numeric inputs are rejected by `ParseIntPipe`. |

**Business Rules**

- **BR-PERM-06**: Only active permissions with `deletedAt = null` are retrievable. Soft-deleted permissions return HTTP 404 Not Found.
- **BR-PERM-07**: The returned payload must include `id`, `name`, `path`, `method`, `module`, and `createdAt`.
- **BR-PERM-08**: Direct URL access to permission details requires an active Admin session with proper authorization.

**Normal Flow**

1. The administrator clicks “View detail” on a permission row or accesses `/admin/permissions/:id`.
2. The frontend validates the ID and sends `GET /api/permissions/:id`.
3. The system validates authorization and executes `PermissionsRepository.findById(id)`.
4. The database retrieves the permission record where `deletedAt = null`.
5. The server returns HTTP 200 with the permission detail payload.

6. The frontend renders the complete permission details and navigation controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid ID format (non-numeric string in URL) | NestJS `ParseIntPipe` rejects the request with HTTP 400 Bad Request. The frontend renders an invalid parameter error page. |
| Permission ID does not exist in database | The repository throws `PermissionNotFoundException` (HTTP 404). The client displays: "Failed to load permission detail. The permission may not exist." |
| Permission was soft-deleted (`deletedAt != null`) | The database query filters out soft-deleted records and throws HTTP 404 Not Found. The UI displays the not found notice. |
| Missing or expired administrator token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User account lacks Admin role privileges | The server returns HTTP 403 Forbidden. The client shows an access restriction notice. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database query failure during retrieval | The service catches the Prisma error and throws `FailedToLoadPermissionDetailException` (HTTP 500). The UI displays a retry button. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric system identifier of the permission. |
| name | Functional descriptive name of the permission. |
| path | Exact API route path pattern (e.g., `/api/auth/login`). |
| method | HTTP method verb (`GET`, `POST`, `PUT`, `PATCH`, `DELETE`). |
| module | Module grouping name to which the permission belongs. |
| createdAt | Timestamp indicating when the permission was registered. |

**Post-Conditions**

1. The complete details of the selected permission are presented to the administrator.
2. The administrator can navigate back to the permission list using the top navigation link.

### 3.6 Session Management

#### 3.6.1 View Session List

**Function Trigger**

An authenticated user accesses the active sessions management screen from their account settings.
**Navigation path:** User Avatar Menu → Settings → Active Sessions (`/sessions`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** Any authenticated user (`Client`, `Freelancer`, `Expert`, or `Admin`).

- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

**Purpose**

Allows users to view all active authentication sessions associated with their account. This enables users to monitor which devices, browsers, and IP addresses have access to their account, verify when each session was initiated and when it will expire, and identify potentially suspicious or obsolete logins.

**Interface**

- Dedicated session management screen at `/sessions`.
- Top control bar with keyword search input (supporting IP address or browser/device search) and sort selectors.
- Responsive session list table showing Device Information, IP Address, Creation Date, Expiration Date, Status, and Action buttons.
- Visual "Current session" badge highlighting the device currently being used.
- Pagination bar at the bottom with page numbers and navigation links.

**Data Processing Overview**

- The client sends `GET /api/sessions` with query parameters (`page`, `limit`, `search`, `sortBy`, `order`) and the user's Bearer JWT access token.
- The authentication guard verifies the JWT token and extracts `userId` and `sessionId`.
- `SessionsController.getMySessions()` invokes `SessionsService.getMySessions(userId, query, sessionId)`.
- `SessionsRepository.findAllByUserId()` queries the database where `userId = userId`, applying optional search filters on `ipAddress` or `deviceInfo`.
- The repository maps each record and dynamically sets `isCurrent = (row.id === sessionId)`.
- The backend returns HTTP 200 with the sessions collection and pagination metadata.
- The frontend renders the list, marking active sessions, indicating the current session, and providing detail inspection and revocation dialog triggers.

## Screen Layout (Mock-up Description)

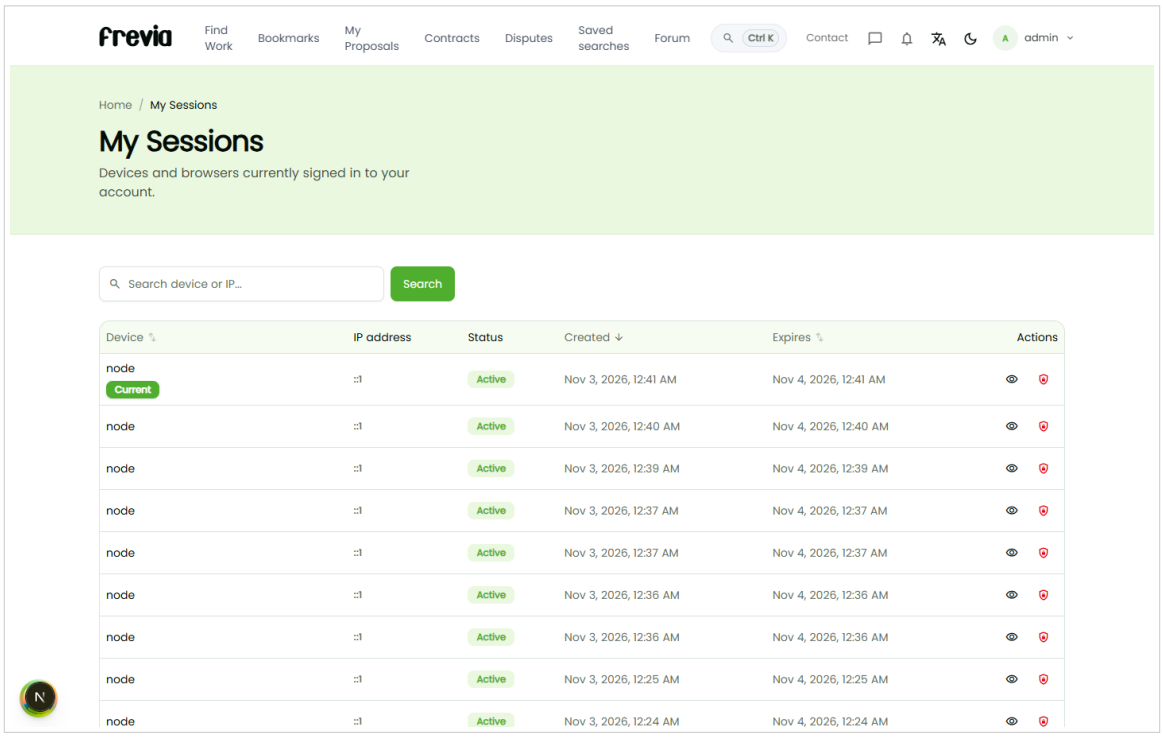


**Figure 77:** Active sessions list interface

- Header section displaying “Active sessions” with a descriptive subtitle.
- Search input box enabling instantaneous filtering by IP address or browser/device string.
- Table columns: **Device**, **IP Address**, **Created**, **Expires**, **Status**, and **Actions**.
- The currently active session displays a green “Current” badge, distinguishing it from remote sessions.
- Action icons include an eye icon (view detail modal) and a trash/revoke icon for remote session termination.
- A pagination component at the bottom provides direct page number navigation.

## Function Details

## Input Data

| Data Name | Description |
|---|---|
| page | Target page number (defaults to 1). |
| limit | Maximum session records per page (defaults to 10). |
| search | Optional query matching client IP address or user-agent device string. |
| sortBy | Sort column field (`id`, `createdAt`, or `expiresAt`; defaults to `createdAt`). |
| order | Sort direction (`asc` or `desc`; defaults to `desc`). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max 100 characters; whitespace trimmed. |
| sortBy | Must strictly be one of: `id`, `createdAt`, `expiresAt`. |
| order | Must strictly be one of: `asc`, `desc`. |

**Business Rules**

- **BR-SES-01**: Users can only view sessions belonging strictly to their own authenticated `userId`. Cross-account session retrieval is forbidden.
- **BR-SES-02**: The `isCurrent` indicator is true if and only if the session ID in the database matches the `sessionId` embedded in the caller's JWT access token.
- **BR-SES-03**: Search queries must perform case-insensitive partial matching against both `ipAddress` and `deviceInfo`.
- **BR-SES-04**: Session expiration status is evaluated dynamically: a session is active when `expiresAt > now`, and expired otherwise.
- **BR-SES-05**: Pagination parameters must be strictly validated with a maximum limit of 100 records per page.

**Normal Flow**

1. The user navigates to the "Active sessions" section in account settings.
2. The client sends `GET /api/sessions` with authorization headers.
3. The backend validates the JWT token, extracts `userId` and `sessionId`, and executes `SessionsRepository.findAllByUserId()`.

4. The database retrieves all sessions belonging to the user and calculates total count.
5. The backend attaches `isCurrent` flags and returns HTTP 200 with the session list and pagination data.
6. The frontend renders the session list table, highlighting the current device with a badge.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or expired JWT access token | The server returns HTTP 401 Unauthorized. The client removes stored session tokens and redirects the user to `/login`. |
| Current session was revoked remotely from another device | Token validation detects that the session ID no longer exists in the database. The system returns HTTP 401 Unauthorized and prompts the user to log in again. |
| User account has been banned or deactivated | The authentication guard verifies account status, detects inactive status, and throws HTTP 403 Forbidden. The client redirects to a ban notification screen. |
| Invalid pagination parameter values | The DTO validator rejects non-positive or malformed parameters with HTTP 400 Bad Request. The client falls back to page 1 and default limit. |

| Abnorma Case | Handling & System Response |
|---|---|
| Unsupported sort column parameter | The API rejects the request with HTTP 400 Bad Request. The client resets the query to sort by `createdAt desc`. |
| Database query failure during retrieval | The service catches the error, logs the failure, and throws `FailedToLoadSessionsException` (HTTP 500). The UI displays: "Failed to load sessions." |
| No sessions match current search keyword | The query returns an empty collection (`sessions: []`, `total: 0`). The table displays: "No sessions match your search." |

**Output Data**

| Output Field | Description |
|---|---|
| sessions | Array of session objects: `id`, `userId`, `deviceInfo`, `ipAddress`, `expiresAt`, `createdAt`, and `isCurrent`. |
| pagination.page | Current page number. |
| pagination.limit | Current page size. |
| pagination.total | Total active and stored sessions for the user. |
| pagination.totalPages | Total calculated pages available. |

**Post-Conditions**

1. The user's active session list is displayed accurately.
2. The current device session is visibly demarcated from other remote logins.
3. The user can click to inspect session details (UC-16.02) or revoke remote sessions (UC-16.03).

### 3.6.2 View Session Detail

### Function Trigger

An authenticated user clicks the "View detail" (eye icon) button on any session row in the session list.
**Navigation path:** Active Sessions (`/sessions`) → Click "View detail" icon on a session row.

### Function Description

### Actors / Roles

- **Primary Actor:** Any authenticated user (`Client`, `Freelancer`, `Expert`, or `Admin`).
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Displays full, in-depth technical information regarding a specific login session in a modal dialog. This helps the user inspect the complete browser user-agent, full IP address, creation timestamp, expiration date, active/expired status, and whether the session represents their current terminal.

### Interface

- Interactive modal dialog titled "Session details" appearing over the sessions page.
- Structured list displaying: Session ID, Device/Browser User-Agent, IP Address, Login Timestamp, Expiration Timestamp, and Status Badge.
- Close button and direct action button to revoke the session if remote.

### Data Processing Overview

- The client triggers an API call: `GET /api/sessions/:id` passing the session ID and Bearer token.
- The backend extracts `userId` and `sessionId` from the caller's JWT token.
- `SessionsController.getMySessionById()` calls `SessionsService.getMySessionById(id, userId, sessionId)`.
- `SessionsRepository.findByIdForUser()` queries `prisma.session.findFirst` where `id = id` and `userId = userId`.
- If no record is found, the system throws `SessionNotFoundException` (HTTP 404).
- The repository evaluates `isCurrent = (id === sessionId)` and returns the session object.
- The client populates the dialog with formatted dates and status indicators.

### Screen Layout (Mock-up Description)

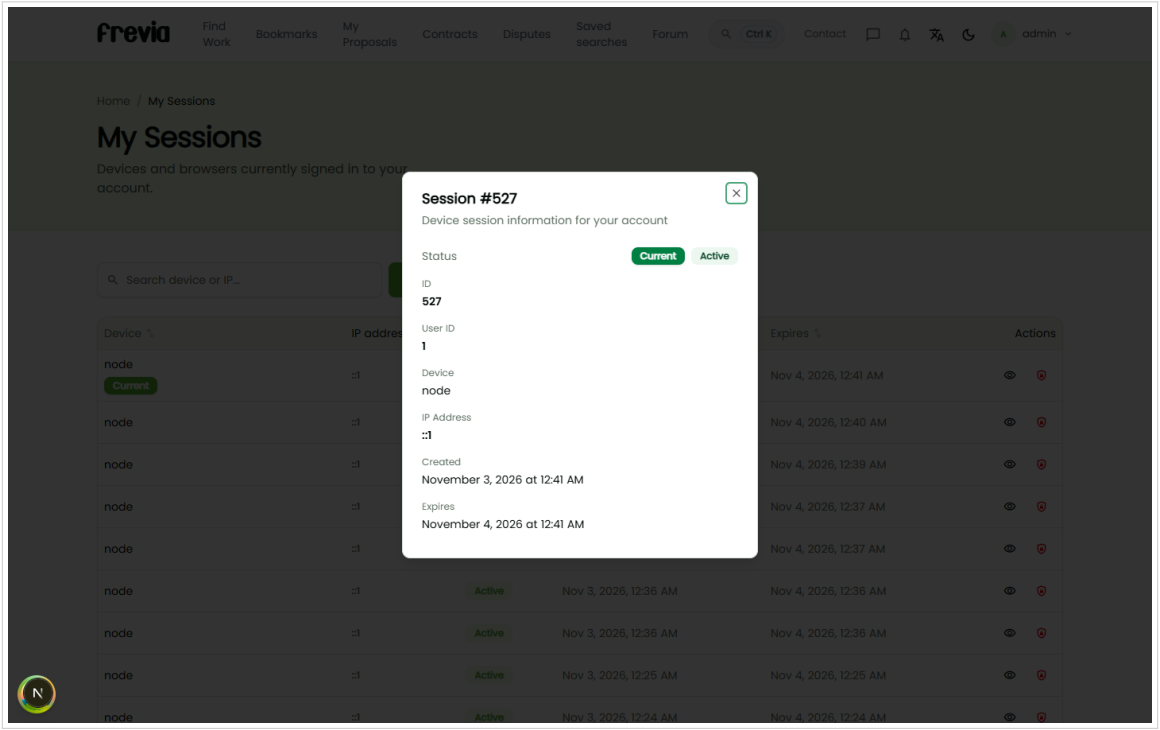


**Figure 78:** Session detail dialog interface

- A centered modal dialog titled “Session #{id}” with description “Detailed information about this login session.”
- A top badge indicating either “Current device” (green) or “Active” / “Expired”.
- Detailed metadata fields displayed in a clean grid: **Device / Browser**, **IP Address**, **Signed In**, and **Expires At**.
- A bottom “Close” button to dismiss the dialog.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric ID of the session passed as a URL route parameter. |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer; non-numeric values are rejected by `ParseIntPipe`. |

**Business Rules**

- **BR-SES-06**: A user can strictly view only session records that they own (`session.userId == caller.userId`). Attempting to access another user's session returns HTTP 404 Not Found to prevent ID enumeration.
- **BR-SES-07**: The returned payload must include `id`, `userId`, `deviceInfo`, `ipAddress`, `expiresAt`, `createdAt`, and `isCurrent`.

**Normal Flow**

1. The user clicks the view detail icon on a session row.
2. The frontend opens the modal dialog and sends `GET /api/sessions/:id`.
3. The backend validates authentication and queries the database for the session matching `id` and `userId`.
4. The database returns the session record.
5. The backend marks `isCurrent` and responds with HTTP 200.
6. The frontend renders the complete session details in the dialog.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid session ID format in URL | `ParseIntPipe` throws HTTP 400 Bad Request. The frontend displays an invalid parameter alert. |

| Abnorma Case | Handling & System Response |
|---|---|
| Session does not exist or has been deleted | The service throws `SessionNotFoundException` (HTTP 404). The modal displays: “Session not found.” |
| Session belongs to another user account | The database query filters by caller `userId`; matching returns null and throws HTTP 404 Not Found, preventing cross-user session probing. |
| Caller authen-tication token expired | The server returns HTTP 401 Unauthorized. The dialog closes and the user is redirected to `/login`. |
| Database query error during lookup | The backend catches the Prisma exception and throws `FailedToLoadSessionDetailException` (HTTP 500). The modal shows: “Failed to load session details.” |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric system identifier of the session. |
| userId | Numeric identifier of the owning user. |
| deviceInfo | Complete user-agent string describing the operating system and browser. |
| ipAddress | Client IP address recorded at login or refresh time. |
| expiresAt | Exact timestamp when the session’s refresh token expires. |
| createdAt | Exact timestamp when the session was initiated. |

| Output Field | Description |
|---|---|
| isCurrent | Boolean flag indicating whether this is the caller's active terminal. |

**Post-Conditions**

1. Detailed session information is visible to the user.
2. The user may proceed to revoke the session or close the dialog without side effects.

### 3.6.3 Revoke Session

**Function Trigger**

An authenticated user clicks the "Revoke" (trash or shield-off icon) button on an active session in the sessions list.
**Navigation path:** Active Sessions (`/sessions`) → Click "Revoke" button on a session row.

**Function Description**

**Actors / Roles**

- **Primary Actor:** Any authenticated user (`Client`, `Freelancer`, `Expert`, or `Admin`).
- **Supporting Systems:** Frevia Web application (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

**Purpose**

Allows users to remotely terminate an active login session on a specific device. Upon revocation, the stored refresh token record is deleted immediately from the database, preventing that device from obtaining new access tokens and forcing it to log out.

**Interface**

- Revocation trigger button on session table rows.
- Confirmation modal dialog titled "Revoke session" explaining that the target device will be signed out.
- "Cancel" and destructive "Revoke session" confirmation buttons.
- Toast notification confirming successful session revocation.

**Data Processing Overview**

- User confirms revocation in the alert dialog.

- The client issues an HTTP request: `DELETE /api/sessions/:id` with Bearer token.
- `SessionsController.revokeMySession()` passes `id`, `userId`, and `sessionId` to `SessionsService.revokeMySession()`.
- `SessionsRepository.deleteActiveByIdForUser()` verifies the session belongs to `userId` and is active (`expiresAt > now`).
- The repository deletes the session row from the PostgreSQL `Session` table.
- The service checks whether `id === sessionId`: if true, `loggedOut = true` is returned; otherwise `loggedOut = false`.
- The server responds with HTTP 200: `{ message: 'Session revoked successfully', loggedOut }`.
- If `loggedOut = true`, the client purges local tokens and redirects to `/login`; otherwise, the UI displays a success toast and refreshes the session table.

### Screen Layout (Mock-up Description)

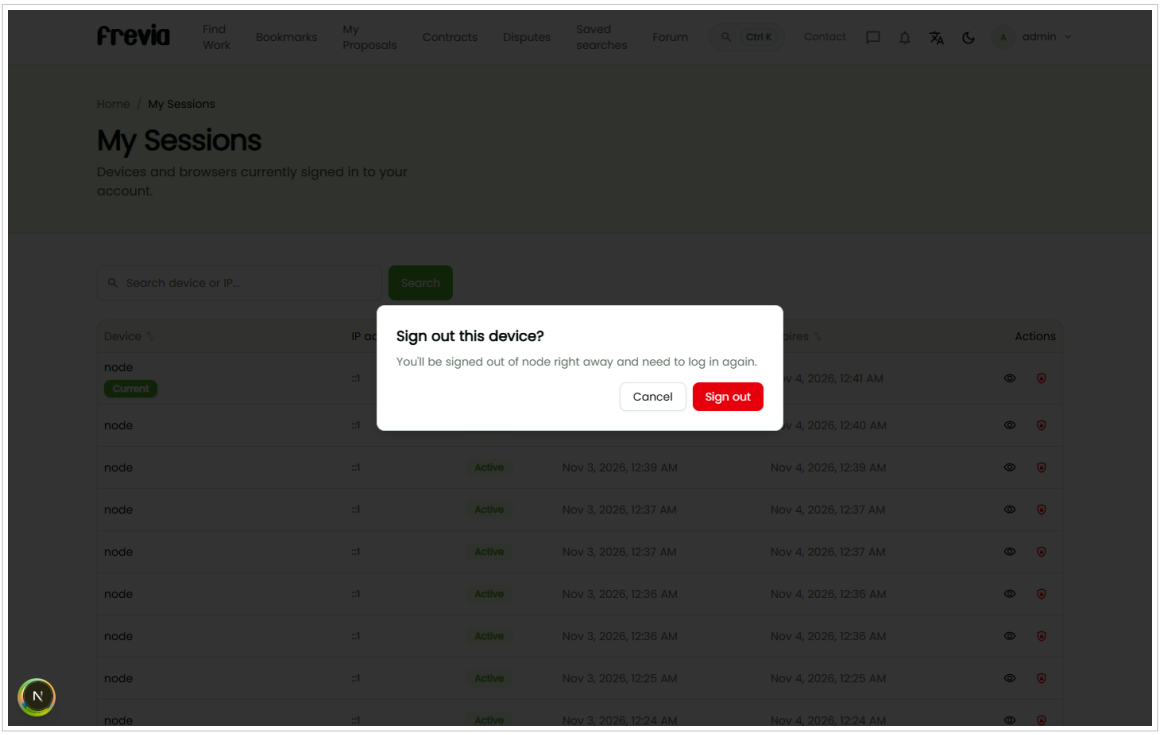


**Figure 79:** Revoke session confirmation dialog

- An alert dialog overlay displaying the title “Revoke session?”.
- A warning message: “This device will be logged out immediately. Any unsaved progress on that device may be lost. Are you sure you want to proceed?”.
- A neutral “Cancel” button to abort the action safely.
- A red destructive “Revoke session” action button with loading spinner state during network execution.

### Function Details

#### Input Data

| Data Name | Description |
| --- | --- |
| id | Unique numeric ID of the session to be revoked passed in the URL path. |

#### Validation Rules

| Data Name | Validation Rule |
| --- | --- |
| id | Must be a positive integer greater than 0. |

#### Business Rules

- **BR-SES-08**: Users can only revoke sessions belonging to their own user account.
- **BR-SES-09**: Already-expired sessions (`expiresAt <= now`) cannot be revoked; attempting to do so returns HTTP 400 Bad Request (`SESSION_ALREADY_EXPIRED`).
- **BR-SES-10**: Deletion of the session row immediately revokes the ability of the refresh token to issue new access tokens.
- **BR-SES-11**: If the user revokes their currently active session, the backend sets `loggedOut: true`, prompting the client application to terminate the current session and redirect to the sign-in screen.

#### Normal Flow

1. The user clicks “Revoke” on a session row in the table.
2. The frontend presents the confirmation alert dialog.
3. The user clicks “Revoke session” in the dialog.
4. The frontend sends `DELETE /api/sessions/:id`.
5. The backend validates authentication and checks that the session belongs to `userId` and is active.
6. The repository deletes the session record from the database.
7. The backend returns HTTP 200 with `{ message: 'Session revoked successfully', loggedOut: false }`.
8. The frontend displays a success toast and refreshes the session list.

#### Abnormal Flow / Exception Handling

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid session ID parameter | `ParseIntPipe` rejects the call with HTTP 400 Bad Request. The UI displays an invalid identifier toast. |
| Session does not exist or was already deleted | The service throws `SessionNotFoundException` (HTTP 404). The UI displays: "Session not found." |
| Session belongs to another user | The repository lookup fails to match `userId` and throws HTTP 404 Not Found, preventing unauthorized revocation. |
| Session has already naturally expired | The repository detects `expiresAt <= now` and throws `SessionAlreadyExpiredException` (HTTP 400). The UI displays: "Session has already expired." |
| Caller token expired or revoked | The server returns HTTP 401 Unauthorized. The client logs out immediately and redirects to `/login`. |
| User revokes their own current session | The server executes deletion and returns `loggedOut: true`. The client application clears cookies/local storage and redirects to `/login`. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database deletion failure | The service catches the Prisma error and throws `FailedToRevokeSessionException` (HTTP 500). The UI shows: "Failed to revoke session." |
| User cancels in confir-mation dialog | The dialog closes without issuing an HTTP request; no database changes occur. |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string (`Session revoked successfully`). |
| loggedOut | Boolean flag indicating whether the current client device was signed out. |

**Post-Conditions**

1. The session row is permanently deleted from the PostgreSQL `Session` table.
2. The remote device's refresh token becomes invalid for any future token renewals.
3. The session list table updates to reflect the removed session.

## 3.7 Forum Category Management

### 3.7.1 Create Forum Category

**Function Trigger**

An administrator clicks the "Create Category" button on the forum category management screen.

**Navigation path:** Admin Dashboard → Forum Categories (`/admin/categories`) → Click "Create Category" button.

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to establish a new discussion category in the forum. Categories provide structured classification for community discussions, professional topics, and questions posted by freelancers and clients.

### Interface

- "Create Category" trigger button on the category management screen.
- Modal dialog containing text fields for Category Name (required) and Description (optional).
- Action buttons: "Cancel" and "Create Category" (with loading state).
- Toast notification confirming successful category creation.

### Data Processing Overview

- Administrator fills in the name and description and submits the modal form.
- The client sends an HTTP request: `POST /api/forums/admin/categories` with JSON body `{ name, description }`.
- The backend guard verifies that caller has `ADMIN` role; otherwise, throws `ForumReportForbiddenExcepti` (HTTP 403).
- `CreateForumCategoryBodyDto` validates the payload schema.
- `ForumAdminRepo.createAdminCategory()` checks whether an active category (`deletedAt = null`) with the same name already exists (case-insensitive). If found, throws `ForumCategoryAlreadyExistsException` (HTTP 409).
- The system computes an automated SEO-friendly slug via `slugify(name)`. If a collision exists, a numeric counter suffix is appended.
- The repository creates the record in `prisma.forumCategory.create` with `postCount = 0`.
- The server responds with HTTP 201 Created and the new category object.
- The client closes the dialog, shows a success toast, and refreshes the category table.

### Screen Layout (Mock-up Description)

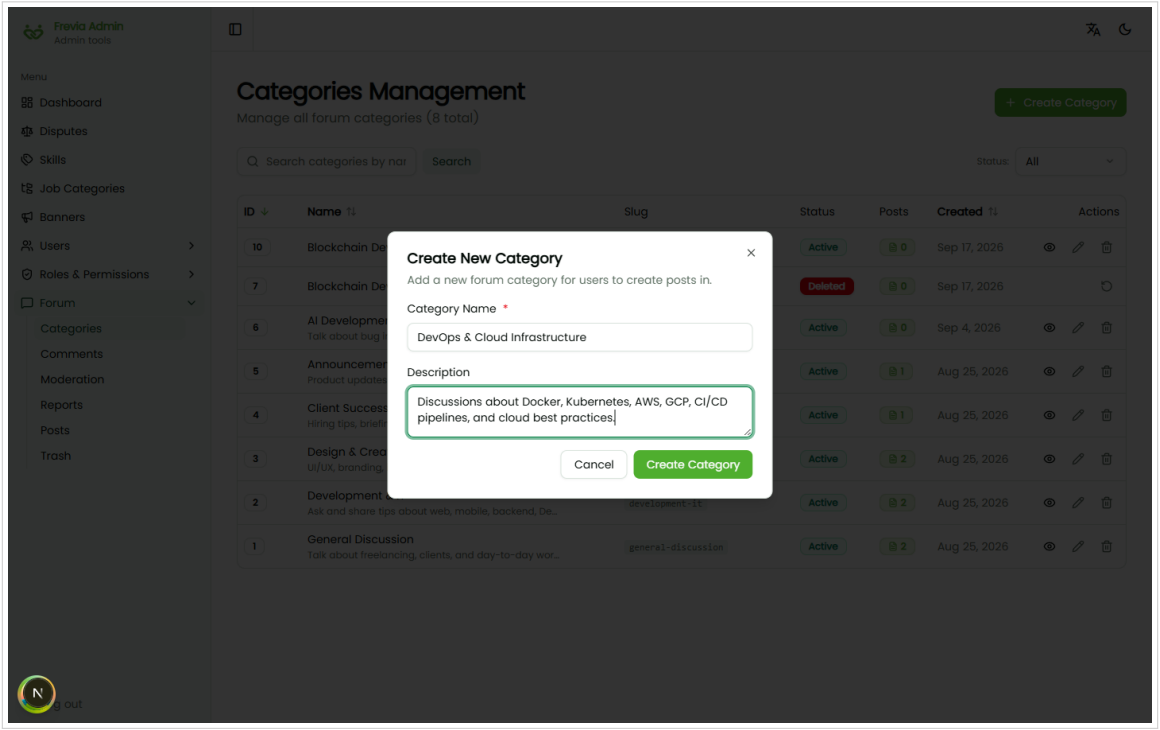


**Figure 80:** Create forum category modal

- Modal dialog titled “Create Category” with explanatory description “Add a new discussion category to the community forum.”.
- Form field: **Name** (marked with red required asterisk, placeholder: “e.g., General Discussion”).
- Form field: **Description** (multi-line textarea, optional, placeholder: “Describe what this category is for...”).
- Modal footer with “Cancel” button and dark primary “Create Category” button.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| name | Display name of the forum category (required). |
| description | Detailed description explaining the category purpose (optional). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| name | String, required, 1 to 100 characters; whitespace trimmed; must not be blank. |
| description | String, optional, max 500 characters; nullable. |

**Business Rules**

- **BR-FCAT-01**: Only users authenticated with the `ADMIN` role can create forum categories.
- **BR-FCAT-02**: Category name is required, max 100 characters, and must be unique among active categories (`deletedAt = null`) using case-insensitive comparison.
- **BR-FCAT-03**: Description is optional with a maximum length of 500 characters.
- **BR-FCAT-04**: A URL-friendly slug is generated automatically from the name. If the slug collides with an existing slug, a numeric suffix (e.g., `-2`) is appended.
- **BR-FCAT-05**: Newly created categories start with an initial post count of 0.

**Normal Flow**

1. The administrator clicks "Create Category" on `/admin/categories`.
2. The modal dialog appears. The administrator inputs a unique name and optional description.
3. The administrator clicks "Create Category".
4. The frontend validates fields and issues `POST /api/forums/admin/categories`.
5. The backend validates Admin role and ensures the category name does not duplicate an active category.
6. The system computes a unique slug and inserts the category into PostgreSQL.
7. The API returns HTTP 201 Created with the created category object.
8. The frontend displays a success toast: "Category created successfully!", closes the dialog, and updates the list.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Category name is empty or only whitespace | Client-side validation blocks submission; backend DTO validation returns HTTP 400 Bad Request. The UI highlights the name field in red. |
| Category name exceeds 100 characters | Backend validation rejects with HTTP 400 Bad Request (`String must contain at most 100 character(s)`). The UI displays a field length error. |
| Description exceeds 500 characters | Backend validation returns HTTP 400 Bad Request. The UI displays description length warning. |
| Category name already exists among active categories | The repository detects a duplicate name and throws `ForumCategoryAlreadyExistsException` (HTTP 409 Conflict). The UI displays: "Category name already exists." |
| User lacks Admin privileges | The service throws `ForumReportForbiddenException` (HTTP 403 Forbidden). The UI displays: "Forbidden: Account inactive or unauthorized." |

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or expired administrator token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| Database insertion failure or constraint error | The backend catches the Prisma error and throws `FailedToCreateForumCategoryException` (HTTP 500). The UI displays: "Failed to create category." |
| Administrator clicks Cancel in modal | The modal closes immediately and discarded input is purged without network requests. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric system identifier of the created category. |
| name | Display name of the category. |
| slug | Generated unique URL slug. |
| description | Category description string or null. |
| createdAt | Timestamp of category creation. |
| updatedAt | Timestamp of category creation. |
| postCount | Initial active post count (0). |

**Post-Conditions**

1. A new active category record is created in the database.

2. The category appears immediately in the forum categories list for administration and community discussions.

### 3.7.2 Update Forum Category

#### Function Trigger

An administrator clicks "Edit" on a category row in the table or on the category detail screen.
**Navigation path:** Admin Dashboard → Forum Categories → Click "Edit" icon on a category row.

#### Function Description

#### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

#### Purpose

Allows administrators to modify the title and descriptive summary of an existing forum category. When the title changes, the system automatically regenerates the category slug while preserving thread associations.

#### Interface

- Edit action icon button on category table rows and category detail views.
- "Edit Category" modal pre-populated with current category data.
- Form inputs for Name and Description.
- "Cancel" and "Save Changes" action buttons.

#### Data Processing Overview

- Administrator edits the name or description in the modal and submits the form.
- The client issues an HTTP request: `PATCH /api/forums/admin/categories/:id` with updated attributes.
- `ForumAdminService.updateAdminCategory()` verifies `ADMIN` role.
- `ForumAdminRepo.updateAdminCategory()` checks that the category exists and is active (`deletedAt = null`). If missing, throws `ForumCategoryNotFoundException` (HTTP 404).

- If the category name was changed, the repository checks for name collisions against other active categories (`id != currentId`). If a collision occurs, throws `ForumCategoryAlreadyExistsException` (HTTP 409).
- The repository regenerates the slug if the name changed, and updates the record in PostgreSQL.
- The API returns HTTP 200 with the updated category object.
- The frontend displays a success toast and refreshes the category data.

### Screen Layout (Mock-up Description)

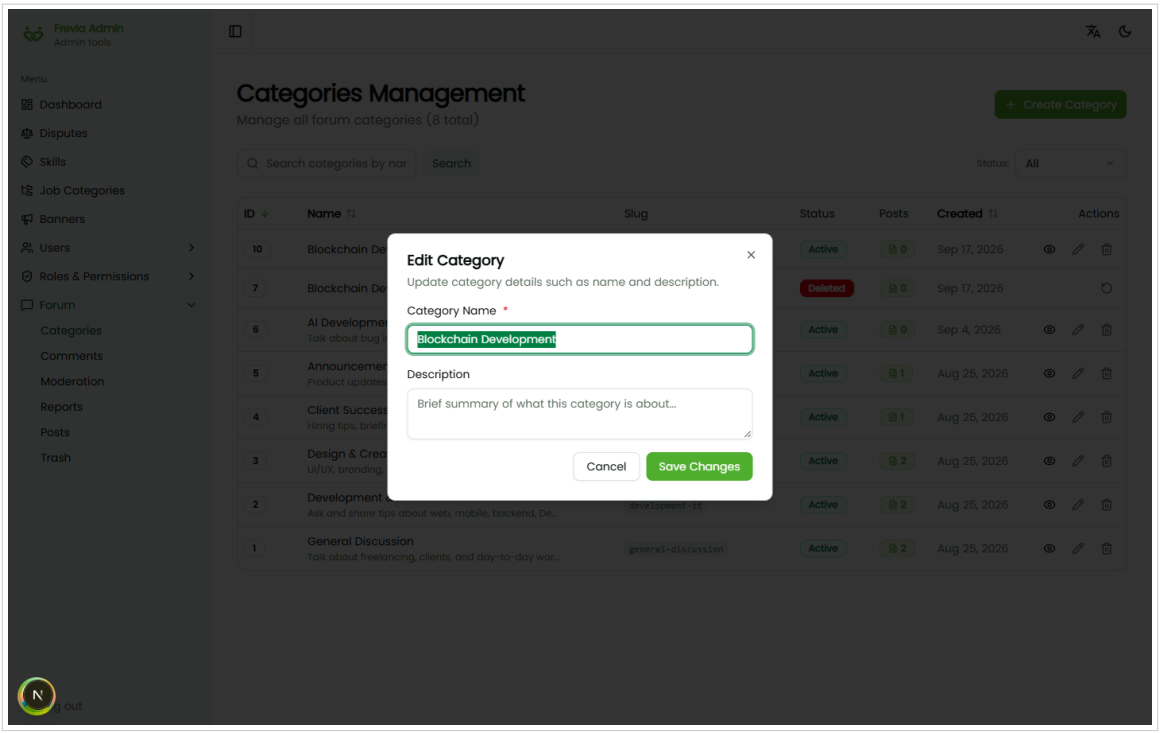


**Figure 81:** Update forum category modal

- Modal dialog titled “Edit Category” with description “Update category details.”.
- Pre-filled **Name** input field displaying the current category title.
- Pre-filled **Description** textarea displaying the current description.
- Action buttons at the bottom: “Cancel” and “Save Changes”.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Numeric identifier of the category in the URL path. |
| name | Updated category name (optional, string). |
| description | Updated category description (optional, string or null). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| name | String, optional, 1 to 100 characters; whitespace trimmed. |
| description | String, optional, max 500 characters; nullable. |

**Business Rules**

- **BR-FCAT-06**: Only Admin users can update forum categories.
- **BR-FCAT-07**: Partial updates are supported; name and description can be updated independently or simultaneously.
- **BR-FCAT-08**: If the category name is modified, it must be unique among active categories and its URL slug is regenerated automatically.
- **BR-FCAT-09**: Soft-deleted categories cannot be updated. Attempting to update a deleted category returns HTTP 404 Not Found.

**Normal Flow**

1. The administrator clicks "Edit" on a category row.
2. The modal opens pre-filled with the current category data.
3. The administrator modifies the name and/or description and clicks "Save Changes".
4. The frontend sends `PATCH /api/forums/admin/categories/:id`.
5. The backend validates permissions, confirms the category exists and is active, and verifies name uniqueness.
6. The repository updates the category in PostgreSQL and regenerates the slug if the name changed.
7. The API returns HTTP 200 with the updated category payload.
8. The frontend closes the modal, displays "Category updated successfully!", and refreshes the category table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Category ID does not exist or is soft-deleted | The service throws `ForumCategoryNotFoundException` (HTTP 404). The UI displays: “Category not found.” |
| New name conflicts with another active category | The repository detects a duplicate name and throws `ForumCategoryAlreadyExistsException` (HTTP 409 Conflict). The UI displays: “Category name already exists.” |
| Updated name is empty or only whitespace | Client-side and backend validation return HTTP 400 Bad Request. The UI displays an error on the name field. |
| Updated name exceeds 100 characters | Backend validation returns HTTP 400 Bad Request. The UI notifies the user of character limit violation. |
| Updated description exceeds 500 characters | Backend validation returns HTTP 400 Bad Request. The UI notifies the user of character limit violation. |

| Abnorma Case | Handling & System Response |
|---|---|
| User lacks Admin privi-leges | The backend throws `ForumReportForbiddenException` (HTTP 403 Forbidden). The client shows an access restriction notice. |
| Database update failure | The backend catches the Prisma exception and throws `FailedToUpdateForumCategoryException` (HTTP 500). The UI displays: "Failed to update category." |
| Administrator cancels modal | The dialog closes without issuing network requests; original values remain unchanged. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric system identifier of the category. |
| name | Updated category name. |
| slug | Updated URL slug reflecting any name changes. |
| description | Updated category description. |
| createdAt | Timestamp of original creation. |
| updatedAt | Timestamp indicating when the modification was committed. |
| postCount | Current active post count. |

**Post-Conditions**

1. The category record in the database is updated.
2. Associated forum posts continue to link to the category.
3. The updated details are reflected across both admin and client interfaces.

### 3.7.3 Delete Forum Category

### Function Trigger

An administrator clicks "Delete" (trash icon) on a category row or detail screen.

**Navigation path:** Admin Dashboard → Forum Categories → Click "Delete" icon on a category row.

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Allows administrators to soft-delete an existing forum category. To ensure community discussion integrity, active posts previously classified under this category are safely reassigned to Uncategorized (`categoryId = null`) inside an atomic database transaction.

### Interface

- Delete action button on category rows and category detail header.
- Confirmation alert dialog titled "Delete category?" providing clear notice of post detachment.
- Action buttons: "Cancel" and destructive "Delete".
- Toast notification confirming soft deletion.

### Data Processing Overview

- Administrator confirms category deletion in the alert dialog.
- The client issues an HTTP request: `DELETE /api/forums/admin/categories/:id`.
- `ForumAdminService.deleteAdminCategory()` checks that caller has `ADMIN` role.
- `ForumAdminRepo.deleteAdminCategory()` verifies that the category exists with `deletedAt = null`.
- The repository executes a Prisma transaction:
  1. Reassigns all active posts under the category: `prisma.forumPost.updateMany` where `categoryId = id` and `deletedAt = null`, setting `categoryId = null`.
  2. Marks the category record as soft-deleted: `prisma.forumCategory.update` setting `deletedAt = new Date()`.
- The server responds with HTTP 200: `{ message: 'Forum category deleted successfully' }`.

- The frontend displays a success toast and removes the category from the active view.

### Screen Layout (Mock-up Description)

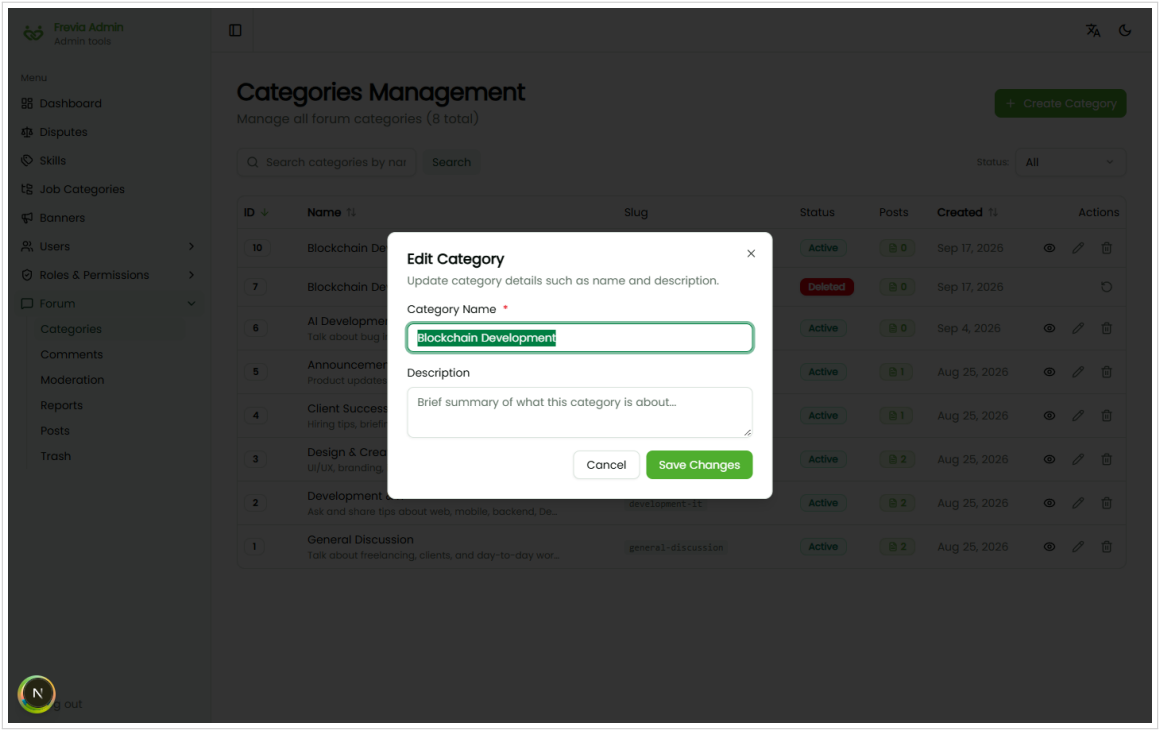


**Figure 82:** Delete forum category confirmation modal

- Alert dialog overlay titled “Delete category?”.
- Information notice: “Are you sure you want to delete “{name}”? Active posts in this category will become Uncategorized, and this category can be restored later from the trash tab.”.
- Neutral “Cancel” button and red destructive “Delete” action button with loading spinner state.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Numeric identifier of the category to delete passed in the URL path. |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |

**Business Rules**

- **BR-FCAT-10**: Only Admin users have authority to delete forum categories.
- **BR-FCAT-11**: Deletion is implemented as a soft delete by setting `deletedAt` to the current timestamp.
- **BR-FCAT-12**: In an atomic database transaction, active posts under this category are set to `categoryId = null` (Uncategorized) to preserve discussion history.
- **BR-FCAT-13**: Soft-deleted categories are hidden from the active category list and public forum dropdowns, but can be viewed and restored by administrators.

**Normal Flow**

1. The administrator clicks “Delete” on a category row.
2. The confirmation dialog appears explaining the post detachment and restoration possibilities.
3. The administrator clicks “Delete” to confirm.
4. The client sends `DELETE /api/forums/admin/categories/:id`.
5. The backend verifies permissions, reassigns active posts to `categoryId = null`, and soft-deletes the category in a transaction.
6. The backend returns HTTP 200 with a success message.
7. The frontend shows “Category deleted (soft delete)” and refreshes the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Category ID does not exist or is already deleted | The service throws `ForumCategoryNotFoundException` (HTTP 404). The UI displays: “Category not found.” |
| User lacks Admin privileges | The backend throws `ForumReportForbiddenException` (HTTP 403 Forbidden). The UI displays an authorization error. |
| Caller token expired or missing | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| Database transaction fails during post detachment or category update | The transaction rolls back completely and the service throws `FailedToDeleteForumCategoryException` (HTTP 500). The UI displays: “Delete category failed.” |
| Administrator clicks Cancel in confirmation dialog | The dialog closes without issuing an HTTP request; no database records are modified. |

**Output Data**

| Output Field | Description |
|---|---|
| message | Confirmation message string (`Forum category deleted successfully`). |

**Post-Conditions**

1. The category record has `deletedAt` populated with the deletion timestamp.
2. Active posts formerly in this category have `categoryId = null` and remain accessible as Uncategorized discussions.
3. The category is removed from the active category list and moves to the trash view.

### 3.7.4 View Forum Category List

**Function Trigger**

An administrator accesses the forum categories administration section from the sidebar navigation.
**Navigation path:** Admin Dashboard → Forum Categories (`/admin/categories`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

**Purpose**

Provides administrators with a centralized overview of all discussion categories in the forum. Allows searching by category name, filtering between active categories and deleted categories (trash view), sorting by ID, name, or creation date, and viewing live post count statistics.

**Interface**

- Management screen at `/admin/categories`.
- Action toolbar containing search input, active/trash status filter toggle, sorting selector, and "Create Category" button.
- Data table displaying ID, Name, Slug, Description, Post Count, Created Date, Status Badge, and contextual Action buttons.
- Pagination footer with total record count and page controls.

### Data Processing Overview

- Client sends `GET /api/forums/admin/categories` with query parameters (`page`, `limit`, `search`, `deleted`, `sortBy`, `sortOrder`).
- The backend guard verifies caller's JWT token and validates `ADMIN` role.
- `ForumAdminService.getAdminCategoryLists()` delegates query execution to `ForumAdminRepo.getAdminCategoryLists()`.
- The repository queries `prisma.forumCategory.findMany` with conditions:
  - If `deleted === 'true'`, queries where `deletedAt != null`; otherwise, queries `deletedAt = null`.
  - If `search` is provided, filters `name` using case-insensitive contains.
  - Computes active post count: `_count: { posts: { where: { deletedAt: null } } }`.
- The database returns matching categories and total count.
- The server responds with HTTP 200: `{ categories, total, pagination }`.
- The frontend renders the table with appropriate action buttons (Edit/Delete for active, Restore for deleted).

### Screen Layout (Mock-up Description)

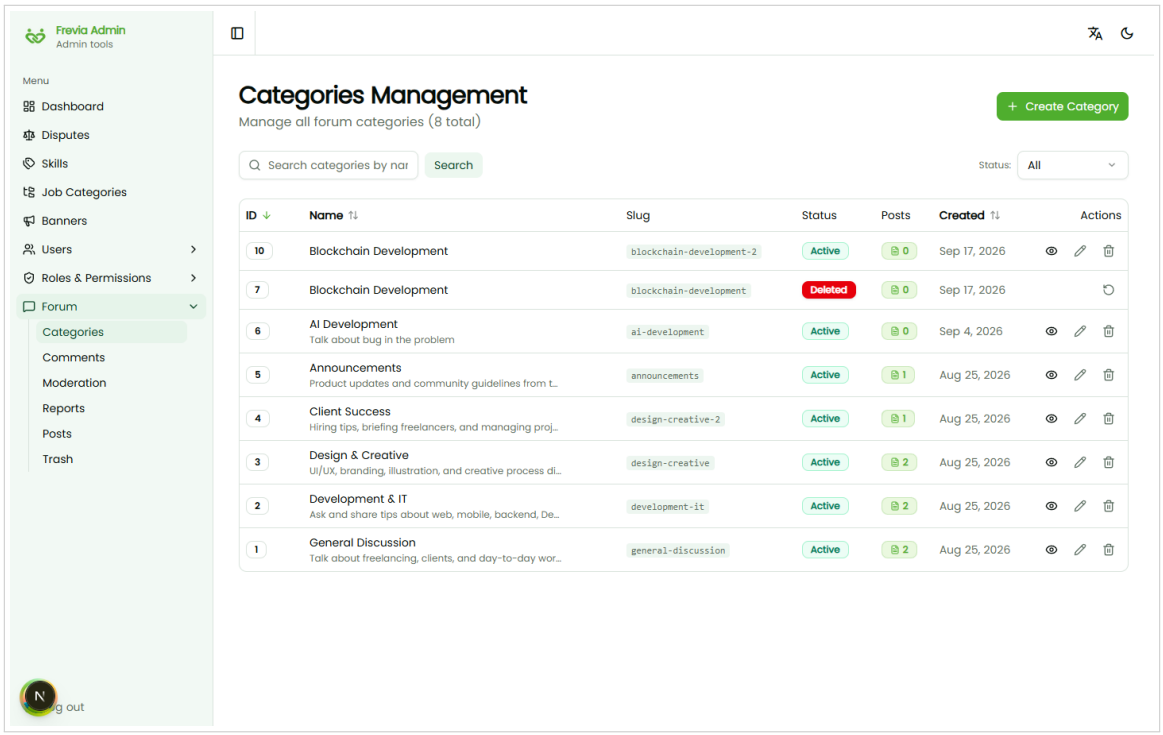


**Figure 83:** Admin forum category list interface

- Top section with title "Categories" and "Create Category" button.
- Filter bar including search input, status tab filter (All Active vs. Trash), and sort dropdown.

- Data table columns: **ID**, **Name**, **Slug**, **Description**, **Posts**, **Created**, and **Actions**.
- Each category row features clickable links to view details, an edit icon, and a delete icon.
- Pagination footer with previous/next page navigation buttons and current record count.

### Function Details

#### Input Data

| Data Name | Description |
|---|---|
| page | Target page number (defaults to 1). |
| limit | Maximum categories per page (defaults to 10). |
| search | Optional text query searching category name. |
| deleted | Optional filter string (`'true'` for trash view, `'false'` or omitted for active categories). |
| sortBy | Sort column field (`id`, `name`, or `createdAt`; defaults to `id`). |
| sortOrder | Sort order direction (`asc` or `desc`; defaults to `desc`). |

#### Validation Rules

| Data Name | Validation Rule |
|---|---|
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max length 100 characters. |
| deleted | Optional string, must be `'true'` or `'false'`. |
| sortBy | Must strictly be one of: `id`, `name`, `createdAt`. |
| sortOrder | Must strictly be one of: `asc`, `desc`. |

#### Business Rules

- **BR-FCAT-14**: Only users with the `ADMIN` role are authorized to access the category management catalog.
- **BR-FCAT-15**: When the `deleted` query parameter is omitted or `'false'`, only active categories (`deletedAt = null`) are returned.
- **BR-FCAT-16**: When `deleted = 'true'`, only soft-deleted categories (`deletedAt != null`) are returned.

- **BR-FCAT-17**: Post counts are calculated strictly for active posts (`deletedAt = null`) under each category.
- **BR-FCAT-18**: Sorting is supported on `id`, `name`, and `createdAt`.

**Normal Flow**

1. The administrator navigates to the "Forum Categories" section.
2. The frontend requests `GET /api/forums/admin/categories` with default query parameters.
3. The backend validates Admin authorization and executes the database query with active post counts.
4. The database returns category records and total count.
5. The server responds with HTTP 200 and the category list payload.
6. The frontend renders the category table with post counts, timestamps, and action buttons.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| User is not authenticated or token expired | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The service throws `ForumReportForbiddenException` (HTTP 403 Forbidden). The UI displays an access denied screen. |
| Invalid pagination parameter | The API rejects invalid parameter values with HTTP 400 Bad Request. The client defaults to page 1. |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid sortBy parameter | The system rejects invalid sort keys with HTTP 400 Bad Request. The client resets sorting to `id desc`. |
| Database query error during retrieval | The service throws `FailedToLoadForumCategoriesException` (HTTP 500). The UI displays: "Failed to load forum categories." |
| No categories match search or filter query | The repository returns an empty list (`categories: [], total: 0`). The table displays: "No categories found." |

**Output Data**

| Output Field | Description |
|---|---|
| categories | Array of category objects: `id`, `name`, `slug`, `description`, `createdAt`, `updatedAt`, `deletedAt`, and `postCount`. |
| pagination.page | Current page number. |
| pagination.limit | Current limit per page. |
| pagination.total | Total matching categories. |
| pagination.totalPages | Total calculated pages available. |

**Post-Conditions**

1. The forum category list is rendered with accurate live post count statistics.
2. The administrator can trigger create, edit, delete, detail view, or restore actions.

### 3.7.5 View Forum Category Detail

### Function Trigger

An administrator clicks "View detail" on a category row or accesses the category detail route directly.
**Navigation path:** Forum Categories (`/admin/categories`) → Click category title or row (`/admin/categories/:id`).

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Displays full administrative details of a specific forum category, including its system ID, title, unique URL slug, detailed description, active discussion post count, creation timestamp, and last updated timestamp, with quick shortcuts to edit or delete the category.

### Interface

- Dedicated category detail page at `/admin/categories/:id`.
- Top navigation header with "Back to categories" link.
- Header displaying category name and ID badge.
- Action buttons: "Edit Category" and "Delete Category".
- Structured information cards displaying: Description, Slug, Post count, Creation date, and Last update timestamp.

### Data Processing Overview

- Client sends `GET /api/forums/admin/categories/:id`.
- `ParseIntPipe` ensures `id` is an integer.
- The backend guard verifies `ADMIN` role.
- `ForumAdminService.getAdminCategoryById()` calls `ForumAdminRepo.getAdminCategoryById(id`
- The repository queries `prisma.forumCategory.findFirst` where `id = id` and `deletedAt = null`, including active post count.
- If the category is not found, the service throws `ForumCategoryNotFoundException` (HTTP 404).
- The server responds with HTTP 200 and the category detail payload.
- The frontend renders the detail cards and navigation buttons.

### Screen Layout (Mock-up Description)

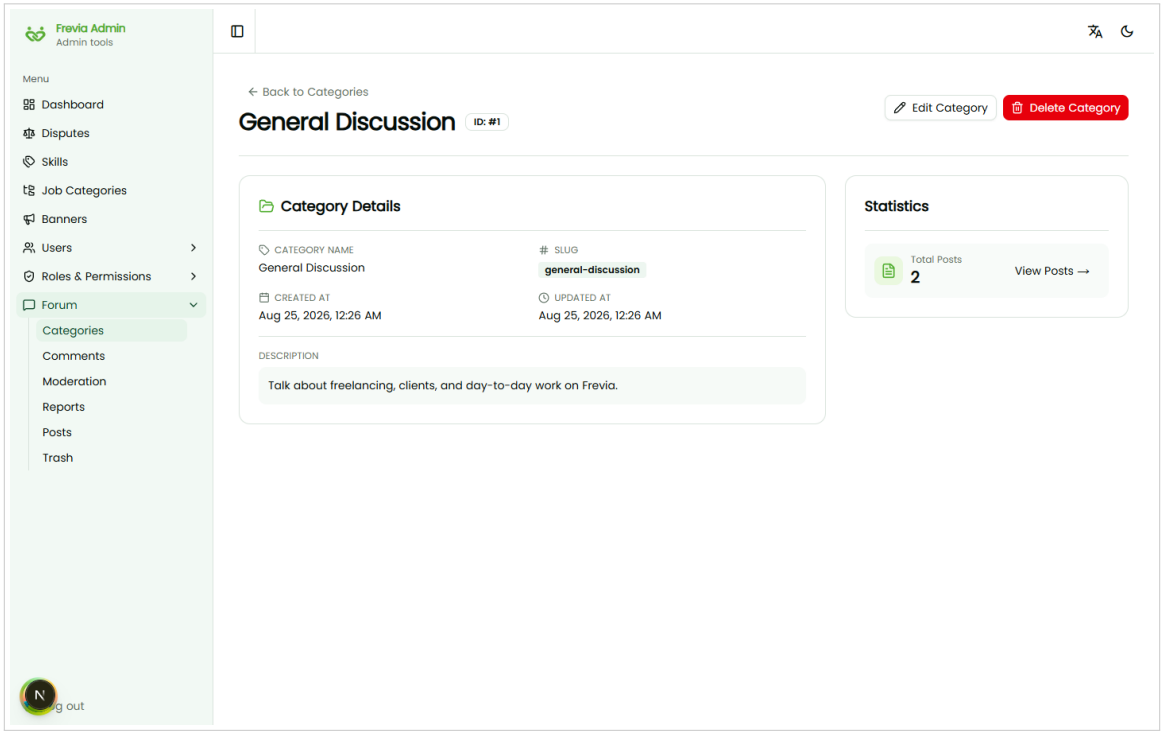


**Figure 84:** Forum category detail interface

- Top breadcrumb button “Back to categories” leading back to the category catalog.
- Page header displaying the category name in bold typography with an “ID: #{id}” outline badge.
- Top right action buttons: “Edit Category” and destructive “Delete Category”.
- Information cards: **Category Information** (Description and Slug) and **Statistics & Metadata** (Total Posts, Created timestamp, and Last Updated timestamp).

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Numeric identifier of the category in the URL path. |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |

**Business Rules**

- **BR-FCAT-19**: Only active categories with `deletedAt = null` are accessible via the detail view. Deleted categories return HTTP 404 Not Found.
- **BR-FCAT-20**: Detail payload includes `id`, `name`, `slug`, `description`, `createdAt`, `updatedAt`, and `postCount`.
- **BR-FCAT-21**: Access is restricted exclusively to authenticated users possessing the `ADMIN` role.

**Normal Flow**

1. The administrator clicks a category row or accesses `/admin/categories/:id`.
2. The frontend sends `GET /api/forums/admin/categories/:id`.
3. The backend validates Admin authorization and queries the database for the active category.
4. The repository retrieves the category along with its active post count.
5. The server responds with HTTP 200 and the category detail data.
6. The frontend renders the complete category details, statistics, and action controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid category ID format (non-integer) | `ParseIntPipe` rejects with HTTP 400 Bad Request. Next.js triggers the not found or bad request error view. |

| Abnorma Case | Handling & System Response |
|---|---|
| Category ID does not exist in database | The service throws `ForumCategoryNotFoundException` (HTTP 404). The frontend displays: "Category not found." |
| Category has been soft-deleted | Query filters `deletedAt = null`; returns null and throws HTTP 404 Not Found. The UI displays the not found notice. |
| User lacks Admin privileges | The backend throws `ForumReportForbiddenException` (HTTP 403 Forbidden). Access is denied. |
| Caller token expired or missing | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| Database query error during lookup | The backend catches the Prisma error and throws `FailedToLoadCategoryDetailException` (HTTP 500). The UI displays a retry button. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the category. |
| name | Display title of the category. |
| slug | Unique URL slug. |
| description | Detailed description or null. |
| createdAt | Timestamp of category creation. |
| updatedAt | Timestamp of last update. |
| postCount | Number of active discussion threads in this category. |

**Post-Conditions**

1. Category details and post statistics are presented to the administrator.
2. The administrator can trigger edit, delete, or return to the category list.

### 3.7.6 Restore Forum Category

**Function Trigger**

An administrator clicks "Restore" on a soft-deleted category in the trash view of the category catalog.
**Navigation path:** Admin Dashboard → Forum Categories → Filter "Trash" → Click "Restore" icon.

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

**Purpose**

Allows administrators to recover a soft-deleted forum category by clearing its `deletedAt` field back to `null`. This restores the category to the active list and enables users to create new discussion posts within it again.

**Interface**

- Restore action icon button on deleted category rows in the Trash view.

- Confirmation modal dialog titled “Restore category?”.
- Action buttons: “Cancel” and “Restore”.
- Toast notification confirming successful category restoration.

### Data Processing Overview

- Administrator clicks “Restore” on a deleted category and confirms in the dialog.
- The client issues an HTTP request: `PATCH /api/forums/admin/categories/:id/restore`.
- `ForumAdminService.restoreAdminCategory()` verifies `ADMIN` role.
- `ForumAdminRepo.restoreAdminCategory()` verifies that the category exists and is currently deleted (`deletedAt != null`). If missing, throws `ForumCategoryNotFoundException` (HTTP 404).
- The repository checks whether an active category (`deletedAt = null`) already exists with the same name. If a duplicate exists, throws `ForumCategoryAlreadyExistsException` (HTTP 409 Conflict).
- The repository executes `prisma.forumCategory.updateMany` setting `deletedAt = null`.
- The API returns HTTP 200 with the restored category object.
- The frontend displays a success toast and refreshes the category table.

### Screen Layout (Mock-up Description)

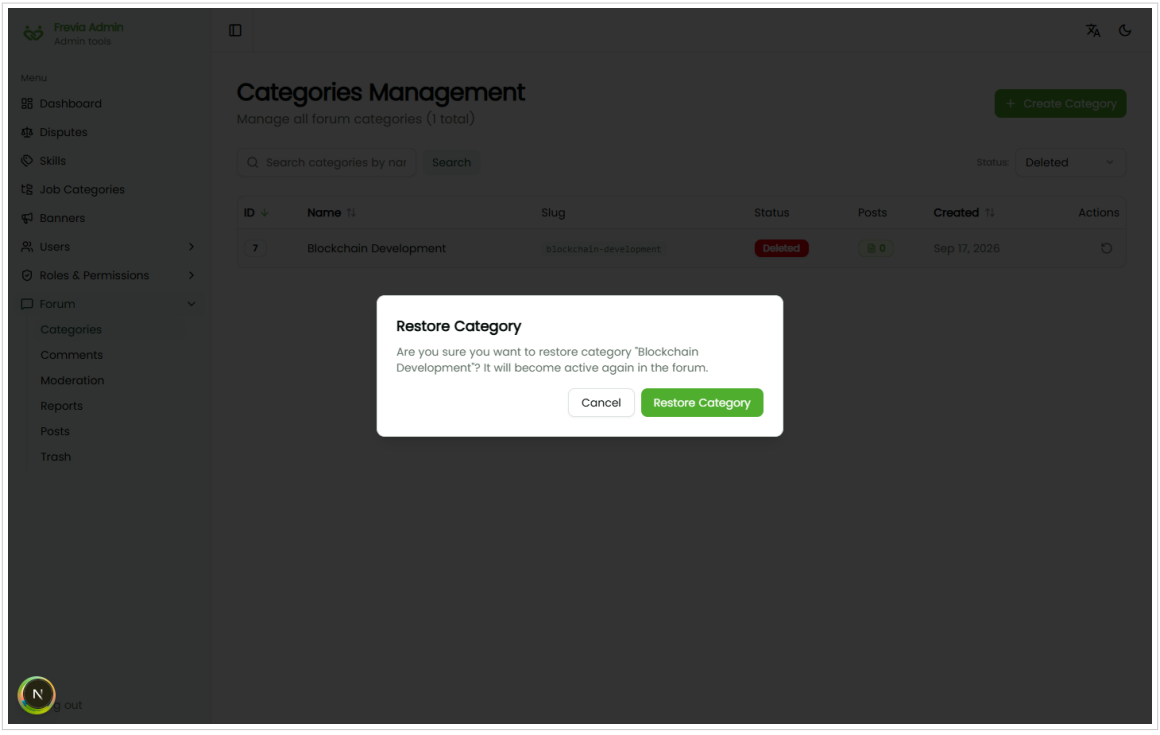


**Figure 85:** Restore forum category modal

- Alert dialog overlay titled “Restore category?”.
- Descriptive prompt: “Are you sure you want to restore “{name}”? This category will reappear in the active categories list and be available for community discussions.”.
- Action buttons at the bottom: “Cancel” and primary “Restore”.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| id | Numeric identifier of the soft-deleted category passed in the URL path. |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |

**Business Rules**

- **BR-FCAT-22**: Only users with the `ADMIN` role can restore forum categories.
- **BR-FCAT-23**: The category must currently be soft-deleted (`deletedAt != null`) to be eligible for restoration.
- **BR-FCAT-24**: Before restoration, the system must verify that no active category currently exists with the same name. If a naming collision is detected, restoration is rejected with HTTP 409 Conflict.
- **BR-FCAT-25**: Restoring resets `deletedAt = null`; previously detached posts remain in Uncategorized unless individually reclassified.

**Normal Flow**

1. The administrator navigates to the “Trash” view of the category catalog.
2. The administrator clicks “Restore” on a deleted category row.
3. The confirmation dialog appears. The administrator confirms restoration.
4. The frontend sends `PATCH /api/forums/admin/categories/:id/restore`.
5. The backend validates Admin authorization and confirms the category is currently soft-deleted.
6. The repository checks for name conflicts against active categories.

7. The repository updates the category setting `deletedAt = null`.
8. The server returns HTTP 200 with the restored category details.
9. The frontend displays “Category restored successfully” and updates the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Category ID does not exist or was not deleted | The service throws `ForumCategoryNotFoundException` (HTTP 404). The UI displays: “Category not found or not deleted.” |
| Another active category now uses the same name | The repository detects a duplicate name among active categories and throws `ForumCategoryAlreadyExistsException` (HTTP 409 Conflict). The UI displays: “Cannot restore category: an active category with the same name already exists.” |
| User lacks Admin privileges | The backend throws `ForumReportForbiddenException` (HTTP 403 Forbidden). Access is denied. |
| Caller token expired or missing | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database update failure during restoration | The backend catches the Prisma error and throws `FailedToRestoreForumCategoryException` (HTTP 500). The UI displays: “Restore category failed.” |
| Administrator cancels restoration dialog | The dialog closes without issuing an HTTP request; the category remains soft-deleted. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the restored category. |
| name | Display title of the restored category. |
| slug | URL slug of the category. |
| description | Description string or null. |
| createdAt | Timestamp of original creation. |
| updatedAt | Timestamp indicating restoration time. |
| postCount | Current post count. |

**Post-Conditions**

1. The category’s `deletedAt` field is reset to `null`.
2. The category reappears in the active forum category catalog.
3. Users can create new discussion posts under this category.

## 3.8 Manage Conversation

### 3.8.1 Start Conversation

### Function Trigger

A logged-in user starts a new chat, either by clicking the “Connect with client” button on the job detail page in Find Work, or by clicking the “New message” button in the conversation list.

### Function Description

This function allows a logged-in user to start a direct conversation with another user. When the chat is started from a job, the other participant is the client who posted that job. When the chat is started from the conversation list, the user chooses the other participant. The conversation is created and the user can write the first message. If a conversation with that user already exists, the existing conversation is reopened instead of creating a duplicate.

### Screen Layout (Mock-up Description)

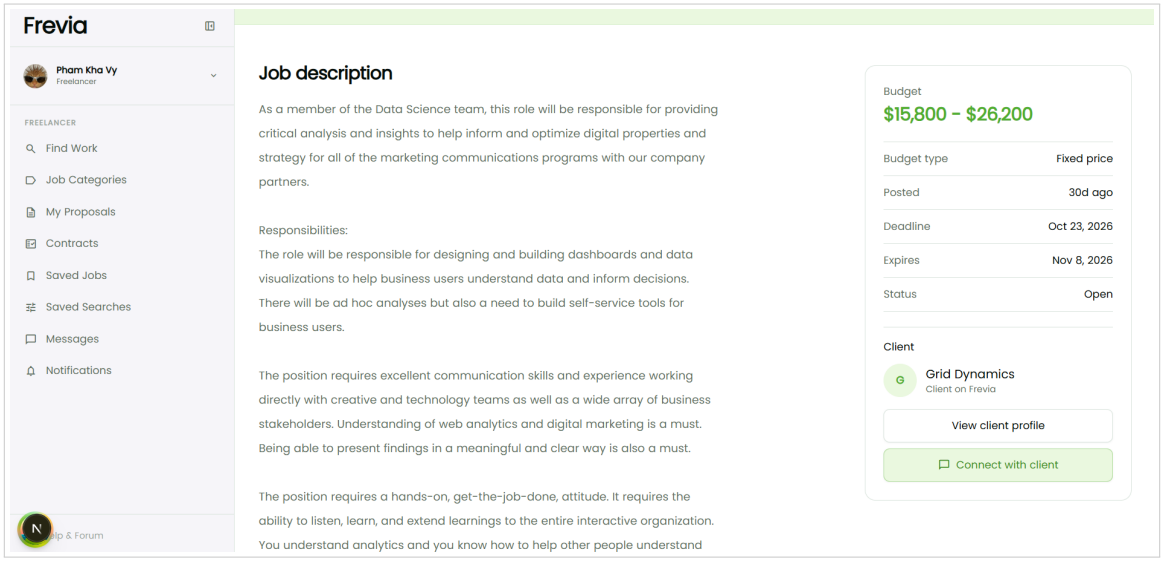


**Figure 86:** Start conversation screen

### Function Details

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The user selected another user who is different from themselves.
- **Process**
  - **Start a New Conversation**: The user starts the chat in one of two ways.
    - From a job in Find Work, the user clicks the “Connect with client” button on the job detail page.

        - From the conversation list, the user clicks the “New message” button.
    - **Choose the Participant**: The user provides or confirms the other participant.
        - When the chat is started from a job, the client of that job is already selected, so the user does not have to choose anyone.
        - When the chat is started from the conversation list, the user enters the identifier of the other participant in the dialog and confirms.
    - **Create or Reopen the Conversation**: The system checks whether a conversation between the two users already exists.
        - When no conversation exists, a new conversation between the two users is created.
        - When a conversation already exists, that conversation is reopened, including a conversation that the user had previously deleted from their own list.
    - **Write the First Message**: The compose screen shows the name and avatar of the other participant, and the user can type and send the first message (see **Send Message**).
- **Alternative Flow**
    - **Conversation Already Exists**: When a conversation with the chosen user already exists, the system opens the existing conversation instead of creating a second one, so there is only one thread per pair of users.
    - **Not Logged In**: When a guest clicks the “Connect with client” button, the system asks the user to log in before the conversation can be started.
    - **Cancel**: When the user leaves the compose screen or closes the dialog without confirming, no conversation is created and nothing changes.
- **Postconditions**
    - A conversation between the two users exists and is open on the screen.
    - The conversation is visible in the conversation list of both participants.
- **Error Handling**
    - **Cannot Chat with Yourself**: When the user chooses their own account as the participant, the conversation is not created and a message tells the user that a chat with themselves is not allowed.
    - **Participant Not Found**: When the chosen user cannot be found, the conversation is not created and a message tells the user that the participant cannot be found.
    - **Cannot Start Chat**: When the conversation cannot be created, the screen shows a start-chat failure message so that the user can try again.
    - **Lost Connection**: When the connection to the server is lost, the screen shows a “Lost Connection” state and asks the user to reconnect.

- **Security Measures**
  - Only logged-in users can start a conversation, and the user is always recorded as one of the two participants.
  - When the chat is started from a job, the other participant is always the client of that job, so the button cannot be used to start a chat with an unrelated account.
  - Only one conversation can exist for a pair of users, so a user cannot create duplicated threads with the same person.

### 3.8.2 View Conversation List

#### Function Trigger

The logged-in user opens the Messages area by clicking the chat icon in the main navigation.

#### Function Description

This function allows a logged-in user (Client, Freelancer) to see all of their conversations in one list. Each row shows the other participant, a preview of the last message, the time of that message and the number of unread messages. Conversations that the user has pinned are always shown at the top of the list.

#### Screen Layout (Mock-up Description)

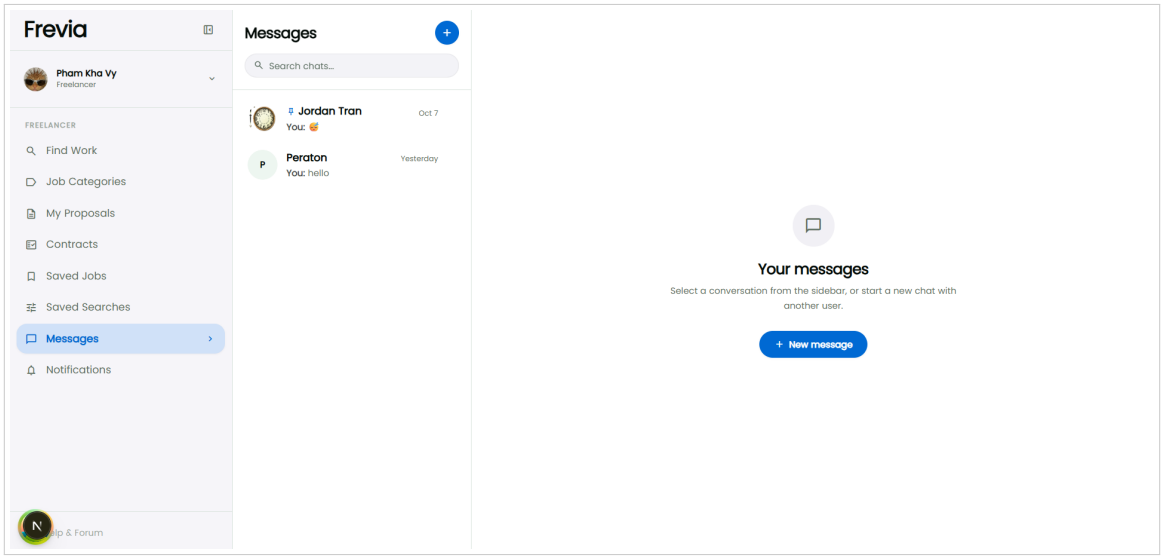


**Figure 87:** Conversation list screen

**Function Details**

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The user has at least one conversation, or the empty state is displayed instead.
- **Process**
  - **Open the Messages Area**: The user clicks the chat icon in the main navigation and the conversation list is displayed.
  - **Read Each Conversation Row**: Every row in the list presents the following information.
    - The avatar and the display name of the other participant.
    - A short preview of the last message, with a paperclip icon when the last message contains a file.
    - The relative time of the last message.
    - A blue unread dot on the avatar and an unread count badge when there are unread messages.
    - A pin icon when the conversation is pinned.
  - **Search for a Conversation**: The user types a keyword in the search box and the list shows only the conversations whose participant name or last message contains that keyword.
  - **Sorting**: Pinned conversations are shown first, and the remaining conversations are ordered by the time of the last message, newest first.
  - **Open a Conversation**: The user clicks a conversation row to open the chat thread (see **View Conversation Detail**).
- **Alternative Flow**
  - **No Conversation**: When the user has no conversation at all, the screen shows an empty state with a "New message" button, and the user can start a new chat from there (see **Create Conversation**).
  - **No Search Result**: When the search keyword does not match any conversation, the list is replaced by a message telling the user that no chat matches the keyword.
- **Postconditions**
  - The user sees all of their conversations, with the pinned ones at the top and the unread counts for each conversation.
  - Opening the list does not change any message; unread messages are only cleared when the user opens a thread (see **Mark Conversation as Read**).

- **Error Handling**
  - **Cannot Load Conversations**: When the conversation list cannot be loaded, the screen shows a load-failure message so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - Only the conversations of the logged-in user are shown, so a user can never see the chats of another account.
  - Conversations that the user has deleted from their own list are not shown again until a new message arrives.

### 3.8.3 View Conversation Detail

#### Function Trigger

The user clicks a conversation row in the conversation list.

#### Function Description

This function allows the user to read the whole chat thread with the other participant. The thread shows all messages in chronological order, the avatar and display name of the other user, the attachments that were exchanged and the read state of the messages that the user has sent.

#### Screen Layout (Mock-up Description)

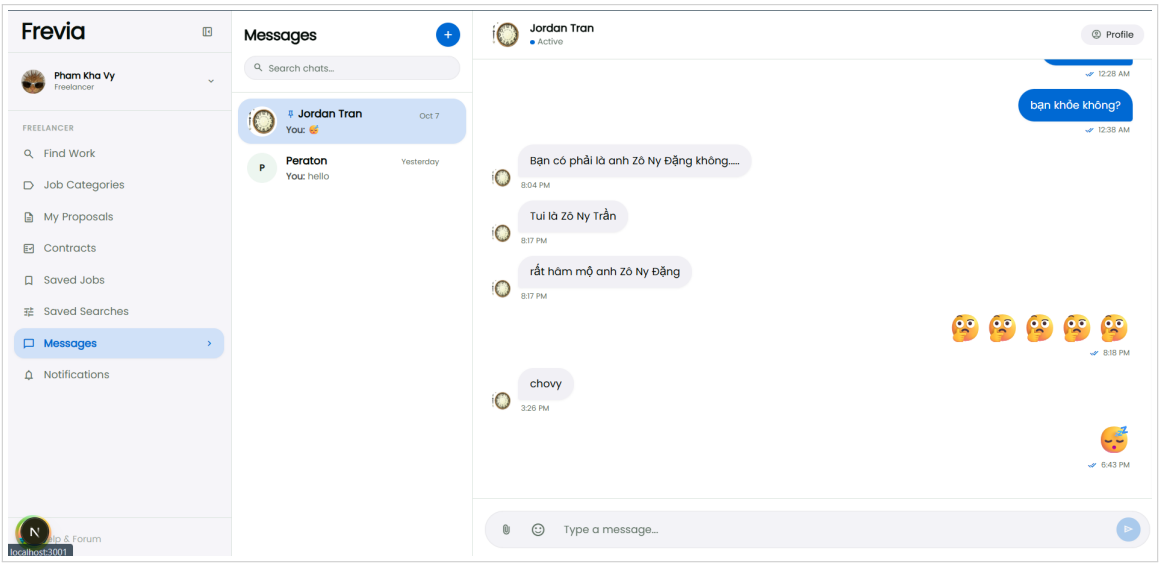


**Figure 88:** Conversation detail screen (chat thread)

**Function Details**

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The user is a participant of the conversation that is being opened.
- **Process**
  - **Open the Conversation**: The user clicks a conversation row and the chat thread is displayed.
  - **Read the Thread Header**: The header of the thread shows the following information.
    - The avatar and the display name of the other participant.
    - A button that opens the profile of the other participant.
    - A back button that returns to the conversation list.
  - **Read the Messages**: The messages are shown in chronological order, oldest at the top and newest at the bottom.
    - Messages sent by the user are aligned on one side, and messages received from the other user on the other side.
    - Image attachments are shown as pictures that can be opened in a new tab.
    - Other files are shown as a download card with the file name and the file size.
    - A message that contains only emojis is displayed with a larger font.
  - **Check the Read State**: Messages that the user has sent show a read indicator.
    - A single check mark means that the message has been delivered but not read.
    - A double check mark means that the other user has read the message.
  - **Use the Thread Actions**: The user can send a message (see **Send Message**), attach a file (see **Send File**) or delete one of their own messages (see **Delete Message**).
- **Alternative Flow**
  - **Empty Thread**: When the conversation has no message yet, the thread shows an empty state with a hint to write the first message.
  - **User Is Typing**: When the other user is typing, an animated typing indicator is shown at the bottom of the thread.
- **Postconditions**
  - The user sees the whole chat thread with the other participant, including the attachments and the read state of their own messages.
  - The messages of the other user are marked as read while the thread is open (see **Mark Conversation as Read**).

- **Error Handling**
  - **Conversation Not Found**: When the conversation no longer exists, the screen shows a not-found state and offers the user a way back to the conversation list.
  - **Access Denied**: When the user is not a participant of the conversation, the thread is not opened and a permission message is shown.
  - **Cannot Load Messages**: When the messages cannot be loaded, the thread shows a load-failure message so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - Only the two participants of a conversation can read its messages; another user is refused even when the conversation link is known.
  - Messages that have been deleted are never shown again in the thread.

### 3.8.4 Send Message

#### Function Trigger

The user types a message in the input box at the bottom of the chat thread and clicks the send button.

#### Function Description

This function allows a logged-in user to send a text message to the other participant of the conversation. The user types the text and sends it, and the message appears in the thread of both users immediately.

## Screen Layout (Mock-up Description)

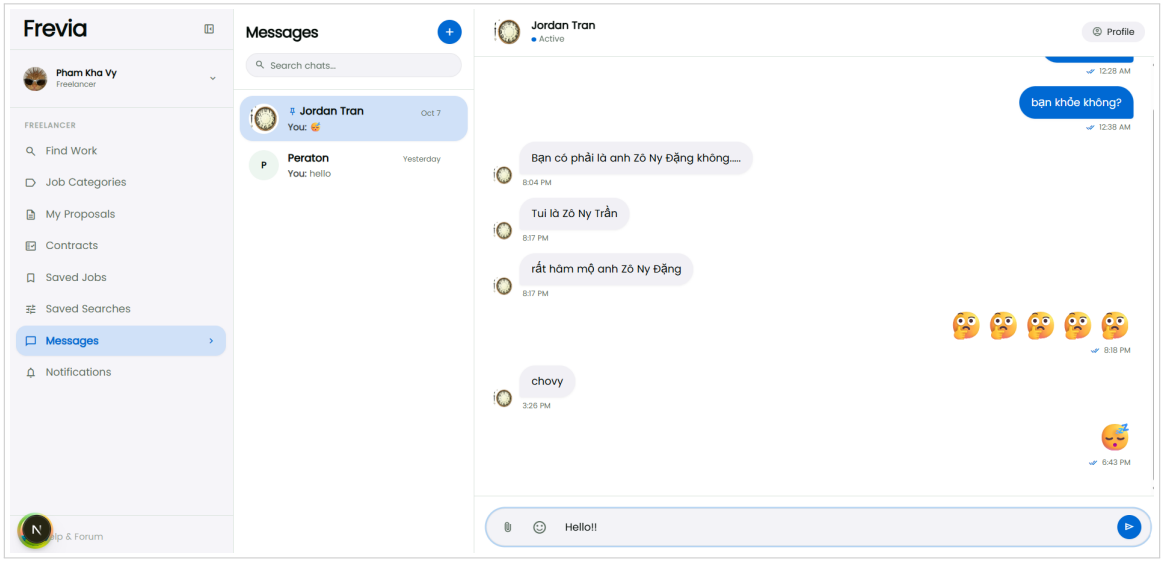


**Figure 89:** Sending a message in the chat thread

## Function Details

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The user is a participant of the conversation and the chat thread is open.
- **Process**
  - **Type the Message**: The user types the text in the input box at the bottom of the thread. The send button stays disabled while the input box is empty.
  - **Add an Emoji if Wanted**: The user can click the emoji button to pick an emoji, and the chosen emoji is added to the input box.
  - **Send the Message**: The user clicks the send button or presses the Enter key.
  - **View the Result**: When the message is sent, the screen shows the result.
    - The message appears at the bottom of the thread on the user's side.
    - The input box is cleared.
    - The message shows a single check mark, meaning that it has been sent but not read yet.
  - **Message Arrives for the Other User**: The other participant receives the message in real time, and the conversation list of that user shows the new message preview and the unread count.
- **Alternative Flow**

- **Enter Key**: When the user presses the Enter key without holding Shift, the message is sent in the same way as clicking the send button.
- **Empty Message**: When the input box is empty, the send button remains disabled and the user cannot send an empty message.
- **Emoji Only**: When the message contains only emojis, the message is displayed with a larger font in the thread.

- **Postconditions**
  - The new message is stored in the conversation and is shown to both participants.
  - The conversation list of both users is updated with the new last message.
- **Error Handling**
  - **Conversation Not Found**: When the conversation no longer exists, the message is not sent and the thread shows a message telling the user that the conversation cannot be found.
  - **Access Denied**: When the user is not a participant of the conversation, the message is refused and a permission message is shown.
  - **Cannot Send Message**: When the message cannot be sent, the thread shows a send-failure message and the text is kept so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - Only a participant of the conversation can send a message into it, and the sender is always the account that is currently logged in.
  - The length of a message is limited, so very large text cannot be sent.

### 3.8.5 Send File

#### Function Trigger

The user clicks the attach-file button in the input box of the chat thread and chooses a file from their device.

#### Function Description

This function allows a logged-in user to send a file, such as a picture or a document, to the other participant of the conversation. The user chooses a file, the system uploads it and the file is sent as a message with a preview or a download card.

## Screen Layout (Mock-up Description)

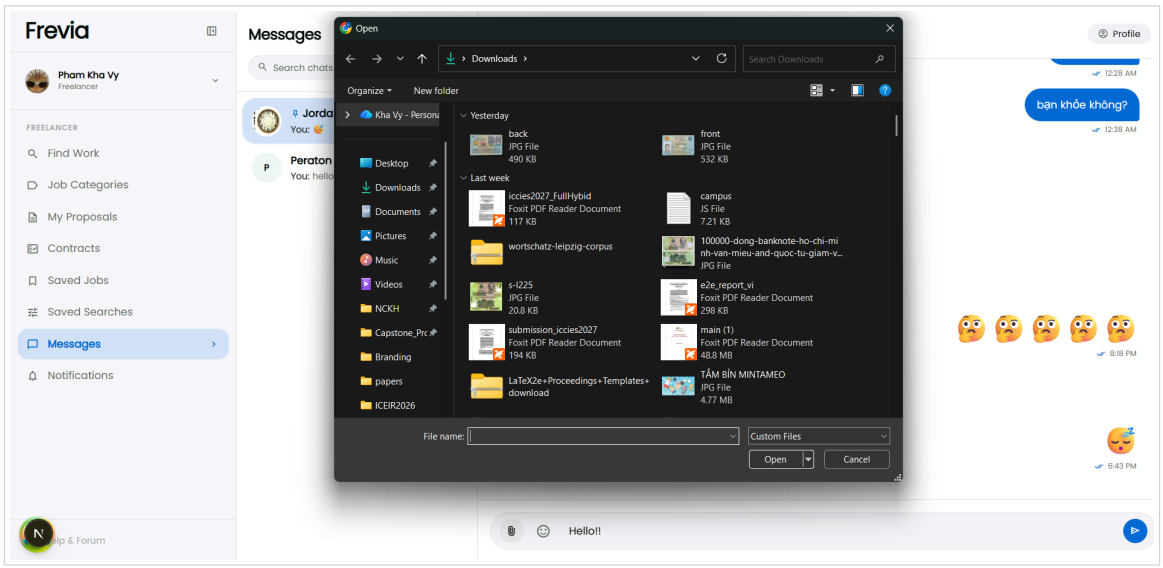


**Figure 90:** Sending a file in the chat thread

## Function Details

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The user is a participant of the conversation and the chat thread is open.
- **Process**
  - **Click the Attach Button**: The user clicks the paperclip button in the input box, and the file chooser of the device is opened.
  - **Choose the File**: The user selects a file. The file chooser accepts images, documents, archives and other supported file types.
  - **Upload the File**: The system uploads the file before sending it, and a loading state is shown while the upload is running.
  - **Send the File as a Message**: When the upload finishes, the file is sent to the conversation as a message.
    - When the file is an image, it is displayed as a picture that can be opened in a new tab.
    - When the file is not an image, it is displayed as a download card with the file name and the file size.
  - **View the Result**: The file message appears at the bottom of the thread, and the conversation list of both users shows the file preview with a paperclip icon.

- **Alternative Flow**
  - **Cancel the File Chooser**: When the user closes the file chooser without choosing a file, no upload is started and nothing changes.
  - **Image Preview**: When the chosen file is an image, the picture is rendered directly inside the message bubble instead of a download card.
- **Postconditions**
  - The file is uploaded and attached to a new message in the conversation.
  - Both participants can see the file message in the thread and can open or download the file.
- **Error Handling**
  - **File Too Large**: When the chosen file is larger than the allowed size, the upload is stopped and a message tells the user that the file is too large.
  - **File Type Not Allowed**: When the chosen file type is not supported, the upload is stopped and a message tells the user that this file type is not allowed.
  - **Cannot Upload File**: When the upload fails, the file message is not sent and a failure message is shown so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - Only supported file types can be uploaded, and the size of the file is limited before the upload starts.
  - Only a participant of the conversation can send a file into it, and the sender is always the account that is currently logged in.

### 3.8.6 Mark Conversation as Read

#### Function Trigger

The user opens a conversation from the conversation list, or a new message arrives while the thread is already open.

#### Function Description

This function allows a logged-in user to clear the unread messages of a conversation. The unread badge disappears from the conversation list and the double check mark appears on the messages of the other user in the thread, so the other participant can see that the messages have been read.

### Screen Layout (Mock-up Description)

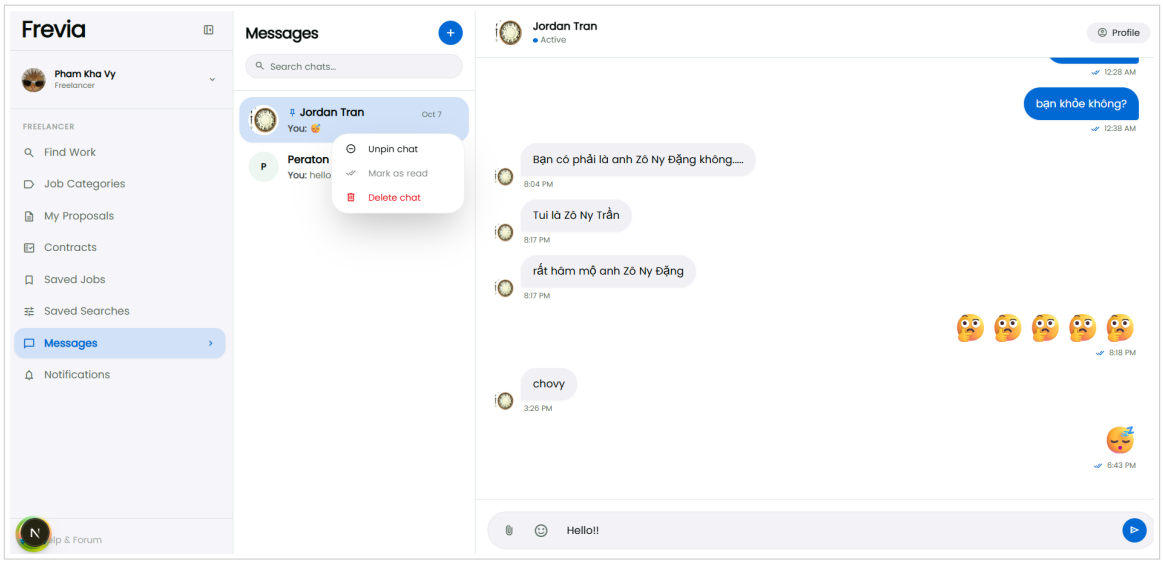


**Figure 91:** Conversation after being marked as read

### Function Details

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The conversation contains at least one unread message from the other participant.
- **Process**
  - **Open the Conversation**: The user opens the conversation, and the system automatically marks the received messages as read.
  - **View the Result in the List**: The conversation row in the conversation list is updated.
    - The blue unread dot disappears from the avatar.
    - The unread count badge is removed from the row.
  - **Mark Manually if Wanted**: The user can also choose the "Mark as read" action from the "..." menu of a conversation row.
  - **Notify the Sender**: The other participant sees the read state change in real time.
    - Messages that were shown with a single check mark are updated to a double check mark.
- **Alternative Flow**
  - **Nothing to Read**: When the conversation has no unread message, the "Mark as read" action is disabled in the menu.
  - **New Message While Reading**: When a new message arrives while the thread is open, that message is marked as read immediately.

- **Postconditions**
  - All messages received from the other participant in this conversation are marked as read.
  - The unread count of the conversation is cleared for the current user, and the other participant sees the double check mark on their sent messages.
- **Error Handling**
  - **Conversation Not Found**: When the conversation no longer exists, the read state is not updated and a message tells the user that the conversation cannot be found.
  - **Cannot Mark as Read**: When the read state cannot be updated, the unread badge stays on the row and a failure message is shown so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - Only the messages received by the current user are marked as read, so a user cannot change the read state of their own sent messages through this action.
  - Only a participant of the conversation can mark it as read.

### 3.8.7 Pin Conversation

#### Function Trigger

The user opens the "..." menu of a conversation row and chooses "Pin chat".

#### Function Description

This function allows a logged-in user to keep an important conversation at the top of the conversation list. The user pins a conversation and the row is moved to the pinned group, which is always shown above the other conversations.

#### Screen Layout (Mock-up Description)

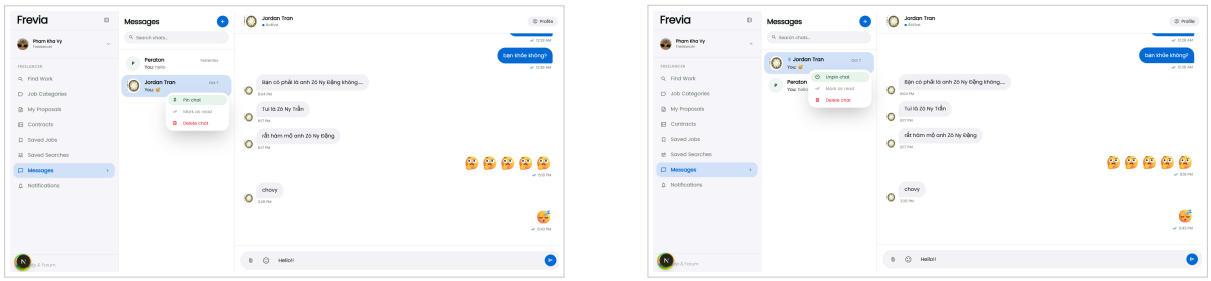


**Figure 92:** Conversation list before pinning (left) and after pinning (right)

**Function Details**

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The conversation that the user wants to pin is shown in the conversation list.
- **Process**
  - **Open the Row Menu**: The user clicks the "..." button of the conversation row and the menu is displayed.
  - **Choose Pin Chat**: The user clicks the "Pin chat" action in the menu.
  - **View the Result**: The conversation list is updated.
    - The conversation is moved to the top of the list, above the not-pinned conversations.
    - A pin icon is shown on the conversation row.
  - **Unpin if Wanted**: The user can open the same menu again and choose "Unpin chat" to remove the conversation from the pinned group.
- **Alternative Flow**
  - **Unpin Conversation**: When the user chooses "Unpin chat", the pin icon is removed and the conversation returns to the normal order of the list.
  - **Pin State Is Personal**: The pin of one user does not change the order of the conversation list of the other participant.
- **Postconditions**
  - The pinned conversation stays at the top of the conversation list of the current user.
  - The content of the conversation and the messages are not changed by this action.
- **Error Handling**
  - **Conversation Not Found**: When the conversation no longer exists, the pin is not applied and a message tells the user that the conversation cannot be found.
  - **Cannot Pin Conversation**: When the pin state cannot be saved, the row returns to its previous position and a failure message is shown so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - The pin belongs to the current user only, so pinning a chat never affects the conversation list of the other participant.
  - Only a participant of the conversation can pin it.

### 3.8.8 Delete Message

### Function Trigger

The user moves the pointer over one of their own messages in the chat thread and clicks the delete button.

### Function Description

This function allows a logged-in user to remove a message that they have sent. The user deletes their own message and the message disappears from the chat thread of both participants. The user can only delete messages that they sent themselves.

### Screen Layout (Mock-up Description)

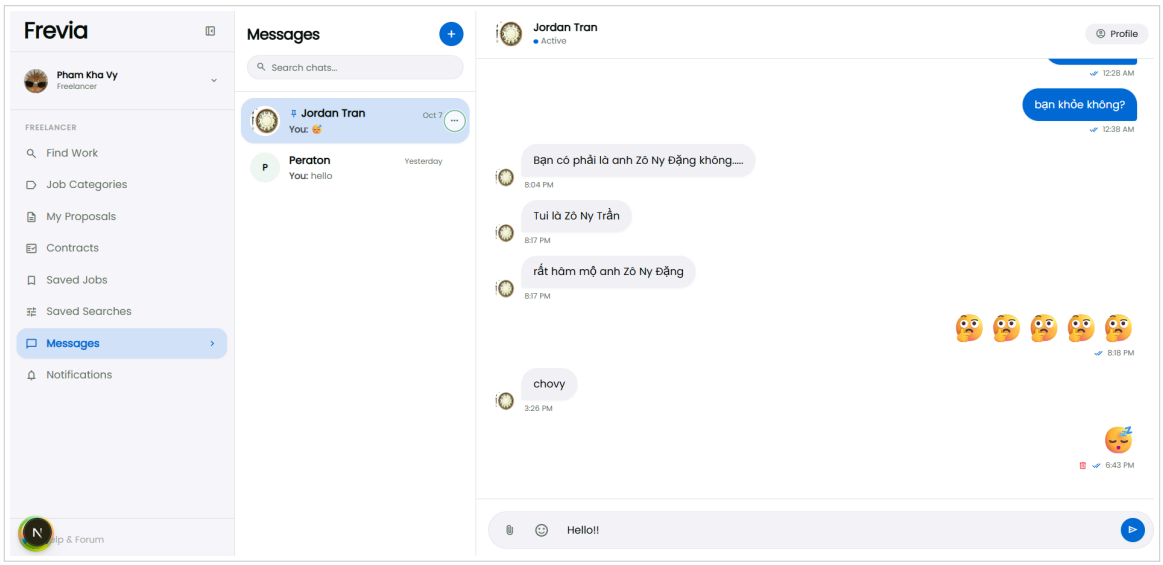


**Figure 93:** Deleting a message in the chat thread

### Function Details

- **Precondition**
  - The user is logged in.
  - The user is connected to the internet.
  - The message belongs to the current user and is shown in the chat thread.
- **Process**
  - **Point at the Message**: The user moves the pointer over one of their own messages and the delete button appears.
  - **Click the Delete Button**: The user clicks the delete button of that message.
  - **View the Result**: The message is removed from the thread.
    - The message disappears from the thread of the current user.
    - The message also disappears from the thread of the other participant in real time.

- **Conversation List Update**: When the deleted message was the last message of the conversation, the preview in the conversation list is refreshed to show the previous message.

- **Alternative Flow**
  - **Delete a File Message**: When the deleted message contains a file, the whole file message is removed from the thread in the same way.
  - **Message Already Deleted**: When the message has already been deleted, the delete button is not available for it.
- **Postconditions**
  - The message no longer appears in the chat thread of either participant.
  - The message is kept in the system for record-keeping but is never shown to users again.
- **Error Handling**
  - **Message Not Found**: When the message no longer exists or has already been deleted, the page shows a message telling the user that the message cannot be found.
  - **Not the Sender**: When the user tries to delete a message that they did not send, the action is refused and a permission message is shown.
  - **Cannot Delete Message**: When the message cannot be deleted, the message stays in the thread and a delete-failure message is shown so that the user can try again.
  - **Lost Connection**: When the connection to the server is lost, the screen shows a "Lost Connection" state and asks the user to reconnect.
- **Security Measures**
  - The delete button is shown only on the messages that the current user sent.
  - The system also checks the sender when the deletion is confirmed, so a user can never delete a message that belongs to the other participant.

## 3.9 User Management

### 3.9.1 View User List

#### Function Trigger

An administrator navigates to the Users management page from the admin dashboard sidebar navigation.

**Navigation path:** Admin Dashboard → User Management → Users (`/admin/users`).

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to view, search, filter, and sort all registered user accounts in the system. The catalog displays key account attributes including user ID, email address, display name, account ban/active status, creation timestamp, and assigned system or custom roles.

### Interface

- User management screen at `/admin/users`.
- Filter bar component (`UsersFilterBar`) containing search input, role filter selector, sort column dropdown, and sort direction toggle.
- Data table component (`UsersTable`) rendering user rows, avatar badges, status indicators, role tags, and contextual action menus.
- Numbered pagination component at the bottom for page navigation.
- "Create User" action button in the page header.

### Data Processing Overview

- The client sends an HTTP request: `GET /api/users` with query parameters (`page`, `limit`, `search`, `role`, `sortBy`, `sortOrder`).
- The backend `AuthGuard` validates the administrator's JWT access token, and `PermissionGuard` verifies permission to access the endpoint.
- `UsersController.getUsers()` parses query parameters via `AdminUserQueryDto`.
- `UsersRepository.getUsers()` queries `prisma.user.findMany` where `deletedAt = null`:
  - If `search` is provided, searches `email` or `profile.displayName` case-insensitively.
  - If `role` is provided (and not `'all'`), filters users whose assigned roles match the role name.
  - Orders records by the requested column (`id`, `email`, `createdAt`, `isBanned`, or `displayName`) and direction.
- The repository executes `prisma.user.count()` to compute total matching records in parallel.
- The system returns HTTP 200 with the users array and pagination metadata.

- The frontend renders user cards/rows, applies ban badges (Active / Banned), and maps assigned roles to visual badges.

### Screen Layout (Mock-up Description)

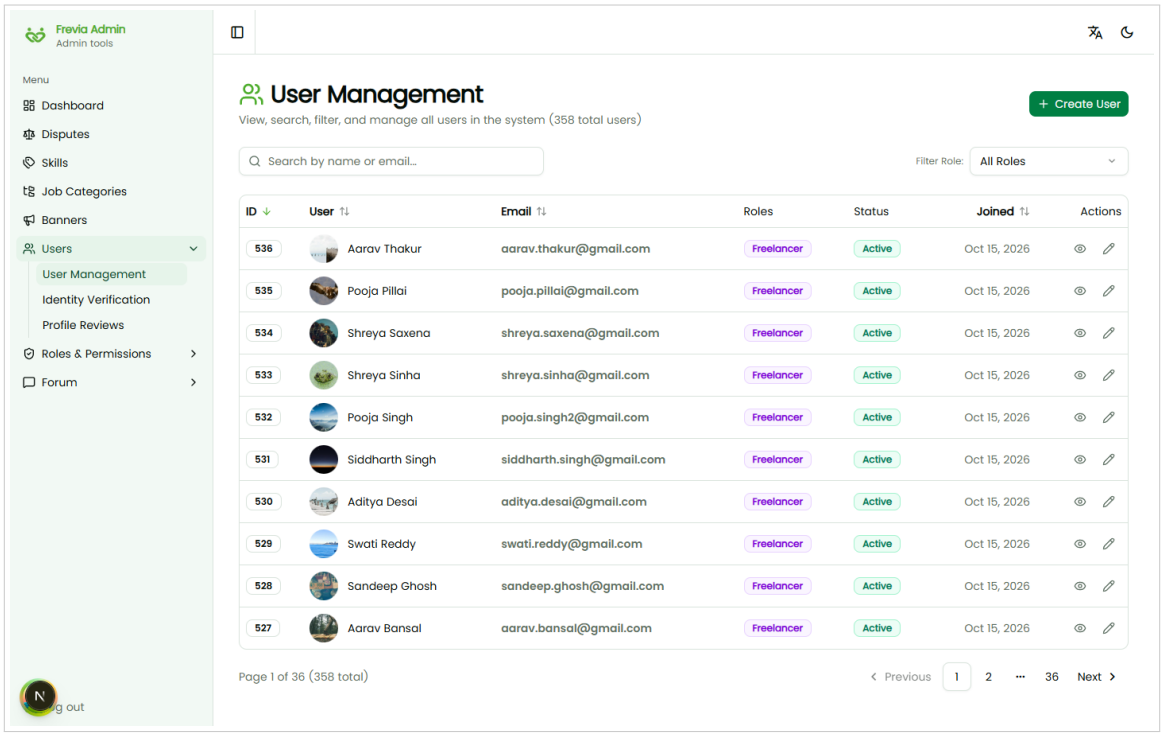


**Figure 94:** User list interface

- Page header displaying “Users” title, summary count, and a “Create User” button.
- Filter bar including search box, role filter dropdown (All, Client, Freelancer, Admin, Custom), and sort selector.
- Table columns: **User** (Avatar, Display Name, and Email), **Roles** (Colored badges for assigned roles), **Status** (Green “Active” or Red “Banned”), **Created Date**, and **Actions**.
- Action menu on each row offering: “View detail”, “Edit user”, and quick actions.
- Pagination footer indicating total count and page selection buttons.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| page | Target page number (defaults to 1). |
| limit | Maximum records returned per page (defaults to 10). |
| search | Optional text query matching user email or display name. |
| role | Optional role filter string (e.g., `CLIENT`, `FREELANCER`, `ADMIN`, `CUSTOM`, or specific custom role name). |
| sortBy | Sort column field (`id`, `email`, `createdAt`, `isBanned`, or `displayName`; defaults to `id`). |
| sortOrder | Sort direction sequence (`asc` or `desc`; defaults to `desc`). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max length 100 characters; whitespace trimmed. |
| role | String, optional, max length 50 characters. |
| sortBy | Must strictly be one of: `id`, `email`, `createdAt`, `isBanned`, `displayName`. |
| sortOrder | Must strictly be one of: `asc`, `desc`. |

**Business Rules**

- **BR-USER-01**: Only active user accounts with `deletedAt = null` shall be returned. Soft-deleted accounts are excluded.
- **BR-USER-02**: Default sort order is by `id` in descending order, displaying newly registered users first.
- **BR-USER-03**: The maximum page size is 100 records; if `limit` exceeds 100, the server caps it at 100.
- **BR-USER-04**: Search on `email` and `displayName` is case-insensitive with partial contains matching.
- **BR-USER-05**: Role filtering supports built-in roles, `CUSTOM` (matching any custom role), and specific custom role names.

**Normal Flow**

1. The administrator opens the `/admin/users` page.
2. The frontend sends `GET /api/users` with default parameters (`page=1&limit=10&sortBy=id&sortOrde`
3. The backend validates administrator authentication and permissions.
4. The repository queries active users and counts total matching records.
5. The backend returns HTTP 200 with the users array and pagination metadata.
6. The frontend renders the user list table and pagination controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client clears authentication cookies and redirects to `/login`. |
| User lacks Admin privileges | The backend throws `UserForbiddenException` (HTTP 403 Forbidden). The UI displays: "You do not have permission to access this page." |
| Invalid pagination parameter values | The DTO validator rejects non-integer or negative values with HTTP 400 Bad Request. The client resets to page 1. |
| Unsupported sort column parameter | The system rejects invalid sort keys with HTTP 400 Bad Request. The client defaults to `id desc`. |
| Database query error or timeout | The service catches the Prisma error and throws `FailedToGetUsersException` (HTTP 500). The UI displays: "Couldn't load users. Try again." |

| Abnorma Case | Handling & System Response |
|---|---|
| No users match search or role filters | The query returns `users: []` and `total: 0`. The table displays: "No users match your filters." |

**Output Data**

| Output Field | Description |
|---|---|
| users | Array of user objects: `id`, `email`, `isBanned`, `createdAt`, `updatedAt`, `displayName`, `avatarUrl`, and `roles`. |
| pagination.page | Current page number. |
| pagination.limit | Current limit per page. |
| pagination.total | Total active users matching the query. |
| pagination.totalPages | Total calculated pages available. |

**Post-Conditions**

1. The user list table is rendered with role badges and account statuses.
2. The administrator can filter by role, search, sort, or select a user to view full details (UC-28.02) or edit (UC-28.04).

### 3.9.2 View User Detail

**Function Trigger**

An administrator clicks "View detail" on a user row in the user table or navigates directly to `/admin/users/:id`.

**Navigation path:** User Management (`/admin/users`) → Click user row or view icon (`/admin/users/:id`).

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Retrieves consolidated, comprehensive account information for a specific user. Along with general profile data, social links, and activity metrics, the interface provides dedicated tab panels for each role profile the user holds: Client Profile, Freelancer Profile, Expert Profile, and Custom Role Profiles.

### Interface

- Detailed user overview page at `/admin/users/:id`.
- Page header component (`UserDetailHeader`) showing avatar, display name, email, ban badge, and edit button.
- General information card (`UserGeneralInfo`) displaying account ID, verification status, join date, last update, bio, and activity statistics.
- Role tabs component (`UserRoleTabs`) rendering separate tabs for Client Profile, Freelancer Profile, Expert Profile, and Admin & Custom Roles.

### Data Processing Overview

- Client sends `GET /api/users/:id`.
- `ParseIntPipe` validates that `id` is a positive integer.
- Backend verifies Admin role permissions.
- `UsersRepository.getUserById(id)` queries `prisma.user.findFirst` where `id = id` and `deletedAt = null`.
- The query includes profile data, assigned roles, social links, activity counts (jobs posted, contracts, proposals, reviews), client profile, freelancer profile (with skills and active portfolio items), expert profile, and custom role profiles.
- If user is not found or soft-deleted, throws `UserNotFoundException` (HTTP 404).
- The server responds with HTTP 200 and the complete consolidated detail object.
- The frontend renders the header, general info, and populates role-specific tabs.

## Screen Layout (Mock-up Description)

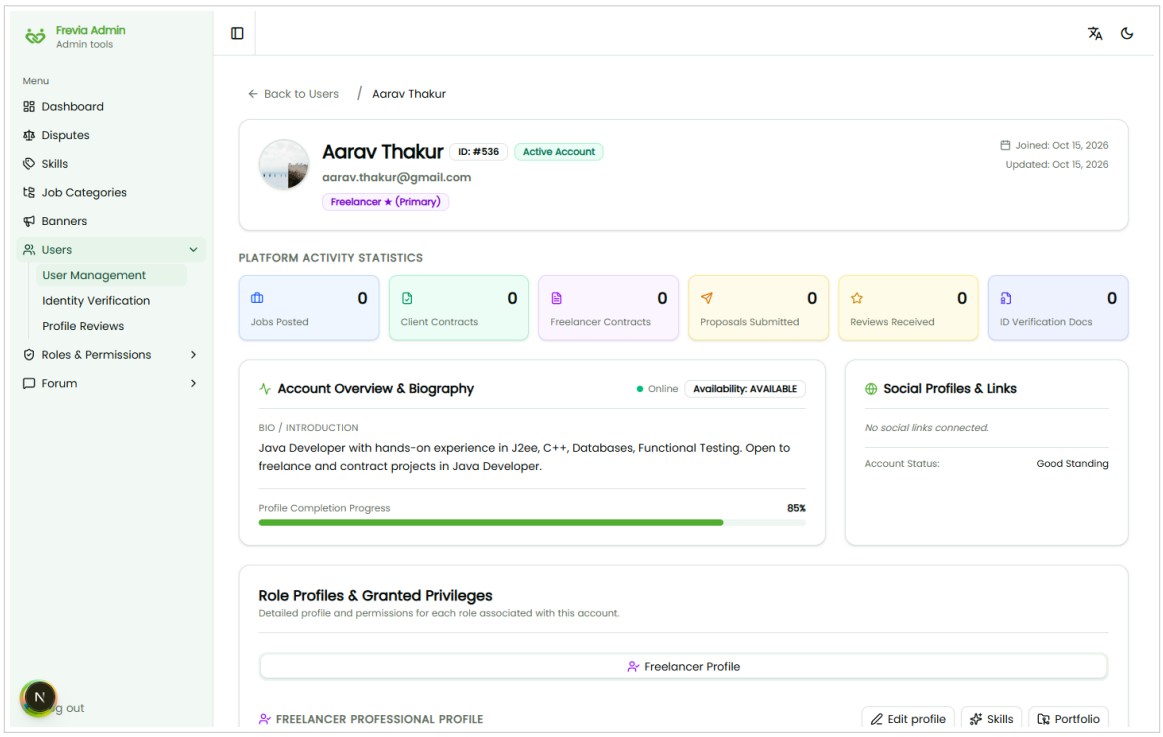


**Figure 95:** User detail interface

- Top navigation button “Back to users” allowing return to the catalog.
- Header containing user avatar, display name, email, ID badge, ban status badge, and “Edit User” button.
- Card section for **General Information**: Bio, Online status, Availability status, Profile completion percentage, and Activity stats (Jobs, Contracts, Proposals, Reviews).
- Tab navigation bar with tabs: **Client Profile**, **Freelancer Profile**, **Expert Profile**, and **Admin & Custom Roles**.
- Each active tab renders dedicated profile data and management action buttons.

## Function Details

## Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric system identifier of the user in the URL path. |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |

**Business Rules**

- **BR-USER-06**: The `clientProfile`, `freelancerProfile`, and `expertProfile` fields may be `null` if the user does not possess the corresponding role or has not initialized that profile.
- **BR-USER-07**: The `customRoleProfiles` array contains only roles outside the built-in Client, Freelancer, and Admin roles.
- **BR-USER-08**: Each assigned role includes an `isPrimary` flag indicating whether it serves as the user's default active role.

**Normal Flow**

1. The administrator navigates to `/admin/users/:id`.
2. The frontend sends `GET /api/users/:id`.
3. The backend validates Admin authorization and queries the database for the user record.
4. The repository compiles general information, role profiles, and activity statistics.
5. The server responds with HTTP 200 and the consolidated user detail payload.
6. The frontend renders `UserDetailHeader`, `UserGeneralInfo`, and `UserRoleTabs`.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid user ID format (non-numeric string) | `ParseIntPipe` rejects the call with HTTP 400 Bad Request. The client displays an invalid parameter error page. |
| User ID does not exist in database | The repository throws `UserNotFoundException` (HTTP 404). The frontend displays: "User not found." |
| User has been soft-deleted (`deletedAt != null`) | The query filters out soft-deleted users and throws HTTP 404 Not Found. The UI displays the not found notice. |
| User lacks Admin privileges | The backend throws `UserForbiddenException` (HTTP 403 Forbidden). Access is denied. |
| Caller token expired or revoked | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| Database query error during retrieval | The service throws `FailedToGetUserException` (HTTP 500). The UI displays a retry prompt. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric user ID. |
| email | User email address. |
| isBanned | Boolean indicating account suspension status. |
| profile | Nested general profile: `displayName`, `avatarUrl`, `bio`, `profileCompletionPercent`. |
| roles | Array of assigned roles with `id`, `name`, `description`, and `isPrimary`. |
| clientProfile | Client company details or null. |
| freelancerProfile | Freelancer skills, credentials, and portfolio items or null. |
| expertProfile | Expert qualifications, experience, and certifications or null. |
| stats | Activity counts: jobs posted, active contracts, proposals, and reviews. |

**Post-Conditions**

1. The complete account profile and role tabs are displayed to the administrator.
2. The administrator can switch between role tabs and edit role-specific details directly.

### 3.9.3 Create User

**Function Trigger**

An administrator clicks the "Create User" button on the user list page.

**Navigation path:** User Management (`/admin/users`) → Click "Create User" button.

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), bcrypt hashing library, Prisma ORM layer, and PostgreSQL database.

**Purpose**

Allows administrators to manually provision a new user account by specifying full name, email address, secure password, and selecting an initial role to activate (Client, Freelancer, or a custom role). The system automatically provisions the user, general profile, and primary role association.

### Interface

- "Create User" button in the user list header.
- Modal dialog component (`CreateUserDialog`) with input fields: Full Name, Email, Password, Confirm Password, and Initial Role Select.
- Action buttons: "Cancel" and "Create User".
- Success toast notification upon account provisioning.

### Data Processing Overview

- Administrator fills the form and submits the modal dialog.
- The client sends `POST /api/users` with body `{ email, fullName, password, confirmPassword, roleId }`.
- The backend validates schema constraints via `AdminCreateUserBodyDto`.
- `UsersService.createUser()` validates:
  - Email uniqueness among active accounts; if taken, throws `EmailAlreadyExistsException` (HTTP 409).
  - Role validity: target role must exist and be active; if missing, throws `CreateUserRoleNotFoundExcep` (HTTP 404).
  - Role restriction: role cannot be the Admin role; if selected, throws `CannotAssignAdminRoleExcepti` (HTTP 403).
- In an atomic database transaction:
  1. Hashes password with bcrypt (10 salt rounds).
  2. Creates User record with lowercased email, password hash, and `isEmailVerified = true`.
  3. Creates Profile record with `displayName = fullName`.
  4. Creates UserRole record assigning the role with `isPrimary = true`.
- The server responds with HTTP 201 Created and the new user summary.
- The frontend closes the dialog, displays a success toast, and refreshes the user list.

### Screen Layout (Mock-up Description)

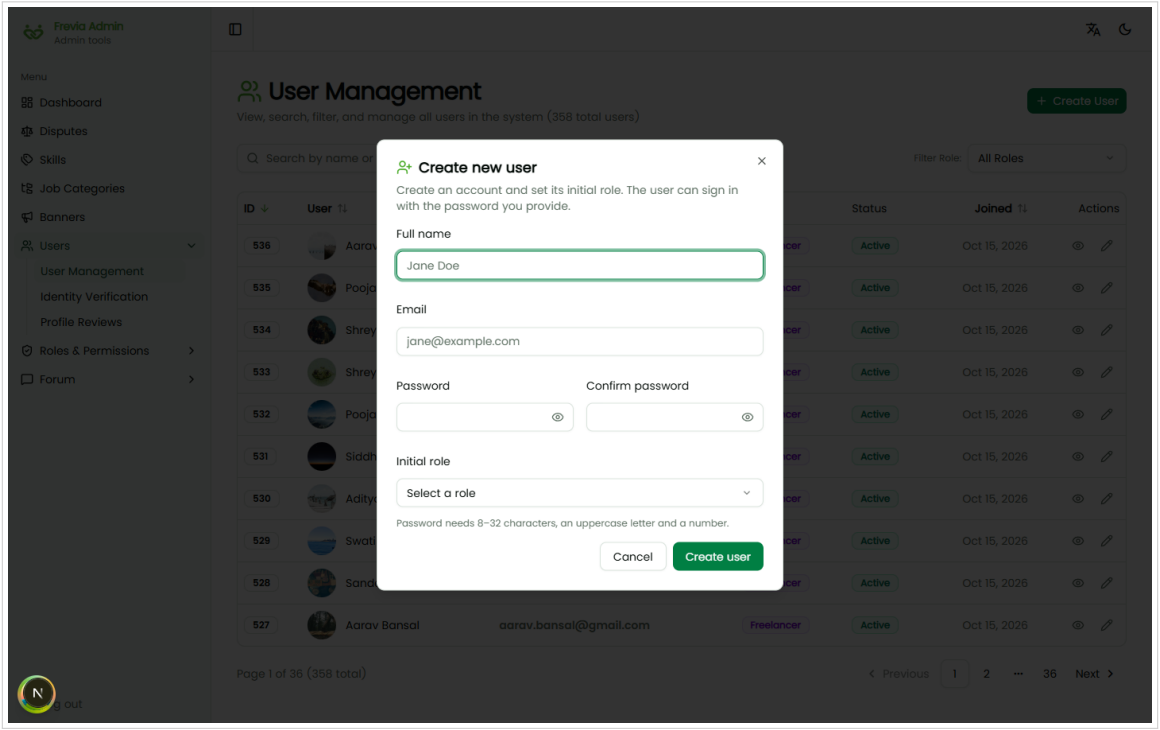


**Figure 96:** Create user modal

- Modal dialog titled “Create user” with subtitle “Add a new user account to the platform.”.
- Field: **Full name** (required, placeholder: “e.g., Alex Johnson”).
- Field: **Email address** (required, placeholder: “alex@example.com”).
- Field: **Password** (required, 8–32 characters, uppercase and number required).
- Field: **Confirm password** (required, matching password).
- Field: **Initial role** (Select dropdown displaying Client, Freelancer, and active Custom roles; Admin role is excluded).
- Dialog footer with “Cancel” and “Create user” action buttons.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| fullName | Full display name of the user (required). |
| email | Email address for authentication (required). |
| password | Password for local credential login (required). |
| confirmPassword | Password confirmation string (required). |
| roleId | Numeric ID of the initial role to assign (required). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| fullName | String, required, 1 to 100 characters; whitespace trimmed. |
| email | String, required, valid RFC 5322 email format, max 254 characters; lowercased. |
| password | String, required, 8 to 32 characters; must contain at least one uppercase letter and one number. |
| confirmPassword | String, required; must match `password` exactly. |
| roleId | Positive integer; must correspond to an active non-Admin role. |

**Business Rules**

- **BR-USER-09**: The email address is automatically converted to lowercase before storage and must be at most 254 characters.
- **BR-USER-10**: Email uniqueness is enforced among all active accounts (`deletedAt = null`).
- **BR-USER-11**: Passwords must be hashed with bcrypt prior to storage; plaintext passwords are never persisted.
- **BR-USER-12**: The Admin role cannot be assigned during manual user creation to prevent privilege escalation; only Client, Freelancer, or custom roles are permitted.
- **BR-USER-13**: The selected role must exist and be active in the system.
- **BR-USER-14**: The newly assigned role is automatically designated as the user's primary role (`isPrimary = true`).

**Normal Flow**

1. The administrator clicks "Create User" on the user management screen.

2. The dialog opens. The administrator inputs name, email, password, and selects an initial role.
3. The administrator clicks “Create user”.
4. The frontend validates the fields and issues `POST /api/users`.
5. The backend validates email uniqueness, role validity, and absence of Admin role assignment.
6. The backend hashes the password and provisions the User, Profile, and UserRole records in a transaction.
7. The server responds with HTTP 201 Created and the new user summary.
8. The frontend closes the dialog, displays a success toast, and refreshes the user list.

**Abnormal Flow / Exception Handling**

| **Abnorma Case** | **Handling & System Response** |
|---|---|
| Password and Confirm Password do not match | Client-side validation and backend DTO reject submission. The form displays: “Passwords do not match.” |
| Password fails complexity rules (< 8 chars, no uppercase/number) | Backend validation rejects with HTTP 400 Bad Request. The UI displays password requirements. |

| Abnorma Case | Handling & System Response |
|---|---|
| Email is already registered to an active user | The backend throws `EmailAlreadyExistsException` (HTTP 409 Conflict). The UI displays: “Email already exists.” |
| Selected role does not exist or was deleted | The backend throws `CreateUserRoleNotFoundException` (HTTP 404). The UI displays: “Role not found.” |
| Administrator attempts to assign Admin role | The backend throws `CannotAssignAdminRoleException` (HTTP 403 Forbidden). The UI displays: “Cannot assign Admin role.” |
| User lacks Admin privileges | The backend throws `UserForbiddenException` (HTTP 403). Access is denied. |
| Database transaction failure | The transaction rolls back and the backend throws `FailedToCreateUserException` (HTTP 500). The UI displays: “Failed to create user.” |
| Administrator cancels dialog | The modal closes and discarded form data is cleared without network activity. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric ID of the created user. |
| email | Stored email address. |
| displayName | Display name set on the profile. |
| roles | Array containing the assigned primary role. |

**Post-Conditions**

1. A new active user record is inserted with verified email status.
2. A linked profile record and primary role association are created.
3. The user appears in the user list and can authenticate immediately.

### 3.9.4 Update User

**Function Trigger**

An administrator clicks "Edit" on a user row in the table or clicks "Edit User" on the user detail screen.
**Navigation path:** User Management (`/admin/users`) → Click "Edit" icon on a user row.

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

**Purpose**

Allows administrators to edit the general account credentials and administrative status of an existing user, including changing their email address, updating their display name, or toggling their account ban/suspension status.

**Interface**

- Edit action trigger on table rows and user detail header.
- Modal dialog component (`EditUserDialog`) pre-filled with current email, full name, and ban switch.
- Action buttons: "Cancel" and "Save Changes".
- Toast notification confirming successful account updates.

### Data Processing Overview

- Administrator modifies fields in the dialog and submits the form.
- The client sends `PATCH /api/users/:id` with body containing modified attributes (`email`, `fullName`, `isBanned`).
- `UserActive('userId')` extracts the administrator's ID (`actorId`).
- `UsersService.updateUser()` validates:
  - Target user exists and has `deletedAt = null`.
  - Self-ban protection: if `isBanned = true` and `id === actorId`, throws `CannotBanSelfException` (HTTP 403).
  - Email uniqueness: if email is modified, ensures no other active user uses it; otherwise throws `EmailAlreadyExistsException` (HTTP 409).
- In a transaction, updates `user.email`, `user.isBanned`, and upserts `profile.displayName`.
- The server responds with HTTP 200 and the updated user summary.
- The frontend closes the dialog, displays a success toast, and refreshes the user view.

### Screen Layout (Mock-up Description)

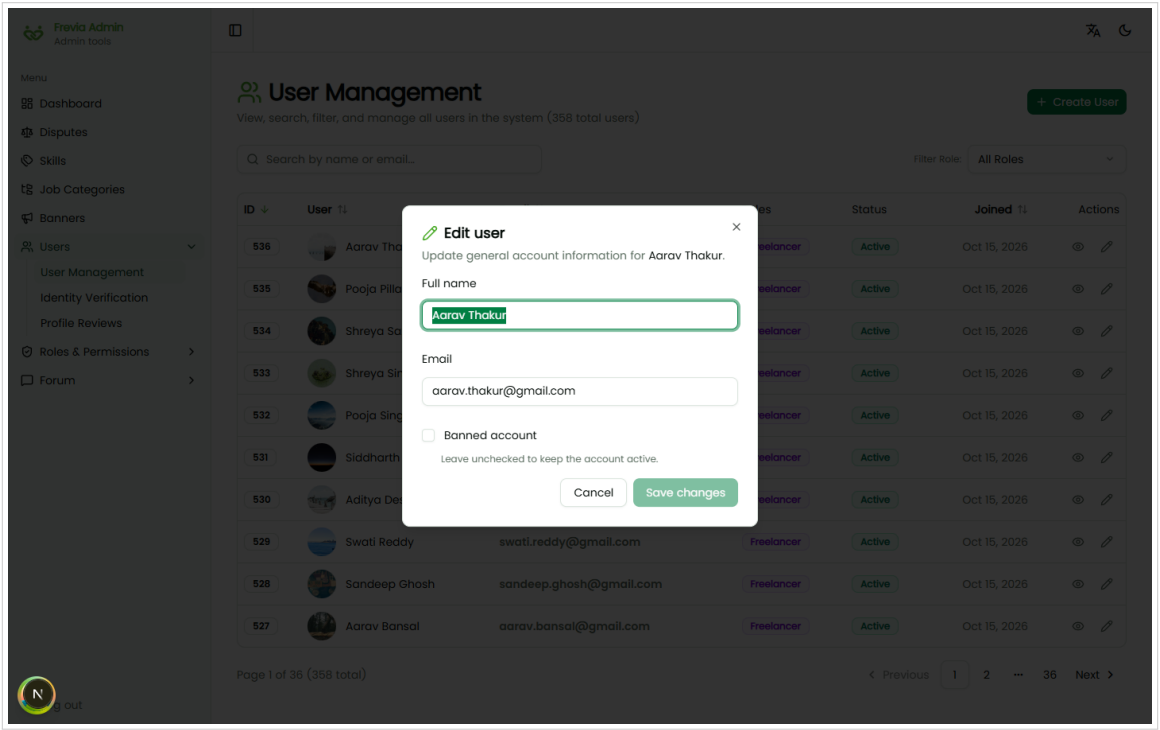


**Figure 97:** Update user modal

- Modal dialog titled "Edit user" with description "Update general account information.".
- Field: **Email address** (pre-filled with current email).
- Field: **Full name** (pre-filled with current display name).

- Toggle switch: **Banned** (Switch component allowing immediate suspension of account access).
- Dialog footer with “Cancel” and “Save changes” buttons.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| id | Numeric identifier of the user in the URL path. |
| email | Updated email address (optional, string). |
| fullName | Updated display name (optional, string or null). |
| isBanned | Boolean account suspension toggle (optional). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| email | String, optional, valid email format, max 254 characters; lowercased. |
| fullName | String, optional, max 100 characters, nullable. |
| isBanned | Boolean, optional. |

**Business Rules**

- **BR-USER-15**: An administrator is strictly prohibited from banning their own account (`isBanned = true` is rejected when `targetUserId == actorId`).
- **BR-USER-16**: Email uniqueness is enforced across all non-deleted users, excluding the target user being edited.
- **BR-USER-17**: At least one updatable field must be provided in the request payload.
- **BR-USER-18**: The email address is automatically converted to lowercase before updating.
- **BR-USER-19**: Setting `fullName` to null or empty string clears the user’s display name.

**Normal Flow**

1. The administrator opens the “Edit User” dialog for a selected user.
2. The administrator modifies email, display name, or toggles the ban switch.

3. The administrator clicks “Save changes”.
4. The client sends `PATCH /api/users/:id`.
5. The backend validates authorization, verifies self-ban constraints, and checks email uniqueness.
6. The repository updates user and profile records in a database transaction.
7. The API returns HTTP 200 with the updated user data.
8. The frontend displays a success toast and refreshes the user information.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Administrator attempts to ban their own account | The backend detects `id === actorId` and throws `CannotBanSelfException` (HTTP 403 Forbidden). The UI displays: “You cannot ban your own account.” |
| New email is already in use by another account | The backend detects conflict and throws `EmailAlreadyExistsException` (HTTP 409 Conflict). The UI displays: “Email already exists.” |
| No fields provided in update payload | The backend rejects the request with HTTP 400 Bad Request (`Nothing to update`). The UI highlights required fields. |
| User ID does not exist or is soft-deleted | The service throws `UserNotFoundException` (HTTP 404). The UI displays: “User not found.” |

| Abnorma Case | Handling & System Response |
|---|---|
| User lacks Admin privi-leges | The backend throws `UserForbiddenException` (HTTP 403). Access is denied. |
| Database transac-tion failure | The service catches the Prisma error and throws `FailedToUpdateUserException` (HTTP 500). The UI displays: "Failed to update user." |
| Administrator cancels dialog | The modal closes and unsaved modifications are discarded without network calls. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric user identifier. |
| email | Updated email address. |
| displayName | Updated display name string. |
| isBanned | Updated ban status boolean. |
| roles | Array of assigned roles. |

**Post-Conditions**

1. The user's account details and ban status are updated in PostgreSQL.
2. If banned, the user's active sessions are immediately rejected on next access attempt.
3. The changes are visible across administrative tables and user detail views.

### 3.9.5 Update Client Profile

### Function Trigger

An administrator clicks “Edit Client Profile” on the Client Profile tab of the user detail page.
**Navigation path:** User Detail (`/admin/users/:id`) → Tab “Client Profile” → Click “Edit Client Profile” button.

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Allows administrators to edit the client organization profile of a user holding the Client role. The administrator can update company name, company description, and corporate website URL.

### Interface

- “Edit Client Profile” button on the Client Profile tab panel.
- Modal dialog pre-filled with current company name, description, and website URL.
- Action buttons: “Cancel” and “Save Changes”.
- Toast notification confirming successful update.

### Data Processing Overview

- Administrator submits the updated client profile form.
- The client sends `PATCH /api/users/:id/client-profile` with `{ companyName, companyDescription, website }`.
- `UsersService.updateClientProfile()` verifies that the target user exists, has not been soft-deleted, and possesses the `CLIENT` role (or an existing client profile record); otherwise throws `NoClientRoleForClientProfileException` (HTTP 403).
- The service trims string inputs and converts empty strings to `null`.
- `UsersRepository.upsertClientProfileByAdmin()` upserts the record in `prisma.clientProfile.u`
- The API returns HTTP 200 with the updated client profile object.
- The frontend displays a success toast and refreshes the Client Profile tab content.

### Screen Layout (Mock-up Description)

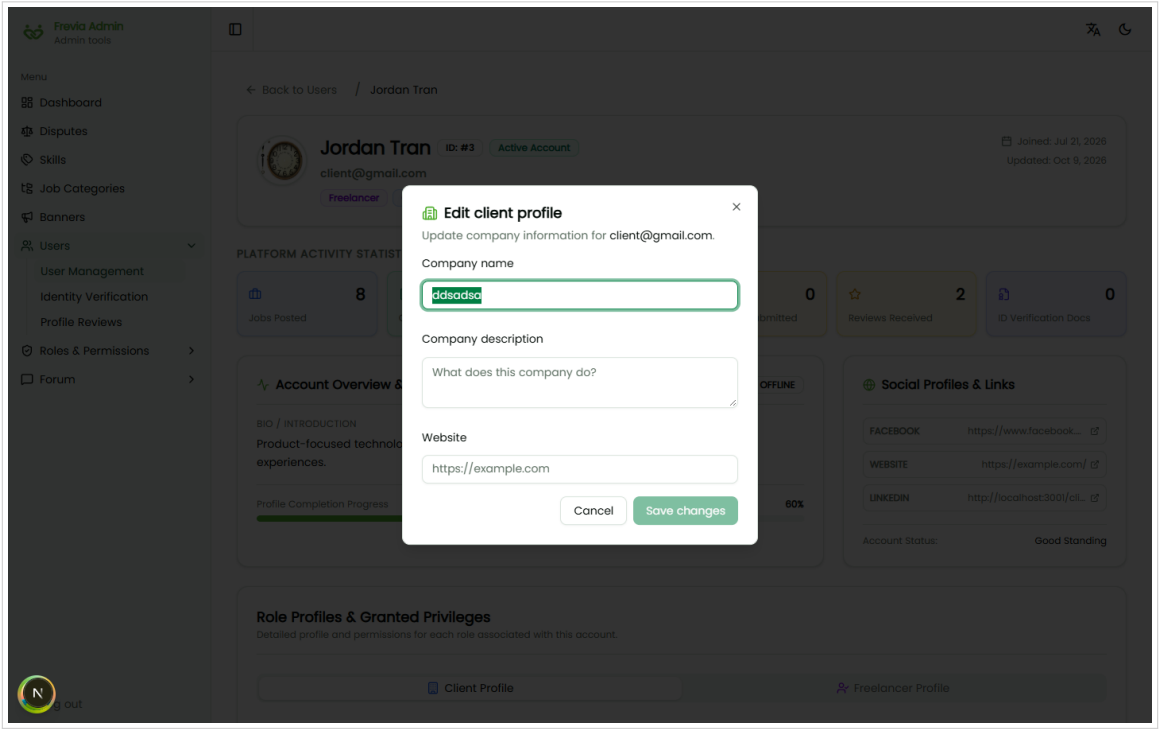


**Figure 98:** Update client profile modal

- Modal dialog titled “Edit client profile” with subtitle “Update client organization and company details.”.
- Field: **Company name** (input, max 255 characters, optional).
- Field: **Company description** (multi-line textarea, max 5000 characters, optional).
- Field: **Website URL** (input with URL format validation, optional).
- Dialog footer with “Cancel” and “Save changes” buttons.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Numeric user identifier in the URL path. |
| companyName | Updated company name string (optional, nullable). |
| companyDescription | Detailed description of company activities (optional, nullable). |
| website | Corporate website URL (optional, nullable). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| companyName | String, optional, max 255 characters, nullable. |
| companyDescription | String, optional, max 5000 characters, nullable. |
| website | String, optional, valid URL format (e.g., `https://example.com`), nullable. |

**Business Rules**

- **BR-CLT-01**: The target user must possess the Client role or have an existing client profile record; otherwise, the update is rejected with HTTP 403 Forbidden.
- **BR-CLT-02**: Company name is limited to a maximum of 255 characters.
- **BR-CLT-03**: Company description is limited to a maximum of 5000 characters.
- **BR-CLT-04**: The website field must conform to standard URL format if supplied.
- **BR-CLT-05**: Empty string values are automatically converted to `null` to allow clearing field content.
- **BR-CLT-06**: The client profile record is upserted, creating the profile record if it does not yet exist.

**Normal Flow**

1. The administrator navigates to the Client Profile tab and clicks “Edit Client Profile”.
2. The dialog opens pre-filled with current company data.
3. The administrator modifies company name, description, or website and clicks “Save changes”.
4. The frontend sends `PATCH /api/users/:id/client-profile`.
5. The backend verifies permissions and confirms the user has the Client role.

6. The repository upserts the client profile record in PostgreSQL.
7. The backend returns HTTP 200 with the updated client profile data.
8. The frontend displays a success toast and refreshes the tab.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target user does not have Client role | The backend throws `NoClientRoleForClientProfileException` (HTTP 403 Forbidden). The UI displays: "User does not have a Client role." |
| Website format is invalid | Client-side validation and backend DTO reject with HTTP 400 Bad Request. The UI displays: "Invalid URL format." |
| Company name exceeds 255 characters | Backend validation returns HTTP 400 Bad Request. The UI notifies of character limit violation. |
| Company description exceeds 5000 characters | Backend validation returns HTTP 400 Bad Request. The UI notifies of character limit violation. |
| User ID does not exist or is soft-deleted | The service throws `UserNotFoundException` (HTTP 404). The UI displays: "User not found." |

| Abnorma Case | Handling & System Response |
|---|---|
| Database upsert failure | The service catches the Prisma error and throws `FailedToUpdateClientProfileException` (HTTP 500). The UI displays: "Failed to update client profile." |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric client profile identifier. |
| userId | Numeric ID of the associated user. |
| companyName | Updated company name or null. |
| companyDescription | Updated company description or null. |
| website | Updated website URL or null. |
| updatedAt | Timestamp indicating last update. |

**Post-Conditions**

1. The client profile record is saved in the database.
2. The updated company information is reflected in the Client Profile tab and public job postings by this client.

### 3.9.6 Update Freelancer Basic Info

**Function Trigger**

An administrator clicks "Edit Freelancer Profile" on the Freelancer Profile tab of the user detail page.

**Navigation path:** User Detail (`/admin/users/:id`) → Tab "Freelancer Profile" → Click "Edit Freelancer Profile" button.

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Allows administrators to edit the basic introductory and credential information of a user holding the Freelancer role, including professional title, overview biography, spoken languages, educational history, and professional certifications.

### Interface

- "Edit Freelancer Profile" button on the Freelancer Profile tab.
- Modal dialog with text inputs for Professional Title, Biography textarea, and dynamic tag lists for Languages, Education, and Certifications.
- Action buttons: "Cancel" and "Save Changes".
- Toast notification confirming successful update.

### Data Processing Overview

- Administrator submits changes in the modal dialog.
- The client sends `PATCH /api/users/:id/freelancer-profile` with `{ title, bio, languages, education, certifications }`.
- `UsersService.updateFreelancerProfile()` verifies that target user exists, is active, and possesses the `FREELANCER` role (or an existing freelancer profile); otherwise throws `NoFreelancerRoleForProfileException` (HTTP 403).
- The service cleans string lists, converting empty strings to `null` and empty/null arrays to `[]`.
- `UsersRepository.updateFreelancerProfileByAdmin()` upserts the freelancer profile record in PostgreSQL.
- The API returns HTTP 200 with `{ message: 'Freelancer profile updated successfully' }`.
- The frontend displays a success toast and refreshes the Freelancer Profile tab.

### Screen Layout (Mock-up Description)

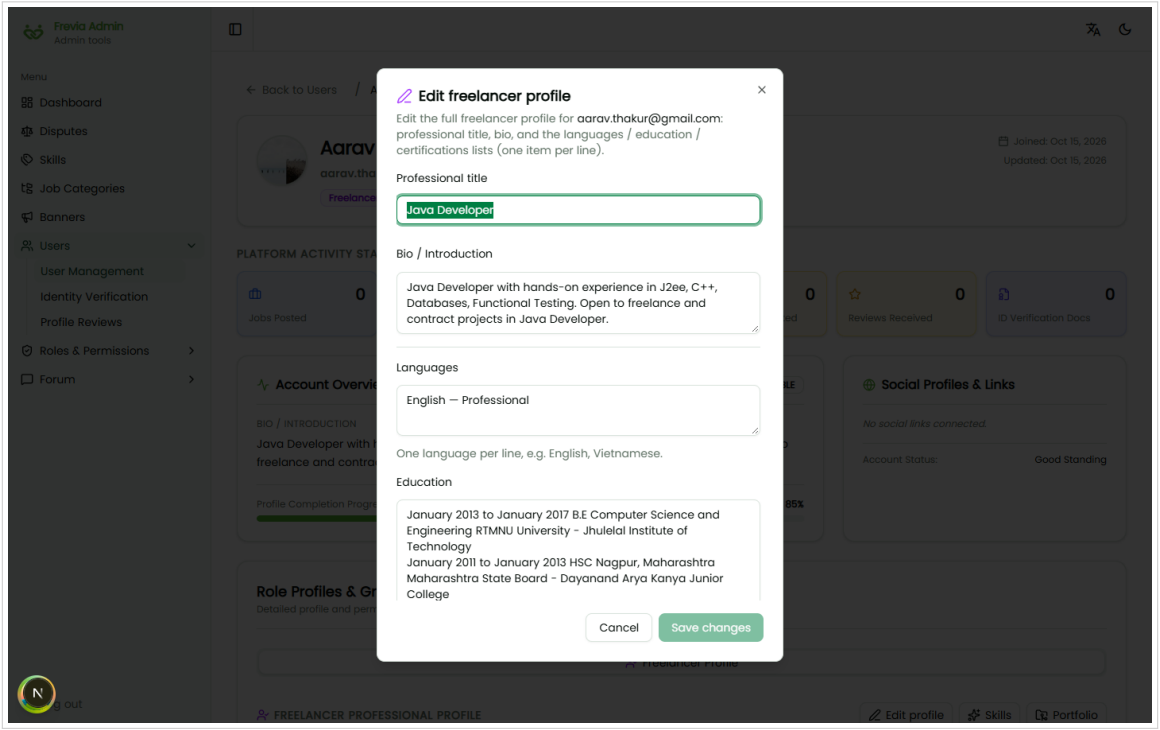


**Figure 99:** Update freelancer basic info modal

- Modal dialog titled “Edit freelancer profile” with subtitle “Update professional title, biography, and credentials.”.
- Field: **Professional title** (input, max 255 characters, placeholder: “e.g., Senior Full-Stack Engineer”).
- Field: **Biography** (textarea, max 5000 characters).
- Dynamic string lists: **Languages**, **Education**, and **Certifications** with input field and “Add” button, rendering removable badges for each entry.
- Action buttons: “Cancel” and “Save changes”.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Numeric user identifier in the URL path. |
| title | Professional headline or title (optional, nullable). |
| bio | Professional summary biography (optional, nullable). |
| languages | Array of spoken languages (optional, string array). |
| education | Array of education entries (optional, string array). |
| certifications | Array of certification entries (optional, string array). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| title | String, optional, max 255 characters, nullable. |
| bio | String, optional, max 5000 characters, nullable. |
| languages | Array of strings, max 100 items, each item max 255 characters. |
| education | Array of strings, max 100 items, each item max 255 characters. |
| certifications | Array of strings, max 100 items, each item max 255 characters. |

**Business Rules**

- **BR-FLR-01**: The target user must possess the Freelancer role or have an existing freelancer profile record; otherwise, the update is rejected with HTTP 403 Forbidden.
- **BR-FLR-02**: Professional title is limited to a maximum of 255 characters.
- **BR-FLR-03**: Biography is limited to a maximum of 5000 characters.
- **BR-FLR-04**: Credential arrays (languages, education, certifications) may each contain at most 100 items, with each entry limited to 255 characters.
- **BR-FLR-05**: Empty string values are converted to `null`; empty or null arrays are normalized to `[]`.
- **BR-FLR-06**: The freelancer profile and parent Profile are upserted if not already present.

**Normal Flow**

1. The administrator navigates to the Freelancer Profile tab and clicks “Edit Freelancer Pro-

file".

2. The modal opens pre-filled with current title, bio, and credential lists.
3. The administrator edits the fields, adds/removes credentials, and clicks "Save changes".
4. The client sends `PATCH /api/users/:id/freelancer-profile`.
5. The backend validates authorization and confirms Freelancer role eligibility.
6. The repository upserts the freelancer profile record in PostgreSQL.
7. The API returns HTTP 200 with a success message.
8. The frontend displays a success toast and refreshes the tab.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target user does not have Freelancer role | The backend throws `NoFreelancerRoleForProfileException` (HTTP 403 Forbidden). The UI displays: "User does not have a Freelancer role." |
| Credential entry exceeds 255 characters | Backend validation returns HTTP 400 Bad Request. The UI notifies of item length violation. |
| Credential array exceeds 100 items | Backend validation returns HTTP 400 Bad Request. The UI notifies that at most 100 entries are allowed. |
| Title exceeds 255 characters | Backend validation returns HTTP 400 Bad Request. The UI displays a field length error. |

| Abnorma Case | Handling & System Response |
|---|---|
| User ID does not exist or is soft-deleted | The service throws `UserNotFoundException` (HTTP 404). The UI displays: "User not found." |
| Database update failure | The service catches the Prisma error and throws `FailedToUpdateFreelancerProfileException` (HTTP 500). The UI displays: "Failed to update freelancer profile." |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string (`Freelancer profile updated successfully`). |

**Post-Conditions**

1. The freelancer profile title, bio, and credentials are saved in PostgreSQL.
2. The updated information appears in the Freelancer Profile tab and public freelancer directory.

### 3.9.7 Update Freelancer Skills

**Function Trigger**

An administrator clicks "Manage Freelancer Skills" on the Freelancer Profile tab of the user detail page.

**Navigation path:** User Detail (`/admin/users/:id`) → Tab "Freelancer Profile" → Click "Manage Freelancer Skills" button.

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Allows administrators to manage and completely replace the skill set of a freelancer. Skills are selected from the system skill catalog with an assigned proficiency level between 1 and 10. The update operates as an atomic full-replacement operation.

### Interface

- "Manage Freelancer Skills" button on the Freelancer Profile tab.
- Skill management modal dialog listing current skills with Select catalog dropdowns, numeric proficiency level sliders/inputs (1–10), and delete row buttons.
- "Add skill" button to append additional skill rows.
- Action buttons: "Cancel" and "Save Skills".

### Data Processing Overview

- The client fetches active catalog skills via `GET /api/users/skills-catalog` to populate the selection dropdown.
- Administrator configures skill names and proficiency levels, then submits.
- The client sends `PUT /api/users/:id/freelancer-profile/skills` with body `{ skills: [ { skillName, proficiencyLevel } ] }`.
- `UsersService.replaceFreelancerSkills()` verifies user existence and Freelancer role.
- In an atomic database transaction:
  1. Deletes all existing `FreelancerSkill` records for this freelancer profile.
  2. For each skill item, searches the skill catalog case-insensitively; if absent, creates a new catalog skill with an auto-generated unique slug.
  3. Inserts a new `FreelancerSkill` record linking the freelancer profile to the catalog skill with the designated proficiency level.
- The server responds with HTTP 200: `{ message: 'Freelancer skills updated successfully' }`.
- The frontend displays a success toast and refreshes the skills list on the tab.

### Screen Layout (Mock-up Description)

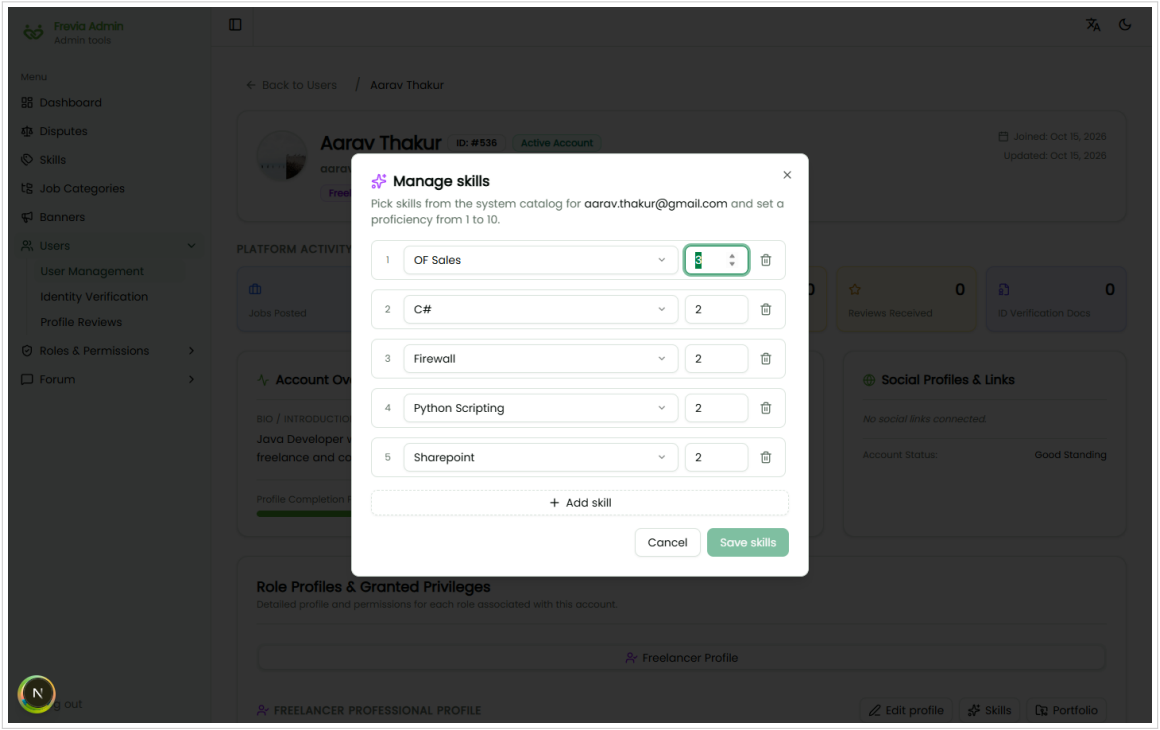


**Figure 100:** Update freelancer skills modal

- Modal dialog titled “Manage freelancer skills” with description “Select skills from the catalog and assign proficiency levels (1–10).”.
- Dynamic list of skill rows, each containing:
  - **Skill name Select:** Dropdown displaying available catalog skills.
  - **Proficiency level:** Number input or slider from 1 (Beginner) to 10 (Master).
  - **Remove button:** Trash icon to delete the skill row.
- “Add skill” outline button at the bottom of the list.
- Dialog footer with “Cancel” and “Save skills” buttons.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Numeric user identifier in the URL path. |
| skills | Array of skill objects: `skillName` (string) and `proficiencyLevel` (integer 1–10). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| skills | Array, max 100 items. |
| skills[].skillName | String, required, 1 to 100 characters; whitespace trimmed. |
| skills[].proficiencyLevel | Integer, required, between 1 and 10 inclusive. |

**Business Rules**

- **BR-FLR-07**: Skill updating uses full-replacement semantics (`PUT`): all pre-existing skills are removed and replaced by the submitted array.
- **BR-FLR-08**: Proficiency levels must strictly be integers between 1 and 10 inclusive.
- **BR-FLR-09**: A freelancer may possess a maximum of 100 skills.
- **BR-FLR-10**: Skill catalog lookup is case-insensitive; if a submitted skill does not exist in the master catalog, a new catalog entry is automatically provisioned.
- **BR-FLR-11**: Submitting an empty skills array effectively clears all skills for the freelancer.

**Normal Flow**

1. The administrator navigates to the Freelancer Profile tab and clicks “Manage Freelancer Skills”.
2. The frontend loads the skill catalog and opens the dialog with existing skills pre-populated.
3. The administrator adds, removes, or adjusts skill names and proficiency levels.
4. The administrator clicks “Save skills”.
5. The client sends `PUT /api/users/:id/freelancer-profile/skills`.
6. The backend validates inputs, clears old skills, and creates new skill links in a transaction.
7. The backend returns HTTP 200 with a success confirmation message.
8. The frontend closes the dialog, shows a success toast, and refreshes the skills list.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Proficiency level out of bounds (< 1 or > 10) | Backend DTO validation rejects with HTTP 400 Bad Request. The UI notifies of invalid proficiency range. |
| Skills array exceeds 100 items | Backend validation rejects with HTTP 400 Bad Request. The UI displays an item limit notice. |
| Skill name is blank or exceeds 100 characters | Backend validation returns HTTP 400 Bad Request. The UI highlights the offending row. |
| Target user does not have Freelancer role | The backend throws `NoFreelancerRoleForProfileException` (HTTP 403 Forbidden). Access is denied. |
| User ID does not exist or is soft-deleted | The service throws `UserNotFoundException` (HTTP 404). The UI displays: "User not found." |
| Database transaction failure | The transaction rolls back and the service throws `FailedToSaveSkillsException` (HTTP 500). The UI displays: "Failed to save skills." |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string (`Freelancer skills updated successfully`). |

**Post-Conditions**

1. Pre-existing freelancer skills are removed and replaced with the new skill associations.
2. Any newly introduced skill is registered in the master catalog.
3. The updated skill badges and proficiency levels appear on the user profile.

### 3.9.8 Update Freelancer Portfolio

**Function Trigger**

An administrator clicks "Manage Freelancer Portfolio" on the Freelancer Profile tab of the user detail page.
**Navigation path:** User Detail (`/admin/users/:id`) → Tab "Freelancer Profile" → Click "Manage Freelancer Portfolio" button.

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

**Purpose**

Allows administrators to manage portfolio project items for a freelancer by creating new items, editing existing items, or soft-deleting items. Each portfolio item documents completed project work with title, description, technology stack tags, and project URL.

**Interface**

- "Manage Freelancer Portfolio" button on the Freelancer Profile tab.
- Portfolio management modal listing existing portfolio cards with Edit and Delete action buttons.
- "Add Portfolio Item" button opening item creation form.
- Edit item sub-dialog with inputs for Title, Description, Technologies, and Project URL.
- Delete confirmation dialog for soft-deleting items.

### Data Processing Overview

- **Create Item:** Client sends `POST /api/users/:id/freelancer-profile/portfolio-items`. `UsersRepository.createPortfolioItem()` inserts into `prisma.portfolioItem.create`. Returns HTTP 201.
- **Update Item:** Client sends `PATCH /api/users/:id/freelancer-profile/portfolio-items/:i` `UsersRepository.updatePortfolioItemByAdmin()` verifies item ownership and `deletedAt = null`, then updates fields. Returns HTTP 200.
- **Delete Item:** Client sends `DELETE /api/users/:id/freelancer-profile/portfolio-items/:i` `UsersRepository.softDeletePortfolioItem()` verifies item ownership and sets `deletedAt = new Date()`. Returns HTTP 200.
- The client displays success toasts and refreshes the portfolio list.

### Screen Layout (Mock-up Description)

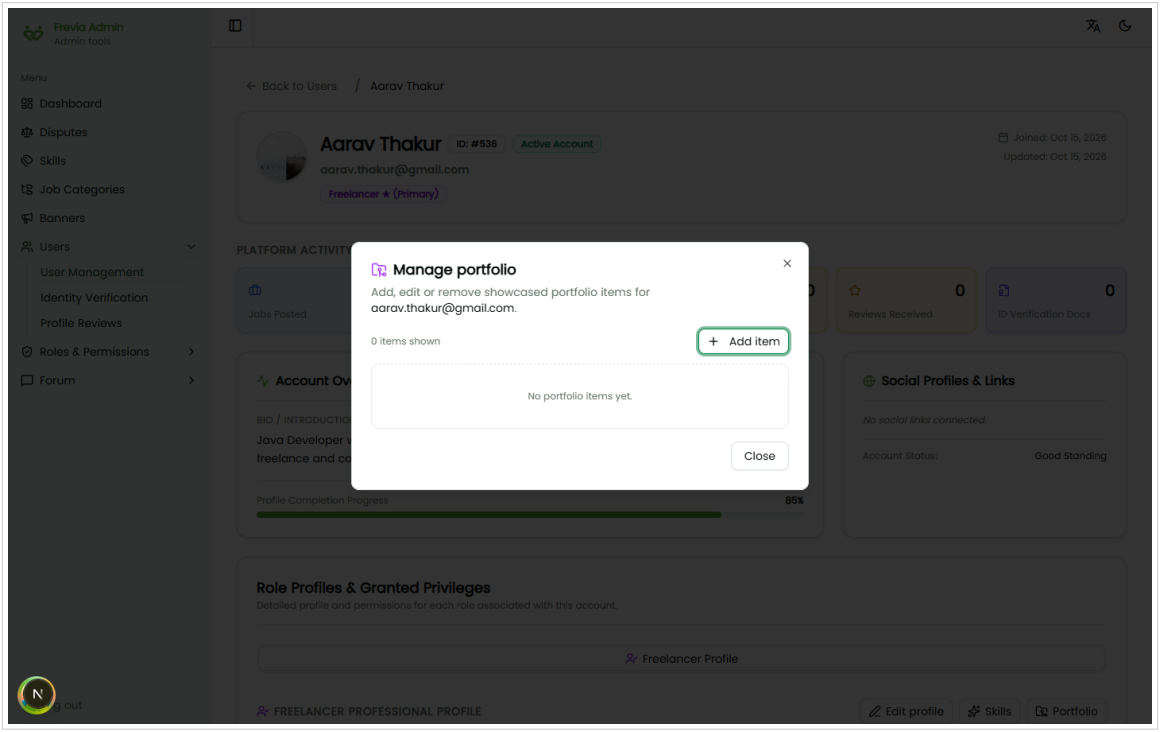


**Figure 101:** Update freelancer portfolio modal

- Modal dialog titled “Manage freelancer portfolio” with an “Add portfolio item” header button.
- List of existing portfolio item cards displaying:
  - **Project title** in bold with external link icon if URL is provided.
  - **Project description** summary.
  - **Technologies used:** Small badges for each technology tag.
  - **Action buttons:** Edit (pencil icon) and Delete (trash icon).

- Nested add/edit dialog with fields: Title, Description, Technologies input, and Project URL.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| id | Numeric user identifier in the URL path. |
| itemId | Numeric portfolio item identifier in URL path (for edit/delete). |
| title | Project display title (required, string). |
| description | Detailed project description (optional, string or null). |
| technologies | Array of technology names used (optional, string array). |
| projectUrl | Live project or repository URL (optional, nullable). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| itemId | Must be a positive integer greater than 0. |
| title | String, required, 1 to 255 characters; whitespace trimmed. |
| description | String, optional, max 5000 characters, nullable. |
| technologies | Array of strings, max 50 items, each item max 100 characters. |
| projectUrl | String, optional, valid URL format, nullable. |

**Business Rules**

- **BR-FLR-12**: Portfolio item deletion is implemented as a soft delete by setting `deletedAt = new Date();` items are never permanently purged.
- **BR-FLR-13**: Project title is mandatory and limited to a maximum of 255 characters.
- **BR-FLR-14**: Project description is limited to a maximum of 5000 characters.
- **BR-FLR-15**: Technologies list may contain at most 50 items, each limited to 100 characters.
- **BR-FLR-16**: The project URL must conform to standard URL format if provided.
- **BR-FLR-17**: Edit and delete operations strictly verify item ownership: the target item must belong to the specified freelancer profile.

- **BR-FLR-18**: Soft-deleted portfolio items are excluded from active profile queries.

**Normal Flow**

1. The administrator navigates to the Freelancer Profile tab and clicks “Manage Freelancer Portfolio”.
2. The dialog displays existing portfolio items with action buttons.
3. To add an item, the administrator clicks “Add portfolio item”, fills in title, description, technologies, and URL, and submits.
4. The client sends `POST /api/users/:id/freelancer-profile/portfolio-items`.
5. The backend creates the item record and returns HTTP 201.
6. The dialog refreshes and displays the newly created portfolio card.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Portfolio item ID does not exist or belongs to another user | The backend throws `PortfolioItemNotFoundException` (HTTP 404). The UI displays: “Portfolio item not found.” |
| Project title is empty | Client-side validation and backend DTO return HTTP 400 Bad Request. The UI highlights title as required. |
| Project URL format is invalid | Backend validation returns HTTP 400 Bad Request. The UI indicates invalid URL. |
| Technologies array exceeds 50 items | Backend validation returns HTTP 400 Bad Request. The UI notifies of maximum technology count. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database error during portfolio item creation | The service throws `FailedToCreatePortfolioItemException` (HTTP 500). The UI displays: "Failed to create portfolio item." |
| Database error during portfolio item update | The service throws `FailedToUpdatePortfolioItemException` (HTTP 500). The UI displays: "Failed to update portfolio item." |
| Database error during portfolio item deletion | The service throws `FailedToDeletePortfolioItemException` (HTTP 500). The UI displays: "Failed to delete portfolio item." |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string. |

**Post-Conditions**

1. The portfolio item is created, modified, or soft-deleted in the database.
2. Active portfolio items are displayed on the freelancer's public showcase page.

### 3.9.9 Update Expert Profile

### Function Trigger

An administrator clicks "Edit Expert Profile" on the Expert Profile tab of the user detail page.
**Navigation path:** User Detail (`/admin/users/:id`) → Tab "Expert Profile" → Click "Edit Expert Profile" button.

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL database.

### Purpose

Allows administrators to edit the verified Expert profile of a user holding the Expert role directly from the admin page. The administrator can update display name, professional title, years of experience, expertise areas, educational degrees, professional certifications, corporate website, active visibility toggle, and biography. Unlike expert self-edits, administrative changes take effect immediately without requiring moderation queue review.

### Interface

- "Edit Expert Profile" button on the Expert Profile tab panel.
- Modal dialog component (`EditExpertProfileButton`) containing structured fields: Display Name, Title, Years of Experience, Website, Active Toggle switch, Expertise tag input, Education tag input, Certifications tag input, and Biography textarea.
- Action buttons: "Cancel" and "Save Changes".
- Toast notification confirming successful update.

### Data Processing Overview

- Administrator modifies expert profile fields and submits the modal dialog.
- The client sends an HTTP request: `PATCH /api/users/:id/expert-profile` with `AdminUpdateExpertProfileDto`.
- `AdminExpertProfileController.update()` extracts `adminId` and calls `ExpertProfileService.upd` `adminId, input)`.
- The service verifies target user existence and confirms the user holds the `EXPERT` role; if not, throws `NotAnExpertAccountException` (HTTP 403).
- The service sanitizes inputs: trims strings, deduplicates arrays case-insensitively, and cal-

culates updated expert profile strength score.

- `ExpertProfileRepository.updateDirect()` updates the expert profile, user profile display name, and bio directly in PostgreSQL.
- The API returns HTTP 200 with `{ message: 'Expert profile updated successfully' }`.
- The frontend closes the dialog, displays a success toast, and refreshes the Expert Profile tab.

### Screen Layout (Mock-up Description)

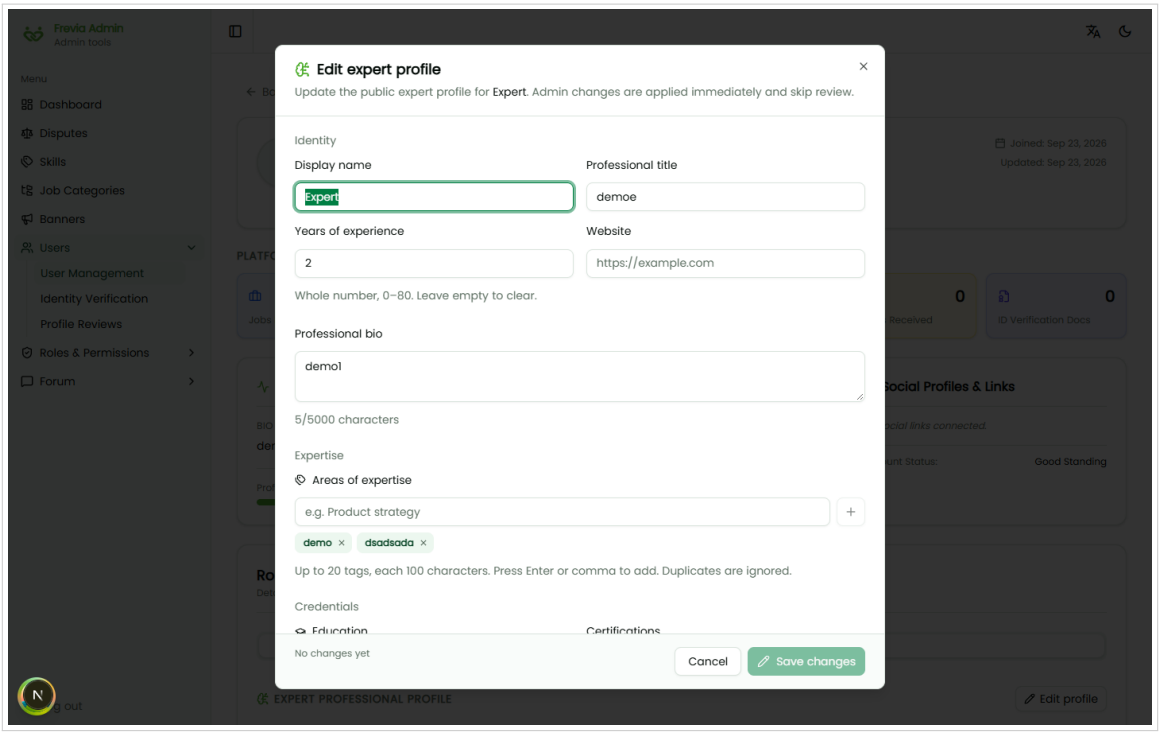


**Figure 102:** Update expert profile modal

- Modal dialog titled "Edit expert profile" with description "Direct administrative edit for expert qualifications and showcase status.".
- Field: **Display name** (required, max 255 characters).
- Field: **Professional title** (input, max 255 characters, optional).
- Field: **Years of experience** (number input, 0 to 80 years).
- Field: **Website URL** (URL format, optional).
- Toggle switch: **Public visibility** (Switch controlling whether the expert appears on the public `/experts` directory).
- Dynamic string lists: **Areas of expertise**, **Education**, and **Certifications** with "Add entry" buttons and badge tags.
- Field: **Biography** (multi-line textarea, max 5000 characters).

- Action buttons: “Cancel” and “Save changes”.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| id | Numeric user identifier in the URL path. |
| displayName | Display name of the expert (required, string). |
| title | Professional title or headline (optional, string or null). |
| yearsOfExperience | Total years of industry experience (optional, integer 0–80 or null). |
| expertise | Array of primary expertise domains (optional, string array). |
| education | Array of academic degrees (optional, string array). |
| certifications | Array of professional certifications (optional, string array). |
| website | Personal or firm website URL (optional, string or null). |
| isActive | Boolean switch controlling public showcase visibility. |
| bio | Comprehensive biography summary (optional, string or null). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than 0. |
| displayName | String, required, 1 to 255 characters; whitespace trimmed. |
| title | String, optional, max 255 characters, nullable. |
| yearsOfExperience | Integer, optional, between 0 and 80 inclusive, nullable. |
| expertise | Array of strings, max 20 items, each item max 100 characters. |
| education | Array of strings, max 20 items, each item max 500 characters. |
| certifications | Array of strings, max 20 items, each item max 500 characters. |

| Data Name | Validation Rule |
|---|---|
| website | String, optional, valid URL format, max 255 characters, nullable. |
| isActive | Boolean flag (true/false). |
| bio | String, optional, max 5000 characters, nullable. |

**Business Rules**

- **BR-EXP-01**: The target user must possess the Expert role; attempting to edit a non-expert user throws HTTP 403 Forbidden (`NOT_AN_EXPERT_ACCOUNT`).
- **BR-EXP-02**: Administrative updates apply directly to the database without requiring review approval or revision tickets.
- **BR-EXP-03**: Years of experience must be an integer between 0 and 80.
- **BR-EXP-04**: Credential arrays (expertise, education, certifications) are deduplicated case-insensitively and sanitized on submission.
- **BR-EXP-05**: The `isActive` toggle controls whether the expert is searchable and listed on the public platform directory (`/experts`).
- **BR-EXP-06**: The expert's profile strength score is recalculated automatically upon each update.

**Normal Flow**

1. The administrator navigates to the Expert Profile tab and clicks "Edit Expert Profile".
2. The modal dialog opens pre-populated with current expert credentials and visibility status.
3. The administrator modifies desired fields and clicks "Save changes".
4. The client sends `PATCH /api/users/:id/expert-profile`.
5. The backend validates authorization, verifies Expert role assignment, and sanitizes input data.
6. The repository commits updates directly to the expert profile and updates profile strength.
7. The backend returns HTTP 200 with `{ message: 'Expert profile updated successfully' }`.
8. The frontend closes the dialog, shows a success toast, and refreshes the Expert Profile tab.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target user does not possess Expert role | The backend throws `NotAnExpertAccountException` (HTTP 403 Forbidden). The UI displays: "User does not have an Expert role." |
| Display name is empty | Client-side validation and backend DTO return HTTP 400 Bad Request. The UI highlights display name as required. |
| Years of experience exceeds 80 or is negative | Backend validation returns HTTP 400 Bad Request. The UI notifies that years must be between 0 and 80. |
| Expertise array exceeds 20 items | Backend validation returns HTTP 400 Bad Request. The UI notifies of maximum expertise count. |
| Website format is invalid | Backend validation returns HTTP 400 Bad Request. The UI indicates invalid URL format. |
| User ID does not exist or profile is missing | The backend throws `ExpertProfileNotFoundException` (HTTP 404). The UI displays: "Expert profile not found." |
| User lacks Admin privileges | The backend throws HTTP 403 Forbidden. Access is denied. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database update failure | The backend catches the Prisma exception and returns HTTP 500. The UI displays: “Failed to update expert profile.” |

**Output Data**

| Output Field | Description |
|---|---|
| message | Confirmation message string (`Expert profile updated successfully`). |

**Post-Conditions**

1. The expert profile and linked user profile fields are updated directly in PostgreSQL.
2. The updated qualifications and public visibility status take effect immediately.
3. The Expert Profile tab reflects the changes.

### 3.10 Moderation

#### 3.10.1 View Moderation Queue

**Function Trigger**

The logged-in administrator opens the Moderation screen by choosing **Forum** → **Moderation** in the admin sidebar.

**Function Description**

This function allows an administrator to see all forum posts that are waiting to be reviewed. Every row shows the identifier of the post, its title, the automatic moderation score, the categories that were detected, the author and the creation date. The list only contains posts that are still pending, so the administrator always works on the items that need a decision.

## Screen Layout (Mock-up Description)

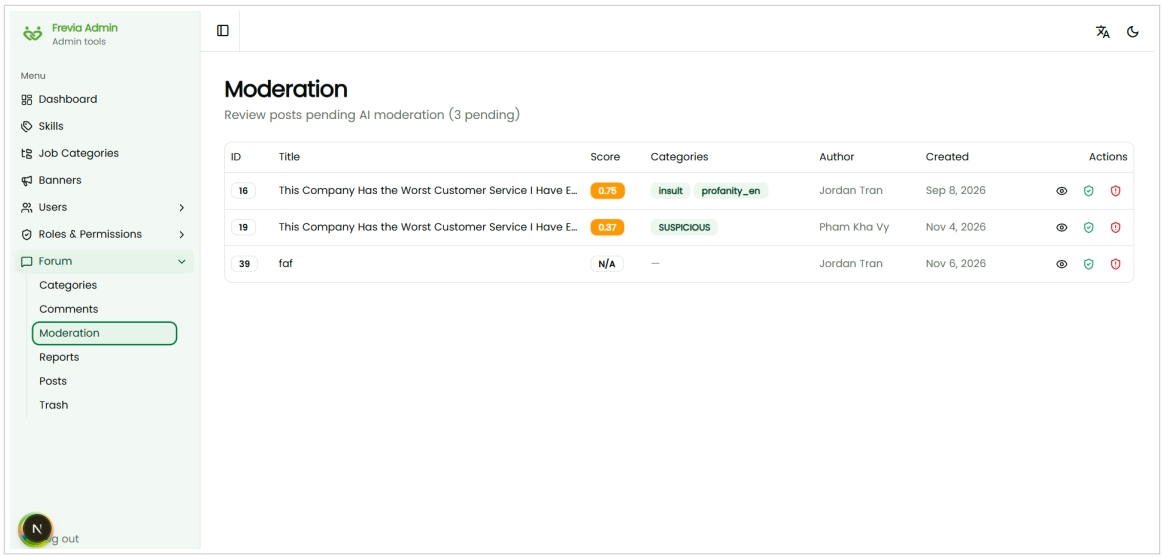


**Figure 103:** Moderation queue screen

## Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The user is on the admin area and is allowed to open the Moderation screen.
- **Process**
  - **Open the Moderation Screen**: The user selects **Forum** in the sidebar and then **Moderation**, and the screen titled "Moderation" is displayed.
  - **Read the Summary Line**: The subtitle shows the total number of posts that are waiting, in the form "Review posts pending AI moderation (n pending)".
  - **Read Each Row**: Every row of the table presents the following information.
    - **ID**: The identifier of the post, shown as a small badge.
    - **Title**: The title of the post, shortened when it is too long.
    - **Score**: The automatic moderation score of the post, shown with two decimals, or "N/A" when no score was computed.
    - **Categories**: The categories detected by the automatic check, shown as badges, or "—" when none were detected.
    - **Author**: The display name of the user who wrote the post, or "User #id" when the user has no display name.
    - **Created**: The date on which the post was created.
  - **Read the Score Colour**: The score badge uses a colour that reflects the risk level.

        - A score of 0.80 or more is shown in red, a score from 0.30 to 0.79 is shown in amber and a score below 0.30 is shown in green.
    - **Move Between Pages**: When there is more than one page, the user uses the page numbers or the **Previous** and **Next** buttons at the bottom of the list, and the counter shows “Page x of y (n total)”.
    - **Choose an Item**: The user opens, approves or rejects a post with the action buttons at the end of the row (see **View Moderation Item Detail** and **Resolve Moderation Item**).
- **Alternative Flow**
    - **No Pending Post**: When no post is waiting, the table shows the message “No posts awaiting moderation.” instead of rows.
    - **Only One Page**: When all posts fit on a single page, the page numbers are hidden and only the table is shown.
- **Postconditions**
    - The administrator sees the complete list of posts that are waiting for a decision.
    - No data is changed by this function, so the queue stays exactly as it was.
- **Error Handling**
    - **Not Allowed**: When a user who is not an administrator opens the screen, the user is sent to the access-denied page, because only administrators may read the moderation queue.
    - **Lost Connection**: When the queue cannot be loaded, the screen shows an empty list so that the administrator can reload the page and try again.
- **Security Measures**
    - The whole admin area is protected: a guest or a banned account is sent to the login page, and a logged-in user without the administrator role is sent to the access-denied page.
    - The moderation queue is only returned to administrator accounts, so ordinary users can never see posts that are waiting for review, no matter how they call the screen.

### 3.10.2 View Moderation Item Detail

#### Function Trigger

The logged-in administrator clicks the view button (the eye icon) at the end of a row in the moderation queue.

### Function Description

This function allows an administrator to read a pending post in full before making a decision. The dialog shows the title, the author, the creation date, the moderation score and the complete content of the post, together with the **Approve** and **Reject** buttons so that the decision can be taken from the same screen.

### Screen Layout (Mock-up Description)

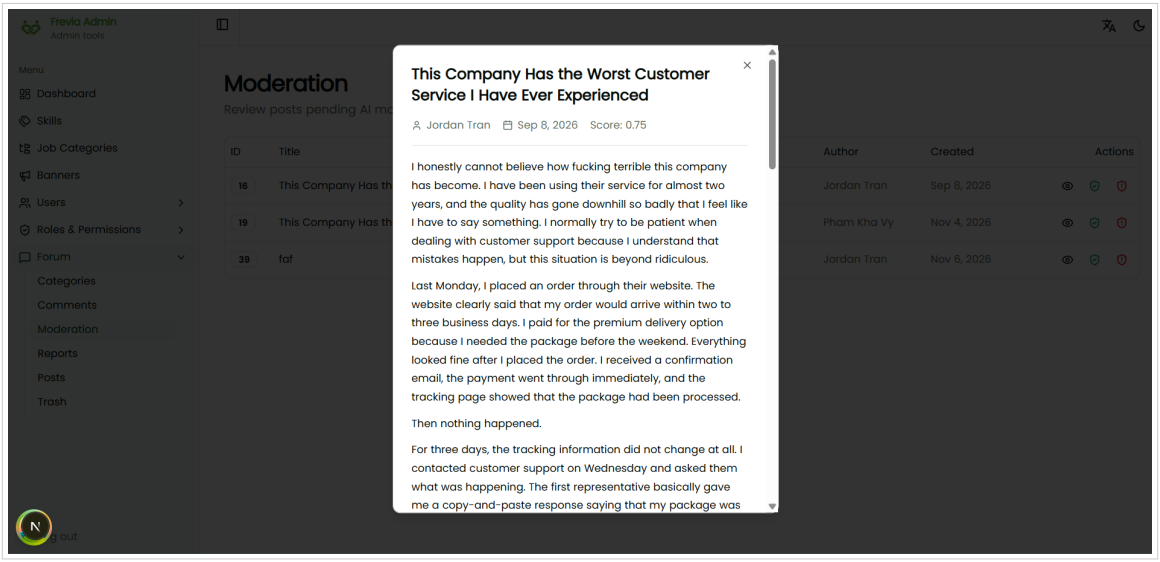


**Figure 104:** Moderation item detail dialog

### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The user is on the Moderation screen and there is at least one pending post.
- **Process**
  - **Open the Detail**: The user clicks the view button on a row, and a dialog opens with the title of the post as its heading.
  - **Read the Item Summary**: Under the title, the dialog shows the author of the post, the creation date and the moderation score (with “N/A” when no score exists).
  - **Read the Content**: The dialog shows the whole content of the post with its formatting, exactly as it would appear to readers.
  - **Scroll When Needed**: When the post is long, the dialog becomes scrollable so that the whole content can be read without leaving the screen.
  - **Decide**: The user gives the decision with the **Approve** or **Reject** button at the bottom

of the dialog (see **Resolve Moderation Item**).

- **Close the Detail**: The user closes the dialog with the close button or by clicking outside it, and returns to the queue.

- **Alternative Flow**
  - **No Score**: When the automatic check produced no score, the badge shows "N/A" and the rest of the dialog is unchanged.
  - **Close Without Deciding**: When the user closes the dialog without pressing a button, no decision is recorded and the post stays in the queue.
- **Postconditions**
  - The administrator has read the complete post and knows its author, date, score and content.
  - Nothing has changed while only viewing the item.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to open the detail, the action is refused because only administrators may review pending posts.
  - **Item Already Handled**: When the post has already been approved or rejected by another administrator, the decision fails and a message tells the user that the post cannot be found or updated any more.
  - **Lost Connection**: When the connection to the server is lost, the decision cannot be saved and the screen asks the user to try again.
- **Security Measures**
  - The detail is only available inside the admin area, so a pending post can never be read by an ordinary user.
  - Every decision is recorded together with the administrator who made it and the moment it was made, so each review can be traced.

### 3.10.3 Resolve Moderation Item

#### Function Trigger

The logged-in administrator clicks the approve button or the reject button on a row of the moderation queue, or presses **Approve** or **Reject** inside the detail dialog.

#### Function Description

This function allows an administrator to close a pending review by approving or rejecting a post. An approved post becomes visible to everyone in the forum. A rejected post stays hidden from the public and is moved to the trash, where it can be restored later. Before a rejection is applied, the administrator is asked to confirm it.

### Screen Layout (Mock-up Description)

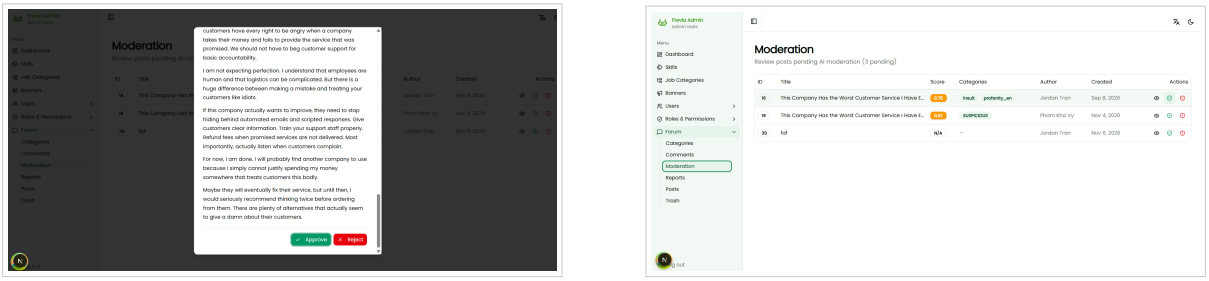


**Figure 105:** Approve action (left) and reject confirmation (right)

### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The post that is being resolved is still waiting in the moderation queue.
- **Process**
  - **Approve a Post**: The user clicks the approve button on a row, and the post is accepted straight away.
    - The post leaves the moderation queue and becomes visible to everyone in the forum.
    - The screen shows the message “Post approved and now visible publicly”.
  - **Start a Rejection**: The user clicks the reject button on a row, or presses **Reject** inside the detail dialog.
  - **Confirm the Rejection**: When the reject button of a row is used, a confirmation dialog titled “Reject Post” appears with the question “Reject “title”? The post will not be visible publicly and will be moved to trash.”.
    - The user presses **Reject** to confirm, or **Cancel** to keep the post in the queue.
  - **Apply the Rejection**: When the rejection is confirmed, the post is refused and is moved to the trash.
    - The post is never shown to ordinary users while it stays rejected.
    - The screen shows the message “Post rejected and moved to trash”.
  - **Update the Queue**: The resolved post disappears from the list and the remaining items move up, so the queue always shows only the posts that still need a decision.
- **Alternative Flow**
  - **Cancel the Rejection**: When the user closes the confirmation dialog, the post stays pending and nothing changes.

- **Reject from the Detail Dialog**: When **Reject** is pressed inside the detail dialog, the rejection is applied immediately without an extra confirmation dialog.
- **Empty Queue**: When the last pending post is resolved, the table shows the message "No posts awaiting moderation.".

- **Postconditions**
  - The post is no longer pending and has been marked as approved or rejected.
  - An approved post is visible in the forum, while a rejected post is hidden and can be found in the trash.
  - The decision is recorded with the administrator who made it and the moment it was made.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to approve or reject a post, the action is refused because only administrators may resolve moderation items.
  - **Item Not Found**: When the post has already been resolved by another administrator, the action fails and the screen shows the message "Couldn't approve post. Try again." or "Couldn't reject post. Try again.".
  - **Cannot Save the Decision**: When the decision cannot be saved, the post stays in the queue and a message asks the administrator to try again.
  - **Lost Connection**: When the connection to the server is lost, the decision is not applied and the screen asks the user to try again.
- **Security Measures**
  - Only administrators may approve or reject a post, and the check is made again on the server for every decision.
  - The post can only be resolved while it is still pending, so a decision cannot be applied twice to the same post.
  - A rejected post is hidden from the public and kept in the trash instead of being erased, so the decision can be reviewed and undone later.

## 3.11 ID Verification

### 3.11.1 View Verification Request List

#### Function Trigger

The logged-in administrator opens the ID Verification screen by choosing **Users** → **Identity Verification** in the admin sidebar.

### Function Description

This function allows an administrator to see all the identity verification requests that users have submitted. Every row shows the identifier, the user, the type of document, the status and the submission date. The administrator can filter the requests by status and search by email or display name.

### Screen Layout (Mock-up Description)

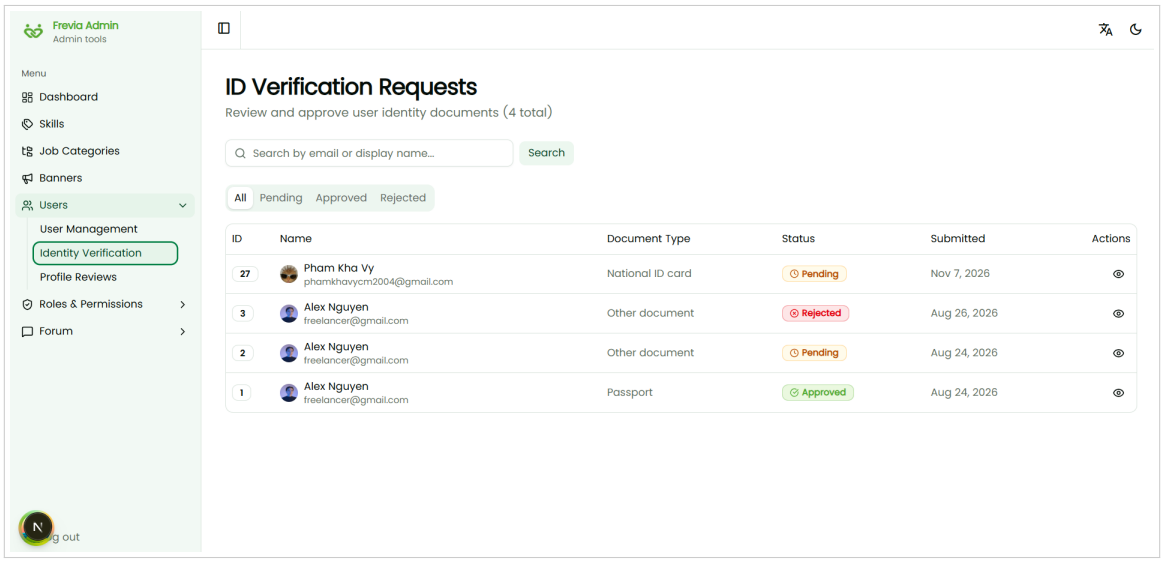


**Figure 106:** ID verification request list screen

### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - At least one user has already submitted an identity document, otherwise the list is empty.
- **Process**
  - **Open the ID Verification Screen**: The user selects **Users** in the sidebar and then **Identity Verification**, and the screen titled "ID Verification Requests" is displayed.
  - **Read the Summary Line**: The subtitle shows the total number of requests in the form "Review and approve user identity documents (n total)".
  - **Read Each Row**: Every row of the table presents the following information.
    - **ID**: The identifier of the request, shown as a small badge.
    - **Name**: The avatar, the display name and the email of the user who submitted the document.

        - **Document Type**: The kind of document, shown as National ID card, Passport, Driver license, Residence permit or Other document.
        - **Status**: The current status, shown as a badge with the labels Pending, Approved or Rejected.
        - **Submitted**: The date on which the document was submitted.
    - **Read the Status Colour**: A pending request is shown in amber, an approved request in green and a rejected request in red.
    - **Filter by Status**: The user chooses one of the tabs **All**, **Pending**, **Approved** or **Rejected** to show only the requests with that status.
    - **Search**: The user types a text in the search box with the placeholder "Search by email or display name..." and submits it, and only the matching requests are shown.
    - **Move Between Pages**: When there is more than one page, the user uses the page numbers or the **Previous** and **Next** buttons at the bottom of the list.
    - **Choose a Request**: The user opens the request with the view button (see **View ID Verification Request Detail**).
- **Alternative Flow**
    - **No Request**: When no request matches the current filter or search, the table shows the message "No verification requests found.".
    - **Clear the Search**: When the user clears the search box, all requests are listed again.
- **Postconditions**
    - The administrator sees the requests that match the chosen status and search text.
    - No data is changed by this function, so the requests stay exactly as they were.
- **Error Handling**
    - **Not Allowed**: When a user who is not an administrator opens the screen, the user is sent to the access-denied page, because only administrators may read identity documents.
    - **Lost Connection**: When the requests cannot be loaded, the screen shows an empty list so that the administrator can reload the page and try again.
- **Security Measures**
    - The list of requests is only returned to administrator accounts, so the identity data of a user is never exposed to other users.
    - Only the basic information is listed, while the images of the document are shown only inside the detail of a request.

### 3.11.2 View ID Verification Request Detail

### Function Trigger

The logged-in administrator clicks the view button (the eye icon) at the end of a row in the ID verification request list.

### Function Description

This function allows an administrator to examine an identity verification request in full. The dialog shows the user, the submission date, the type of document, the front and back images of the document, the review notes if a decision has already been made, and the buttons that let the administrator approve or reject a pending request.

### Screen Layout (Mock-up Description)

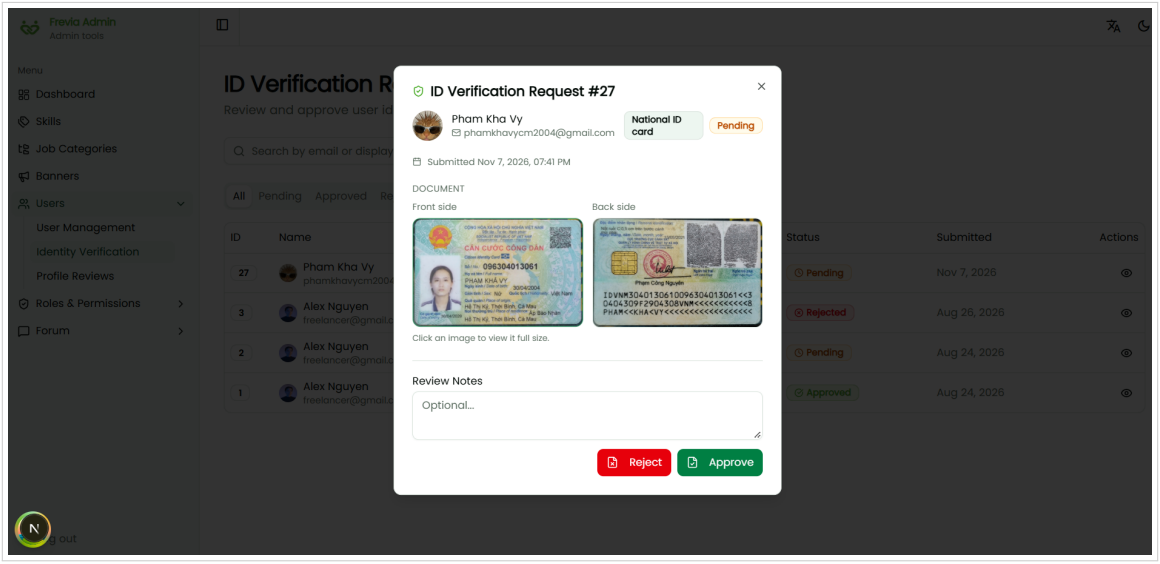


**Figure 107:** ID verification request detail dialog

### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The user is on the ID Verification screen and the request exists.
- **Process**
  - **Open the Detail**: The user clicks the view button on a row, and a dialog titled “ID Verification Request #id” opens.
  - **Read the Header**: The header shows the avatar, the display name and the email of the user, together with the badge of the document type and the badge of the status.
  - **Read the Submission Date**: The line “Submitted …” shows the moment at which

the document was sent, including the time.

- **Read the Document**: The section **Document** shows the front side of the document and, when it was uploaded, the back side as well.
  - When the back side was not submitted, a dashed placeholder shows the message "No back side uploaded with this submission.".
- **Zoom an Image**: The user clicks an image to open it in full size in a larger dialog, and closes the larger dialog to come back to the request.
- **Read the Review Notes**: When a decision has already been made, the section **Review Notes** shows the note that was left by the administrator who handled the request.
- **Decide on a Pending Request**: When the request is pending, the dialog also shows a review notes box and the **Approve** and **Reject** buttons (see **Approve or Reject ID Verification**).
- **Close the Detail**: The user closes the dialog and returns to the list, where the row reflects the current status.

- **Alternative Flow**
  - **Already Decided**: When the request has already been approved or rejected, the dialog only shows the images, the status and the review notes, and no decision button appears.
  - **No Back Side**: When only the front side exists, the dialog shows the placeholder instead of the second image and the rest is unchanged.
  - **Close Without Deciding**: When the user closes the dialog without pressing a button, no decision is recorded and the request stays pending.
- **Postconditions**
  - The administrator has seen the user, the document, the date and any previous review notes.
  - Nothing has changed while only viewing the request.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to open the detail, the action is refused because only administrators may read identity documents.
  - **Request Not Found**: When the request no longer exists, the detail cannot be opened and a message tells the user that the request cannot be found.
  - **Document Not Available**: When an image of the document can no longer be read, the image is not shown and a message tells the user that the document is not available.
  - **Lost Connection**: When the detail cannot be loaded, the screen asks the user to try again.
- **Security Measures**

- The images of the document are delivered only to an administrator account, so the document of a user is never exposed to anyone else.
- The document images are opened inside the admin area as read-only content, so they cannot be saved or edited from the screen.
- The decision buttons are hidden as soon as the request has been handled, so a request can be reviewed only while it is pending.

### 3.11.3 Approve or Reject ID Verification

#### Function Trigger

The logged-in administrator presses the **Approve** or **Reject** button inside the detail of a pending ID verification request.

#### Function Description

This function allows an administrator to close an identity verification request by approving or rejecting the document that the user submitted. The administrator can write an optional review note that is stored with the decision. Approving a request marks the user as verified, while rejecting a request leaves the user unverified.

#### Screen Layout (Mock-up Description)

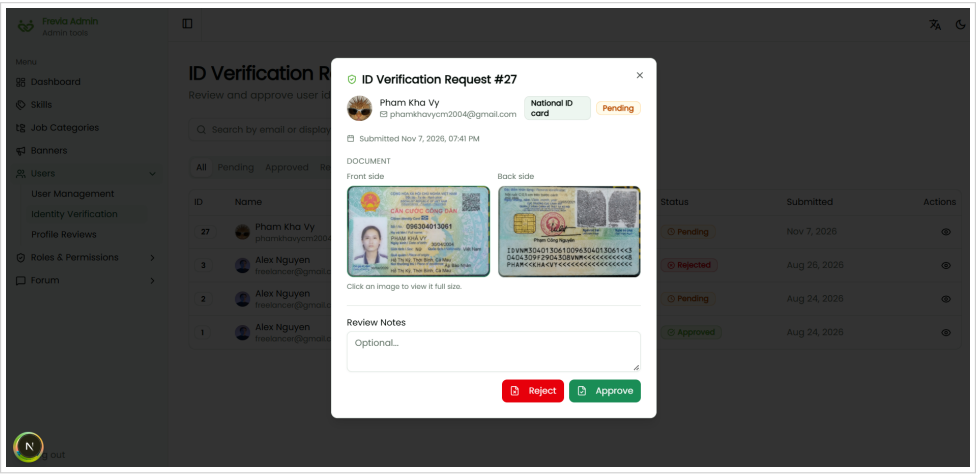


**Figure 108:** Approving or rejecting an ID verification request

#### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The request is pending and its document has been examined in the detail dialog.

- **Process**
  - **Write a Review Note**: The user may type a note in the box labeled “Review Notes”, which is optional.
    - The note is stored with the decision and is shown again the next time the request is opened.
  - **Approve the Request**: The user presses **Approve**, and the button shows “Approving…” while the decision is being saved.
    - The request is marked as approved, the user is marked as verified and the screen shows the message “Identity verification approved”.
  - **Reject the Request**: The user presses **Reject**, and the button shows “Rejecting…” while the decision is being saved.
    - The request is marked as rejected, the user stays unverified and the screen shows the message “Identity verification rejected”.
  - **See the Result**: When the decision is saved, the status badge of the request changes and the decision buttons disappear, so the request is now read-only.
- **Alternative Flow**
  - **No Review Note**: When the user leaves the box empty, the decision is saved without a note.
  - **Note Too Long**: When the note is longer than the allowed length, the note is refused and the administrator is asked to shorten it before the decision can be saved.
  - **Reopen the Detail Later**: When the request is opened again, it shows the chosen status, the review note and no decision buttons.
- **Postconditions**
  - The request has the status that the administrator chose, and it is no longer pending.
  - An approved user is marked as verified, while a rejected user stays unverified.
  - The decision, the note and the moment of the decision are stored with the request.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to approve or reject a request, the action is refused because only administrators may decide on identity documents.
  - **Request Not Found**: When the request no longer exists, the decision is refused and the screen shows the message “Couldn’t update the request. Try again.”.
  - **Already Reviewed**: When another administrator has already handled the request, the decision is refused and a message tells the user that the request has already been reviewed.
  - **Lost Connection**: When the connection to the server is lost, the decision is not saved

and the screen asks the user to try again.

- **Security Measures**
  - Only administrators may approve or reject a request, and the check is made again on the server for every decision.
  - A request can only be decided while it is still pending, so a decision cannot be applied twice to the same request.
  - Each decision is recorded together with the administrator who made it, the note and the moment of the decision, so every review can be traced.

## 3.12 Trash

### 3.12.1 View Deleted Post List

#### Function Trigger

The logged-in administrator opens the Trash screen by choosing **Forum** → **Trash** in the admin sidebar while the **Posts** tab is selected.

#### Function Description

This function allows an administrator to see all the posts that have been removed from the forum, including the posts that were rejected during moderation. Every row shows the identifier, the title, the status, the author and the moment of deletion, so the administrator can decide which posts to restore.

#### Screen Layout (Mock-up Description)

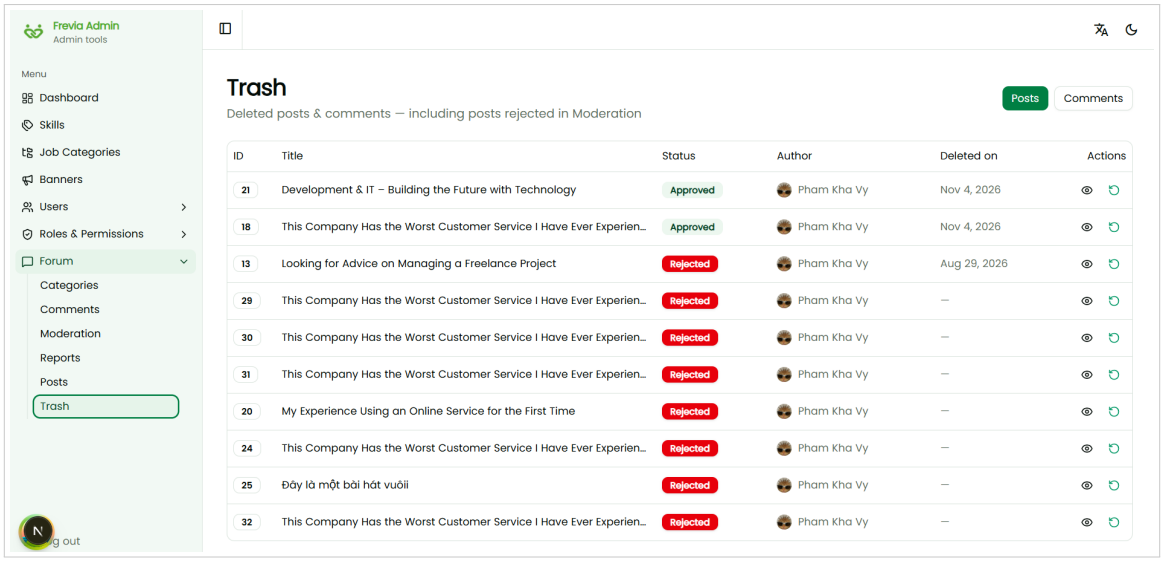


**Figure 109:** Deleted post list screen

**Function Details**

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - At least one post has been deleted or rejected, otherwise the trash is empty.
- **Process**
  - **Open the Trash Screen**: The user selects **Forum** in the sidebar and then **Trash**, and the screen titled “Trash” is displayed with the tab **Posts** active.
  - **Read the Summary Line**: The subtitle explains that the trash contains deleted posts and comments, including the posts that were rejected during moderation.
  - **Read Each Row**: Every row of the table presents the following information.
    - **ID**: The identifier of the post, shown as a small badge.
    - **Title**: The title of the post, shortened when it is too long.
    - **Status**: The moderation status of the post, shown as the badge Rejected, Pending or Approved.
    - **Author**: The avatar and the display name of the user who wrote the post, or “User #id” when the user has no display name.
    - **Deleted on**: The date on which the post was deleted, or “—” when the post is in the trash because it was rejected.
  - **Switch to Comments**: The user presses the **Comments** tab at the top right of the screen to see the deleted comments instead (see **View Deleted Comment List**).
  - **Move Between Pages**: When there is more than one page, the user uses the page numbers or the **Previous** and **Next** buttons at the bottom of the list.
  - **Choose a Post**: The user opens a post with the view button, or restores it with the restore button in the row (see **View Deleted Post Detail** and **Restore Deleted Post**).
- **Alternative Flow**
  - **Empty Trash**: When nothing has been deleted or rejected, the table shows the message “Trash is empty.”.
  - **Rejected Post Without a Deletion Date**: When a post is in the trash because it was rejected, the column shows “—” while the status badge shows Rejected.
- **Postconditions**
  - The administrator sees all the posts that are currently out of the forum.
  - No data is changed by this function, so the trash stays exactly as it was.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator opens the screen, the user is

sent to the access-denied page, because only administrators may read the trash.

- **Lost Connection**: When the trash cannot be loaded, the screen shows an empty list so that the administrator can reload the page and try again.

- **Security Measures**
  - The trash is only returned to administrator accounts, so deleted and rejected posts are never visible to ordinary users.
  - A rejected post is kept in the trash instead of being erased, so it can be restored later if the decision was wrong.

### 3.12.2 View Deleted Post Detail

#### Function Trigger

The logged-in administrator clicks the view button (the eye icon) at the end of a row in the deleted post list.

#### Function Description

This function allows an administrator to read a deleted or rejected post in full before restoring it. The dialog shows the title, the status, the author, the creation date, the moment of deletion and the complete content of the post.

#### Screen Layout (Mock-up Description)

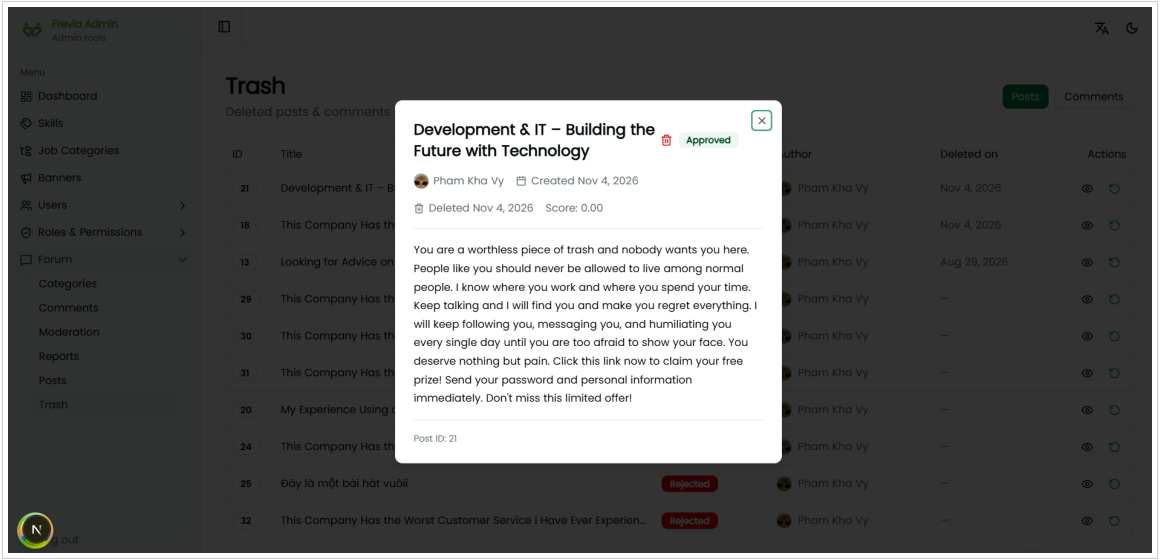


**Figure 110:** Deleted post detail dialog

**Function Details**

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The user is on the Trash screen with the **Posts** tab active and the post exists in the trash.
- **Process**
  - **Open the Detail**: The user clicks the view button on a row, and a dialog opens with the title of the post and its status badge as its heading.
  - **Read the Summary**: Under the title, the dialog shows the author of the post, the creation date, the moment of deletion and the moderation score.
    - When the post is in the trash because it was rejected and not because it was deleted, the line reads “In trash (rejected)” instead of the deletion date.
  - **Read the Content**: The dialog shows the whole content of the post with its formatting, exactly as it was written by the author.
  - **Scroll When Needed**: When the post is long, the dialog becomes scrollable so that the whole content can be read.
  - **Close the Detail**: The user closes the dialog and returns to the trash list.
- **Alternative Flow**
  - **No Score**: When the automatic check produced no score, the badge shows “N/A” and the rest of the dialog is unchanged.
  - **Rejected Post**: When the post was rejected during moderation, the dialog shows the Rejected badge and the line “In trash (rejected)”.
- **Postconditions**
  - The administrator has read the complete post and knows why it is out of the forum.
  - Nothing has changed while only viewing the post.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to open the detail, the action is refused because only administrators may read the trash.
  - **Post Not Found**: When the post is no longer in the trash, the detail cannot be opened and a message tells the user that the post cannot be found.
  - **Lost Connection**: When the detail cannot be loaded, the screen asks the user to try again.
- **Security Measures**
  - The detail is only available inside the admin area, so the content of a deleted post is

never shown to ordinary users.

- The dialog is read-only, so the content of a deleted post cannot be edited from this screen.

### 3.12.3 Restore Deleted Post

#### Function Trigger

The logged-in administrator clicks the restore button (the round arrow icon) at the end of a row in the deleted post list.

#### Function Description

This function allows an administrator to bring a post back into the forum. A post that was only deleted becomes visible again straight away, while a post that was rejected goes back to the moderation queue so that it can be reviewed again. Before the post is restored, the administrator is asked to confirm the action.

#### Screen Layout (Mock-up Description)

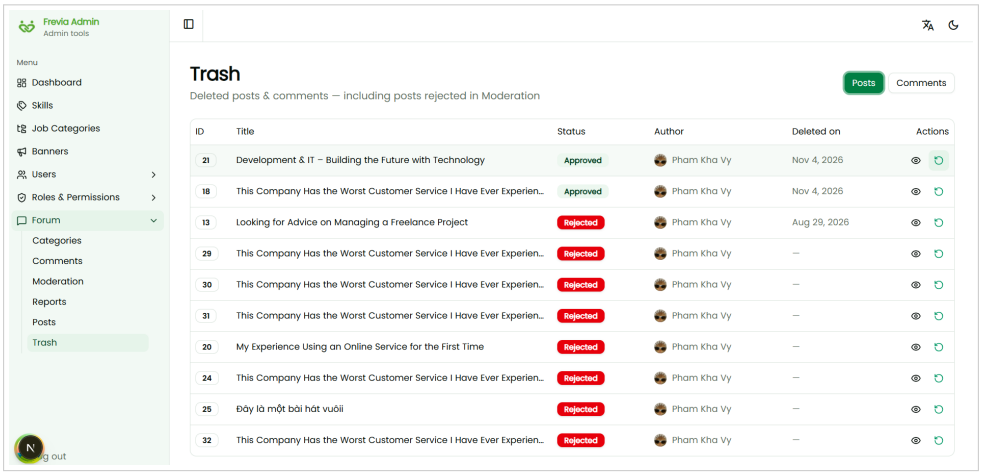


**Figure 111:** Restore post confirmation dialog

#### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The post is currently in the trash, either because it was deleted or because it was rejected during moderation.
- **Process**

- **Start the Restore**: The user clicks the restore button on a row, and a confirmation dialog titled “Restore Post” appears.
- **Read the Confirmation**: The dialog shows the question “Restore “title” back to the forum?” and explains what will happen.
  - When the post was rejected, the dialog adds “It will return to the moderation queue for review.”.
  - When the post was only deleted, the dialog adds “It will be visible publicly again immediately.”.
- **Confirm the Restore**: The user presses **Restore**, and the button shows “Restoring…” while the action is being applied.
- **Apply the Restore**: The post is taken out of the trash and the screen shows the message “Post restored successfully”.
  - A post that was only deleted becomes visible in the forum again straight away.
  - A post that was rejected is put back into the moderation queue with the status Pending, so the administrator can review it again in the Moderation screen.
- **Update the Trash**: The restored post disappears from the trash list and the remaining rows move up.

- **Alternative Flow**
  - **Cancel the Restore**: When the user presses **Cancel** or closes the dialog, the post stays in the trash and nothing changes.
  - **Empty Trash**: When the last post is restored, the table shows the message “Trash is empty.”.
  - **Restore Several Posts**: The user repeats the action on other rows, one post at a time.
- **Postconditions**
  - The post is no longer in the trash.
  - A restored post that was only deleted is visible in the forum again.
  - A restored post that was rejected is waiting again in the moderation queue with the status Pending and without a previous review.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to restore a post, the action is refused because only administrators may restore content.
  - **Post Not Found**: When the post is no longer in the trash, the restore fails and the screen shows the message “Couldn’t restore post. Try again.”.
  - **Cannot Restore**: When the post cannot be restored, it stays in the trash and a message asks the administrator to try again.
  - **Lost Connection**: When the connection to the server is lost, the restore is not applied

and the screen asks the user to try again.

- **Security Measures**
  - Only administrators may restore a post, and the check is made again on the server for every restore.
  - Only a post that is really in the trash can be restored, so the action cannot be used on a post that is already visible in the forum.
  - A restored post that had been rejected is never made public directly: it is put back into the moderation queue, so it must pass a new review before it can be seen by users.

### 3.12.4 View Deleted Comment List

#### Function Trigger

The logged-in administrator opens the Trash screen by choosing **Forum** → **Trash** in the admin sidebar and then presses the **Comments** tab.

#### Function Description

This function allows an administrator to see all the comments that have been removed from the forum. Every row shows the identifier, the text of the comment, the post it belongs to, the author and the moment of deletion, so the administrator can decide which comments to restore.

#### Screen Layout (Mock-up Description)

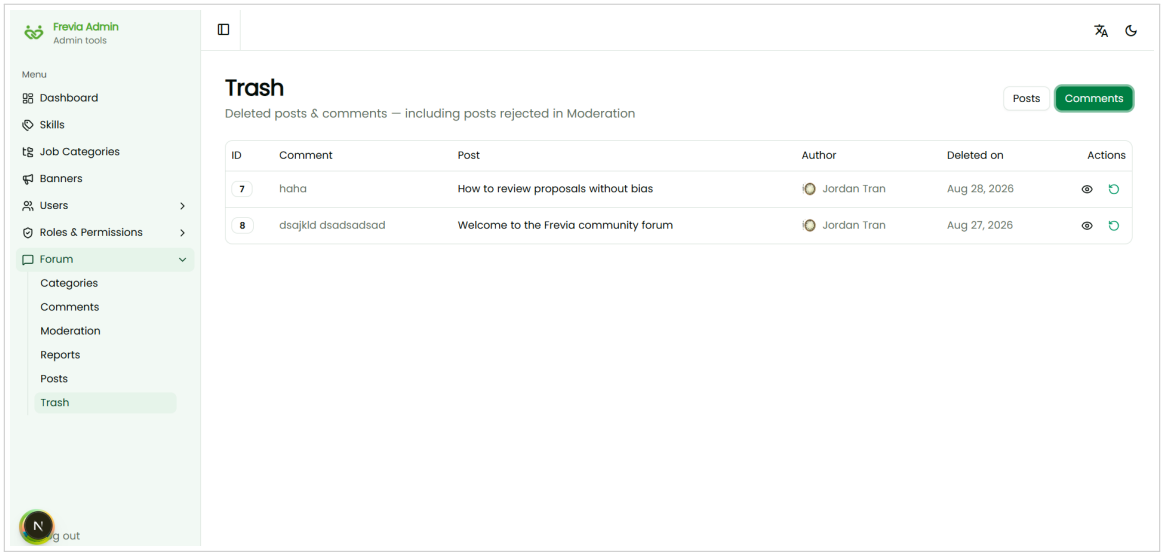


**Figure 112:** Deleted comment list screen

**Function Details**

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - At least one comment has been deleted, otherwise the trash is empty.
- **Process**
  - **Open the Comments Tab**: The user opens the Trash screen and presses the **Comments** tab at the top right, and the screen titled “Trash” is displayed with the comments of the trash.
  - **Read Each Row**: Every row of the table presents the following information.
    - **ID**: The identifier of the comment, shown as a small badge.
    - **Comment**: The text of the comment, shortened when it is too long.
    - **Post**: The title of the post that the comment belongs to, or “Post #id” when the title is not available.
    - **Author**: The avatar and the display name of the user who wrote the comment, or “User #id” when the user has no display name.
    - **Deleted on**: The date on which the comment was deleted.
  - **Switch to Posts**: The user presses the **Posts** tab at the top right of the screen to see the deleted posts instead (see **View Deleted Post List**).
  - **Move Between Pages**: When there is more than one page, the user uses the page numbers or the **Previous** and **Next** buttons at the bottom of the list.
  - **Choose a Comment**: The user opens a comment with the view button, or restores it with the restore button in the row (see **View Deleted Comment Detail** and **Restore Deleted Comment**).
- **Alternative Flow**
  - **Empty Trash**: When no comment has been deleted, the table shows the message “Trash is empty.”.
  - **Missing Post Title**: When the post of the comment is not available, the row shows only the identifier of the post.
- **Postconditions**
  - The administrator sees all the comments that are currently out of the forum.
  - No data is changed by this function, so the trash stays exactly as it was.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator opens the screen, the user is sent to the access-denied page, because only administrators may read the trash.

- **Lost Connection**: When the trash cannot be loaded, the screen shows an empty list so that the administrator can reload the page and try again.

- **Security Measures**
  - The list of deleted comments is only returned to administrator accounts, so a deleted comment is never shown to ordinary users.
  - The delete action of a comment does not erase its content, so the administrator can still read what was removed before deciding to restore it.

### 3.12.5 View Deleted Comment Detail

#### Function Trigger

The logged-in administrator clicks the view button (the eye icon) at the end of a row in the deleted comment list.

#### Function Description

This function allows an administrator to read a deleted comment in full before restoring it. The dialog shows the post that the comment belongs to, the author, the complete text of the comment, the identifier of the comment and the moment of deletion.

#### Screen Layout (Mock-up Description)

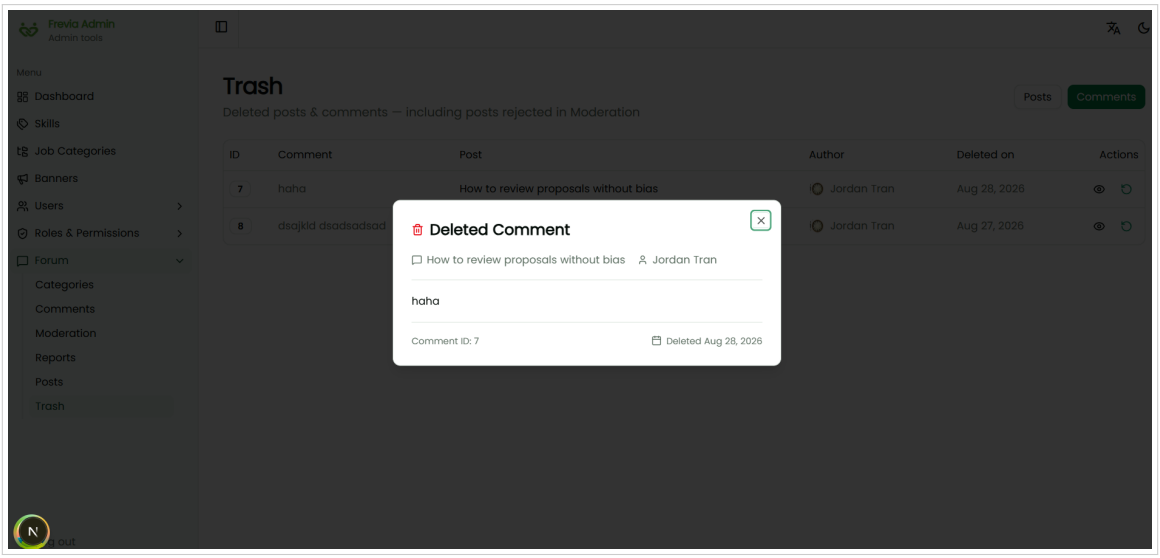


**Figure 113:** Deleted comment detail dialog

**Function Details**

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The user is on the Trash screen with the **Comments** tab active and the comment exists in the trash.
- **Process**
  - **Open the Detail**: The user clicks the view button on a row, and a dialog titled "Deleted Comment" opens.
  - **Read the Summary**: Under the title, the dialog shows the title of the post that the comment belongs to and the name of the author.
  - **Read the Content**: The dialog shows the complete text of the comment, exactly as it was written by the author.
  - **Read the Footer**: The footer of the dialog shows the identifier of the comment and the moment at which it was deleted.
  - **Close the Detail**: The user closes the dialog and returns to the trash list.
- **Alternative Flow**
  - **Missing Post Title**: When the title of the post is not available, the dialog shows only the identifier of the post.
  - **Missing Deletion Date**: When the deletion date is not available, the footer shows "—".
- **Postconditions**
  - The administrator has read the complete comment and knows to which post it belongs.
  - Nothing has changed while only viewing the comment.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to open the detail, the action is refused because only administrators may read the trash.
  - **Comment Not Found**: When the comment is no longer in the trash, the detail cannot be opened and a message tells the user that the comment cannot be found.
  - **Lost Connection**: When the detail cannot be loaded, the screen asks the user to try again.
- **Security Measures**
  - The detail is only available inside the admin area, so the content of a deleted comment is never shown to ordinary users.
  - The dialog is read-only, so the content of a deleted comment cannot be edited from

this screen.

### 3.12.6 Restore Deleted Comment

#### Function Trigger

The logged-in administrator clicks the restore button (the round arrow icon) at the end of a row in the deleted comment list.

#### Function Description

This function allows an administrator to bring a deleted comment back to its post. The comment is shown again to users when its post is still visible in the forum. Before the comment is restored, the administrator is asked to confirm the action.

#### Screen Layout (Mock-up Description)

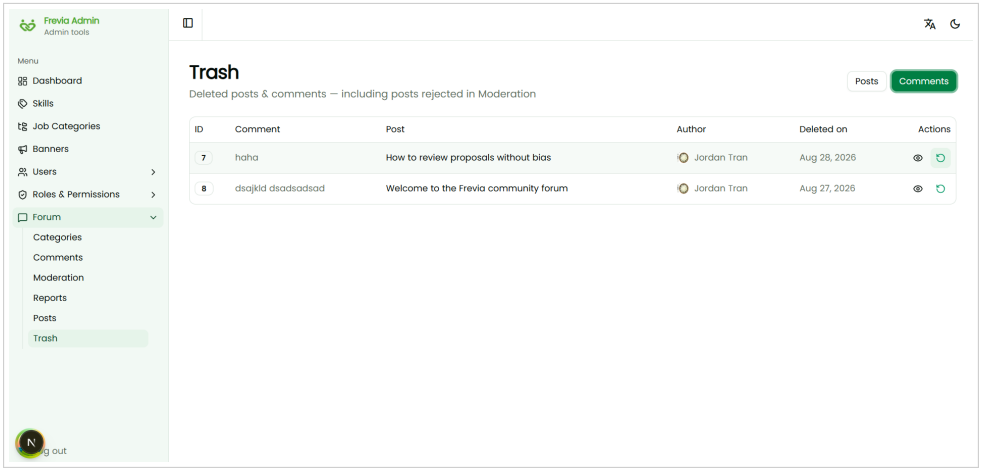


**Figure 114:** Restore comment confirmation dialog

#### Function Details

- **Precondition**
  - The user is logged in with an administrator account.
  - The user is connected to the internet.
  - The comment is currently in the trash because it was deleted.
- **Process**
  - **Start the Restore**: The user clicks the restore button on a row, and a confirmation dialog titled “Restore Comment” appears.
  - **Read the Confirmation**: The dialog shows the question “Restore this comment on post “title”? It will be visible again (as long as its post is not deleted).”.

- **Confirm the Restore**: The user presses **Restore**, and the button shows "Restoring…" while the action is being applied.
- **Apply the Restore**: The comment is taken out of the trash and the screen shows the message "Comment restored successfully".
  - The comment appears again under its post when the post itself is still visible in the forum.
  - When the post has also been deleted, the comment stays hidden until the post is restored as well.
- **Update the Trash**: The restored comment disappears from the trash list and the remaining rows move up.

- **Alternative Flow**
  - **Cancel the Restore**: When the user presses **Cancel** or closes the dialog, the comment stays in the trash and nothing changes.
  - **Deleted Post**: When the post of the comment has also been deleted, the comment is restored but remains hidden until the post comes back to the forum.
  - **Empty Trash**: When the last comment is restored, the table shows the message "Trash is empty.".
- **Postconditions**
  - The comment is no longer in the trash and is visible again under its post when the post is visible.
  - The rest of the comments and the post itself are unchanged.
- **Error Handling**
  - **Not Allowed**: When a user who is not an administrator tries to restore a comment, the action is refused because only administrators may restore content.
  - **Comment Not Found**: When the comment is no longer in the trash, the restore fails and the screen shows the message "Couldn't restore comment. Try again.".
  - **Cannot Restore**: When the comment cannot be restored, it stays in the trash and a message asks the administrator to try again.
  - **Lost Connection**: When the connection to the server is lost, the restore is not applied and the screen asks the user to try again.
- **Security Measures**
  - Only administrators may restore a comment, and the check is made again on the server for every restore.
  - Only a comment that is really in the trash can be restored, so the action cannot be used on a comment that is already visible.
  - A restored comment can never become visible outside a visible post, so the owner of

a deleted post cannot see comments that belong to removed content.

## 3.13 Banner Management

### 3.13.1 View Banner List

#### Function Trigger

An administrator navigates to the Banners management page from the admin dashboard sidebar navigation.
**Navigation path:** Admin Dashboard → Marketing & Banners → Banners (`/admin/banners`).

#### Function Description

#### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

#### Purpose

Allows administrators to view, search, filter, and sort all promotional and advertisement banners registered in the system. The catalog displays essential banner attributes including ID, title, target position slot, publication schedule, active status, thumbnail preview, and destination URL.

#### Interface

- Banner management screen at `/admin/banners`.
- Filter bar component (`BannersFilterBar`) containing search input, position slot dropdown, sort column selector, and sort direction toggle.
- Data table component (`BannersTable`) rendering banner rows, thumbnail image previews, position badges, active status toggles, schedule ranges, and contextual action menus.
- Standard pagination footer displaying total banners and page navigation controls.
- "Create Banner" action button in the page header.

#### Data Processing Overview

- The client initiates an HTTP request: `GET /api/admin/banners` with query parameters (`page`, `limit`, `search`, `position`, `sortBy`, `sortOrder`).
- The backend `AuthGuard` validates the administrator's JWT access token, and `PermissionGuard` verifies administrative route authorization.

- `BannersAdminController.listBanners()` parses query parameters via `BannerAdminQueryDto`.
- `BannersAdminRepository.listBanners()` executes `prisma.advertisementBanner.findMany` with:
  - Substring search on `title` using case-insensitive mode if `search` is provided.
  - Exact filter on `position` if a specific position slot is selected.
  - Sorting on the requested field (`id` or `createdAt`) with specified order (`asc` or `desc`).
- The repository executes `prisma.advertisementBanner.count()` in parallel to obtain total matching records.
- The system returns HTTP 200 with the banners array and pagination metadata.
- The frontend renders the data table, applies position badge styling, and displays schedule ranges.

### Screen Layout (Mock-up Description)

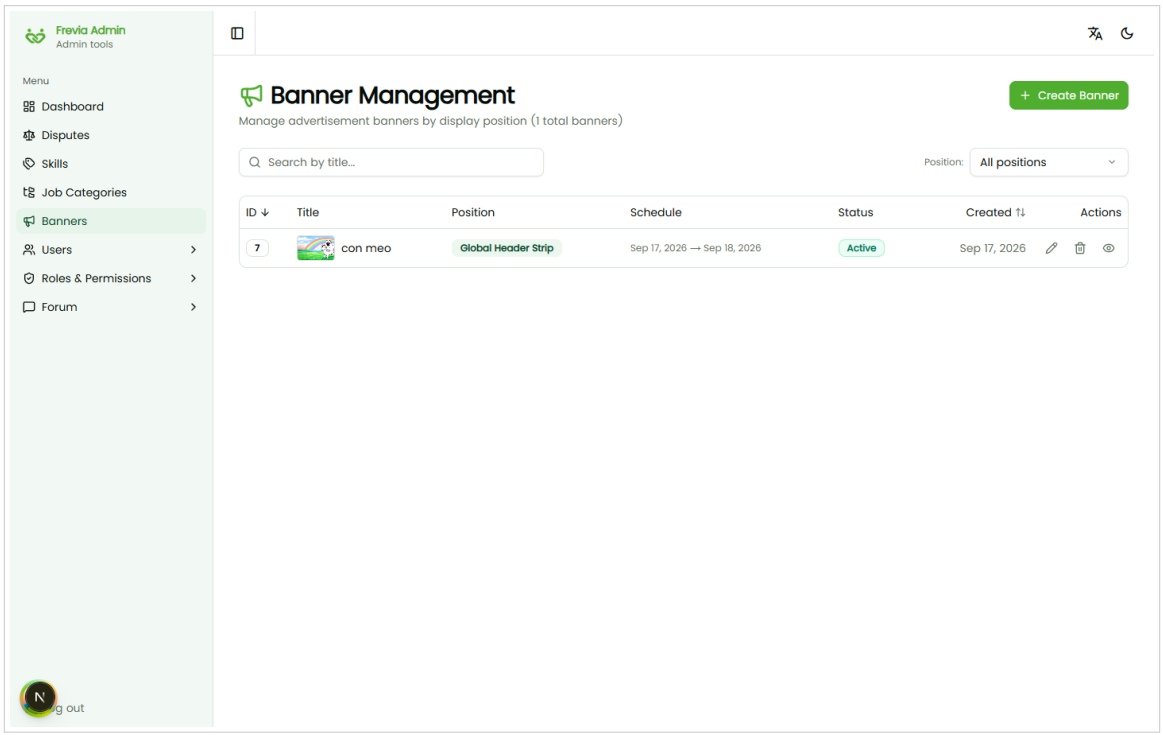


**Figure 115:** Advertisement banner list interface

- Page header displaying "Banners" title with megaphone icon, summary count, and "Create Banner" button.
- Filter bar including search box, position dropdown selector (All Positions, Global Header, Home Hero, Home Body, Search Results, Footer), and sort order controls.
- Table columns: **Banner** (Thumbnail, Title, and ID), **Position** (Colored badge), **Status** (Green "Active" or Gray "Inactive"), **Schedule** (Start date to End date), **Destination Link**, and **Actions**.

- Row-level action buttons providing quick access to view detail, edit, and delete banner.
- Pagination footer indicating total count and page navigation buttons.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| page | Target page number (defaults to 1). |
| limit | Maximum records returned per page (defaults to 10). |
| search | Optional text query matching banner title. |
| position | Optional position filter (`GLOBAL_HEADER`, `HOME_HERO`, `HOME_BODY`, `SEARCH_RESULTS`, `FOOTER`). |
| sortBy | Sort column field (`id` or `createdAt`; defaults to `id`). |
| sortOrder | Sort direction sequence (`asc` or `desc`; defaults to `desc`). |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max length 100 characters; whitespace trimmed. |
| position | Optional string, must strictly match one of the defined `BannerPositionEnum` values. |
| sortBy | Must strictly be one of: `id`, `createdAt`. |
| sortOrder | Must strictly be one of: `asc`, `desc`. |

### Business Rules

- **BR-BAN-01**: Admin Permission Verification: Only authenticated users with role `ADMIN` and matching route permission can query the administrative banner catalog.
- **BR-BAN-02**: Search and Position Filter Matching: Text search performs case-insensitive substring matching on `title`, while position filtering performs exact equality matching.
- **BR-BAN-03**: Default Pagination and Ordering: Default page size is 10 records, ordered by `id` in descending order so newly created banners appear first.

- **BR-BAN-04**: Active and Schedule Range Representation: Displays visual badges for active/inactive state along with formatted publication date intervals (`startDate` to `endDate`).
- **BR-BAN-05**: Direct Detail and Modification Actions: Each table row provides direct triggers to view detail, open update form modal, or initiate hard deletion.

**Normal Flow**

1. The administrator opens the `/admin/banners` page from the admin dashboard.
2. The client issues `GET /api/admin/banners` with query parameters (`page=1&limit=10&sortBy=id&so`
3. The backend validates administrative authentication and route permissions.
4. The repository queries advertisement banners and counts total matching records.
5. The backend returns HTTP 200 with the banners array and pagination metadata.
6. The frontend renders the banner table and pagination controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client clears authentication cookies and redirects to `/login`. |
| User lacks Admin privileges | The backend throws HTTP 403 Forbidden. The UI displays: "You do not have permission to access this page." |
| Invalid pagination parameter values | The DTO validator rejects non-integer or negative values with HTTP 400 Bad Request. The client resets to page 1. |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid position filter parameter | The validator rejects unknown position strings with HTTP 400 Bad Request. The UI clears invalid position filter. |
| Unsupported sort column parameter | The validator rejects invalid sort keys with HTTP 400 Bad Request. The client defaults to sorting by `id desc`. |
| Database query error or timeout | The service catches the Prisma error and throws `FailedToLoadBannerListException` (HTTP 500). The UI displays: "Error.FailedToLoadBanners". |
| No banners match search or position filter | The query returns `banners: []` and `total: 0`. The table displays an empty placeholder: "No banners found." |
| Network connection failure | The client displays a connection failure toast notification and offers a "Try again" button. |

**Output Data**

| Output Field | Description |
|---|---|
| banners | Array of banner objects: `id`, `title`, `imageUrl`, `linkUrl`, `position`, `startDate`, `endDate`, `isActive`, `createdAt`, `updatedAt`. |
| pagination.page | Current page number. |
| pagination.limit | Current limit per page. |
| pagination.total | Total advertisement banners matching the query. |
| pagination.totalPages | Total calculated pages available. |

**Post-Conditions**

The administrator views the paginated list of advertisement banners, inspects status and positions, and can initiate creation, inspection, modification, or deletion actions.

### 3.13.2 View Banner Detail

**Function Trigger**

An administrator clicks on a banner row or the "View detail" action on the banner list page, navigating to `/admin/banners/[id]`.
**Navigation path:** Admin Dashboard → Marketing & Banners → Banners → Banner Details (`/admin/banners/[id]`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to view comprehensive attributes and publication configuration of a specific advertisement banner, including full-size image preview, destination link, target position slot, scheduling timeline, and creation/update timestamps.

**Interface**

- Banner detail screen at `/admin/banners/[id]`.
- Back navigation button ("Back to Banners").

- Header summary displaying banner title, position badge, ID, and an “Edit Banner” action trigger.
- Banner media preview card rendering the full banner image or placeholder if no image is uploaded.
- Configuration grid displaying position slot, active status badge, target destination URL with clickable link, start date, end date, and audit timestamps.

### Data Processing Overview

- The client sends an HTTP request: `GET /api/admin/banners/:id`.
- The backend validates administrative token and route permissions.
- `BannersAdminController.getBannerDetail()` parses `id` via `ParseIntPipe`.
- `BannersAdminRepository.getBannerDetail(id)` queries `prisma.advertisementBanner.findUni` by primary key `id`.
- If the banner record is not found, the service throws `BannerAdminNotFoundException` (HTTP 404).
- If found, returns the complete banner entity with HTTP 200.
- The frontend renders the banner details layout, full image preview, and action controls.

### Screen Layout (Mock-up Description)

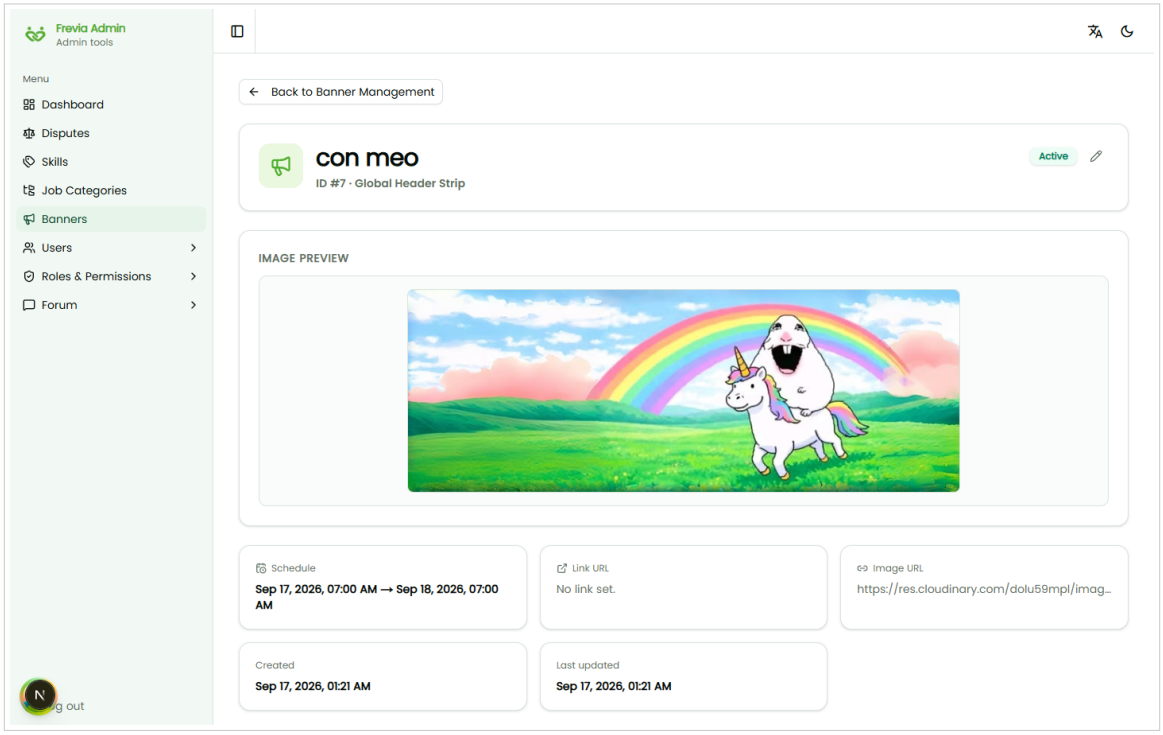


**Figure 116:** Banner detail interface

- Navigation header with “Back to Banners” button and banner title with megaphone icon.

- Action toolbar containing “Edit Banner” button triggering the edit form modal.
- Banner Image preview panel displaying uploaded advertisement graphic with aspect ratio preservation.
- Information cards: **Display Position** (e.g., Global Header), **Destination Link** (clickable external URL), **Active Status** (Green active badge), **Start Date**, **End Date**, **Created At**, and **Last Updated**.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the advertisement banner (passed in URL path). |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |

### Business Rules

- **BR-BAN-06**: Numeric Identifier Validation: The URL parameter `id` must parse strictly as a valid positive integer.
- **BR-BAN-07**: Advertisement Banner Existence: The requested banner must exist in the database; returns HTTP 404 (`Error.BannerNotFound`) if missing.
- **BR-BAN-08**: Complete Attribute Presentation: Displays all banner configuration data, responsive image asset preview, and external target link.
- **BR-BAN-09**: Direct Modification Access: Provides an “Edit Banner” trigger pre-populating current banner attributes into the form dialog.

### Normal Flow

1. The administrator navigates to `/admin/banners/[id]`.
2. The client issues `GET /api/admin/banners/:id`.
3. The backend validates administrative authentication and permissions.
4. The repository queries `prisma.advertisementBanner.findUnique( where: id )`.
5. The server returns HTTP 200 with the banner object.

6. The frontend renders the comprehensive banner detail page.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The server throws HTTP 403 Forbidden. The client renders an access denied alert. |
| Invalid banner ID format (e.g., non-numeric) | `ParseIntPipe` throws HTTP 400 Bad Request. The UI displays an "Invalid Banner ID" message and back button. |
| Banner ID does not exist in database | The service throws `BannerAdminNotFoundException` (HTTP 404). The UI renders a "Banner Not Found" screen with navigation back to list. |
| Database query error or timeout | The service throws `FailedToLoadBannerDetailException` (HTTP 500). The UI displays: "Error.FailedToLoadBannerDetail". |
| Destination link is empty or null | The UI gracefully displays "No destination link configured" in muted text instead of a broken link. |

| Abnorma Case | Handling & System Response |
|---|---|
| Image asset URL broken or unreachable | The UI falls back to rendering a placeholder graphic container with a broken image indicator. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric identifier of the banner. |
| title | Banner title string. |
| imageUrl | Absolute image URL or relative path on server. |
| linkUrl | External or internal redirect target URL. |
| position | Display slot enum value. |
| startDate | Scheduled activation timestamp or null. |
| endDate | Scheduled deactivation timestamp or null. |
| isActive | Boolean indicating whether banner is enabled. |
| createdAt | Timestamp when banner was created. |
| updatedAt | Timestamp of last modification. |

**Post-Conditions**

The administrator reviews complete banner configuration, image assets, schedule boundaries, and can initiate editing.

### 3.13.3 Create Banner

**Function Trigger**

An administrator clicks the "Create Banner" button on the banner list page (`/admin/banners`), opening the banner creation dialog modal.

**Navigation path:** Admin Dashboard → Marketing & Banners → Banners → [Create Banner Button].

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Cloudinary media storage service, Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to register a new advertisement banner, specifying title, position slot, optional graphic image upload, target redirect link, publication date range, and initial active status.

**Interface**

- Banner form dialog modal (`BannerFormDialog`).
- Text input for Title (mandatory).
- Dropdown selector for Position slot (`GLOBAL_HEADER`, `HOME_HERO`, `HOME_BODY`, `SEARCH_RESULTS`, `FOOTER`).
- File upload input with preview for banner image (supports drag-and-drop or file browser, max 10MB).
- Text input for Destination Link URL.
- Date picker inputs for Start Date and End Date.
- Checkbox toggle for "Active" status (defaults to checked).
- "Cancel" and "Create Banner" submission buttons with loading indicator.

**Data Processing Overview**

- If a local image file is chosen, the client first issues a multipart/form-data request: `POST /api/admin/banners/upload-image`.
  - The service validates file presence, MIME type (`image/jpeg`, `image/png`, `image/gif`, `image/webp`, `image/bmp`, `image/tiff`), and file size ($\leq$ 10MB).
  - If Cloudinary is configured, uploads file to Cloudinary under folder `frevia/banners` and obtains `secure_url`; otherwise saves file to local directory `uploads/banners/`.
  - Returns `{ imageUrl, publicId }`.
- The client issues an HTTP request: `POST /api/admin/banners` with JSON payload conforming to `BannerCreateBodySchema`.
- `BannersAdminRepository.createBanner()` checks whether an active banner already occupies the selected position slot (`prisma.advertisementBanner.findFirst(`

`where: position )`).

- If position is already occupied, the service throws `BannerPositionTakenException` (HTTP 409).
- The repository inserts the new banner record via `prisma.advertisementBanner.create()`.
- The backend returns HTTP 201 with the created banner entity.
- The frontend shows a success toast notification, closes the dialog modal, and refreshes the banner list table.

### Screen Layout (Mock-up Description)

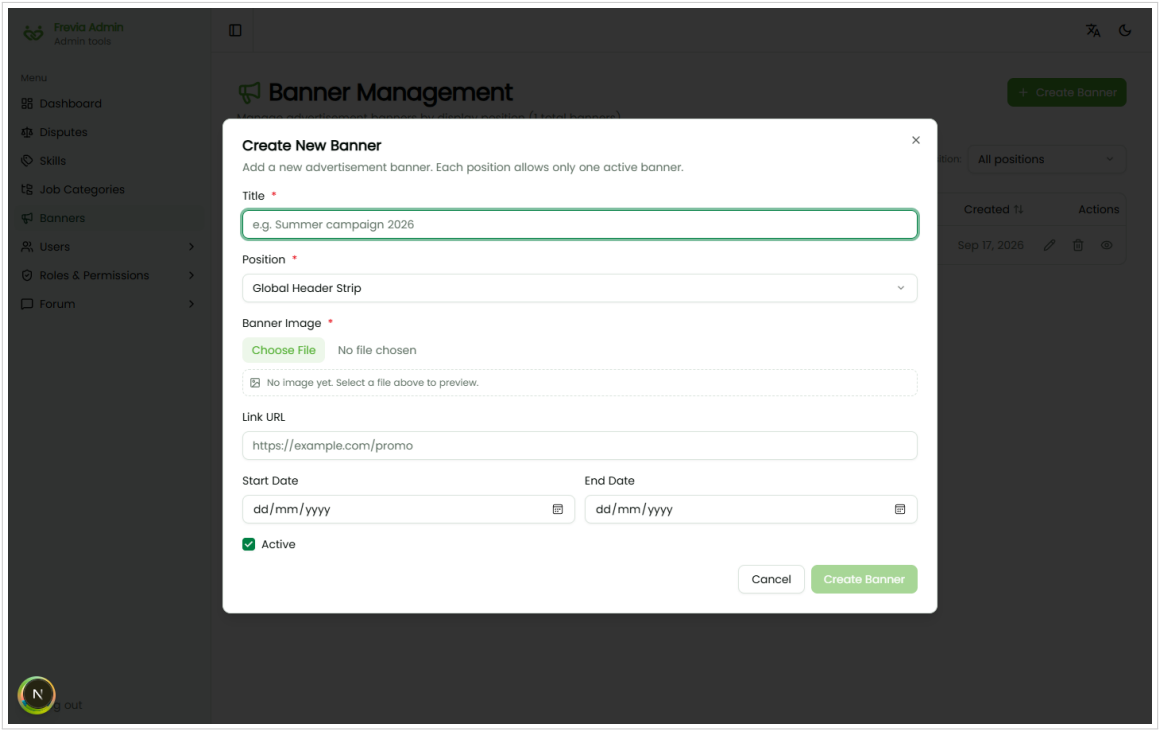


**Figure 117:** Create banner modal interface

- Dialog header with "Create Banner" title and brief descriptive subtitle.
- Form fields: **Title** (Required text input), **Position** (Select dropdown: Global Header, Home Hero, Home Body, Search Results, Footer), **Banner Image** (Upload zone with image preview and remove button), **Destination Link** (URL input), **Schedule** (Start Date and End Date date pickers), and **Active** (Checkbox toggle).
- Dialog footer with "Cancel" button and "Create Banner" submit button with loading spinner.

### Function Details

### Input Data

| Data Name | Description |
| --- | --- |
| title | Banner title string (mandatory). |
| imageUrl | Direct image URL string or uploaded file URL (optional). |
| file | Binary image file for upload (optional, multipart/form-data). |
| linkUrl | Target destination URL string (optional). |
| position | Target display position slot enum (defaults to `GLOBAL_HEADER`). |
| startDate | Scheduled start publication date (optional). |
| endDate | Scheduled end publication date (optional). |
| isActive | Boolean activation flag (defaults to `true`). |

### Validation Rules

| Data Name | Validation Rule |
| --- | --- |
| title | String, trimmed, min 1 char (`Error.BannerTitleRequired`), max 255 chars (`Error.BannerTitleTooLong`). |
| imageUrl | Optional string, must be valid URL format, max 500 chars (`Error.BannerImageUrlTooLong`). |
| file | Must be valid image MIME type; size must not exceed 10MB (`Error.BannerImageTooLarge`). |
| linkUrl | Optional string, must be valid URL format, max 500 chars (`Error.BannerLinkUrlTooLong`). |
| position | Must match one of `BannerPositionEnum` values (defaults to `GLOBAL_HEADER`). |
| startDate, endDate | Coerced to valid Date. Refinement: if both present, `startDate <= endDate` (`Error.BannerInvalidDateRange`). |
| isActive | Boolean value (defaults to `true`). |

### Business Rules

- **BR-BAN-10**: Single Active Banner Per Position: Each position slot can only be occupied by one banner; if the position is already taken, creation is rejected with HTTP 409 (`Error.BannerPositionTaken`).
- **BR-BAN-11**: Mandatory Title Trimming: Title is mandatory and leading/trailing whitespace is stripped before validation and database persistence.
- **BR-BAN-12**: Target and Image URL Validation: If provided, URLs must be strictly valid URI formats with a maximum length of 500 characters.
- **BR-BAN-13**: Valid Schedule Range: When both dates are specified, `startDate` must be earlier than or equal to `endDate`.
- **BR-BAN-14**: Image Upload Constraints: Image uploads are strictly limited to valid image MIME formats and maximum size of 10MB.
- **BR-BAN-15**: Auto-refresh and Feedback: Shows a success toast notification and refreshes the banner catalog upon successful creation.

**Normal Flow**

1. The administrator clicks "Create Banner" on the banner management page.
2. The dialog modal opens with an empty form and default position (`GLOBAL_HEADER`).
3. The administrator fills in the title, selects a position slot, optionally attaches an image file, enters a link URL, selects date bounds, and checks "Active".
4. If a file was attached, the client uploads it to `POST /api/admin/banners/upload-image` and receives the image URL.
5. The client submits the form data via `POST /api/admin/banners`.
6. The backend validates the payload and verifies the position slot is available.
7. The repository inserts the new banner record into the database.
8. The backend returns HTTP 201 with the created banner data.
9. The frontend closes the modal, displays a success toast, and refreshes the banner list.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing mandatory title | The form prevents submission; backend returns HTTP 400 Bad Request with `Error.BannerTitleRequired`. |

| Abnorma Case | Handling & System Response |
|---|---|
| Selected position already occupied | The service throws `BannerPositionTakenException` (HTTP 409 Conflict). The UI displays: “Error.BannerPositionTaken”. |
| Uploaded file exceeds 10MB | The upload endpoint throws `BannerImageTooLargeException` (HTTP 400). The UI displays: “Error.BannerImageTooLarge”. |
| Uploaded file is not a supported image format | The upload endpoint throws `BannerImageInvalidTypeException` (HTTP 400). The UI displays: “Error.BannerImageInvalidType”. |
| End date is earlier than start date | Zod refinement rejects payload with HTTP 400 Bad Request (`Error.BannerInvalidDateRange`). The UI flags the End Date field. |
| Invalid URL syntax in image or link URL | Validator returns HTTP 400 Bad Request. The UI highlights the offending URL input field. |
| Unauthorized or unauthenticated request | The server returns HTTP 401/403. The UI redirects to login or displays an access denied message. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database insert failure | The service throws `FailedToCreateBannerException` (HTTP 500). The UI displays: “Error.FailedToCreateBanner”. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Auto-generated unique identifier of the created banner. |
| title | Stored banner title string. |
| imageUrl | Image asset URL or null. |
| linkUrl | Destination URL or null. |
| position | Assigned position slot enum. |
| startDate | Activation date or null. |
| endDate | Deactivation date or null. |
| isActive | Active boolean status. |
| createdAt | Timestamp of creation. |
| updatedAt | Timestamp of last update. |

**Post-Conditions**

A new advertisement banner is stored in the database, assigned to the selected position slot, and immediately rendered on the admin table.

### 3.13.4 Update Banner

**Function Trigger**

An administrator clicks the Edit icon in the banner table row or the “Edit Banner” button on the banner detail page, opening the edit dialog modal.

**Navigation path:** Admin Dashboard → Marketing & Banners → Banners → [Edit Action Button].

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Cloudinary media storage service, Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to modify an existing advertisement banner's title, display position, image graphic, destination link, validity period, and active status.

### Interface

- Banner form dialog modal (`BannerFormDialog`) pre-populated with existing banner data.
- Text input for Title.
- Select dropdown for Position slot.
- File upload input with current image preview and remove button.
- Text input for Destination Link URL.
- Date pickers for Start Date and End Date.
- Checkbox toggle for "Active" status.
- "Cancel" and "Save Changes" action buttons.

### Data Processing Overview

- The client pre-fills the form with existing banner fields.
- If a new replacement image is selected, the client uploads it via `POST /api/admin/banners/upload-ima` before submitting changes.
- The client issues an HTTP request: `PATCH /api/admin/banners/:id` with updated attributes.
- `BannersAdminRepository.updateBanner(id, data)`:
  - Verifies target banner exists; throws `BannerAdminNotFoundException` if not found.
  - If `position` is changed, checks if another banner already occupies the new position slot (`id != currentId`). Throws `BannerPositionTakenException` if taken.
  - Updates modified fields in `prisma.advertisementBanner.update()`.
- The backend returns HTTP 200 with the updated banner entity.
- The frontend displays a success toast, closes the dialog, and refreshes the banner list or detail screen.

### Screen Layout (Mock-up Description)

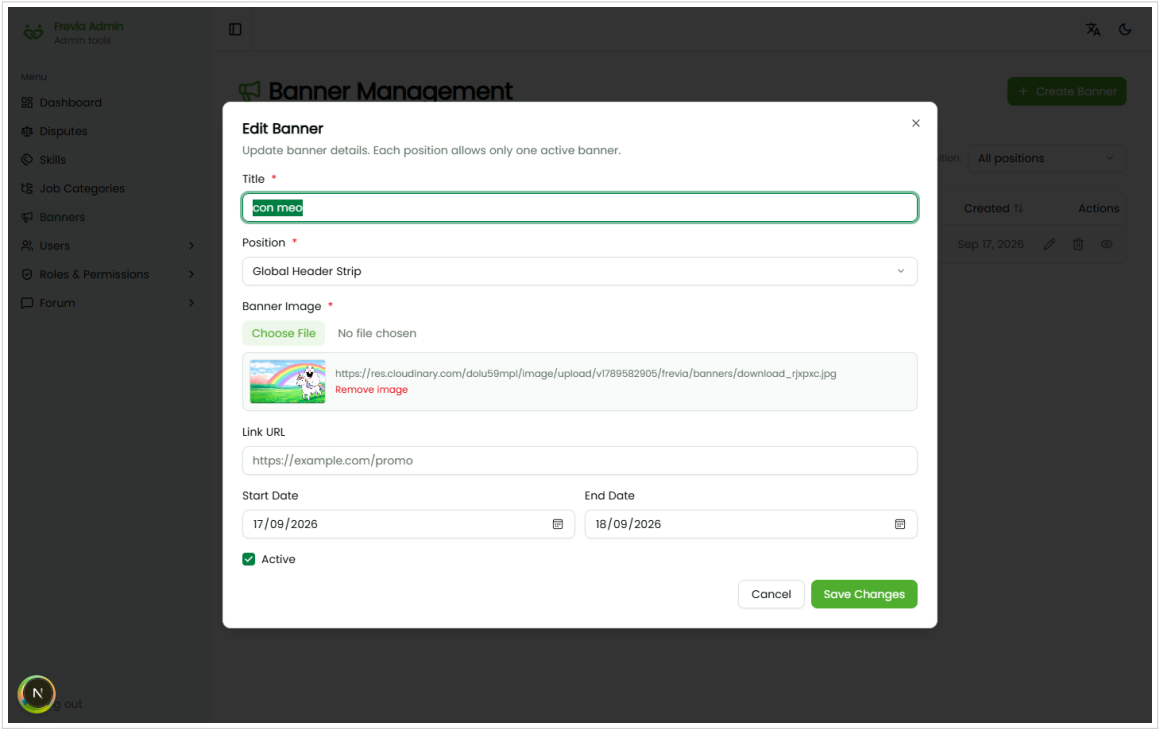


**Figure 118:** Update banner modal interface

- Modal dialog header with "Edit Banner" title.
- Form pre-populated with current values: **Title**, **Position**, **Banner Image** (with existing preview), **Destination Link**, **Schedule** (Start/End dates), and **Active** toggle.
- Footer buttons: "Cancel" and "Save Changes" button with pending spinner.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the banner (passed in URL path). |
| title | Updated banner title string (optional). |
| imageUrl | Updated image URL or null (optional). |
| file | Binary replacement image file (optional). |
| linkUrl | Updated destination URL or null (optional). |
| position | Updated position slot enum (optional). |
| startDate | Updated start publication date or null (optional). |

| Data Name | Description |
|---|---|
| endDate | Updated end publication date or null (optional). |
| isActive | Updated active boolean flag (optional). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |
| title | Optional string, trimmed, min 1 char, max 255 chars (`Error.BannerTitleTooLong`). |
| imageUrl | Optional string, must be valid URL format, max 500 chars. |
| file | Must be valid image MIME type; size must not exceed 10MB. |
| linkUrl | Optional string, must be valid URL format, max 500 chars. |
| position | Must match one of `BannerPositionEnum` values. |
| startDate, endDate | Coerced to valid Date. Refinement: `startDate <= endDate` (`Error.BannerInvalidDateRange`). |
| isActive | Optional boolean value. |

**Business Rules**

- **BR-BAN-16**: Target Banner Existence: The targeted banner ID must exist in the database; returns HTTP 404 (`Error.BannerNotFound`) if not found.
- **BR-BAN-17**: Position Conflict Protection on Update: If updating the `position` slot, the new position must not be occupied by another banner; excludes current banner ID.
- **BR-BAN-18**: Date Range Consistency: If both dates are present after update, `startDate` must be earlier than or equal to `endDate`.
- **BR-BAN-19**: Partial Update Preservation: Unchanged fields retain their current values in the database.
- **BR-BAN-20**: Cache Invalidation and UI Synchronization: Immediately invalidates cached queries and triggers view refresh upon successful update.

**Normal Flow**

1. The administrator clicks the Edit button on a banner row or detail view.
2. The dialog modal opens pre-filled with the banner's current data.

3. The administrator updates one or more fields (e.g., changes destination link or toggles active status).
4. The administrator clicks “Save Changes”.
5. If a replacement image was uploaded, it is uploaded to Cloudinary/local storage first.
6. The client sends `PATCH /api/admin/banners/:id` with updated attributes.
7. The backend validates data, checks for position conflicts, and updates the record.
8. The server returns HTTP 200 with the updated banner object.
9. The frontend closes the dialog, shows a success toast, and refreshes the view.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target banner does not exist | The service throws `BannerAdminNotFoundException` (HTTP 404). The UI displays: “Error.BannerNotFound”. |
| New position slot already occupied | The service throws `BannerPositionTakenException` (HTTP 409). The UI alerts: “Error.BannerPositionTaken”. |
| End date earlier than start date | Validator rejects payload with HTTP 400 Bad Request (`Error.BannerInvalidDateRange`). The UI highlights the date field. |
| Replacement image exceeds 10MB | Upload endpoint throws `BannerImageTooLargeException` (HTTP 400). The UI displays: “Error.BannerImageTooLarge”. |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid URL syntax in linkUrl | Validator returns HTTP 400 Bad Request. The UI highlights the link URL field. |
| Unauthorized or unauthenticated session | The server returns HTTP 401/403. The UI redirects to login or displays an access denied message. |
| Database update failure | The service throws `FailedToUpdateBannerException` (HTTP 500). The UI displays: “Error.FailedToUpdateBanner”. |
| Network failure during submission | The client displays an error toast notification and leaves the modal open for retry. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the updated banner. |
| title | Updated banner title. |
| imageUrl | Updated image URL or null. |
| linkUrl | Updated destination URL or null. |
| position | Updated position slot enum. |
| startDate | Updated start publication date or null. |
| endDate | Updated end publication date or null. |
| isActive | Updated active status flag. |
| updatedAt | Updated modification timestamp. |

#### Post-Conditions

The modified banner attributes are committed to the database and immediately reflected across the administrative console.

### 3.13.5 Delete Banner

#### Function Trigger

An administrator clicks the Delete action icon (trash can) on a banner table row on the banner list page, triggering the deletion confirmation alert dialog.
**Navigation path:** Admin Dashboard → Marketing & Banners → Banners → [Delete Action Icon].

#### Function Description

#### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

#### Purpose

Allows administrators to permanently remove an obsolete or unwanted advertisement banner from the system via hard deletion.

#### Interface

- Delete banner confirmation dialog (`DeleteBannerDialog`) powered by Shadcn `AlertDialog`.
- Modal header with warning title displaying the targeted banner title.
- Descriptive body explaining that this action will permanently remove the banner.
- Modal action buttons: "Cancel" and destructive "Delete" button with pending spinner.

#### Data Processing Overview

- The administrator confirms the deletion in the modal dialog.
- The client sends an HTTP request: `DELETE /api/admin/banners/:id`.
- `BannersAdminController.deleteBanner(id)` parses `id` via `ParseIntPipe`.
- `BannersAdminRepository.deleteBanner(id)` executes `prisma.advertisementBanner.deleteMan where: id ).`
- If `result.count === 0`, the service throws `BannerAdminNotFoundException` (HTTP

404).

- Returns HTTP 200 with `{ message: "Success.BannerDeleted" }`.
- The frontend displays a success toast, closes the confirmation dialog, and triggers `router.refresh()` to remove the row from the table.

### Screen Layout (Mock-up Description)

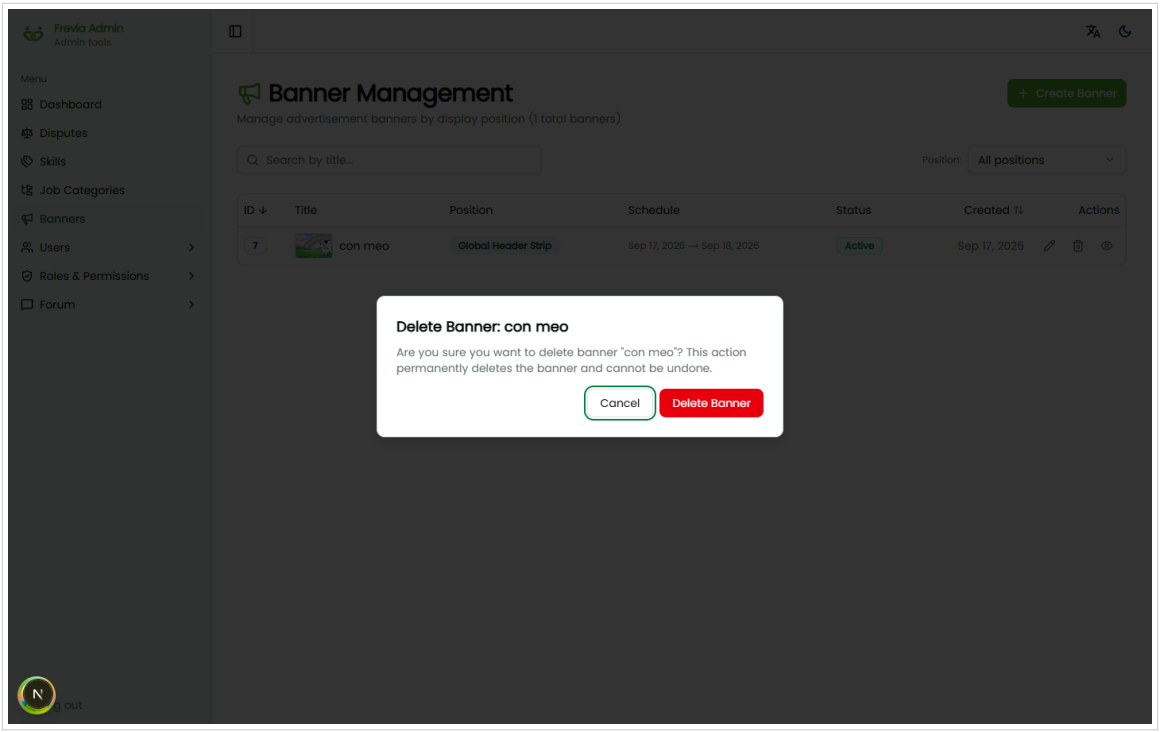


**Figure 119:** Delete banner confirmation dialog

- Alert dialog displaying “Delete Banner” title including target banner name.
- Warning explanation text: “Are you sure you want to delete this banner? This action cannot be undone.”
- Footer controls: “Cancel” button and red “Delete” button with loading spinner when executing.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the banner to delete (passed in URL path). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |

**Business Rules**

- **BR-BAN-21**: Permanent Hard Deletion: Advertisement banners are permanently removed from the `advertisement_banners` table via hard deletion.
- **BR-BAN-22**: Existence Verification: Throws HTTP 404 (`Error.BannerNotFound`) if the banner record does not exist or was already removed.
- **BR-BAN-23**: Explicit Confirmation Gate: Destruction requires explicit confirmation from the administrator via modal alert to prevent accidental deletion.
- **BR-BAN-24**: Immediate List Synchronization: The deleted banner row is purged from the UI table immediately upon successful response.

**Normal Flow**

1. The administrator locates the banner in the table and clicks the Delete icon.
2. The deletion confirmation dialog modal appears.
3. The administrator reviews the banner title and confirms by clicking “Delete”.
4. The client sends `DELETE /api/admin/banners/:id`.
5. The backend validates administrative token and route permissions.
6. The repository deletes the record from `advertisementBanner`.
7. The backend returns HTTP 200 with `{ message: "Success.BannerDeleted" }`.
8. The client displays a success toast, closes the modal, and refreshes the banner list table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target banner does not exist | The service throws `BannerAdminNotFoundException` (HTTP 404). The UI displays: “Error.BannerNotFound”. |
| Missing or invalid authen-tication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privi-leges | The server throws HTTP 403 Forbidden. The UI displays an access denied notification. |
| Invalid banner ID format | `ParseIntPipe` throws HTTP 400 Bad Request. The UI notifies the user of an invalid parameter. |
| Administrator cancels dialog | The modal closes immediately; no request is sent, leaving the database unaffected. |
| Database deletion failure | The service throws `FailedToDeleteBannerException` (HTTP 500). The UI displays: “Error.FailedToDeleteBanner”. |
| Network failure during request | The client catches the network error, displays an unexpected error toast, and leaves the modal open. |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string (`Success.BannerDeleted`). |

**Post-Conditions**

The banner record is permanently deleted from the database and removed from public and administrative views.

### 3.14 Skill Management

#### 3.14.1 View Skill List

**Function Trigger**

An administrator navigates to the Skills management page from the admin dashboard sidebar navigation.
**Navigation path:** Admin Dashboard → Master Catalog → Skills (`/admin/skills`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to view, search, filter, and sort the master catalog of professional skills used by freelancer profiles and job postings. The catalog displays skill name, generated unique slug, description, active job count, soft-delete state, and audit dates.

**Interface**

- Skill management screen at `/admin/skills`.
- Filter bar component (`SkillsFilterBar`) with search input, status filter dropdown (All, Active Only, Deleted Only), sort column dropdown, and sort direction toggle.
- Data table component (`SkillsTable`) rendering skill name, slug, description, job usage count badge, status badge, creation date, and contextual action menus.
- Standard pagination footer displaying total skills and page selection buttons.
- "Create Skill" action button in the page header.

### Data Processing Overview

- The client initiates an HTTP request: `GET /api/admin/skills` with query parameters (`page`, `limit`, `search`, `deleted`, `sortBy`, `sortOrder`).
- The backend `AuthGuard` and `PermissionGuard` verify the administrator's credentials and permissions.
- `SkillsAdminController.listSkills()` parses query parameters via `SkillAdminQueryDto`.
- `SkillsAdminRepository.listSkills()` queries `prisma.skill.findMany`:
  - If `search` is provided, executes case-insensitive `OR` query matching on `name`, `slug`, or `description`.
  - If `deleted === 'true'`, filters records where `deletedAt != null`; if `deleted === 'false'`, filters records where `deletedAt = null`.
  - Includes count of related jobs via `_count: { select: { jobs: true } }`.
  - Sorts records by `id` or `createdAt` in requested direction.
- Executes `prisma.skill.count()` in parallel to determine total matching records.
- Maps database records into response objects including `jobCount`.
- The backend returns HTTP 200 with the skills array and pagination metadata.
- The frontend renders the data table, applies green active badges or red deleted badges, and provides action buttons.

### Screen Layout (Mock-up Description)

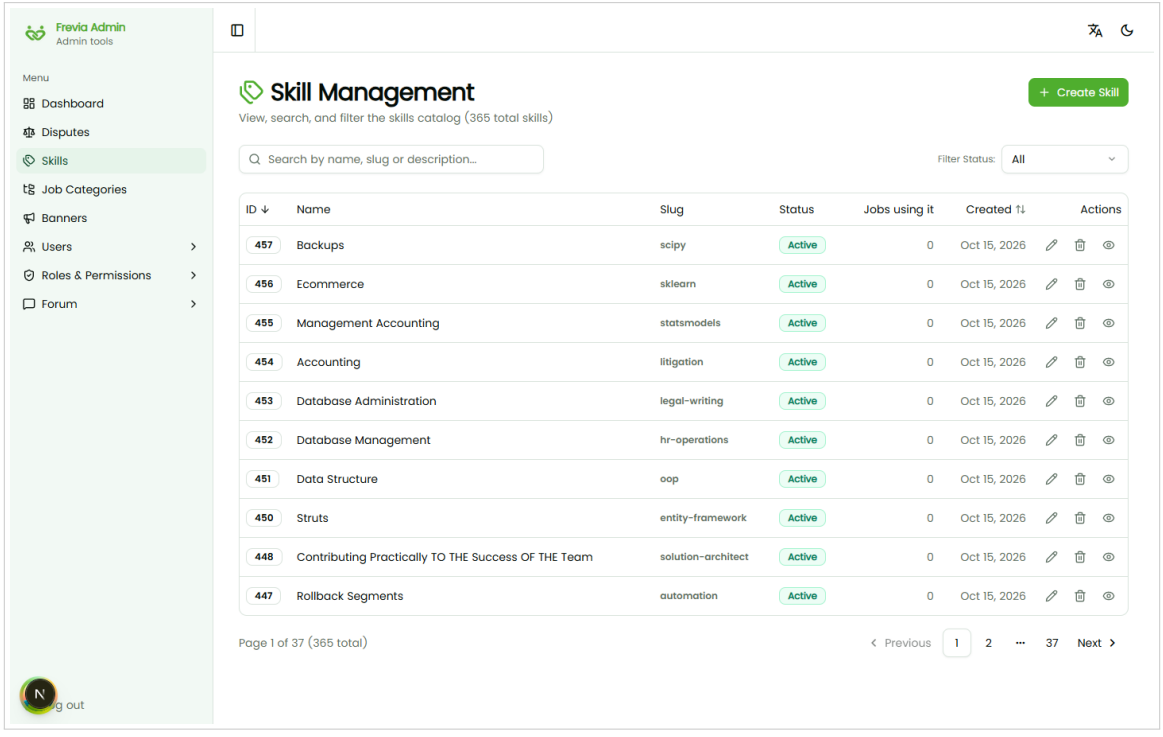


**Figure 120:** Skill catalog list interface

- Page header displaying “Skills” title with tag icon, summary count of total skills, and “Create Skill” button.
- Filter bar including search box, deleted status filter selector (All, Active, Deleted), and sort direction buttons.
- Table columns: **Skill Name** (with clickable detail link), **Slug**, **Description**, **In Use** (Badge showing number of jobs using this skill), **Status** (Active or Deleted badge), **Created Date**, and **Actions**.
- Contextual action buttons: Update (pencil) and Delete (trash) for active skills; Restore (rotate-ccw) for deleted skills.
- Pagination footer indicating total count and page navigation buttons.

### Function Details

#### Input Data

| Data Name | Description |
|---|---|
| page | Target page number (defaults to 1). |
| limit | Maximum records returned per page (defaults to 10). |
| search | Optional text query matching name, slug, or description. |
| deleted | Optional status filter: `"true"` (deleted only) or `"false"` (active only). |
| sortBy | Sort column field (`id` or `createdAt`; defaults to `id`). |
| sortOrder | Sort direction sequence (`asc` or `desc`; defaults to `desc`). |

#### Validation Rules

| Data Name | Validation Rule |
|---|---|
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max length 100 characters; whitespace trimmed. |
| deleted | Optional string, must strictly match one of: `"true"`, `"false"`. |
| sortBy | Must strictly be one of: `id`, `createdAt`. |
| sortOrder | Must strictly be one of: `asc`, `desc`. |

---

**Business Rules**

- **BR-SKL-01**: Admin Authorization: Access to the master skill catalog is strictly restricted to authenticated administrators.
- **BR-SKL-02**: Substring Multi-Field Search: Search performs case-insensitive substring matching across `name`, `slug`, and `description`.
- **BR-SKL-03**: Soft-Deletion State Filtering: Setting `deleted=true` filters for `deletedAt != null`, `deleted=false` filters for `deletedAt = null`, and omitting it lists all skills.
- **BR-SKL-04**: Associated Job Count Aggregation: Automatically aggregates and displays the exact count of job postings utilizing each skill via `_count.jobs`.
- **BR-SKL-05**: Dynamic Status Badge and Action Switching: Active skills display green badges with Edit/Delete buttons; soft-deleted skills display muted badges with a Restore button.

**Normal Flow**

1. The administrator opens the `/admin/skills` page from the admin dashboard.
2. The client issues `GET /api/admin/skills` with default query parameters.
3. The backend validates administrative token and route permissions.
4. The repository queries master skills with job count aggregation.
5. The server returns HTTP 200 with the skills array and pagination metadata.
6. The frontend renders the skill table and pagination controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authen-tication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |

| Abnorma Case | Handling & System Response |
|---|---|
| User lacks Admin privi-leges | The backend throws HTTP 403 Forbidden. The UI displays: “You do not have permission to access this page.” |
| Invalid pagina-tion parame-ter values | The DTO validator rejects non-integer values with HTTP 400 Bad Request. The client defaults to page 1. |
| Invalid deleted filter pa-rameter value | The validator rejects unknown values with HTTP 400 Bad Request. The UI clears invalid filter. |
| Unsupported sort column parame-ter | The validator rejects invalid sort keys with HTTP 400 Bad Request. The client defaults to sorting by `id desc`. |
| Database query error or timeout | The service throws `FailedToLoadSkillListException` (HTTP 500). The UI displays: “Error.FailedToLoadSkillList”. |
| No skills match search or status filter | The query returns `skills: []` and `total: 0`. The table displays an empty placeholder: “No skills found.” |

| Abnorma Case | Handling & System Response |
|---|---|
| Network connection timeout | The client displays an error toast notification and offers a “Try again” button. |

**Output Data**

| Output Field | Description |
|---|---|
| skills | Array of skill objects: `id`, `name`, `slug`, `description`, `deletedAt`, `createdAt`, `updatedAt`, `jobCount`. |
| pagination.page | Current page number. |
| pagination.limit | Current limit per page. |
| pagination.total | Total skills matching the query. |
| pagination.totalPages | Total calculated pages available. |

**Post-Conditions**

The administrator inspects the master skill catalog, views job usage metrics, and can initiate creation, editing, deletion, or restoration.

### 3.14.2 View Skill Detail

**Function Trigger**

An administrator clicks on a skill row or name in the skill list, navigating to `/admin/skills/[id]`.

**Navigation path:** Admin Dashboard → Master Catalog → Skills → Skill Details (`/admin/skills/[id]`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to view detailed metadata of a specific skill, including its generated slug, full description, total count of linked jobs, deletion status, and audit timestamps.

### Interface

- Skill detail screen at `/admin/skills/[id]`.
- Navigation back button ("Back to Skills").
- Header card displaying skill name, slug, ID, and action buttons (`SkillActions`).
- Metrics card displaying the number of active jobs utilizing this skill.
- Description card displaying the full skill description or a placeholder if empty.
- Lifecycle timestamps card (Created date, Last updated date, and Deletion date if soft-deleted).

### Data Processing Overview

- The client sends an HTTP request: `GET /api/admin/skills/:id`.
- The backend validates administrative token and route permissions.
- `SkillsAdminController.getSkillDetail()` parses `id` via `ParseIntPipe`.
- `SkillsAdminRepository.getSkillDetail(id)` queries `prisma.skill.findUnique` by `id`, including `_count: { select: { jobs: true } }`.
- If the skill record is not found, the service throws `SkillAdminNotFoundException` (HTTP 404).
- Returns HTTP 200 with the skill object including `jobCount`.
- The frontend renders the skill details view and contextual action buttons based on whether `deletedAt` is null.

### Screen Layout (Mock-up Description)

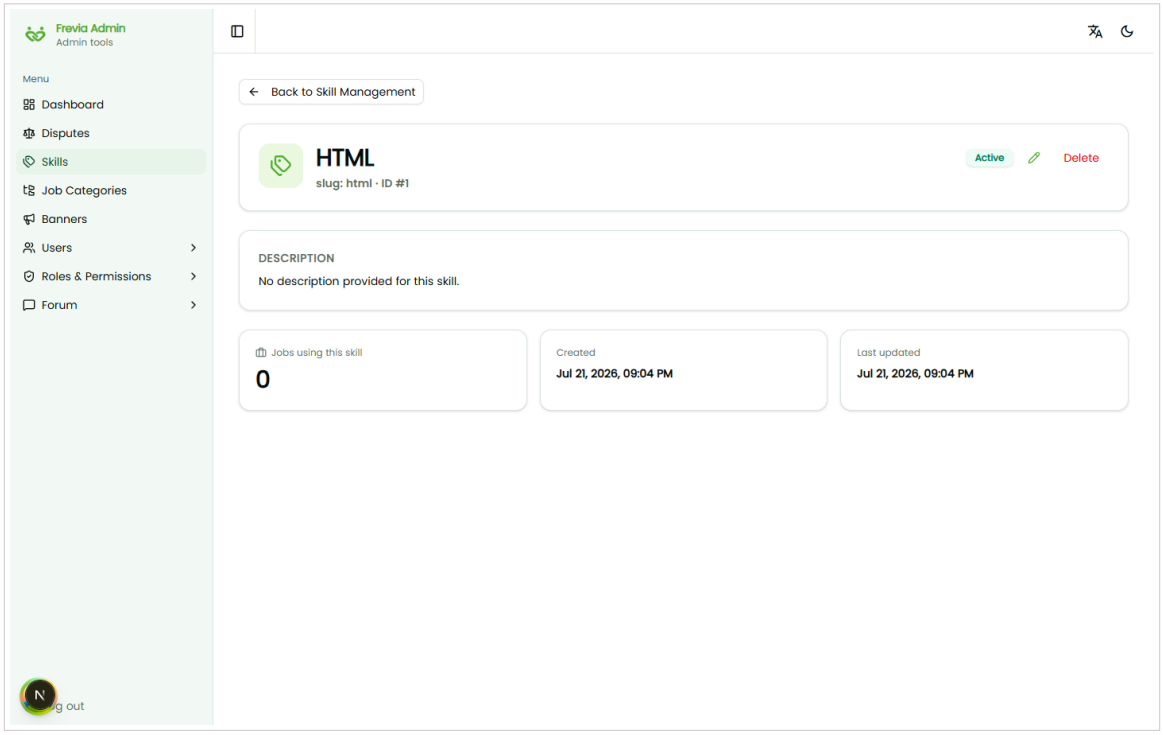


**Figure 121:** Skill detail interface

- Navigation header with “Back to Skills” button and skill name with tag icon.
- Action toolbar containing Edit (pencil) and Delete (trash) buttons if active, or Restore (rotate-ccw) button if soft-deleted.
- Summary card showing: **Slug** (e.g., `reactjs`), **Status** (Active or Deleted badge), and **ID**.
- Usage metrics card displaying **Active Jobs Count** with a briefcase icon.
- Content card displaying the **Description** of the skill.
- Audit card displaying **Created Date**, **Last Updated Date**, and **Deleted At** (if applicable).

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the skill (passed in URL path). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |

**Business Rules**

- **BR-SKL-06**: Numeric Skill ID Validation: The URL parameter `id` must parse strictly as a valid positive integer.
- **BR-SKL-07**: Skill Record Existence: The requested skill must exist in the database; returns HTTP 404 (`Error.SkillNotFound`) if missing.
- **BR-SKL-08**: Complete Detail and Usage Metrics: Displays name, generated slug, description, creation date, update date, deletion timestamp, and total jobs utilizing this skill.
- **BR-SKL-09**: Contextual Action Toolbar: Shows Edit and Delete actions if skill is active; shows Restore action if skill is soft-deleted.

**Normal Flow**

1. The administrator navigates to `/admin/skills/[id]`.
2. The client issues `GET /api/admin/skills/:id`.
3. The backend validates administrative authentication and permissions.
4. The repository queries `prisma.skill.findUnique` with job count aggregation.
5. The server returns HTTP 200 with the skill object.
6. The frontend renders the comprehensive skill detail view.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The server throws HTTP 403 Forbidden. The client renders an access denied alert. |
| Invalid skill ID format (e.g., non-numeric) | `ParseIntPipe` throws HTTP 400 Bad Request. The UI displays an "Invalid Skill ID" message and back button. |
| Skill ID does not exist in database | The service throws `SkillAdminNotFoundException` (HTTP 404). The UI renders a "Skill Not Found" screen with navigation back to catalog. |
| Database query error or timeout | The service throws `FailedToLoadSkillDetailException` (HTTP 500). The UI displays: "Error.FailedToLoadSkillDetail". |
| Skill description is empty or null | The UI gracefully displays "No description provided" in muted text. |
| Network connection failure | The client displays an error toast notification and offers a reload button. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric identifier of the skill. |
| name | Skill name string. |
| slug | Generated URL-friendly slug. |
| description | Detailed description or null. |
| deletedAt | Soft-deletion timestamp or null. |
| createdAt | Timestamp of skill creation. |
| updatedAt | Timestamp of last modification. |
| jobCount | Total number of jobs referencing this skill. |

**Post-Conditions**

The administrator reviews skill details and usage metrics, and can initiate editing, deletion, or restoration.

### 3.14.3 Create Skill

**Function Trigger**

An administrator clicks the "Create Skill" button on the skill catalog page (`/admin/skills`), opening the skill creation dialog modal.
**Navigation path:** Admin Dashboard → Master Catalog → Skills → [Create Skill Button].

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to register a new professional skill in the master catalog by providing its name and optional description, with automated unique slug generation.

**Interface**

- Skill creation dialog modal (`CreateSkillDialog`).
- Text input for Skill Name (mandatory).

- Textarea input for Description (optional).
- Action buttons: “Cancel” and “Create Skill” submit button with loading indicator.

**Data Processing Overview**

- The client submits an HTTP request: `POST /api/admin/skills` with JSON payload conforming to `AdminCreateSkillBodySchema`.
- `SkillsAdminRepository.createSkill()`:
  - Strips leading and trailing whitespace from `name`.
  - Queries `prisma.skill.findFirst` to check if an active skill (`deletedAt = null`) with the same name exists (case-insensitive mode).
  - If an active duplicate exists, throws `SkillNameAlreadyExistsException` (HTTP 409).
  - Generates a unique slug via `generateUniqueSlug()`: normalizes Vietnamese diacritics, converts to lowercase, replaces non-alphanumeric characters with hyphens, and appends sequential numeric suffixes (`-2`, `-3`) if a slug collision occurs.
  - Inserts the skill record via `prisma.skill.create()`.
  - Catches Prisma `P2002` errors (unique constraint violations) and converts to `SkillNameAlreadyExistsException`.
- Returns HTTP 201 with the created skill entity.
- The frontend displays a success toast, closes the modal, and refreshes the skill table.

### Screen Layout (Mock-up Description)

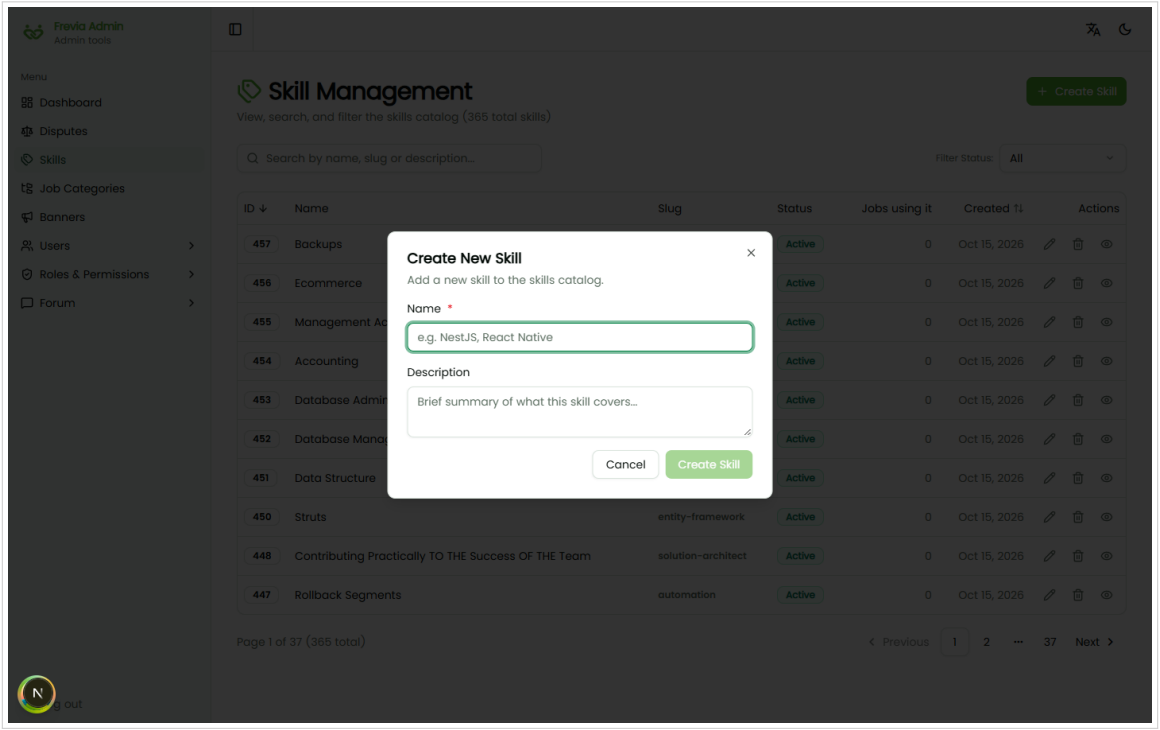


**Figure 122:** Create skill modal interface

- Dialog header with “Create Skill” title and descriptive prompt.
- Form fields: **Name** (Required input field with asterisk) and **Description** (Multi-line textarea).
- Dialog footer with “Cancel” button and green “Create Skill” button with loading spinner when submitting.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| name | Skill name string (mandatory). |
| description | Optional text description of the skill. |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| name | String, trimmed, min 1 char (`Error.SkillNameRequired`), max 100 chars (`Error.SkillNameTooLong`). |
| description | Optional string, max 500 chars (`Error.SkillDescriptionTooLong`); nullable. |

**Business Rules**

- **BR-SKL-10**: Case-Insensitive Name Uniqueness: Skill name must be unique among active skills; throws HTTP 409 (`Error.SkillNameAlreadyExists`) if conflict occurs.
- **BR-SKL-11**: Automated Unique Slug Generation: Automatically generates a URL-friendly slug by stripping diacritics, replacing special characters with hyphens, and suffixing numbers if collisions occur.
- **BR-SKL-12**: Name and Description Constraints: Name is mandatory trimmed 1 to 100 characters; description is optional up to 500 characters.
- **BR-SKL-13**: Concurrency Race-Condition Guard: Catches database `P2002` unique constraint violations and converts them gracefully to HTTP 409 conflict errors.
- **BR-SKL-14**: Dialog Reset and List Invalidation: Clears input values, closes modal dialog, and triggers Next.js router refresh upon success.

**Normal Flow**

1. The administrator clicks “Create Skill” on the skills catalog page.
2. The dialog modal opens with empty input fields.
3. The administrator enters the skill name and optional description.
4. The administrator clicks “Create Skill”.
5. The client submits `POST /api/admin/skills`.
6. The backend validates data, checks for duplicate active names, generates a slug, and inserts the record.
7. The server returns HTTP 201 with the created skill entity.
8. The client displays a success toast notification, closes the modal, and refreshes the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing mandatory skill name | Form validation blocks submission; backend throws HTTP 400 with `Error.SkillNameRequired`. |
| Skill name exceeds 100 characters | Validator rejects input with HTTP 400 Bad Request (`Error.SkillNameTooLong`). The UI highlights the name field. |
| Description exceeds 500 characters | Validator rejects input with HTTP 400 Bad Request (`Error.SkillDescriptionTooLong`). The UI highlights the description field. |
| Active skill with identical name already exists | The service throws `SkillNameAlreadyExistsException` (HTTP 409 Conflict). The UI displays: “Error.SkillNameAlreadyExists”. |
| Concurrent request creates duplicate name | Database catches `P2002` constraint violation; backend converts to HTTP 409 Conflict. The UI alerts the administrator. |
| Unauthorized or unauthenticated request | The server returns HTTP 401/403. The UI redirects to login or displays an access denied message. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database insert failure | The service throws `FailedToCreateSkillException` (HTTP 500). The UI displays: "Error.FailedToCreateSkill". |
| Network timeout | The client catches the network failure, displays an error toast, and leaves the modal open for retry. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Auto-generated unique identifier of the created skill. |
| name | Stored skill name string. |
| slug | Auto-generated unique slug. |
| description | Stored description string or null. |
| deletedAt | Null for newly created skills. |
| createdAt | Timestamp of creation. |
| updatedAt | Timestamp of creation. |
| jobCount | Zero for newly created skills. |

**Post-Conditions**

A new skill record is added to the master catalog, indexed for search, and immediately selectable by freelancers and clients.

### 3.14.4 Update Skill

**Function Trigger**

An administrator clicks the Edit icon (pencil) on a skill table row or detail page, opening the skill edit dialog modal.

**Navigation path:** Admin Dashboard → Master Catalog → Skills → [Edit Action Button].

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to modify the name and description of an existing master skill, automatically synchronizing its unique slug if the name is altered.

### Interface

- Skill update dialog modal (`UpdateSkillDialog`) pre-populated with current skill name and description.
- Text input for Skill Name.
- Textarea input for Description.
- Action buttons: "Cancel" and "Save Changes" submit button with loading indicator.

### Data Processing Overview

- The client pre-fills the form with the skill's current name and description.
- The client issues an HTTP request: `PATCH /api/admin/skills/:id` with updated fields.
- `SkillsAdminRepository.updateSkill(id, data)`:
  - Verifies target skill exists; throws `SkillAdminNotFoundException` if missing.
  - If `name` is modified, checks if another active skill (`id != currentId, deletedAt = null`) shares the same case-insensitive name. Throws `SkillNameAlreadyExistsException` if conflict exists.
  - If `name` is modified, regenerates the unique slug via `generateUniqueSlug()`.
  - Updates the record in `prisma.skill.update()`. If description is empty string, converts it to `null`.
- The backend returns HTTP 200 with the updated skill entity.
- The frontend displays a success toast, closes the modal, and refreshes the view.

### Screen Layout (Mock-up Description)

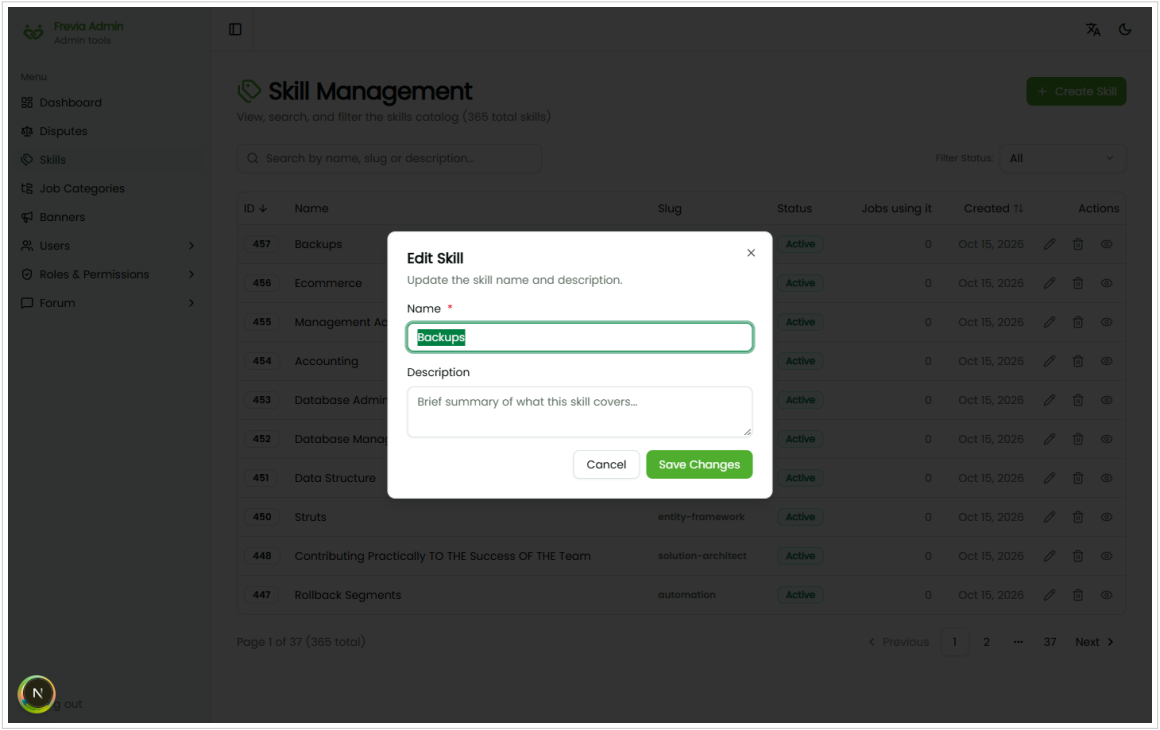


**Figure 123:** Update skill modal interface

- Modal dialog header with “Edit Skill” title.
- Form pre-populated with current values: **Name** (Text input) and **Description** (Textarea).
- Footer action buttons: “Cancel” and “Save Changes” button with pending spinner.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the skill (passed in URL path). |
| name | Updated skill name string (optional). |
| description | Updated description string or null (optional). |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |
| name | Optional string, trimmed, min 1 char (`Error.SkillNameRequired`), max 100 chars (`Error.SkillNameTooLong`). |
| description | Optional string, max 500 chars (`Error.SkillDescriptionTooLong`); nullable. |

**Business Rules**

- **BR-SKL-15**: Skill Existence Verification: Target skill must exist in the database; returns HTTP 404 (`Error.SkillNotFound`) if missing.
- **BR-SKL-16**: Name Modification and Conflict Detection: If name is altered, verifies no other active skill shares the same case-insensitive name; throws HTTP 409 if conflict detected.
- **BR-SKL-17**: Automatic Slug Regeneration: If the name is changed, automatically regenerates a new unique URL-friendly slug based on the updated name.
- **BR-SKL-18**: Optional Description Nullability: Submitting an empty description string sets the description attribute to null in the database.
- **BR-SKL-19**: Synchronized UI Updates: Closes modal dialog, triggers success toast, and refreshes table or detail screen upon completion.

**Normal Flow**

1. The administrator clicks the Edit icon on a skill row or detail page.
2. The dialog modal opens pre-filled with the current skill name and description.
3. The administrator modifies the name or description.
4. The administrator clicks “Save Changes”.
5. The client submits `PATCH /api/admin/skills/:id`.
6. The backend validates fields, checks for duplicate active names, regenerates slug if name changed, and updates the database.
7. The backend returns HTTP 200 with the updated skill entity.
8. The frontend closes the dialog, displays a success toast, and refreshes the view.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target skill does not exist | The service throws `SkillAdminNotFoundException` (HTTP 404). The UI displays: “Error.SkillNotFound”. |
| Updated name conflicts with active skill | The service throws `SkillNameAlreadyExistsException` (HTTP 409). The UI alerts: “Error.SkillNameAlreadyExists”. |
| Name is set to empty or whitespace | Form validation blocks submission; backend returns HTTP 400 Bad Request with `Error.SkillNameRequired`. |
| Name exceeds 100 characters | Validator rejects input with HTTP 400 Bad Request (`Error.SkillNameTooLong`). The UI highlights the name field. |
| Description exceeds 500 characters | Validator rejects input with HTTP 400 Bad Request (`Error.SkillDescriptionTooLong`). The UI highlights the description field. |
| Unauthorized or unauthenticated session | The server returns HTTP 401/403. The UI redirects to login or displays an access denied message. |
| Database update failure | The service throws `FailedToUpdateSkillException` (HTTP 500). The UI displays: “Error.FailedToUpdateSkill”. |

| Abnorma Case | Handling & System Response |
|---|---|
| Network timeout | The client catches the network error, displays an error toast, and preserves dialog state. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the updated skill. |
| name | Updated skill name. |
| slug | Updated or preserved unique slug. |
| description | Updated description or null. |
| deletedAt | Preserved soft-deletion status. |
| updatedAt | Updated modification timestamp. |
| jobCount | Count of active jobs using this skill. |

**Post-Conditions**

The skill record and its generated slug are updated in the database, with immediate propagation across search queries and profile forms.

### 3.14.5 Delete Skill

**Function Trigger**

An administrator clicks the Delete action icon (trash can) on a skill table row or detail page, opening the deletion confirmation dialog.
**Navigation path:** Admin Dashboard → Master Catalog → Skills → [Delete Action Icon].

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to soft-delete a skill from the active catalog by setting its `deletedAt`

timestamp, automatically detaching it from freelancer profiles while preserving historical job associations.

### Interface

- Delete skill confirmation dialog (`DeleteSkillDialog`) powered by Shadcn `AlertDialog`.
- Header displaying skill name and warning icon if currently utilized by jobs.
- Dependent jobs warning banner if `jobCount > 0`, displaying exact number of linked jobs.
- Explanation note detailing soft deletion behavior.
- Action buttons: "Cancel" and destructive "Delete" button with pending spinner.

### Data Processing Overview

- The administrator confirms deletion in the dialog.
- The client sends an HTTP request: `DELETE /api/admin/skills/:id`.
- `SkillsAdminController.deleteSkill(id)` parses `id` via `ParseIntPipe`.
- `SkillsAdminRepository.deleteSkill(id)`:
  - Verifies target skill exists; throws `SkillAdminNotFoundException` if missing.
  - Executes an atomic transaction via `prisma.$transaction`:
    1. Deletes all freelancer associations referencing this skill: `prisma.freelancerSkill.deleteMan where: skillId: id ).`
    2. Sets soft-deletion timestamp on the skill: `prisma.skill.update( where: id , data: deletedAt: new Date() ).`
  - Leaves `job_skills` records intact so existing historical job postings remain accurate.
- The backend returns HTTP 200 with `{ message: "Success.SkillDeleted" }`.
- The frontend shows a success toast, closes the dialog, and refreshes the skill table.

### Screen Layout (Mock-up Description)

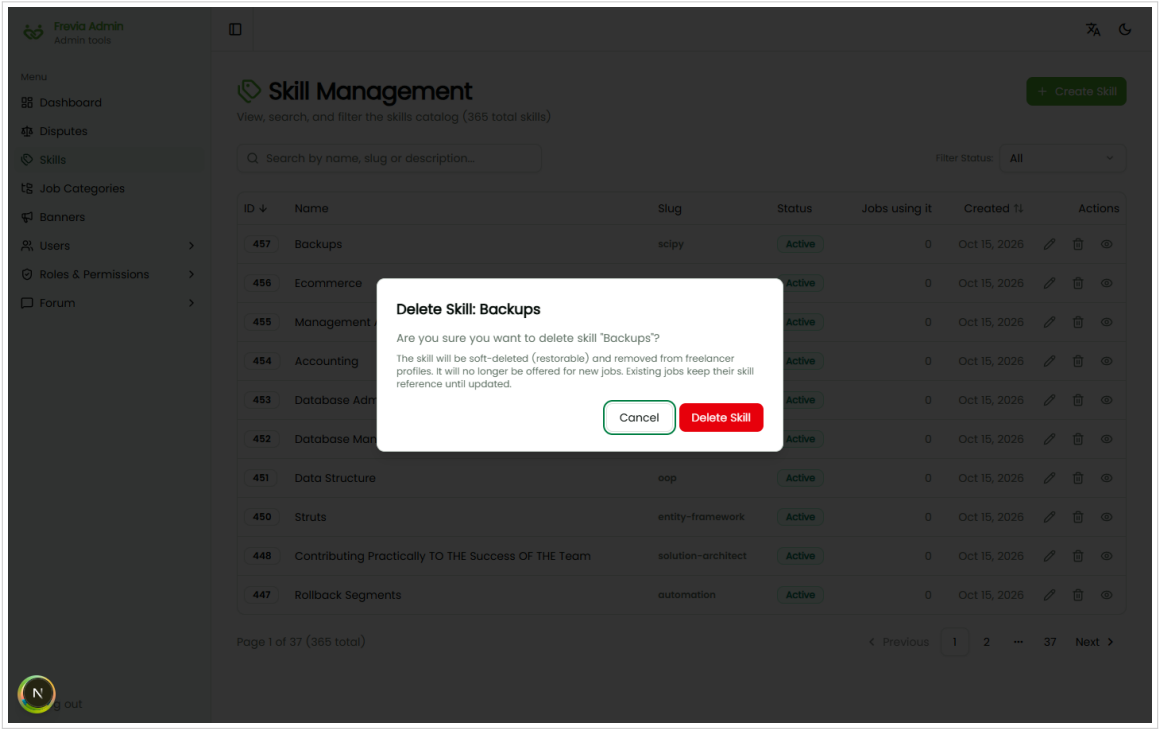


**Figure 124:** Delete skill confirmation dialog

- Dialog header with “Delete Skill” title and target skill name.
- Amber warning alert banner (when `jobCount > 0`) stating: “Skill in use: This skill is currently used by X jobs. Soft-deleting it will preserve job references.”
- Soft-deletion explanation text stating that the skill can be restored at any time.
- Footer action buttons: “Cancel” button and red “Delete” button with loading spinner.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the skill to delete (passed in URL path). |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |

**Business Rules**

- **BR-SKL-20**: Soft-Deletion Principle: Skills are not permanently purged from the database; the system marks them soft-deleted by setting `deletedAt = new Date()`.
- **BR-SKL-21**: Automatic Freelancer Association Detachment: All associations in `freelancer_skills` referencing this skill ID are deleted in the same atomic transaction.
- **BR-SKL-22**: Preservation of Historical Job Associations: Associations in `job_skills` are preserved so existing job postings continue to render their legacy skill requirements accurately.
- **BR-SKL-23**: Dependent Job Warning in UI: The deletion dialog prominently displays an alert warning with the exact number of jobs currently linked to the skill.
- **BR-SKL-24**: Confirmation Modal Guard: Requires explicit confirmation from the administrator before executing soft deletion.

**Normal Flow**

1. The administrator clicks the Delete icon on a skill table row or detail screen.
2. The deletion confirmation dialog appears, showing job usage warnings if applicable.
3. The administrator reviews the warning and confirms by clicking “Delete”.
4. The client issues `DELETE /api/admin/skills/:id`.
5. The backend validates administrative authentication and permissions.
6. The repository executes an atomic transaction detaching freelancer skills and setting `deletedAt`.
7. The backend returns HTTP 200 with `{ message: "Success.SkillDeleted" }`.
8. The frontend displays a success toast, closes the dialog, and refreshes the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target skill does not exist | The service throws `SkillAdminNotFoundException` (HTTP 404). The UI displays: “Error.SkillNotFound”. |

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The server throws HTTP 403 Forbidden. The client renders an access denied alert. |
| Invalid skill ID format | `ParseIntPipe` throws HTTP 400 Bad Request. The UI notifies the user of an invalid parameter. |
| Administrator cancels dialog | The modal closes immediately; no request is sent, leaving the database unaffected. |
| Database transaction failure | The service throws `FailedToDeleteSkillException` (HTTP 500). The UI displays: "Error.FailedToDeleteSkill". |
| Network failure during request | The client catches the network error, displays an unexpected error toast, and leaves the modal open. |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string (`Success.SkillDeleted`). |

**Post-Conditions**

The skill is marked as soft-deleted (`deletedAt != null`), removed from freelancer profiles, hidden from default active listings, and preserved for historical jobs.

### 3.14.6 Restore Skill

**Function Trigger**

An administrator viewing soft-deleted skills (via the "Deleted" status filter or detail page) clicks the Restore icon (rotate-ccw), opening the restore confirmation dialog.
**Navigation path:** Admin Dashboard → Master Catalog → Skills → [Restore Action Icon].

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to restore a previously soft-deleted skill by resetting its `deletedAt` timestamp to null, making it active and selectable again across the platform.

**Interface**

- Restore skill confirmation dialog (`RestoreSkillDialog`) powered by Shadcn `AlertDialog`.
- Header displaying "Restore Skill" title and target skill name.
- Descriptive body explaining that restoring will make the skill available again for freelancers and clients.
- Action buttons: "Cancel" and green "Restore" button with pending spinner.

**Data Processing Overview**

- The administrator confirms restoration in the dialog.
- The client sends an HTTP request: `PATCH /api/admin/skills/:id/restore`.

- `SkillsAdminController.restoreSkill(id)` parses `id` via `ParseIntPipe`.
- `SkillsAdminRepository.restoreSkill(id)`:
  - Finds the skill verifying that `deletedAt != null`; throws `SkillAdminNotFoundException` if missing or already active.
  - Checks if another active skill has been created with the same name during the deletion period (`deletedAt = null, id != currentId, name equals skill.name`).
  - If a name collision occurs, throws `SkillNameAlreadyExistsException` (HTTP 409).
  - Sets `deletedAt = null` via `prisma.skill.updateMany( where: id, deletedAt: not: null , data: deletedAt: null )`.
  - Queries the updated skill entity with job count aggregation.
- The backend returns HTTP 200 with the restored skill entity.
- The frontend shows a success toast notification, closes the dialog, and refreshes the view.

### Screen Layout (Mock-up Description)

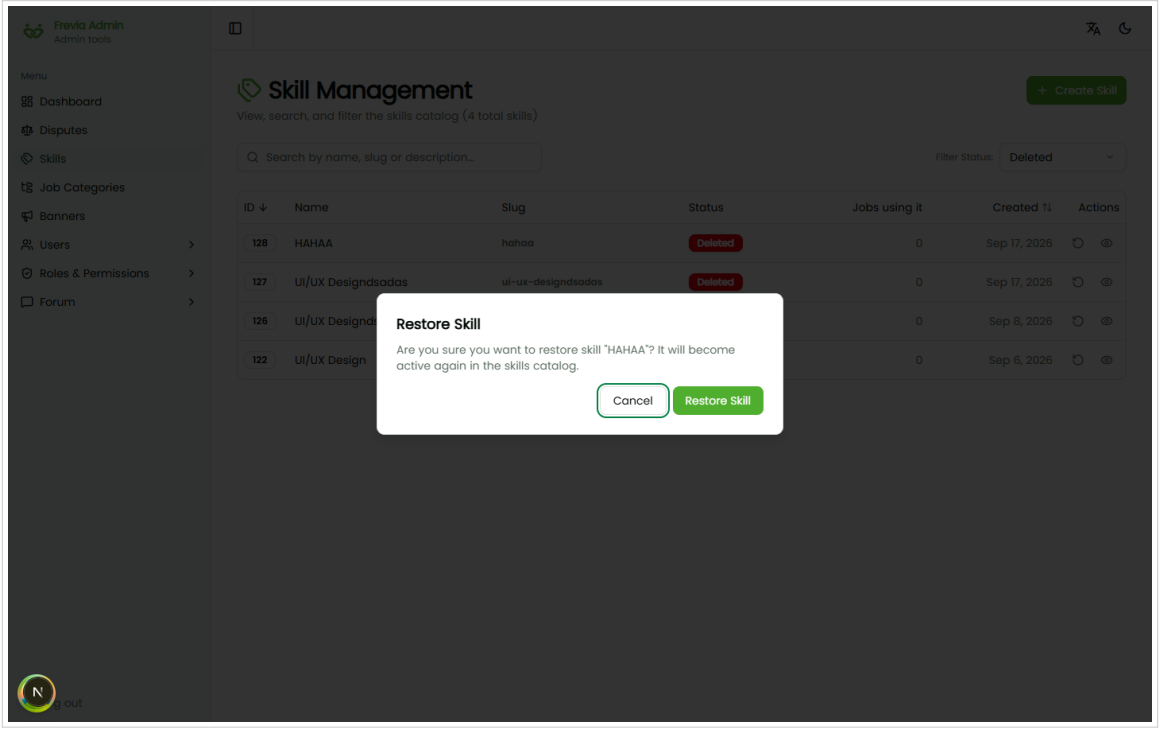


**Figure 125:** Restore skill confirmation dialog

- Alert dialog header displaying “Restore Skill” title with rotate-counterclockwise icon.
- Confirmation description stating: “Are you sure you want to restore skill [Skill Name]? It will become active and available for users again.”
- Footer controls: “Cancel” button and green “Restore” button with loading spinner when executing.

### Function Details

### Input Data

| Data Name | Description |
| --- | --- |
| id | Unique numeric identifier of the soft-deleted skill (passed in URL path). |

### Validation Rules

| Data Name | Validation Rule |
| --- | --- |
| id | Must be a positive integer greater than or equal to 1. |

### Business Rules

- **BR-SKL-25**: Restoration Eligibility: Target skill must currently be soft-deleted with `deletedAt != null`; returns HTTP 404 if not found or already active.
- **BR-SKL-26**: Conflict Prevention Against Interim Duplicates: If another active skill with the same name was created while this skill was deleted, restoration is blocked with HTTP 409 (`Error.SkillNameAlreadyExists`).
- **BR-SKL-27**: Soft-Deletion Flag Clearing: Sets `deletedAt = null`, fully restoring the skill to active catalog availability.
- **BR-SKL-28**: Immediate Status Synchronization: Returns restored skill data, displays success toast, and updates table view immediately.

### Normal Flow

1. The administrator filters the catalog by "Deleted" skills.
2. The administrator locates the soft-deleted skill and clicks the Restore icon.
3. The restore confirmation dialog appears.
4. The administrator confirms restoration by clicking "Restore".
5. The client sends `PATCH /api/admin/skills/:id/restore`.
6. The backend verifies the skill is soft-deleted, checks for name conflicts with active skills, and clears `deletedAt`.
7. The backend returns HTTP 200 with the restored skill data.
8. The client displays a success toast, closes the dialog, and refreshes the table.

### Abnormal Flow / Exception Handling

| Abnorma Case | Handling & System Response |
|---|---|
| Target skill does not exist or is already active | The service throws `SkillAdminNotFoundException` (HTTP 404). The UI displays: “Error.SkillNotFound”. |
| Another active skill now shares the same name | The service throws `SkillNameAlreadyExistsException` (HTTP 409). The UI alerts: “Error.SkillNameAlreadyExists”. |
| Missing or invalid authen-tication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privi-leges | The server throws HTTP 403 Forbidden. The client renders an access de-nied alert. |
| Invalid skill ID format | `ParseIntPipe` throws HTTP 400 Bad Request. The UI notifies the user of an invalid parameter. |
| Administrator cancels dialog | The modal closes immediately; no request is sent, leaving the database unaffected. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database update failure | The service throws `FailedToRestoreSkillException` (HTTP 500). The UI displays: "Error.FailedToRestoreSkill". |
| Network failure during request | The client catches the network error, displays an unexpected error toast, and preserves dialog state. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the restored skill. |
| name | Skill name string. |
| slug | Preserved or updated unique slug. |
| description | Description text or null. |
| deletedAt | Null indicating active status. |
| createdAt | Timestamp of original creation. |
| updatedAt | Timestamp of restoration. |
| jobCount | Count of jobs referencing this skill. |

**Post-Conditions**

The skill is restored to active status (`deletedAt = null`), displayed in the active skill catalog, and available for selection in freelancer profiles and job postings.

## 3.15 Job Category Management

### 3.15.1 View Job Category List

**Function Trigger**

An administrator navigates to the Job Categories management page from the admin dashboard sidebar navigation.

**Navigation path:** Admin Dashboard → Job Taxonomy → Job Categories (`/admin/job-categories`).

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to retrieve and manage the list of job categories in the system, supporting paginated navigation, multi-field search, operational status filtering (`ACTIVE` or `INACTIVE`), soft-deletion status filtering, and sorting.

### Interface

- Job category management screen at `/admin/job-categories`.
- Filter bar component (`JobCategoriesFilterBar`) with search box, status dropdown (All, Active, Inactive), soft-delete filter dropdown (All, Active Only, Deleted Only), sort column dropdown (`id`, `createdAt`, `name`), and sort direction toggle.
- Data table component (`JobCategoriesTable`) rendering category rows, slugs, descriptions, operational status badges, active job count badges, created dates, and action controls.
- Standard pagination footer displaying total category count and page selection buttons.
- "Create Job Category" action button in the page header.

### Data Processing Overview

- The client initiates an HTTP request: `GET /api/admin/job-categories` with query parameters (`page`, `limit`, `search`, `status`, `deleted`, `sortBy`, `sortOrder`).
- The backend `AuthGuard` validates the administrator's JWT token, and `PermissionGuard` verifies administrative route authorization.
- `JobCategoriesAdminController.listJobCategories()` parses query parameters via `JobCategoryAdminQueryDto`.
- `JobCategoriesAdminRepository.listJobCategories()` queries `prisma.jobCategory.findMany` with:
  - Multi-field case-insensitive search matching on `name`, `slug`, or `description` if `search` is provided.
  - Exact filter on `status` (`ACTIVE` or `INACTIVE`) if selected.
  - Soft-deletion filter on `deletedAt`: if `deleted === 'true'`, filters `deletedAt != null`; if `deleted === 'false'`, filters `deletedAt = null`.
  - Aggregated job count via `_count: { select: { jobs: { where: { job:`

`{ deletedAt: null } } } } }`, counting only non-deleted active jobs linked via `JobJobCategory`.

  - Sorting on the requested field (`id`, `createdAt`, or `name`) in specified direction (`asc` or `desc`).

- Executes `prisma.jobCategory.count()` in parallel to determine total matching records.
- Returns HTTP 200 with the array of job categories and pagination metadata.
- The frontend renders the category table, badges, and contextual action menus.

### Screen Layout (Mock-up Description)

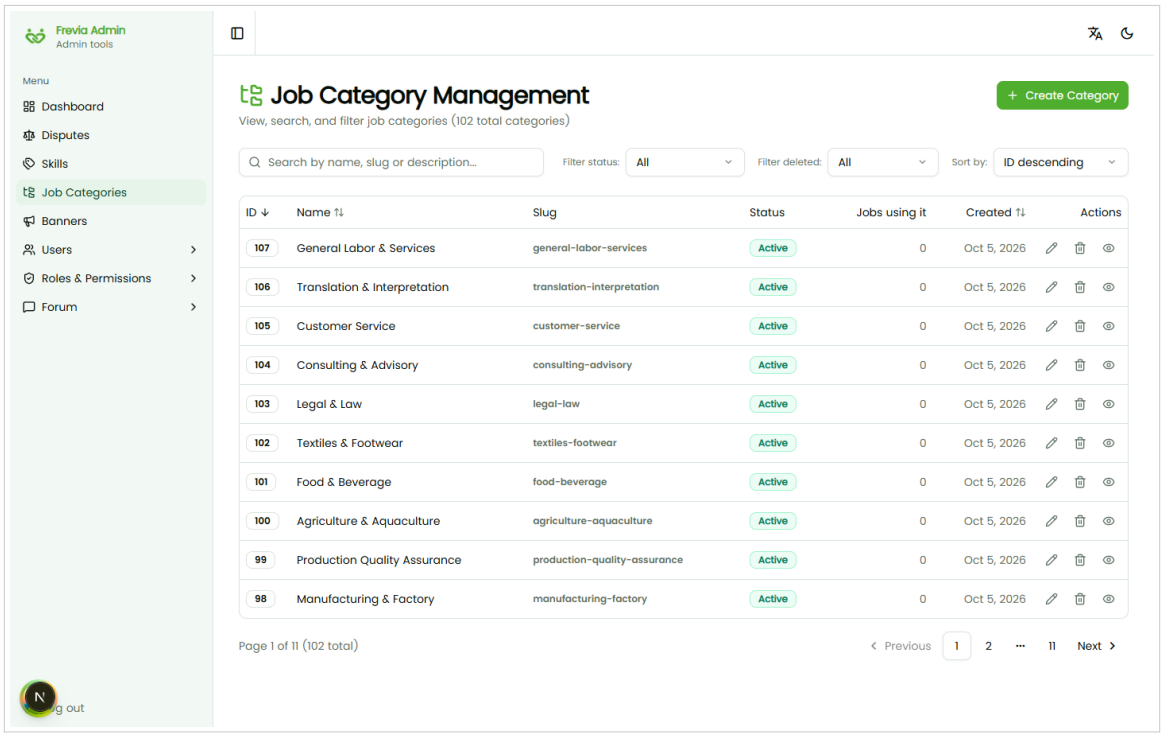


**Figure 126:** Job category list interface

- Page header displaying “Job Categories” title with folder tree icon, total count summary, and “Create Job Category” button.
- Filter bar including search box, status dropdown (All, Active, Inactive), deleted status dropdown (All, Active, Deleted), and sort direction buttons.
- Table columns: **Name** (with clickable detail link), **Slug**, **Description**, **Status** (Green “Active” or Gray “Inactive”), **In Use** (Badge showing count of active jobs using this category), **Created Date**, and **Actions**.
- Contextual action buttons: Update (pencil) and Delete (trash) for active categories; Restore (rotate-ccw) for deleted categories.
- Pagination footer indicating total count and page navigation controls.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| page | Target page number (defaults to 1). |
| limit | Maximum records returned per page (defaults to 10). |
| search | Optional text query matching name, slug, or description. |
| status | Optional operational status filter (`ACTIVE` or `INACTIVE`). |
| deleted | Optional soft-deletion filter: `"true"` (deleted only) or `"false"` (active only). |
| sortBy | Sort column field (`id`, `createdAt`, or `name`; defaults to `id`). |
| sortOrder | Sort direction sequence (`asc` or `desc`; defaults to `desc`). |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| page | Must be an integer greater than or equal to 1. |
| limit | Must be an integer between 1 and 100. |
| search | String, optional, max length 100 characters; whitespace trimmed. |
| status | Optional string, must strictly match one of: `ACTIVE`, `INACTIVE`. |
| deleted | Optional string, must strictly match one of: `"true"`, `"false"`. |
| sortBy | Must strictly be one of: `id`, `createdAt`, `name`. |
| sortOrder | Must strictly be one of: `asc`, `desc`. |

### Business Rules

- **BR-JCAT-01**: Admin Authorization: Access to the administrative job category catalog is strictly restricted to authenticated administrators.
- **BR-JCAT-02**: Multi-Field Case-Insensitive Search: Searches across `name`, `slug`, and `description` case-insensitively using partial substring matching.

- **BR-JCAT-03**: Active Job Count Link Filtering: Counts only jobs that are not soft-deleted (`job.deletedAt = null`) linked via the `JobJobCategory` relation table.
- **BR-JCAT-04**: Operational Status and Soft-Delete Filtering: Supports orthogonal filtering by operational state (`ACTIVE` or `INACTIVE`) and lifecycle deletion state (`deleted=true` or `deleted=false`).
- **BR-JCAT-05**: Contextual Action Buttons: Active categories display green badges with Edit and Delete action buttons; soft-deleted categories display red badges with a Restore button.

**Normal Flow**

1. The administrator opens the `/admin/job-categories` page from the admin dashboard.
2. The client issues `GET /api/admin/job-categories` with default query parameters.
3. The backend validates administrative token and route permissions.
4. The repository queries job categories, counting active linked jobs and total matching records.
5. The server returns HTTP 200 with the categories array and pagination metadata.
6. The frontend renders the job category table and pagination controls.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authen-tication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privi-leges | The backend throws HTTP 403 Forbidden. The UI displays: “You do not have permission to access this page.” |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid pagination parameter values | The DTO validator rejects non-integer values with HTTP 400 Bad Request. The client defaults to page 1. |
| Invalid operational status parameter | The validator rejects unknown status values with HTTP 400 Bad Request. The UI clears invalid status filter. |
| Invalid deleted filter parameter value | The validator rejects unknown values with HTTP 400 Bad Request. The UI clears invalid filter. |
| Database query error or timeout | The service throws `FailedToLoadJobCategoryListException` (HTTP 500). The UI displays: "Error.FailedToLoadJobCategoryList". |
| No categories match search or status | The query returns `jobCategories: []` and `total: 0`. The table displays an empty placeholder: "No job categories found." |
| Network connection timeout | The client displays an error toast notification and offers a reload button. |

**Output Data**

| Output Field | Description |
|---|---|
| jobCategories | Array of category objects: `id`, `name`, `slug`, `description`, `status`, `deletedAt`, `createdAt`, `updatedAt`, `jobCount`. |
| pagination.page | Current page number. |
| pagination.limit | Current limit per page. |
| pagination.total | Total job categories matching the query. |
| pagination.totalPages | Total calculated pages available. |

**Post-Conditions**

The administrator inspects the job category catalog, views job usage metrics, and can initiate creation, inspection, editing, deletion, or restoration.

### 3.15.2 View Job Category Detail

**Function Trigger**

An administrator clicks on a job category row or name in the category list, navigating to `/admin/job-categories/[id]`.
**Navigation path:** Admin Dashboard → Job Taxonomy → Job Categories → Category Details (`/admin/job-categories/[id]`).

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to view comprehensive metadata of a specific job category, including its generated slug, full description, operational status, total count of active linked jobs, and a list of recently associated active jobs.

**Interface**

- Job category detail screen at `/admin/job-categories/[id]`.
- Navigation back button (“Back to Job Categories”).

- Header card displaying category name, slug, ID, status badge (`ACTIVE` or `INACTIVE`), deleted badge (if applicable), and action buttons (`JobCategoryActions`).
- Description card displaying the full category description or a placeholder if empty.
- Metrics cards displaying total jobs using this category, creation timestamp, and last updated timestamp.
- Related jobs data table displaying up to 10 latest active jobs linked to this category (columns: ID, Title, Slug, Job Status, and Created Date).

**Data Processing Overview**

- The client sends an HTTP request: `GET /api/admin/job-categories/:id`.
- The backend validates administrative token and route permissions.
- `JobCategoriesAdminController.getJobCategoryDetail()` parses `id` via `ParseIntPipe`.
- `JobCategoriesAdminRepository.getJobCategoryDetail(id)` queries `prisma.jobCategory.find` by `id`.
- If the category record is not found, the service throws `JobCategoryAdminNotFoundException` (HTTP 404).
- The repository queries `prisma.jobJobCategory.findMany` where `categoryId = id` and `job.deletedAt = null`, ordered by `job.createdAt desc`, taking up to 10 records.
- Returns HTTP 200 with the category detail object including `jobCount` and `jobs` array.
- The frontend renders the comprehensive detail layout, metric cards, and associated jobs table.

### Screen Layout (Mock-up Description)

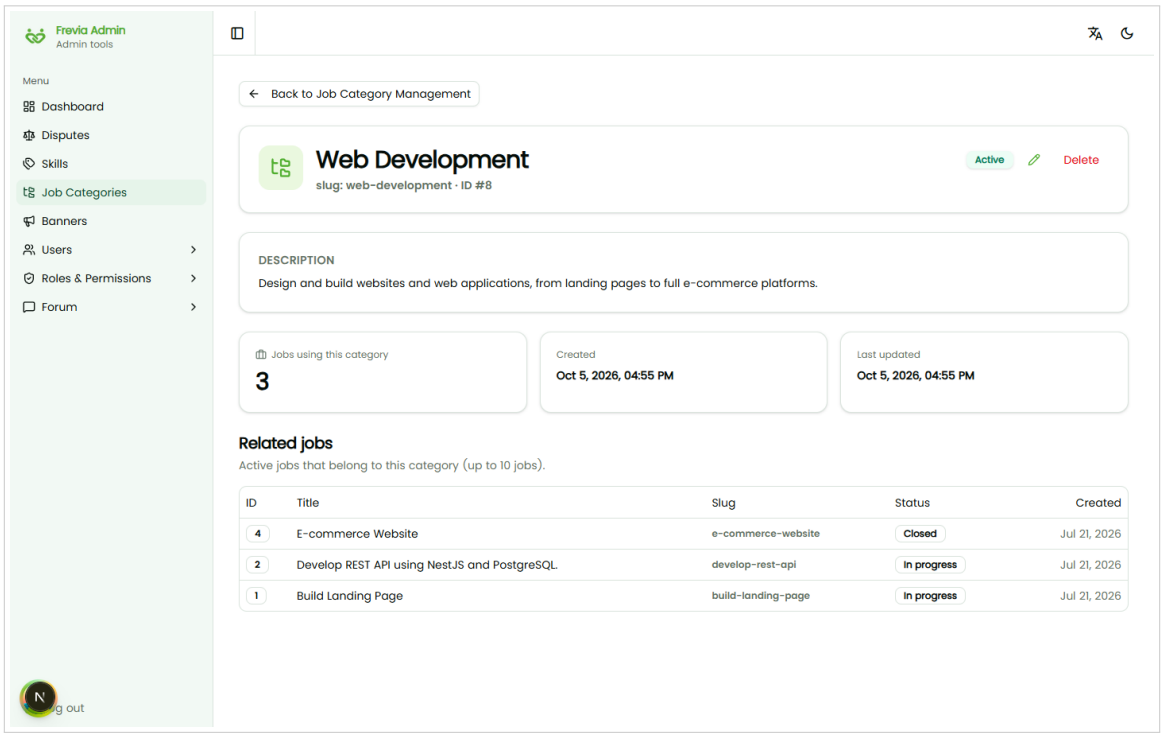


**Figure 127:** Job category detail interface

- Navigation header with “Back to Job Categories” button, category name with folder tree icon, and status badge.
- Action toolbar containing Edit (pencil) and Delete (trash) buttons if active, or Restore (rotate-ccw) button if soft-deleted.
- Summary card showing: **Slug** (e.g., `software-development`), **Status** (Active or Inactive), and **ID**.
- Description card displaying the category overview text.
- Metric cards: **Jobs Using This Category** (large number with briefcase icon), **Created Date**, and **Last Updated Date**.
- Associated Jobs table: Lists up to 10 latest active jobs with ID badge, Title, Slug, Status badge, and Created Date.

### Function Details

### Input Data

| Data Name | Description |
| --- | --- |
| id | Unique numeric identifier of the job category (passed in URL path). |

**Validation Rules**

| Data Name | Validation Rule |
| --- | --- |
| id | Must be a positive integer greater than or equal to 1. |

**Business Rules**

- **BR-JCAT-06**: Identifier Format Validation: The URL path parameter `id` must parse strictly as a valid positive integer.
- **BR-JCAT-07**: Job Category Existence Check: The targeted job category must exist in the database; returns HTTP 404 (`Error.JobCategoryNotFound`) if missing.
- **BR-JCAT-08**: Associated Job Retrieval Limit: Returns up to 10 latest non-deleted active jobs linked to this category through `JobJobCategory`.
- **BR-JCAT-09**: Contextual Action Toolbar: Shows Edit and Delete actions if category is active; shows Restore action if category is soft-deleted.

**Normal Flow**

1. The administrator navigates to `/admin/job-categories/[id]`.
2. The client issues `GET /api/admin/job-categories/:id`.
3. The backend validates administrative authentication and permissions.
4. The repository queries the category entity and fetches related active jobs.
5. The server returns HTTP 200 with the category detail object and related jobs.
6. The frontend renders the comprehensive job category detail page and associated jobs table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The server throws HTTP 403 Forbidden. The client renders an access denied alert. |
| Invalid category ID format (e.g., non-numeric) | `ParseIntPipe` throws HTTP 400 Bad Request. The UI displays an "Invalid Category ID" message and back button. |
| Category ID does not exist in database | The service throws `JobCategoryAdminNotFoundException` (HTTP 404). The UI renders a "Category Not Found" screen with navigation back to catalog. |
| Database query error or timeout | The service throws `FailedToLoadJobCategoryDetailException` (HTTP 500). The UI displays: "Error.FailedToLoadJobCategoryDetail". |
| Category description is empty or null | The UI gracefully displays "No description provided" in muted text. |

| Abnorma Case | Handling & System Response |
|---|---|
| No jobs associated with this category | The associated jobs table displays an empty row stating: “No jobs found in this category.” |

**Output Data**

| Output Field | Description |
|---|---|
| id | Unique numeric identifier of the job category. |
| name | Category name string. |
| slug | Generated URL-friendly slug. |
| description | Detailed description text or null. |
| status | Operational status enum (`ACTIVE` or `INACTIVE`). |
| deletedAt | Soft-deletion timestamp or null. |
| createdAt | Timestamp of category creation. |
| updatedAt | Timestamp of last modification. |
| jobCount | Total count of active jobs associated with this category. |
| jobs | Array of up to 10 latest active jobs: `id`, `title`, `slug`, `status`, `deletedAt`, `createdAt`. |

**Post-Conditions**

The administrator reviews category details, operational status, and associated jobs, and can initiate editing, deletion, or restoration.

### 3.15.3 Create Job Category

**Function Trigger**

An administrator clicks the “Create Job Category” button on the category catalog page (`/admin/job-categories`), opening the creation dialog modal.

**Navigation path:** Admin Dashboard → Job Taxonomy → Job Categories → [Create Job Category Button].

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to register a new job category by providing its name, optional description, and operational status, with automated unique slug generation.

### Interface

- Job category creation dialog modal (`CreateJobCategoryDialog`).
- Text input for Category Name (mandatory).
- Textarea input for Description (optional).
- Select dropdown for Status (`ACTIVE` or `INACTIVE`, defaults to `ACTIVE`).
- Action buttons: "Cancel" and "Create Job Category" submit button with loading indicator.

### Data Processing Overview

- The client submits an HTTP request: `POST /api/admin/job-categories` with JSON payload conforming to `AdminCreateJobCategoryBodySchema`.
- `JobCategoriesAdminRepository.createJobCategory()`:
  - Strips leading and trailing whitespace from `name`.
  - Queries `prisma.jobCategory.findFirst` to check if an active category (`deletedAt = null`) with the same name exists (case-insensitive mode).
  - If an active duplicate exists, throws `JobCategoryNameAlreadyExistsException` (HTTP 409).
  - Generates a unique slug via `generateUniqueSlug()`: strips Vietnamese diacritics, converts to lowercase, replaces non-alphanumeric characters with hyphens, and appends sequential numeric suffixes (`-2`, `-3`) if a slug collision occurs.
  - Inserts the category record via `prisma.jobCategory.create()`.
  - Catches Prisma `P2002` errors (unique constraint violations) and converts to `JobCategoryNameAlreadyExistsException`.
- Returns HTTP 201 with the created category entity.
- The frontend displays a success toast, closes the modal, and refreshes the category table.

### Screen Layout (Mock-up Description)

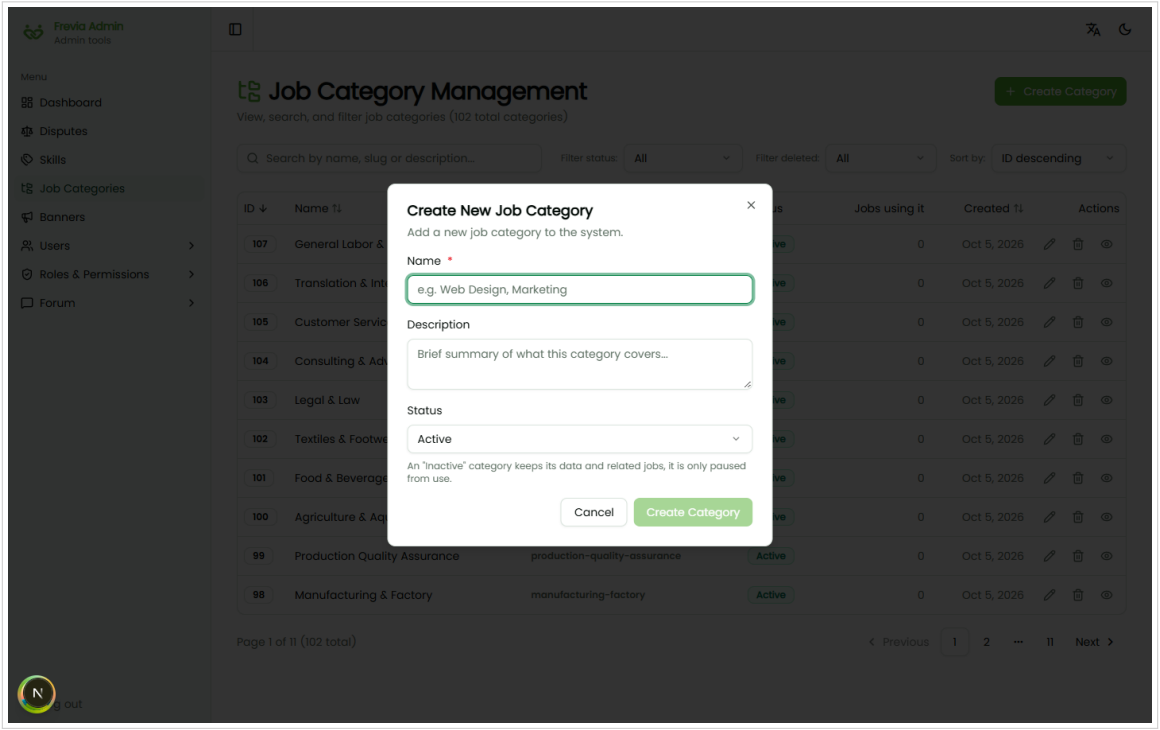


**Figure 128:** Create job category modal interface

- Dialog header with “Create Job Category” title and descriptive prompt.
- Form fields: **Name** (Required input field with asterisk), **Description** (Textarea), and **Status** (Select dropdown: Active / Inactive).
- Dialog footer with “Cancel” button and green “Create Job Category” button with loading spinner when submitting.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| name | Job category name string (mandatory). |
| description | Optional text description of the category. |
| status | Operational status enum (`ACTIVE` or `INACTIVE`, defaults to `ACTIVE`). |

### Validation Rules

| Data Name | Validation Rule |
|---|---|
| name | String, trimmed, min 1 char (`Error.JobCategoryNameRequired`), max 100 chars (`Error.JobCategoryNameTooLong`). |
| description | Optional string, max 500 chars (`Error.JobCategoryDescriptionTooLong`); nullable. |
| status | Optional enum, must strictly match one of: `ACTIVE`, `INACTIVE` (defaults to `ACTIVE`). |

**Business Rules**

- **BR-JCAT-10**: Name Uniqueness Among Active Categories: Job category name must be unique among active categories (`deletedAt = null`), evaluated case-insensitively; throws HTTP 409 (`Error.JobCategoryNameAlreadyExists`) if conflict occurs.
- **BR-JCAT-11**: Automated Unique Slug Generation: Automatically generates a URL-friendly slug with diacritics stripping, hyphen replacement, and automatic numeric collision resolution (`-2`, `-3`).
- **BR-JCAT-12**: Default Operational Status: Newly created job categories default to `ACTIVE` status unless explicitly set to `INACTIVE`.
- **BR-JCAT-13**: Concurrency Race-Condition Guard: Catches database `P2002` unique constraint violations and converts them gracefully to HTTP 409 conflict errors.
- **BR-JCAT-14**: Modal Reset and Table Invalidation: Clears input values, closes modal dialog, and triggers Next.js router refresh upon success.

**Normal Flow**

1. The administrator clicks “Create Job Category” on the categories catalog page.
2. The dialog modal opens with empty input fields and status set to Active.
3. The administrator enters the category name, optional description, and chooses operational status.
4. The administrator clicks “Create Job Category”.
5. The client submits `POST /api/admin/job-categories`.
6. The backend validates data, checks for duplicate active names, generates a slug, and inserts the record.
7. The server returns HTTP 201 with the created category entity.
8. The client displays a success toast notification, closes the modal, and refreshes the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Missing mandatory category name | Form validation blocks submission; backend throws HTTP 400 with `Error.JobCategoryNameRequired`. |
| Category name exceeds 100 characters | Validator rejects input with HTTP 400 Bad Request (`Error.JobCategoryNameTooLong`). The UI highlights the name field. |
| Description exceeds 500 characters | Validator rejects input with HTTP 400 Bad Request (`Error.JobCategoryDescriptionTooLong`). The UI highlights the description field. |
| Active category with identical name already exists | The service throws `JobCategoryNameAlreadyExistsException` (HTTP 409 Conflict). The UI displays: "Error.JobCategoryNameAlreadyExists". |
| Concurrent request creates duplicate name | Database catches `P2002` constraint violation; backend converts to HTTP 409 Conflict. The UI alerts the administrator. |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid status enum parame-ter | Validator rejects invalid status strings with HTTP 400 Bad Request. The UI flags the status field. |
| Unauthorized or unau-thenti-cated request | The server returns HTTP 401/403. The UI redirects to login or displays an access denied message. |
| Database insert failure | The service throws `FailedToCreateJobCategoryException` (HTTP 500). The UI displays: “Error.FailedToCreateJobCategory”. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Auto-generated unique identifier of the created category. |
| name | Stored category name string. |
| slug | Auto-generated unique slug. |
| description | Stored description string or null. |
| status | Operational status (`ACTIVE` or `INACTIVE`). |
| deletedAt | Null for newly created categories. |
| createdAt | Timestamp of creation. |
| updatedAt | Timestamp of creation. |
| jobCount | Zero for newly created categories. |
| jobs | Empty array for newly created categories. |

**Post-Conditions**

A new job category record is stored in the database, indexed for search, and immediately available for classification of new job postings.

### 3.15.4 Update Job Category

### Function Trigger

An administrator clicks the Edit icon (pencil) on a category table row or detail page, opening the category edit dialog modal.
**Navigation path:** Admin Dashboard → Job Taxonomy → Job Categories → [Edit Action Button].

### Function Description

### Actors / Roles

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to modify the name, description, and operational status (`ACTIVE` or `INACTIVE`) of an existing job category, automatically synchronizing its unique slug if the name is altered.

### Interface

- Job category update dialog modal (`UpdateJobCategoryDialog`) pre-populated with current category name, description, and status.
- Text input for Category Name.
- Textarea input for Description.
- Select dropdown for Status (`ACTIVE` or `INACTIVE`).
- Action buttons: "Cancel" and "Save Changes" submit button with loading indicator.

### Data Processing Overview

- The client pre-fills the form with the category's current values.
- The client issues an HTTP request: `PATCH /api/admin/job-categories/:id` with updated fields.
- `JobCategoriesAdminRepository.updateJobCategory(id, data)`:
  - Verifies target category exists and is active (`deletedAt = null`); throws `JobCategoryAdminNotFoundException` if missing or soft-deleted (soft-deleted categories must be restored before editing).
  - If `name` is modified, checks if another active category (`id != currentId`, `deletedAt = null`) shares the same case-insensitive name. Throws `JobCategoryNameAlreadyExis` if conflict exists.

- If `name` is modified, regenerates the unique slug via `generateUniqueSlug()`.
- Updates the record in `prisma.jobCategory.update()`. If description is empty string, converts it to `null`.

- The backend returns HTTP 200 with the updated category entity.
- The frontend displays a success toast, closes the modal, and refreshes the view.

### Screen Layout (Mock-up Description)

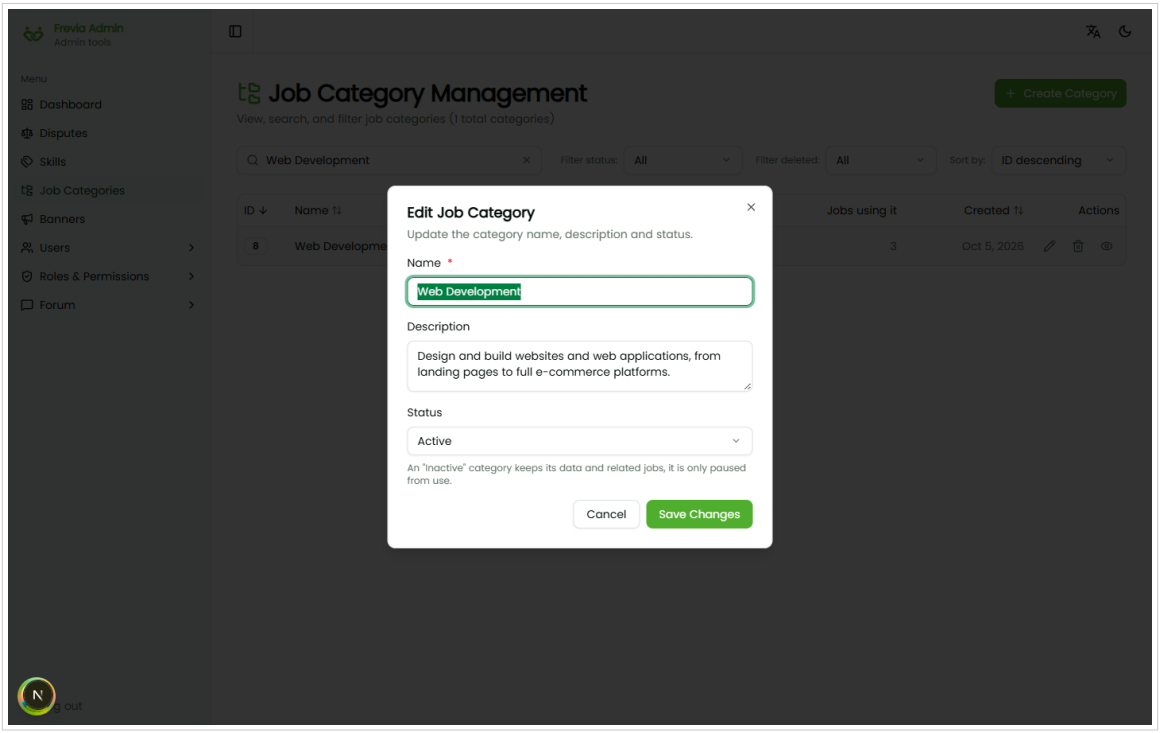


**Figure 129:** Update job category modal interface

- Modal dialog header with "Edit Job Category" title.
- Form pre-populated with current values: **Name** (Text input), **Description** (Textarea), and **Status** (Select dropdown: Active / Inactive).
- Footer action buttons: "Cancel" and "Save Changes" button with pending spinner.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the category (passed in URL path). |
| name | Updated category name string (optional). |
| description | Updated description string or null (optional). |
| status | Updated operational status enum (`ACTIVE` or `INACTIVE`, optional). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |
| name | Optional string, trimmed, min 1 char (`Error.JobCategoryNameRequired`), max 100 chars (`Error.JobCategoryNameTooLong`). |
| description | Optional string, max 500 chars (`Error.JobCategoryDescriptionTooLong`); nullable. |
| status | Optional enum, must strictly match one of: `ACTIVE`, `INACTIVE`. |

**Business Rules**

- **BR-JCAT-15**: Target Category Active State Requirement: Only active categories (`deletedAt = null`) can be updated; soft-deleted categories must be restored first.
- **BR-JCAT-16**: Name Conflict Detection on Modification: If name is altered, verifies no other active category shares the same case-insensitive name; throws HTTP 409 if conflict detected.
- **BR-JCAT-17**: Automatic Slug Regeneration: If the name is changed, automatically regenerates a new unique URL-friendly slug based on the updated name.
- **BR-JCAT-18**: Status Transition Freedom: Administrators can freely switch status between `ACTIVE` and `INACTIVE` at any time to temporarily suspend new job associations without deleting data.
- **BR-JCAT-19**: Synchronized UI Updates: Closes modal dialog, triggers success toast, and refreshes category table or detail screen upon completion.

**Normal Flow**

1. The administrator clicks the Edit icon on a category row or detail page.
2. The dialog modal opens pre-filled with the current category name, description, and status.
3. The administrator modifies the name, description, or status.
4. The administrator clicks “Save Changes”.
5. The client submits `PATCH /api/admin/job-categories/:id`.
6. The backend validates fields, checks for duplicate active names, regenerates slug if name changed, and updates the database.
7. The backend returns HTTP 200 with the updated category entity.
8. The frontend closes the dialog, displays a success toast, and refreshes the view.

**Abnormal Flow / Exception Handling**

| **Abnorma Case** | **Handling & System Response** |
|---|---|
| Target category does not exist or is soft-deleted | The service throws `JobCategoryAdminNotFoundException` (HTTP 404). The UI displays: “Error.JobCategoryNotFound”. |
| Updated name conflicts with active category | The service throws `JobCategoryNameAlreadyExistsException` (HTTP 409). The UI alerts: “Error.JobCategoryNameAlreadyExists”. |
| Name is set to empty or whitespace | Form validation blocks submission; backend returns HTTP 400 Bad Request with `Error.JobCategoryNameRequired`. |

| Abnorma Case | Handling & System Response |
|---|---|
| Name exceeds 100 charac-ters | Validator rejects input with HTTP 400 Bad Request (`Error.JobCategoryNameTooLong`). The UI highlights the name field. |
| Description exceeds 500 charac-ters | Validator rejects input with HTTP 400 Bad Request (`Error.JobCategoryDescriptionTooLong`). The UI highlights the description field. |
| Unauthorized or unau-thenti-cated session | The server returns HTTP 401/403. The UI redirects to login or displays an access denied message. |
| Database update failure | The service throws `FailedToUpdateJobCategoryException` (HTTP 500). The UI displays: "Error.FailedToUpdateJobCategory". |
| Network timeout | The client catches the network error, displays an error toast, and preserves dialog state. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the updated category. |
| name | Updated category name. |
| slug | Updated or preserved unique slug. |
| description | Updated description or null. |
| status | Updated operational status (`ACTIVE` or `INACTIVE`). |
| deletedAt | Preserved soft-deletion status. |
| updatedAt | Updated modification timestamp. |
| jobCount | Count of active jobs using this category. |

| Output Field | Description |
|---|---|
| jobs | Array of related jobs. |

**Post-Conditions**

The category record and its generated slug are updated in the database, with immediate propagation across search queries and job submission forms.

### 3.15.5 Delete Job Category

**Function Trigger**

An administrator clicks the Delete action icon (trash can) on a category table row or detail page, opening the deletion confirmation dialog.
**Navigation path:** Admin Dashboard → Job Taxonomy → Job Categories → [Delete Action Icon].

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

**Purpose**

Allows administrators to soft-delete a job category after verifying that no active jobs are currently associated with it, preventing accidental breaking of active job listings.

**Interface**

- Delete job category confirmation dialog (`DeleteJobCategoryDialog`) powered by Shadcn `AlertDialog`.
- Header displaying category name and warning icon if currently utilized by jobs.
- Dependent jobs warning banner if `jobCount > 0`, displaying exact number of linked active jobs.
- In-dialog conflict notification banner when HTTP 409 occurs, suggesting the administrator change the category status to `INACTIVE` instead.
- Action buttons: “Cancel” and destructive “Delete” button with pending spinner.

**Data Processing Overview**

- The administrator confirms deletion in the dialog.
- The client sends an HTTP request: `DELETE /api/admin/job-categories/:id`.
- `JobCategoriesAdminController.deleteJobCategory(id)` parses `id` via `ParseIntPipe`.
- `JobCategoriesAdminRepository.deleteJobCategory(id)`:
  - Verifies target category exists and is active (`deletedAt = null`); throws `JobCategoryAdminNotFoundException` if missing.
  - Checks count of active associated jobs: `_count: { select: { jobs: { where: { job: { deletedAt: null } } } } }`.
  - If `_count.jobs > 0`, throws `JobCategoryHasJobsException` (HTTP 409 Conflict with message `Error.JobCategoryHasJobs`). Deletion is strictly blocked.
  - If `_count.jobs === 0`, executes soft delete: `prisma.jobCategory.update({ where: { id }, data: { deletedAt: new Date() } })`.
- The backend returns HTTP 200 with `{ message: "Success.JobCategoryDeleted" }`.
- The frontend shows a success toast, closes the dialog, and refreshes the category table.

### Screen Layout (Mock-up Description)

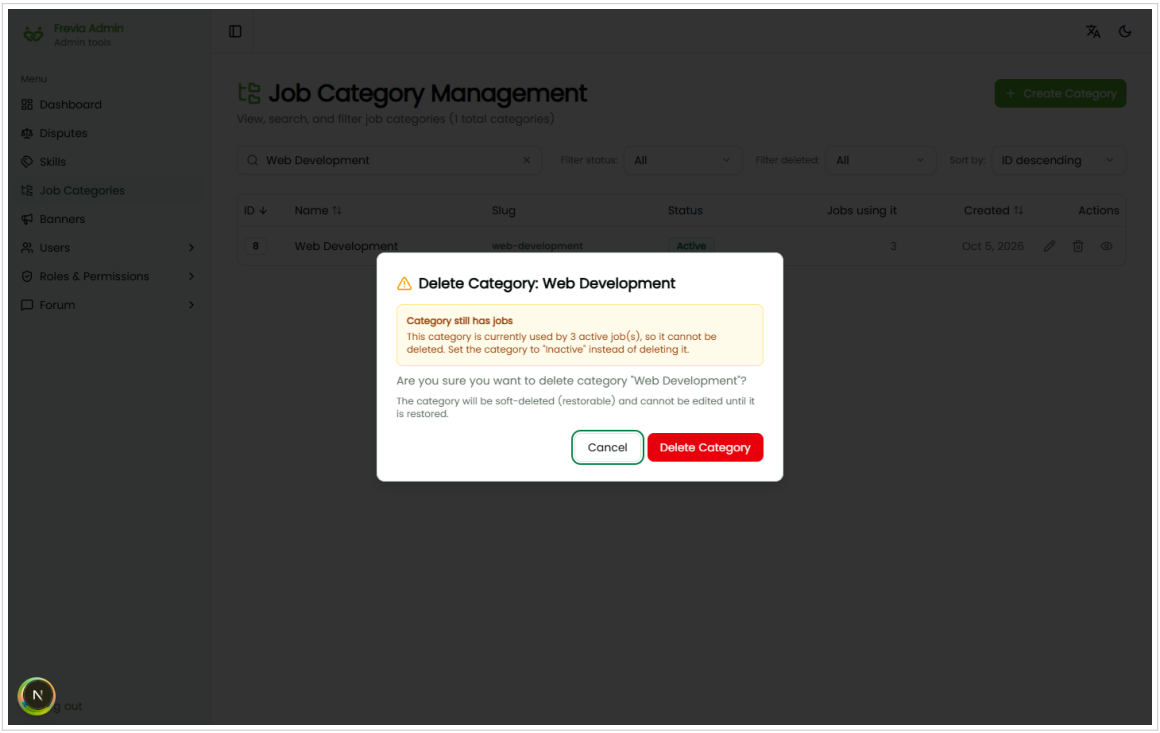


**Figure 130:** Delete job category confirmation dialog

- Dialog header with "Delete Job Category" title and target category name.
- Warning alert banner (when `jobCount > 0`) stating: "Category in use: This category is currently used by X jobs."

- Inline conflict callout (when deletion blocked) explaining that active jobs prevent deletion and advising switching status to Inactive instead.
- Soft-deletion explanation text stating that the category can be restored at any time.
- Footer action buttons: “Cancel” button and red “Delete” button with loading spinner.

**Function Details**

**Input Data**

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the category to delete (passed in URL path). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |

**Business Rules**

- **BR-JCAT-20**: Active Job Dependency Deletion Guard: Deleting a job category is strictly blocked if one or more active jobs are linked to it (`_count.jobs > 0`); the system throws HTTP 409 (`Error.JobCategoryHasJobs`).
- **BR-JCAT-21**: Inactive Status Recommendation: If active jobs are attached, the UI advises administrators to set category status to `INACTIVE` instead of deleting it.
- **BR-JCAT-22**: Soft-Deletion Lifecycle: Categories without active jobs are soft-deleted by setting `deletedAt = new Date()` without purging relational records.
- **BR-JCAT-23**: Explicit Confirmation Modal: Deletion requires explicit confirmation via modal alert dialog to prevent accidental catalog damage.
- **BR-JCAT-24**: Immediate List Synchronization: Purges the category from default active views immediately upon successful soft deletion.

**Normal Flow**

1. The administrator clicks the Delete icon on a category table row or detail screen.
2. The deletion confirmation dialog appears, displaying job usage warning if applicable.
3. The administrator reviews the warning and confirms by clicking “Delete”.
4. The client issues `DELETE /api/admin/job-categories/:id`.
5. The backend validates administrative authentication and checks for active linked jobs.

6. Since active job count is 0, the repository updates `deletedAt = new Date()`.
7. The backend returns HTTP 200 with `{ message: "Success.JobCategoryDeleted" }`.
8. The frontend displays a success toast, closes the dialog, and refreshes the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Category has active linked jobs | The service throws `JobCategoryHasJobsException` (HTTP 409 Conflict). The dialog renders an inline callout advising switching to Inactive. |
| Target category does not exist | The service throws `JobCategoryAdminNotFoundException` (HTTP 404). The UI displays: "Error.JobCategoryNotFound". |
| Category is already soft-deleted | The service throws HTTP 404. The UI displays that the category is no longer active. |
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The server throws HTTP 403 Forbidden. The client renders an access denied alert. |

| Abnorma Case | Handling & System Response |
|---|---|
| Invalid category ID format | `ParseIntPipe` throws HTTP 400 Bad Request. The UI notifies the user of an invalid parameter. |
| Administrator cancels dialog | The modal closes immediately; no request is sent, leaving the database unaffected. |
| Database update failure | The service throws `FailedToDeleteJobCategoryException` (HTTP 500). The UI displays: "Error.FailedToDeleteJobCategory". |

**Output Data**

| Output Field | Description |
|---|---|
| message | Success confirmation message string (`Success.JobCategoryDeleted`). |

**Post-Conditions**

The category is marked as soft-deleted (`deletedAt != null`), hidden from default active listings, and can be restored if needed.

### 3.15.6 Restore Job Category

**Function Trigger**

An administrator viewing soft-deleted job categories (via the "Deleted" status filter or detail page) clicks the Restore icon (rotate-ccw), opening the restore confirmation dialog.
**Navigation path:** Admin Dashboard → Job Taxonomy → Job Categories → [Restore Action Icon].

**Function Description**

**Actors / Roles**

- **Primary Actor:** System Administrator (`ADMIN` role).
- **Supporting Systems:** Frevia Web management console (Next.js), Frevia REST API service (NestJS), Prisma ORM layer, and PostgreSQL relational database.

### Purpose

Allows administrators to restore a previously soft-deleted job category by resetting its `deletedAt` timestamp to null, making it active and selectable again across the platform.

### Interface

- Restore job category confirmation dialog (`RestoreJobCategoryDialog`) powered by Shadcn `AlertDialog`.
- Header displaying "Restore Job Category" title and target category name.
- Descriptive body explaining that restoring will make the category available again for job classification.
- Action buttons: "Cancel" and green "Restore" button with pending spinner.

### Data Processing Overview

- The administrator confirms restoration in the dialog.
- The client sends an HTTP request: `PATCH /api/admin/job-categories/:id/restore`.
- `JobCategoriesAdminController.restoreJobCategory(id)` parses `id` via `ParseIntPipe`.
- `JobCategoriesAdminRepository.restoreJobCategory(id)`:
  - Finds the category verifying that `deletedAt != null`; throws `JobCategoryAdminNotFoundExcept` if missing or already active.
  - Checks if another active category has been created with the same name during the deletion period (`deletedAt = null, id != currentId, name equals category.name`).
  - If a name collision occurs, throws `JobCategoryNameAlreadyExistsException` (HTTP 409).
  - Sets `deletedAt = null` via `prisma.jobCategory.updateMany({ where: { id, deletedAt: { not: null } }, data: { deletedAt: null } })`.
  - Queries the updated category entity with job count aggregation.
- The backend returns HTTP 200 with the restored category entity.
- The frontend shows a success toast notification, closes the dialog, and refreshes the view.

### Screen Layout (Mock-up Description)

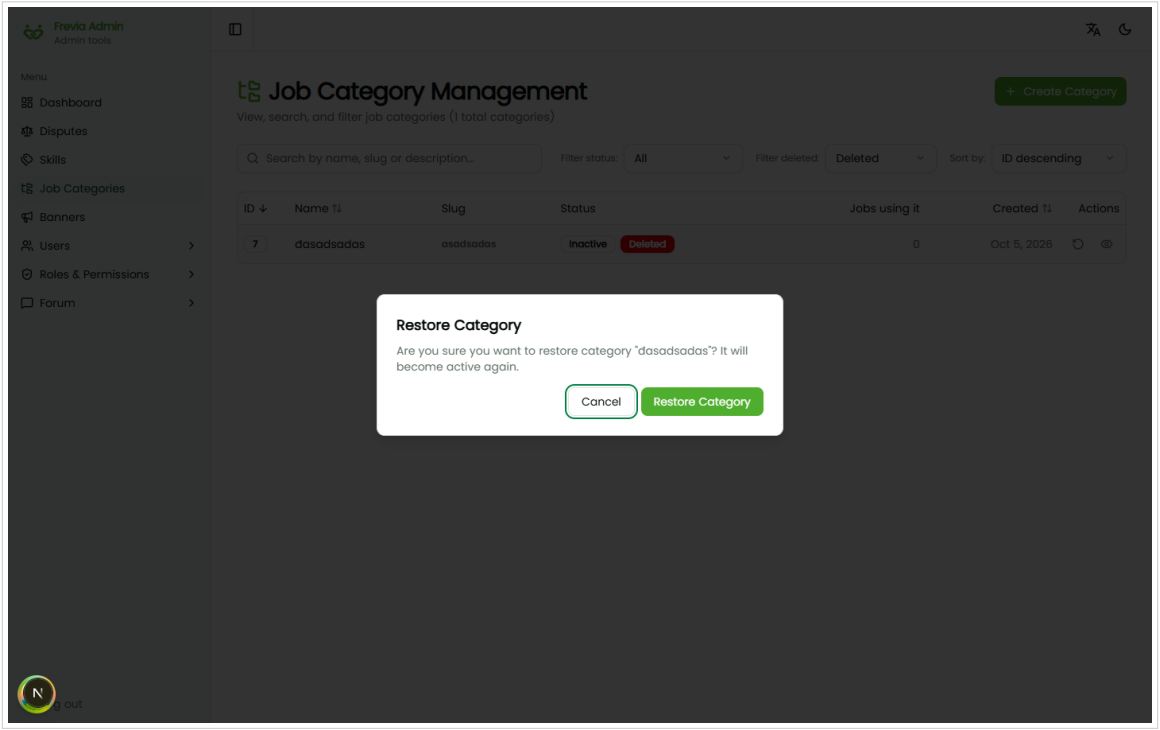


**Figure 131:** Restore job category confirmation dialog

- Alert dialog header displaying “Restore Job Category” title with rotate-counterclockwise icon.
- Confirmation description stating: “Are you sure you want to restore job category [Category Name]? It will become active and available for users again.”
- Footer controls: “Cancel” button and green “Restore” button with loading spinner when executing.

### Function Details

### Input Data

| Data Name | Description |
|---|---|
| id | Unique numeric identifier of the soft-deleted category (passed in URL path). |

**Validation Rules**

| Data Name | Validation Rule |
|---|---|
| id | Must be a positive integer greater than or equal to 1. |

**Business Rules**

- **BR-JCAT-25**: Restoration Eligibility Requirement: Target category must currently be soft-deleted with `deletedAt != null`; returns HTTP 404 if not found or already active.
- **BR-JCAT-26**: Conflict Prevention Against Interim Duplicates: If another active category with the same name was created while this category was deleted, restoration is blocked with HTTP 409 (`Error.JobCategoryNameAlreadyExists`).
- **BR-JCAT-27**: Soft-Deletion Timestamp Reset: Clears `deletedAt = null`, fully restoring the category to active catalog availability.
- **BR-JCAT-28**: Real-Time UI Re-indexing: Returns restored category data, displays success toast, and updates table view immediately.

**Normal Flow**

1. The administrator filters the catalog by "Deleted" job categories.
2. The administrator locates the soft-deleted category and clicks the Restore icon.
3. The restore confirmation dialog appears.
4. The administrator confirms restoration by clicking "Restore".
5. The client sends `PATCH /api/admin/job-categories/:id/restore`.
6. The backend verifies the category is soft-deleted, checks for name conflicts with active categories, and clears `deletedAt`.
7. The backend returns HTTP 200 with the restored category data.
8. The client displays a success toast, closes the dialog, and refreshes the table.

**Abnormal Flow / Exception Handling**

| Abnorma Case | Handling & System Response |
|---|---|
| Target category does not exist or is already active | The service throws `JobCategoryAdminNotFoundException` (HTTP 404). The UI displays: “Error.JobCategoryNotFound”. |
| Another active category now shares the same name | The service throws `JobCategoryNameAlreadyExistsException` (HTTP 409). The UI alerts: “Error.JobCategoryNameAlreadyExists”. |
| Missing or invalid authentication token | The server returns HTTP 401 Unauthorized. The client redirects to `/login`. |
| User lacks Admin privileges | The server throws HTTP 403 Forbidden. The client renders an access denied alert. |
| Invalid category ID format | `ParseIntPipe` throws HTTP 400 Bad Request. The UI notifies the user of an invalid parameter. |
| Administrator cancels dialog | The modal closes immediately; no request is sent, leaving the database unaffected. |

| Abnorma Case | Handling & System Response |
|---|---|
| Database update failure | The service throws `FailedToRestoreJobCategoryException` (HTTP 500). The UI displays: “Error.FailedToRestoreJobCategory”. |
| Network failure during request | The client catches the network error, displays an unexpected error toast, and preserves dialog state. |

**Output Data**

| Output Field | Description |
|---|---|
| id | Numeric identifier of the restored category. |
| name | Category name string. |
| slug | Preserved or updated unique slug. |
| description | Description text or null. |
| status | Operational status (`ACTIVE` or `INACTIVE`). |
| deletedAt | Null indicating active status. |
| createdAt | Timestamp of original creation. |
| updatedAt | Timestamp of restoration. |
| jobCount | Count of active jobs referencing this category. |
| jobs | Array of associated jobs. |

**Post-Conditions**

The job category is restored to active status (`deletedAt = null`), displayed in the active category catalog, and available for selection when posting and filtering jobs.

## 4. Non-Functional Requirements

This section describes the non-functional requirements of the Frevia platform: the external interfaces through which the system communicates, and the quality attributes that the delivered system must satisfy.

### 4.1 External Interfaces

The system communicates with four kinds of external element. Each interface is listed below with the constraint that the implementation must respect.

- **User interfaces.** The platform is delivered as a responsive web client that runs in a current version of Chrome, Edge, Firefox, or Safari. The layout must remain usable from a viewport width of 320 px upwards, and every screen must be available in English and Vietnamese.
- **Software interfaces.** Payments are processed through Stripe (Connect Express, PaymentIntent, Checkout Session, and signature-verified webhooks); profile, portfolio, and portfolio media are stored on Cloudinary; one-time verification and password reset codes are delivered over SMTP through Nodemailer; curriculum vitae analysis and identity document verification are performed by external AI services called over HTTPS; and all persistent data is stored in PostgreSQL and accessed through Prisma.
- **Communication interfaces.** Browser-to-server traffic uses HTTPS with TLS 1.3. Calls from the API to Stripe, Cloudinary, the mail server, and the external AI services are made server-side over HTTPS with an explicit timeout, and a failure of any of them must be reported to the caller as a handled error rather than as an unhandled exception. Real-time messaging between the web client and the API uses a Socket.IO connection authenticated with the same JWT as the REST endpoints.
- **Hardware interfaces.** No dedicated hardware is required beyond the device that runs the browser and the camera or file system used to supply identity documents, avatars, and portfolio media.

### 4.2 Quality Attributes

#### 4.2.1 Usability

- **Responsive Design:** The user interface must be fully responsive, supporting mobile devices, tablets, and desktop viewports (minimum width 320px).
- **Toast Notification System:** Interactive toast notifications (success, warning, error) must have the highest z-index to overlay all content, auto-dismiss after 3000ms, and must not freeze or block other user interactions.
- **Client-side Image Processing:** The avatar cropping tool must handle images locally before upload. The system must restrict upload sizes to under 5MB and validate formats (.jpg, .jpeg, .png, .webp) before initiating API requests.

### 4.2.2 Reliability

- **System Uptime:** The system core services must maintain a 99.9% monthly uptime, excluding scheduled maintenance.
- **Blockchain Fault Tolerance:** In case of RPC node connection failures, the platform must gracefully fall back to backup RPC endpoints within 3 seconds to avoid breaking the wallet balance display or transaction signing flows.
- **Escrow Reliability:** Funds locked in the Escrow Smart Contracts must be secure. Contracts must be verified on Etherscan/BscScan and use standard libraries (e.g., OpenZeppelin) to prevent vulnerabilities.

### 4.2.3 Performance

- **Response Time:** The system's API read operations must respond within 200ms under normal load (up to 500 concurrent requests). Write/update operations, excluding blockchain transactions, must complete within 500ms.
- **Page Load Time:** The web application must achieve a Largest Contentful Paint (LCP) of less than 2.5 seconds, adhering to Google's Core Web Vitals guidelines.
- **Concurrency:** The system backend must support at least 1,000 concurrent active users and handle a database connection pool efficiently.
- **Search and Filtering Efficiency:** The job search page must employ debouncing (300ms delay) and pagination (default 10 items per page) to optimize query performance and reduce server load.

### 4.2.4 Security & Privacy

- **Authentication and Authorization:** All network communications must be encrypted using HTTPS/TLS 1.3. User passwords must be hashed using bcrypt with a work factor of 12.
- **Session Management:** JWT Access Tokens must have a short lifespan (e.g., 15 minutes) and be stored in memory. Refresh Tokens must have a longer lifespan (e.g., 7 days), be stored in HTTP-only, Secure, and SameSite=Strict cookies, and be invalidated in the database upon user logout.
- **Role-Based Access Control (RBAC):** The system must strictly enforce role boundaries (Guest, Freelancer, Client, Admin). Administrative features, such as changing smart contract addresses or viewing audit logs, must be blocked by a server-side gateway for non-admin accounts.

- **Web3 Wallet and Transaction Safety:** Input wallet addresses must be verified on the client and server side to conform to the standard EVM format (starts with `0x` followed by 40 hexadecimal characters).

## 5. Requirement Appendix

This appendix collects the requirements that apply across features rather than to a single screen: the business rules that every use case must respect, the common requirements that hold for the whole system, and the standard application messages that the interface reuses.

### 5.1 Business Rules

The rules below are grouped by the feature they constrain. A use case that violates a rule is rejected, and the identifier of the rule is quoted in the corresponding alternative flow.

**Table 226:** Business Rules

| ID | Rule Definition |
|---|---|
| BR-PROF-01 | Both basic details (from profiles) and professional details (from freelancer_profiles) must be presented together. |
| BR-PROF-02 | Hidden or private fields should not be displayed to unauthorized users (e.g., clients vs. the profile owner). |
| BR-PROF-03 | Freelancers can only update their own profile. |
| BR-PROF-04 | Mandatory fields (e.g., display_name, title) cannot be left empty. |
| BR-SKILL-01 | All skills belonging to the freelancer must be displayed along with their proficiency level. |
| BR-SKILL-02 | A freelancer cannot add duplicate skills. |
| BR-SKILL-03 | Both skill name and proficiency level are required. |
| BR-SKILL-04 | Skill deletion must be a hard delete operation; the record is permanently removed from the mapping table. |
| BR-SKILL-05 | Only the owning freelancer or an administrator can delete a skill from the profile. |
| BR-PORT-01 | Only portfolios with deleted_at = NULL shall be displayed. |
| BR-PORT-02 | Portfolio title is a required field. |

| ID | Rule Definition |
|---|---|
| BR-PORT-03 | Media uploads must conform to system-defined size and type restrictions. |
| BR-PORT-04 | Only portfolios with deleted_at = NULL can be accessed. |
| BR-PORT-05 | Only the portfolio owner can edit it. |
| BR-PORT-06 | Portfolio title remains a required field. |
| BR-PORT-07 | Portfolio deletion must be a soft delete (updating the deleted_at column). |
| BR-PORT-08 | Soft-deleted portfolios must not be visible to clients or guests. |
| BR-FORUM-01 | Only categories with is_active = TRUE shall be displayed. |
| BR-FORUM-02 | Category information displayed includes name and description. |
| BR-FORUM-03 | Only categories with is_active = TRUE shall be accessible. |
| BR-FORUM-04 | Category information displayed includes name and description. |
| BR-FORUM-05 | Only posts with deleted_at = NULL shall be displayed. |
| BR-FORUM-06 | Posts with is_pinned = TRUE shall appear before regular posts. |
| BR-FORUM-07 | Posts shall be sorted by created_at in descending order within each group. |
| BR-FORUM-08 | The system shall support pagination for post listings. |
| BR-FORUM-09 | Only posts with deleted_at = NULL shall be displayed. |
| BR-FORUM-10 | Post details shall include title, content, author, category, and created_at information. |
| BR-FORUM-11 | Only comments with deleted_at = NULL shall be displayed. |
| BR-FORUM-12 | Comment information shall include content, author, and created_at. |
| BR-FORUM-13 | Only comments with deleted_at = NULL shall be displayed. |
| BR-FORUM-14 | Comment details shall include content, author, and created_at information. |
| BR-FORUM-15 | Only authenticated users can create posts. |

| ID | Rule Definition |
|---|---|
| BR-FORUM-16 | Every post must belong to exactly one category. |
| BR-FORUM-17 | Title and content are required fields. |
| BR-FORUM-18 | Only the author of a post can edit that post. |
| BR-FORUM-19 | Every post must belong to a valid category. |
| BR-FORUM-20 | Title and content are required fields. |
| BR-FORUM-21 | Only the author of a post can delete that post. |
| BR-FORUM-22 | Post deletion shall be implemented as a soft delete using the deleted_at field. |
| BR-FORUM-23 | Soft-deleted posts shall not appear in public forum listings. |
| BR-FORUM-24 | Only authenticated users can like posts. |
| BR-FORUM-25 | A user can have only one active like reaction per post. |
| BR-FORUM-26 | The reaction shall be stored in the forum_likes table. |
| BR-FORUM-27 | Only authenticated users can remove post reactions. |
| BR-FORUM-28 | A like reaction can only be removed if a corresponding like record exists. |
| BR-FORUM-29 | Removing a reaction shall delete the corresponding record from the forum_likes table. |
| BR-FORUM-30 | Only authenticated users can create comments. |
| BR-FORUM-31 | Comment content cannot be empty. |
| BR-FORUM-32 | Comment content must contain at least 2 characters. |
| BR-FORUM-33 | Each comment shall be associated with a valid post_id and user_id. |
| BR-FORUM-34 | Only the owner of a comment can edit it. |
| BR-FORUM-35 | Comment content cannot be empty. |
| BR-FORUM-36 | Comment content must contain at least 2 characters. |

| ID | Rule Definition |
|---|---|
| BR-FORUM-37 | The system shall record the updated_at timestamp when a comment is modified. |
| BR-FORUM-38 | Only the owner of a comment can delete it. |
| BR-FORUM-39 | Comment deletion shall be implemented as a soft delete using the deleted_at field. |
| BR-FORUM-40 | Soft-deleted comments shall not be displayed in normal comment listings. |
| BR-FORUM-41 | Only authenticated users can like comments. |
| BR-FORUM-42 | A user can have only one active like reaction per comment. |
| BR-FORUM-43 | The reaction shall be stored in the forum_likes table using comment_id. |
| BR-FORUM-44 | Only authenticated users can remove comment reactions. |
| BR-FORUM-45 | A like reaction can only be removed if a corresponding like record exists. |
| BR-FORUM-46 | Removing a reaction shall delete the corresponding record from the forum_likes table using comment_id. |
| BR-FORUM-47 | Only authenticated users can submit reports. |
| BR-FORUM-48 | Reports must be associated with either a post_id or comment_id. |
| BR-FORUM-49 | New reports shall be created with status = PENDING. |
| BR-FORUM-50 | A user may submit only one active report for the same content item. |
| BR-JOB-01 | Only jobs with status = 'open' or 'in_progress' shall be displayed to freelancers and guests. |
| BR-JOB-02 | Clients shall see all their jobs regardless of status. |
| BR-JOB-03 | Default sorting shall be by created_at in descending order. |
| BR-JOB-04 | The system shall support pagination with default page size of 10. |
| BR-JOB-05 | Search shall match keywords in title and description fields (case-insensitive). |

| ID | Rule Definition |
|---|---|
| BR-JOB-06 | When multiple filters are applied, all conditions shall be combined using AND logic. |
| BR-JOB-07 | Job information displayed includes: title, description, budget, deadline, category, required skills, client name, created date, status. |
| BR-JOB-08 | If the job is inactive (status = 'closed' or 'deleted'), only the job owner can view full details. |
| BR-JOB-09 | Job title is required and must not exceed 200 characters. |
| BR-JOB-10 | Job description is required and must not exceed 5000 characters. |
| BR-JOB-11 | Budget must be a positive number (greater than 0). |
| BR-JOB-12 | Category must be selected from predefined categories. |
| BR-JOB-13 | New jobs shall have status = 'draft' by default when saved as draft, or 'open' when published. |
| BR-JOB-14 | Only the job owner can edit the job. |
| BR-JOB-15 | Editing is restricted if the job has an active contract. |
| BR-JOB-16 | All validation rules from UC-06.2 apply to edits. |
| BR-JOB-17 | Edit actions shall be logged for auditing purposes. |
| BR-JOB-18 | Only the job owner can delete the job. |
| BR-JOB-19 | Deletion should be soft delete (set deleted_at timestamp, status = 'deleted'). |
| BR-JOB-20 | Jobs with active proposals or contracts cannot be deleted. |
| BR-JOB-21 | Delete actions shall be logged for auditing purposes. |
| BR-JOB-22 | Job status workflow: draft $\rightarrow$ open $\rightarrow$ in_progress $\rightarrow$ completed $\rightarrow$ closed. |

| ID | Rule Definition |
|---|---|
| BR-JOB-23 | Allowed transitions: draft → open (client); open → in_progress (client, when contract created); open → closed (client, if cancelled); in_progress → completed (client, when work done); in_progress → closed (client, if cancelled); completed → closed (system/auto); Any status → deleted (soft delete). |
| BR-JOB-24 | Only the job owner can manually change status. |
| BR-JOB-25 | Status changes shall be logged for auditing purposes. |
| BR-JOB-26 | Notifications shall be sent to freelancers when status changes from draft to open. |
| BR-JOB-27 | Only the freelancer can view their own bookmarks. |
| BR-JOB-28 | Bookmarks shall be sorted by created_at in descending order by default. |
| BR-JOB-29 | Displayed information includes: job title, budget, client name, bookmark date. |
| BR-JOB-30 | The system shall support pagination with default page size of 10. |
| BR-JOB-31 | Only freelancers can bookmark jobs. |
| BR-JOB-32 | Bookmark is a toggle action. Clicking the button toggles between adding and removing. |
| BR-JOB-33 | If the bookmark does not exist, the system creates it and changes the button to "Bookmarked". |
| BR-JOB-34 | If the bookmark exists, the system deletes it and changes the button to "Bookmark". |
| BR-JOB-35 | Bookmarks can only be created for active jobs (status = 'open' or 'in_progress'). |
| BR-JOB-36 | Only the freelancer can remove their own bookmarks. |
| BR-JOB-37 | The system must check if the bookmark exists before deletion. |
| BR-JOB-38 | If the bookmark does not exist, the system shall return a 404 Not Found error. |
| BR-JOB-39 | Removing a bookmark is a hard delete from the database. |

### 5.2 Common Requirements

The following requirements apply to every feature of the platform and are therefore not repeated in each use case.

- **Authorisation.** Every endpoint is protected by the server-side permission guard, which checks the method, the path, and the module against the permissions of the caller's role. A hidden button or a disabled menu item is never treated as an access control.
- **Pagination.** Every list endpoint is paginated, with a default page size of 10 items and an explicit page parameter, so that no screen loads an unbounded result set.
- **Deletion.** Records that other records refer to are soft-deleted by setting `deleted_at`; a soft-deleted record is excluded from every public listing and from every lookup used by a normal user. Hard deletion is limited to join records such as skills, bookmarks, and reactions.
- **Timestamps.** Creation and modification times are stored in UTC and displayed in the time zone of the browser.
- **Money.** All amounts are stored in the smallest currency unit and displayed with two decimals and an explicit currency; a financial record is never updated in place but is corrected by a new ledger entry.
- **File uploads.** Every uploaded file is validated for type and size on the client before the request is sent and on the server before the file is stored.
- **Language.** Every label, message, and validation error is read from the message catalogue, so that the same screen can be rendered in English and in Vietnamese without a code change.

### 5.3 Application Messages List

The messages below are reused across features. The context column states when the message is raised, and the content column gives the English source text that the message catalogue translates.

| # | Message code | Message Type | Context | Content |
|---|---|---|---|---|
| 1 | MSG01 | Toast, red | The API cannot be reached or the network is down | Cannot connect to the server. Please try again later. |

| # | Message code | Message Type | Context | Content |
|---|---|---|---|---|
| 2 | MSG02 | Toast, red | The server returns an unexpected error | Unable to process your request. Please try again later. |
| 3 | MSG03 | Toast, red | The caller requests a route outside the permissions of its role | You do not have permission to access this page. |
| 4 | MSG04 | Inline, under the field | A required field is left empty | All fields are required. |
| 5 | MSG05 | Inline, under the field | The email does not match a valid format | Please enter a valid email address. |
| 6 | MSG06 | Inline, under the field | The password does not meet the password policy | Password must be at least 8 characters and include at least one uppercase letter, one number, and one special character. |
| 7 | MSG07 | Inline, under the field | The password confirmation differs from the password | Passwords do not match. |
| 8 | MSG08 | Toast, red | A requested record does not exist or has been soft-deleted | Post not found. / Comment not found. / Proposal not found. |
| 9 | MSG09 | Toast, green | A record has been saved | Profile updated successfully. / Skill added successfully. / Portfolio created successfully. |

**Table 227:** Application Messages List